

## **ANNEX XV RESTRICTION REPORT**

### **PROPOSAL FOR A RESTRICTION**

**SUBSTANCE NAME(S): Octocrilene**

**EC NUMBER(S): 228-250-8**

**CAS NUMBER(S): 6197-30-4**

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## ABOUT THIS REPORT

This report consists of a proposed restriction prepared in accordance with the requirements laid down in Annex XV of REACH Regulation.

It consists of a summary of the proposal, a main report setting out the key evidence justifying the proposed restriction and Appendices with more detailed information and supporting analysis.

This report has been reviewed for confidential information and any such information has been redacted and included in a confidential appendix.

## CHANGE VERSION HISTORY

*The version of the Annex XV report submitted to ECHA should be version 1.0. In case there are editorial/clerical updates following submission, these can be tracked under version 1.1. It is not foreseen that any subsequent versions will be created before the report becomes the Background Document.*

| Version | Change history                                                 | Date         |
|---------|----------------------------------------------------------------|--------------|
| 1.0     | Annex XV restriction report and its appendix submitted to ECHA | 18 JULY 2025 |

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## LIST OF ACRONYMS AND ABBREVIATIONS

| <b>Acronym/<br/>Abbreviation</b> | <b>Meaning</b>                                                                                                                                                             |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AF                               | Assessment Factor                                                                                                                                                          |
| ANSM                             | French National Agency for the Safety of Medicines and Health Products                                                                                                     |
| AOP                              | Adverse Outcome Pathway                                                                                                                                                    |
| ARN                              | Assessment of Regulatory Needs                                                                                                                                             |
| B                                | Bioaccumulation                                                                                                                                                            |
| BAF                              | BioAccumulation Factor                                                                                                                                                     |
| BALF                             | Bronchoalveolar Lavage Fluid                                                                                                                                               |
| BAT                              | Best Available Techniques                                                                                                                                                  |
| BCF                              | BioConcentration Factor                                                                                                                                                    |
| BMDBM                            | Butyl Methoxydibenzoylmethane                                                                                                                                              |
| BMF                              | BioMagnification Factor                                                                                                                                                    |
| BROD                             | Benzyloxyresorufin-O-debenzylase                                                                                                                                           |
| BW                               | Body Weight                                                                                                                                                                |
| CAS                              | Chemical Abstracts Service                                                                                                                                                 |
| CBA                              | Cost-benefit analysis                                                                                                                                                      |
| CCH                              | Compliance Check                                                                                                                                                           |
| CfE                              | Call for Evidence                                                                                                                                                          |
| CLH                              | Harmonised classification and labelling according to the Classification, Labelling and Packaging Regulation (EC) No 1272/2008                                              |
| CLP                              | Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging (CLP) of substances and mixtures |
| CMR                              | Carcinogenic, Mutagenic and Toxic for Reproduction                                                                                                                         |
| CoRAP                            | Community Rolling Action Plan                                                                                                                                              |
| CSR                              | Chemical Safety Report                                                                                                                                                     |
| CV                               | Contingent Valuation                                                                                                                                                       |
| DAD                              | Diode Array Detection                                                                                                                                                      |
| DCE                              | Discrete Choice Experiment                                                                                                                                                 |
| decaBDE                          | Bis(pentabromophenyl) ether                                                                                                                                                |
| DNT                              | Developmental Neurotoxicity                                                                                                                                                |
| DS                               | Dossier Submitter                                                                                                                                                          |
| DT50                             | Disappearance Time                                                                                                                                                         |

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| <b>Acronym/<br/>Abbreviation</b> | <b>Meaning</b>                                                                                                        |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| DU                               | Downstream User                                                                                                       |
| dw                               | Dry Weight                                                                                                            |
| EAS                              | Estrogen, Androgen, Steroidogenesis                                                                                   |
| EC                               | European Commission                                                                                                   |
| EC10                             | Effect Concentration                                                                                                  |
| ECHA                             | European Chemicals Agency                                                                                             |
| ED                               | Endocrine Disruption                                                                                                  |
| EEA                              | European Economic Area, i.e. EU countries + Norway, Liechtenstein, and Iceland                                        |
| EEC                              | European Economic Community                                                                                           |
| EFSA                             | European Food Safety Authority                                                                                        |
| EiF                              | Entry into Force                                                                                                      |
| eMSCA                            | Evaluating Member State Competent Authority                                                                           |
| eNOS                             | endothelial Nitrogen Oxide Synthase                                                                                   |
| EOGRTS                           | Extended One Generation Reproductive Toxicity Study                                                                   |
| EQS                              | Environmental Quality Standards                                                                                       |
| ERC                              | Environmental Release Category                                                                                        |
| ES                               | Emission Scenario                                                                                                     |
| ESI                              | Electrospray Ionisation                                                                                               |
| EU                               | European Union. In this report, the terms 'European Union', 'EU' and 'EU-wide' cover the EEA unless stated otherwise. |
| FCM                              | Food Contact Material                                                                                                 |
| FDA                              | Food and Drug Administration                                                                                          |
| GC                               | Gaseous Chromatography                                                                                                |
| HPLC                             | High-Pressure Liquid Chromatography                                                                                   |
| hs-CRP                           | High Sensitivity C Reactive protein                                                                                   |
| IED                              | Industrial Emission Directive                                                                                         |
| INCI                             | International Nomenclature of Cosmetic Ingredients                                                                    |
| KE                               | Key Event                                                                                                             |
| kg                               | Kilogram                                                                                                              |
| K <sub>oc</sub>                  | Organic carbon-water partition coefficient                                                                            |
| K <sub>ow</sub>                  | Octanol-water partition coefficient                                                                                   |
| LAGDA                            | Larval Amphibian Growth and Development Assay                                                                         |

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| <b>Acronym/<br/>Abbreviation</b> | <b>Meaning</b>                                         |
|----------------------------------|--------------------------------------------------------|
| LOD                              | Limit of Detection                                     |
| LOEC                             | Lowest Observed Effect Concentration                   |
| LOQ                              | Limit of Quantification                                |
| MAP                              | Mediterranean Action Plan                              |
| MCCP                             | Medium-chain chlorinated paraffins                     |
| MDA                              | Malondialdehyde                                        |
| MIE                              | Molecular Initiating Event                             |
| mg                               | Milligram                                              |
| MoA                              | Mode of Action                                         |
| MS                               | Mass Spectrometry                                      |
| MSC                              | Member State Committee                                 |
| MSFD                             | Marine Strategy Framework Directive                    |
| NIS                              | Sodium-iodide Symporter                                |
| NO                               | Nitrogen Oxide                                         |
| NOAEL                            | No Observed Adverse Effect Level                       |
| NOEC                             | No Observed Effect Concentration                       |
| OECD                             | Organisation for Economic Co-operation and Development |
| ORs                              | Outermost Regions                                      |
| PAHs                             | Polycyclic Aromatic Hydrocarbons                       |
| PBT                              | Persistent, Bioaccumulable and Toxic                   |
| PEC                              | Predicted Environmental Concentration                  |
| PLE                              | Pressurized Liquid Extraction                          |
| PNEC                             | Predicted No Effect Concentration                      |
| PROD                             | Pentoxoresorufin-O-depentylase                         |
| QSAR                             | Quantitative Structure-Activity Relationship           |
| OCs                              | Operational Conditions                                 |
| OCTs                             | Overseas Countries and Territories                     |
| OECD                             | Organisation for Economic Co-operation and Development |
| P                                | Persistency                                            |
| PFAS                             | Per- and Polyfluoroalkyl substances                    |
| PFOA                             | Perfluorooctanoic acid                                 |
| PPE                              | Personal Protective Equipment                          |
| PPP                              | Purchasing Power Parity                                |

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| <b>Acronym/<br/>Abbreviation</b> | <b>Meaning</b>                                                                                                                                                                             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PV                               | Present Value                                                                                                                                                                              |
| PVC                              | Polyvinylchloride                                                                                                                                                                          |
| QpQ                              | Quadrupole analyser                                                                                                                                                                        |
| QuEChERS                         | Quick, Easy, Cheap, Effective, Rugged and Safe                                                                                                                                             |
| RAC                              | Risk Assessment Committee                                                                                                                                                                  |
| RCR                              | Risk Characterisation Ratios                                                                                                                                                               |
| REACH                            | Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) |
| RMC                              | Raw Material Cost                                                                                                                                                                          |
| RMM                              | Risk Management Measure                                                                                                                                                                    |
| RMO                              | Risk Management Option                                                                                                                                                                     |
| RMOA                             | Regulatory Management Option Analysis                                                                                                                                                      |
| RNS                              | Reactive Nitrogen Species                                                                                                                                                                  |
| RO                               | Restriction Option                                                                                                                                                                         |
| RoHS                             | Restriction of Hazardous Substances                                                                                                                                                        |
| ROS                              | Reactive Oxygen Species                                                                                                                                                                    |
| RP-HPLC                          | Reversed-Phase High-Pressure Liquid Chromatography                                                                                                                                         |
| SCCS                             | Scientific Committee for Consumer Safety                                                                                                                                                   |
| SDS                              | Safety Data Sheet                                                                                                                                                                          |
| SEAC                             | Committee for Socio-economic Analysis                                                                                                                                                      |
| SEv                              | Substance Evaluation                                                                                                                                                                       |
| SLPs                             | Solid-lipid Particles                                                                                                                                                                      |
| SME                              | Small and Medium-sized Enterprises                                                                                                                                                         |
| SPE                              | Solid-phase Extraction                                                                                                                                                                     |
| SPF                              | Sun Protection Factor                                                                                                                                                                      |
| SPERC                            | Specific Environmental Release Categories                                                                                                                                                  |
| SPME                             | Solid-phase microextraction                                                                                                                                                                |
| STP                              | Sewage Treatment Plant                                                                                                                                                                     |
| SVHC                             | Substance of Very High Concern                                                                                                                                                             |
| TEU                              | Treaty on European Union                                                                                                                                                                   |
| TFEU                             | Treaty on the Functioning of the European Union                                                                                                                                            |
| TiO <sub>2</sub>                 | Titanium dioxide                                                                                                                                                                           |

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| <b>Acronym/<br/>Abbreviation</b> | <b>Meaning</b>                                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------|
| T                                | Toxic                                                                                               |
| TP                               | Transition Period                                                                                   |
| TPA                              | Tonnage per Year                                                                                    |
| TPE                              | Testing Proposal Evaluation                                                                         |
| TPO                              | Thyroid Peroxidase                                                                                  |
| TSH                              | Thyroid Stimulating Hormone                                                                         |
| UAE                              | Ultrasound Assisted Extraction                                                                      |
| UNEP                             | United Nations Environment Programme                                                                |
| UPLC                             | Ultra Performance Liquid Chromatography                                                             |
| UV                               | Ultraviolet                                                                                         |
| UVCB                             | Substances of Unknown or Variable composition, Complex reaction products or of Biological materials |
| vB                               | Very Bioaccumulative                                                                                |
| vP                               | Very Persistent                                                                                     |
| vPvB                             | very Persistent and very Bioaccumulative                                                            |
| WFD                              | Water Framework Directive                                                                           |
| WHO                              | World Health Organisation                                                                           |
| WTP                              | Willingness To Pay                                                                                  |
| ww                               | Wet Weight                                                                                          |
| WWTP                             | Waste Water Treatment Plant                                                                         |
| WWTWs                            | Waste Water Treatment Works                                                                         |
| ZnO                              | Zinc Oxide                                                                                          |

## Executive summary

### Introduction

This Annex XV report addresses the risks to the environment of the use of octocrilene in finished cosmetic products and provides an assessment of the effectiveness, practicability, monitorability and socio-economic impacts of two restriction options (ROs) under REACH as the most suitable risk management option (RMO) to address the identified risks.

Octocrilene is a substance widely used in the European Union (EU) as well as in the rest of the world, in a broad range of applications especially for its UV filter properties. Octocrilene is used in the formulation of finished cosmetic products, laboratory chemicals, pharmaceuticals and photochemicals, sealants, adhesives, coatings, paints, thinners and paint removers. Octocrilene is also used in printing and recorded media reproduction and formulation of mixtures and/or re-packaging for the manufacturing of plastics. The substance is used as substance on its own and/or as constituent in mixtures and (complex) articles for consumer, professional, and industrial uses.

Octocrilene absorbs UV radiation and is therefore used as a UV (mainly UVB) filter in finished cosmetic products offering sun protection. Octocrilene also acts as a general photo-stabilizer, allowing to protect the formula as a whole and specifically ingredients being less photo-stable. Besides, octocrilene is used as a solvent for other crystalline and oil soluble UV filters. Within its cosmetic uses, octocrilene can be used as a UV-B filter at a concentration up to 10% (as acid).

### Concern

Octocrilene enters the environment via emissions during manufacture, the use phase, and the waste stage. As demonstrated from the Dossier Submitter data assessment in this annex XV dossier, **the main driver of the environmental exposure and following risks is the wide dispersive use of octocrilene contained in finished cosmetic products.** The annual use-volumes of octocrilene used in finished cosmetic products is estimated to be of 1752 ton/y. Risks were demonstrated for the environment by applying a tonnage approach or a consumer approach. The Dossier Submitter also demonstrated that risk can occur from the manufacture of the substance when non specific emission scenarios are applied, and from the formulation of plastisol for one registrant when non stringent risk management measures (RMM) are applied. However, these risks can be managed when those measures are applied. As demonstrated during this risk assessment, the main driver for the identified risks is the wide dispersive use of finished cosmetic products. Thus, this is the use taken into account by the Dossier Submitter in the proposed restriction.

Octocrilene is hazardous to several species, among which aquatic species were demonstrated as being the most sensitive (invertebrates), with a no observed effect concentration (NOEC) of 2.66 µg/L on *Daphnia magna*. Hazard data allowed the Dossier Submitter to estimate the risks arising from the use of octocrilene, and it was demonstrated that risk occurred for fresh and marine water, associated sediments and agricultural soil. Moreover, octocrilene meets the classification criteria set out in annex XIII of REACH regulation for persistency and very persistency (vP) and toxicity (T) for the environment but without meeting the classification criteria for bioaccumulation (B). Octocrilene is hazardous for aquatic species (and associated sediments) and soil organisms, and risks for the environment have been identified from the wide dispersive use of finished cosmetic products. An action on a Union-wide basis is warranted to effectively reduce the environmental exposure to octocrilene in the EU, especially as octocrilene is used in dispersive applications, is persistent, can accumulate in the environment and exert adverse effects. An EU-wide action would also limit the potential

for trans-boundary exposure from EU sources, and would avoid trade and competition distortions, thereby ensuring a level playing field in the internal EU market as compared to action undertaken by individual Member States.

### Regulatory risk management options

The risks for the environment identified from the wide dispersive use of finished cosmetic products make it necessary to reduce emissions of octocrilene from these uses to a minimum. Different regulatory risk management options have been considered (e.g., Industrial Emission Directive, RoHS, Biocidal Products Regulation, Product Safety Directive, Waste Directive, REACH authorisation...). A restriction can cover a wide range of uses and can address the risks arising from the manufacture and use of the substances as such as well as in mixtures, including imported mixtures from outside the EU. It was concluded that a REACH restriction would be the most appropriate EU-wide measure to address the risk for the environment arising from the wide dispersive use of octocrilene in finished cosmetic products.

The Dossier Submitter performed the impact assessment of two restriction options (ROs) to restrict the presence of octocrilene in finished cosmetic products with wide dispersive use. The ROs assessed are:

- **RO1:** Restriction on placing octocrilene on the market in consumer and professional finished cosmetic products (concentration limit of 0.001% w/w). A two-year transitional period is implemented after the enforcement of the proposed restriction.
- **RO2:** Restriction on placing octocrilene on the market in consumer and professional finished cosmetic products (concentration limit of 0.001% w/w). A five-year transitional period is implemented after the enforcement of the proposed restriction.

The assessment of the different ROs is underpinned by information on uses, releases, availability of alternatives and socio-economic impacts, and was supported by a call for evidence and multiple interviews with stakeholder associations and companies to explore the impacts of the various ROs on different sectors and gather data. The decision to consider two ROs with transition periods of different durations is motivated by the uncertainty surrounding the ability of industrial stakeholders to anticipate the changes required by the restriction after it has passed the conformity stage within the ECHA committees (RAC and SEAC; see section 3.1).

The Dossier Submitter concluded that both ROs would address the identified risk and would fulfil the REACH Annex XV criteria for a restriction in terms of effectiveness, practicability (including enforceability) and monitorability.

**Compared to the baseline, RO1 and RO2 are expected to reduce releases of octocrilene into the environment by about 13,300 tonnes (10,800 tonnes to soil and 2,500 tonnes to water) and 8,300 tonnes (6,800 tonnes to soil and 1,600 tonnes to water) respectively over the 10-year period considered for the impact assessment.**

The Dossier Submitter also identified uncertainties in the assessment and concluded that the individual and collective uncertainties do not impact the conclusions on the effectiveness of both ROs.

### Socio-economic analysis

The Dossier Submitter estimates the costs and the benefits of the proposed ROs. Thus, if we consider a transitional period of 2 years (RO1), the cost induced by the restriction amounts to 353.69 million euros for the entire study period (2027-2036; annualised cost of 39.08

million euros). Considering a transition period of 5 years (RO2) reduces the cost of the restriction by 53%: the cost induced by the restriction amounts to 165.74 million euros for the entire study period (annualised cost of 18.31 million euros). The majority of these costs correspond to the costs of reformulating and producing sun protection products using alternative substances.

In terms of market value, the annualized cost of RO1 represents 0.04% of total sales in value of cosmetic products in 2023 in Europe and 1.11% of total sales in value for the same year if we focus on sun protection products in Western Europe (0.02 and 0.58% for RO2).

The environmental benefits expected from the restriction arise from the reduction in potential risks associated with the use of octocrilene in cosmetic products for the water (freshwater, marine water, groundwater and sediments) and soil compartments. The Dossier Submitter estimated the benefits of the restriction in monetary terms based on a specific study eliciting the willingness to pay of European households (commissioned by the Dossier Submitter for the specific purpose of this restriction proposal) to avoid the negative impacts of octocrilene used in cosmetic products on marine and freshwater ecosystems. The benefits of the proposed restriction were estimated by considering citizens' willingness to pay (WTP) for improving the state of the aquatic environment, based on the discrete choice experiment (DCE) and contingent valuation (CV) methods. When considering WTP values derived from the CV, the average WTP for improving the state of coastal and freshwater ecosystems (EEA-wide reference value) amounts to approx. €57, €71 and €75 per household per year for a low, medium and high level of environmental improvement respectively. The benefits of the restriction calculated on that basis while transferring WTP values derived from the CV to the whole EEA population (considering 203 102 020 households in the EEA, Eurostat 2023 population data) amount to €49 billion, €39 billion and €26 billion for an 'immediate realisation', a 'moderate' and a 'significant lag' respectively (PV 2024 at 3%).

#### Conclusions on proportionality

The Dossier Submitter was able to discuss the proportionality of the restriction based on a cost-effectiveness and a cost-benefit analysis, as well as a break-even analysis.

Considering avoidance of all releases to water and soil from finished cosmetic products containing octocrilene, **the resulting cost-effectiveness (C/E) ratios amount to approx. €34 and €27 per kg of releases abated for RO1 and RO2 respectively.** When compared with reference values, corresponding to C/E ratios determined for recent proposed restrictions adopted by the European Commission, the C/E ratios established for the present restriction are among the lowest values. On this basis, both restriction options are considered proportionate. This conclusion is robust to the sensitivity analyses carried out, taking into account the costs of the restriction in a worst-case scenario, stricter assumptions in terms of calculating avoided discharges and specific C/E ratios for the different product families concerned (sunscreen products and cosmetic products 'excluding sunscreens').

Comparing benefits and costs within a CBA, it is clear that the benefits of the restriction significantly outweigh the costs. **WTP values derived from the CV at the household (and aggregate) level are well above the minimum WTP that would be required to obtain benefits equal to the costs generated by the restriction (i.e., break even).** Under RO1, a WTP equal or greater than €0.37 per household per year for the next five years would result in benefits greater than the cost for the proposed restriction. Under RO2, this minimum willingness to pay is €0.17 and €5.33 per year per household for the next five years respectively.

**Based on these results, both restriction options are considered proportionate.** In

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contrast with the results of the cost-effectiveness analysis, it should be noted that RO1 performs better than RO2 when considering the results of the CBA, although RO1 is associated with higher costs, these are largely compensated by an earlier materialisation of benefits (allowed by a shorter transition period than RO2). The conclusions of the CBA and the breakeven point analysis are also robust to the sensitivity analyses carried out, considering in particular the costs of the restriction in a worst-case scenario.

Based on these results, both restriction options are considered proportionate. **Besides, in light of the performance of RO1 - earlier risk reduction and good performance in terms of proportionality – the DS recommends implementing OR1 rather than OR2.**

**Proposed restriction - REACH Annex XVII entry (RO1, based on tonnage and risk reduction and performance regarding CBA).**

| Designation of the substance, of the group of substances or of the mixture | Conditions of restriction                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octocrilene                                                                | 1. Shall not be placed on the market, or used, in finished cosmetic products in a concentration equal to or greater than 0.001% w/w.<br><br>2. The restriction shall apply 24 months after its entry into force.<br><br>3. 'finished cosmetic product' is a product as defined by Regulation (EC) No 1223/2009. |
| CAS No 6197-30-4                                                           |                                                                                                                                                                                                                                                                                                                 |
| EC No 228-250-8                                                            |                                                                                                                                                                                                                                                                                                                 |

## Report

### 1. Problem identification

#### 1.1. Background of the restriction

Octocrilene is a substance widely used in the EU as well as in the rest of the world, in a broad range of applications especially for its UV filter properties.

Octocrilene was originally included in the Community rolling action plan (CoRAP) for substance evaluation (SEv) in 2012 pursuant to Article 44(2) of the REACH Regulation in order to clarify concerns about:

- Suspected Persistent, Bioaccumulable and Toxic and/or very persistent very bioaccumulable substance (PBT/vPvB),
- Wide dispersive use,
- High (aggregated) tonnage.

During the evaluation of the Substance, the evaluating Member State Competent Authority (eMSCA) identified additional concerns which were:

- Human health (potential thyroid toxicity, suspected toxicity to reproduction),
- Environment (suspected endocrine activity, exposure of the aquatic environment).

Following the evaluation of the Substance, the eMSCA concluded that further information was required to assess the additional concerns identified. Following discussions in the MSC meeting of 8 to 10 April 2014, a unanimous agreement of the MSC on the decision, as modified at the meeting, was reached on 10 April 2014.

The decision was appealed and the appeal dismissed in its entirety.

Data from the requested information were made available to eMSCA in July 2019. In the course of the SEv and after analysis of the previously requested information, the eMSCA, requested an additional test in March 2020, namely the Larval Amphibian Growth and Development Assay (LAGDA, test method: OECD TG 241, 2015) agreed in the MSC-78 meeting (June 2022) in order to clarify the potential concern related to Endocrine disruption of the substance for the environment. This data was requested to be submitted in June 2024 and are still awaited at the time of preparation of this restriction proposal.

In the meantime, the Dossier Submitter had identified concerns in the course of the Substance Evaluation performed under REACH. Those concerns were related to human health with identified hazards for reproduction, development and presenting endocrine disruptive properties. Moreover, human biomonitoring data available highlighted that population were exposed to octocrilene as it was largely retrieved in human biological samples, such as urine and breast milk (*Schlumpf et al., 2010, Gu et al., 2019, Ao et al., 2018, Bury et al., 2023*). For the environment, octocrilene presented hazards for several trophic species. In combination to hazards, the exposure scenario available in registration dossier were later analysed in a Regulatory Management Option Analysis (RMOA), which concluded that unacceptable risks for the environment exist. This conclusion was arising from the persistent and toxic properties of octocrilene, the bioaccumulation potential and the strong adsorption to organic matter, the potential endocrine disruption properties for the environment. The RMOA also detailed the fact that risks were identified from two specific uses, the formulation of plastisol and the wide dispersive use of cosmetic products. As the risks arising from the

formulation of plastisol can be managed rapidly and quite easily at the manufacturing site, as demonstrated by specific risk management measures (RMMs) put in place from several registrants where the formulation of plastisol does not lead to unacceptable risks for the environment, the Dossier Submitter has decided to focus on a more concerning exposure path, the wide dispersive use in cosmetic products. The most appropriate regulatory management option to control the identified risks was concluded to be a restriction of the substance.

## 1.2. Substance(s) identity, physical and chemical properties

### 1.2.1. Identity of the substance(s)

**Table 1: Identifiers of the substance.**

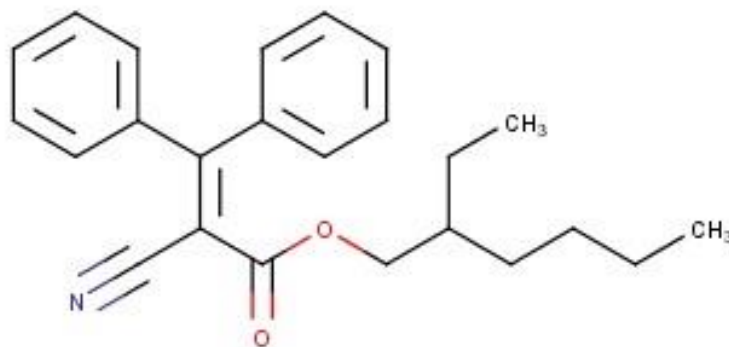
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EC number (public)</b>                             | 228-250-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>CAS number</b>                                     | 6197-30-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>IUPAC name (public)</b>                            | 2-ethylhexyl 2-cyano-3,3-diphenylacrylat<br>2-Ethylhexyl 2-cyano-3,3-diphenylprop-2-enoate<br>2-Ethylhexyl 2-cyano-3,3-diphenyl-2-propenoate<br>2-ethylhexyl 2-cyano-3,3-diphenyl-2-propanoate<br>2-cyano-3,3-diphenyl-acrylic acid 2-ethyl-hexyl ester<br>2-Ethylhexyl 2-cyano-3,3-diphenyl-acrylate<br>(RS)-2-Cyan-3,3-diphenylacrylsäure-2-ethylhexylester<br>2-Propenoic acid, 2-cyano-3,3-diphenyl-, 2-ethylhexyl ester<br>Propen-2-säure, 2-cyano-, 3,3-diphenyl-, 2-ethylhexylester<br>Octocrilene<br>octocrilene<br>Octocrylene |
| <b>INCI Name</b>                                      | Octocrilene                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Index number in Annex VI of the CLP Regulation</b> | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Molecular formula</b>                              | C <sub>24</sub> H <sub>27</sub> NO <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Molecular weight</b>                               | 361.47 g.mol <sup>-1</sup> [25°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Synonyms</b>                                       | Octocrilene                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Type of substance**

☒ Mono-constituent

☐ Multi-constituent

☐ UVCB



**Figure 1: Structural formula of octocrilene.**

### 1.2.2. Physicochemical properties

Table 2: Physico-chemical properties of octocrilene.

| Input                                                   | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reference                               |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Physical form</b>                                    | Yellow, clear, viscous liquid [20°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ECHA dissemination website <sup>1</sup> |
| <b>Melting/freezing point</b>                           | -10°C at 101325 Pa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ECHA dissemination website              |
| <b>Boiling point</b>                                    | 218°C (decomposition of substance before boiling, >300°C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ECHA dissemination website              |
| <b>Relative density</b>                                 | 1.05 [25°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ECHA dissemination website              |
| <b>Vapour Pressure</b>                                  | 3.15E-09 mmHg [25°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MPBPWIN v1.43                           |
| <b>Henry's law constant</b>                             | 3.1E-04 Pa.m <sup>3</sup> .mol <sup>-1</sup> [25°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | HENRYWIN v3.10                          |
| <b>Water solubility</b>                                 | <p>Water Solubility</p> <p>The water solubility of the <u>test item</u> is in the range of 9 - 153 µg/L (Value used for CSA = mean: 40 µg/L +/- 44 µg/L) at T = 20°C +/- 0,5°C.</p> <p><u>Solubility in M4 -water</u></p> <p>The water solubility of the test item in M4 -water is in the range of 21 - 82 µg/L (Value used for CSA = mean: 54 µg/L +/- 21 µg/L) at T = 20°C +/- 0,5°C.</p> <p><u>Solubility in AFSS-media</u></p> <p>The water solubility of the test item in AFSS-media is in the range of 8 - 42 µg/L (Value used for CSA = mean: 19 µg/L +/- 10 µg/L) at T = 20°C +/- 0,5°C.</p> | Unpublished study report (2018b)        |
| <b>Partition coefficient n-octanol/water (Log Kow)</b>  | 6.1 [23°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Unpublished study report (2018d)        |
| <b>Organic carbon/water partition coefficient (Koc)</b> | 48 898 L.kg-1 [20°C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Unpublished study report (2018a)        |
| <b>Flash point</b>                                      | 234°C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ECHA dissemination website              |

<sup>1</sup>[https://chem.echa.europa.eu/100.025.683/dossier-view/8819906f-f574-452a-bc54-2f0185ac3b21/IUC5-19cb9997-f604-48bc-920f-2d15398535a6\\_c67b4123-dfe4-494b-9a40-2d805e665ab0?searchText=octocrilene](https://chem.echa.europa.eu/100.025.683/dossier-view/8819906f-f574-452a-bc54-2f0185ac3b21/IUC5-19cb9997-f604-48bc-920f-2d15398535a6_c67b4123-dfe4-494b-9a40-2d805e665ab0?searchText=octocrilene)

### 1.3. Substance(s) manufacturing, trade and uses overview

#### 1.3.1. Types and sources of data considered

In addition to the information contained in the registration dossiers regarding octocrilene, the Dossier Submitter has gathered additional information on the uses of the substance in cosmetic products *via* a call for evidence and several hearings organised with various stakeholders of the octocrilene supply chain including suppliers, formulators and downstream users.

This section also relies significantly on the analysis of the following datasets:

- Two extensive datasets obtained from the YUKA and CosmEthics applications (YUKA, 2024; CosmEthics, 2023) providing a detailed overview of cosmetic products containing octocrilene at an aggregate level for several EEA countries<sup>2</sup>;
- A dataset extracted from the French National Database of Products and Compositions (BNPC, 2024). These data were cross-referenced with the abovementioned datasets with the same objective of identifying the types of cosmetic products containing octocrilene (e.g., sun protection products, shampoos, etc.). This database also provides information regarding the concentration levels of octocrilene in cosmetic products;
- A dataset collected by the Dossier Submitter itself via a market survey reporting the presence of octocrilene in primary sunscreens<sup>3</sup> in five EEA countries and for three different distribution channels (see Appendix A.2.1.2 for further details on the data collection method and results). These data were collected in particular for cross-referencing with the YUKA and CosmEthics datasets, but also in order to identify potential variability in the share of the supply of primary sunscreens containing octocrilene between EEA countries (see section 3.4.2.1 and Appendix E.4.3.5. regarding the impacts of this assumption);
- Data sets extracted from the Passport database<sup>4</sup> reporting octocrilene use-volumes at the EU-level, absolute and relative use-volumes of octocrilene for each cosmetic product category.<sup>5</sup>

The Table 3 below provides an overview of considered datasets and details their geographical scope as well as the product categories they cover.

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<sup>2</sup> These two databases provided the following information for each category (e.g. Facial care) and sub-category of cosmetic product (e.g. anti-age cream):

- number of product references in the database
- number of product references containing octocrilene
- number of product references not containing octocrilene

See Table 3 below for more details on the characteristics of these databases.

<sup>3</sup> 'Primary sunscreens' or 'sunscreen products' refer to 'any preparation (such as creams, oils, gels, sprays) intended to be placed in contact with the human skin with a view exclusively or mainly to protecting it from UV radiation by absorbing, scattering or reflecting radiation' (European Commission, 2006).

<sup>4</sup> <https://www.euromonitor.com/solutions/passport>

<sup>5</sup> These are the use volumes of octocrilene consumed in the EU through the consumption of cosmetic products containing it. It does not therefore cover the use volumes used by cosmetics manufacturers for formulating their products. On top of these datasets that fed into the description of the uses of the substance, the Dossier Submitter also made use of datasets reporting market size and market shares of the cosmetic industry, sunscreen sales volumes and describing distribution networks.

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**Table 3: Manufacture and uses: characteristics of data sources at the disposal of the Dossier Submitter.**

|                                                                   | CosmEthics                                                                            | YUKA                                                                           | French National Database of Products and Compositions <sup>6</sup>                            | Primary sunscreens Market survey                                                                                           | Passport – Euromonitor International |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Type</b>                                                       | Online database                                                                       | Online database                                                                | Online database                                                                               | Market survey                                                                                                              | Online database                      |
| <b>Extraction date</b>                                            | November 2023                                                                         | 15 March 2024                                                                  | 28 April 2024                                                                                 | 20 December 2023 – 10 February 2024                                                                                        | May 2024 – January 2025              |
| <b>Number of products in database/surveyed at extraction date</b> | 108,423                                                                               | Approximately 2 million                                                        | Approximately 165,000                                                                         | 4,412                                                                                                                      | Not available                        |
| <b>Market</b>                                                     | EU-wide, most widely used in Nordic countries and France                              | Belgium, Germany, France, Italy, Ireland, Luxembourg, Spain                    | France                                                                                        | Finland, France, Germany, Poland, Ireland<br>3 distribution channels: pharmacies, supermarkets and online specialty stores | European Union                       |
| <b>Product categories</b>                                         | Cosmetic and personal care products (product scanning hit rate of approximately 77 %) | Cosmetic and personal care products (minimum product scanning hit rate of 80%) | Cosmetic and personal care products associated with a case handled by a poison control centre | Primary sunscreens                                                                                                         | Cosmetic and personal care products  |
| <b>Period</b>                                                     | Since 2013                                                                            | Since 2017                                                                     | Since 2001                                                                                    | December 20 <sup>th</sup> 2023 - February 10 <sup>th</sup> 2024                                                            | Since 2009                           |

Further details on stakeholder consultation are available in Appendix G.

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<sup>6</sup> The French National Database of Products and Compositions lists more than 260,000 mixtures or substances that could be involved in a poisoning case, including 165,000 mixtures (medicines, cosmetic and personal care products, cleaning and industrial products, plant protection products, etc.), 95,000 substances and 1,500 natural toxins (from animals, plants, fungi). It provides the toxicological information that poison control centres and health authorities need to handle potential poisoning cases (ex: ingestion of a product by a young child).

### 1.3.2. Manufacturing, imports and exports

At least 5 manufacturing sites operate in the European Economic Area (EEA) and hundreds other formulate products with Octocrilene. ECHA disseminated data provides information that there are 9 active registrants (see below).

**Table 4: Active Registrants/suppliers.**

| Company Name / TPR name                                  | Submission date |
|----------------------------------------------------------|-----------------|
| BASF SE                                                  | 21/11/2023      |
| DSM Nutritional Products GmbH- OR2                       | 31/03/2011      |
| DSM Nutritional Products Nederland B.V.                  | 27/04/2020      |
| ERM SA Brussels (L'OREAL DEUTSCHLAND GmbH)               | 09/03/2023      |
| ERM SA Brussels (L'OREAL PRODUITS DE LUXE INTERNATIONAL) | 09/03/2023      |
| Merck S.L.U.                                             | 01/11/2023      |
| REACH24H CONSULTING GROUP                                | 22/01/2021      |
| Symrise AG                                               | 27/02/2020      |
| Wilkinson Sword GmbH                                     | 17/01/2024      |

Source: <https://chem.echa.europa.eu/100.025.683/dossier-list/reach/asset-owner?pageIndex=1&pageSize=100>, September 2024

At the time of preparation of this Annex XV report, there are four inactive registrants/suppliers.

This substance is registered under the REACH Regulation and is manufactured in and / or imported to the EEA, at  $\geq 1\ 000$  to  $< 10\ 000$  tonnes per year (t/y). The joint submission for octocrilene does not fully cover all the tonnage of the substance used in EEA as cosmetic formulation and service life is not covered by the lead registrant. As specific wide dispersive use of cosmetic is targeted by this restriction proposal, the determination of the tonnage dedicated to each use is of first importance. However, there is only rough estimation possible based on registration dossier as not all data are up to date in IUCLID and not all Chemical Safety Report (CSRs) have been updated. Moreover, discrepancies among registrations regarding the qualification of the type of uses and associated quantities do not allow the Dossier Submitter to perform a thorough consolidated estimation on quantities manufactured, imported, exported and used. When compiling the different CSRs available, the estimated annual production and importation of octocrilene in the EEA is comprised between 2500 to 4500 t/y with at least 70 t/y being directly exported. According to the data provided during the Call for Evidence (CfE), the estimated imported amount of octocrilene is even greater, around 5500 t/y (CfE#1758), on which a substantial amount is used for cosmetics.

Moreover, the Dossier Submitter highlights that the real imported tonnage may in reality be higher and not accounted for in CSRs, as tonnages imported below 1 tonne/year of substance per company are not registered.

As an example of these discrepancies, the last estimates of the Dossier Submitter led to an estimated manufactured and imported tonnage of octocrilene, based on CSRs data, comprised between 4100-4500 t/y. Among this tonnage, the **cosmetics uses** (formulation, industrial uses and then the wide dispersives uses resulting products) represents around 40-45% of this estimated tonnage, between **1700-2000 t/y**. Nevertheless, the declared tonnage from

the different exposure scenario for the formulation, industrial use and wide dispersive use (consumer and professionals) of the other uses are solely comprised between 200- 600 t/y (5-15%), and should have been much more, at least 2400 t/y. These discrepancies arise from the methodology applied in REACH CSRs for the risk assessment leading to apply (and declare) a specific maximum tonnage for a representative plant or site of production/use rather than the real amount used. Moreover, the dossiers do not compile the amount really used for every plant in EEA but it compiles solely the maximum possible tonnage per use rendering it difficult to properly assessing the real amount used and the corresponding risks (specificity of MSPERC value).

Due to the discrepancies between individual CSRs as well as between CSRs and CfE, the Dossier Submitter has decided to focus their estimates on the amount of octocrilene declared to be used for cosmetics products from CSRs and compared to the data retrieved from Euromonitor (Euromonitor, 2025) to ensure the robustness of the estimates. Euromonitor database allows to estimate the amount of octocrilene in cosmetic products really put in the market and sold in the European Union (the data available for the considered year of 2023 do not cover Cyprus, Iceland, Luxembourg, Malta, Norway and Liechtenstein). This approach is considered as being the most conservative and closed to the reality of the market. When focusing on quantities used for cosmetics products wide dispersive use, it can be estimated from the different CSRs that **the annual amount of octocrilene used in cosmetic products is of 1752 t/y**, estimation confirmed with the data from Euromonitor (1721 t/y). The Dossier Submitter is aware that this approach is uncommon as usually range estimates are performed and worst-case scenario provided in restriction proposals but the performed estimation is considered as the smallest realistic amount that can be confirmed of octocrilene used in cosmetics products despite the fact that the real tonnage used is certainly higher than the one estimated. This allows the Dossier Submitter to perform risk assessment in a conservative way.

### 1.3.3. Uses overview and technical functions

Octocrilene is manufactured and imported in EEA and is used in several processes in industrial sites, by professional workers and by consumers.

Regarding the uses at industrial sites and by professional workers, octocrilene is used in the formulation of cosmetics, laboratory chemicals, perfumes and fragrances, pharmaceuticals and photochemicals, in sealants, in adhesives, coatings, paints, thinners and paint removers. Octocrilene is also used in printing and recorded media reproduction and formulation of mixtures and/or re-packaging for the manufacturing of plastics. It is used also in the manufacture of rubber products.

Regarding the uses for consumers, octocrilene can be found in various products, especially cosmetics products, and to a very lesser extent in plastics products and soils covers. Octocrilene is used in the manufacture of these consumer products mostly for its UV absorber, UV filter and light stabilizer properties.

Octocrilene performs three main technical functions. Octocrilene absorbs UV radiation efficiently from 290 to 340 nm with a peak absorption at 307 nm (Rai et al., 2012). Octocrilene

is therefore used as a UV (mainly UVB) filter in cosmetic products offering sun protection<sup>7</sup>, i.e., in primary sunscreens as well as in secondary photo-protection products<sup>8</sup>. Octocrilene also acts as a general photo-stabilizer, allowing to protect the formula as a whole and specifically ingredients being less photo-stable. Octocrilene can be used for its photo-stabilizing properties in cosmetic products offering sun protection to stabilise other UV filters (such as Butyl Methoxydibenzoylmethane<sup>9</sup>), but also in other types of cosmetic products (CfE#1756; CfE#1757; CfE#1759; CfE#1760; Cosmed, 2024; L'Oréal, 2024). Finally, octocrilene is used as a solvent for other crystalline and oil soluble UV filters. Secondary functions have also been mentioned.

According to the SPIN Database<sup>10</sup>, the use categories for octocrilene in mixtures reported between 2018 and 2020, besides those for cosmetics products, are the following:

- surface treatment;
- paints, lacquers and varnishes;
- cleaning/washing agents.

Within its cosmetic uses, octocrilene can be used as a UV-B filter in sunscreen products at a concentration up to 10% (as acid) and is often used together with dibenzoylmethane derivatives to stabilise sunscreen products (SCCS, 2021).

Moreover, the PSC Ineris Database<sup>11</sup> and the CosIng –Cosmetic ingredient database<sup>12</sup> confirms that octocrilene is used as a UV filter in cosmetics (sunscreens) and make-up (nail varnish and concealers). It also indicates that octocrilene is an anti-UV for plastics, adhesives and coatings.

There are thus two principal uses for octocrilene: one as UV filter like in cosmetic products and the other as UV stabilisers like in plastisol for the production of plastic films (see appendix A.2.).

Octocrilene is also authorised to be used in the production of plastic articles for being in contact with food in regulation EU 10/2011, allowing a specific migration limit of 0.05 mg/kg. No specific information was retrieved for this use in the different CSRs and as detailed in section 1.4.4, no risks arise from the service life of plastic articles containing octocrilene as an additive. As such, this use is not substantiated more in this restriction dossier. In the

<sup>7</sup> Throughout this document, the Dossier Submitter uses the term 'sun protection products' to refer to both primary sunscreens and secondary photo-protection products. Although the proportion of products containing octocrilene within these product categories is different (see Appendix A.2.1), the claim of sun protection (and therefore the functional importance of Octocrilene) is nevertheless a central common element between these two types of products. Moreover, according to the data provided by Cosmetics Europe, these types of products have similar levels of concentration of octocrilene (see Table 5).

<sup>8</sup> 'Secondary photo-protection products' refer to products with a primary purpose other than sun protection that also contain sun screening agents such as daily face care cream (whose primary function is care, anti-aging, etc.), make up with a Sun Protection Factor (SPF), etc. In this case, the protection against UV radiation is an additional benefit.

<sup>9</sup> Butyl Methoxydibenzoylmethane is the 'INCI name' referring to the UVA filter Avobenzone.

<sup>10</sup> <http://www.spin2000.net/spinmyphp/>

<sup>11</sup> [https://substances.ineris.fr/substance/6197-30-4#usage\\_entity](https://substances.ineris.fr/substance/6197-30-4#usage_entity)

<sup>12</sup> <https://ec.europa.eu/growth/tools-databases/cosing/details/35585>

corresponding CSRs no specific information is available on the use of octocrilene in plastics. As no risks were identified from the preparations containing octocrilene as additive (e.g. paints, adhesives...), the production of plastic articles, inclusion into a matrix, the wide dispersive use of plastics and the service life of plastic, these uses are not substantiated anymore in the dossier.

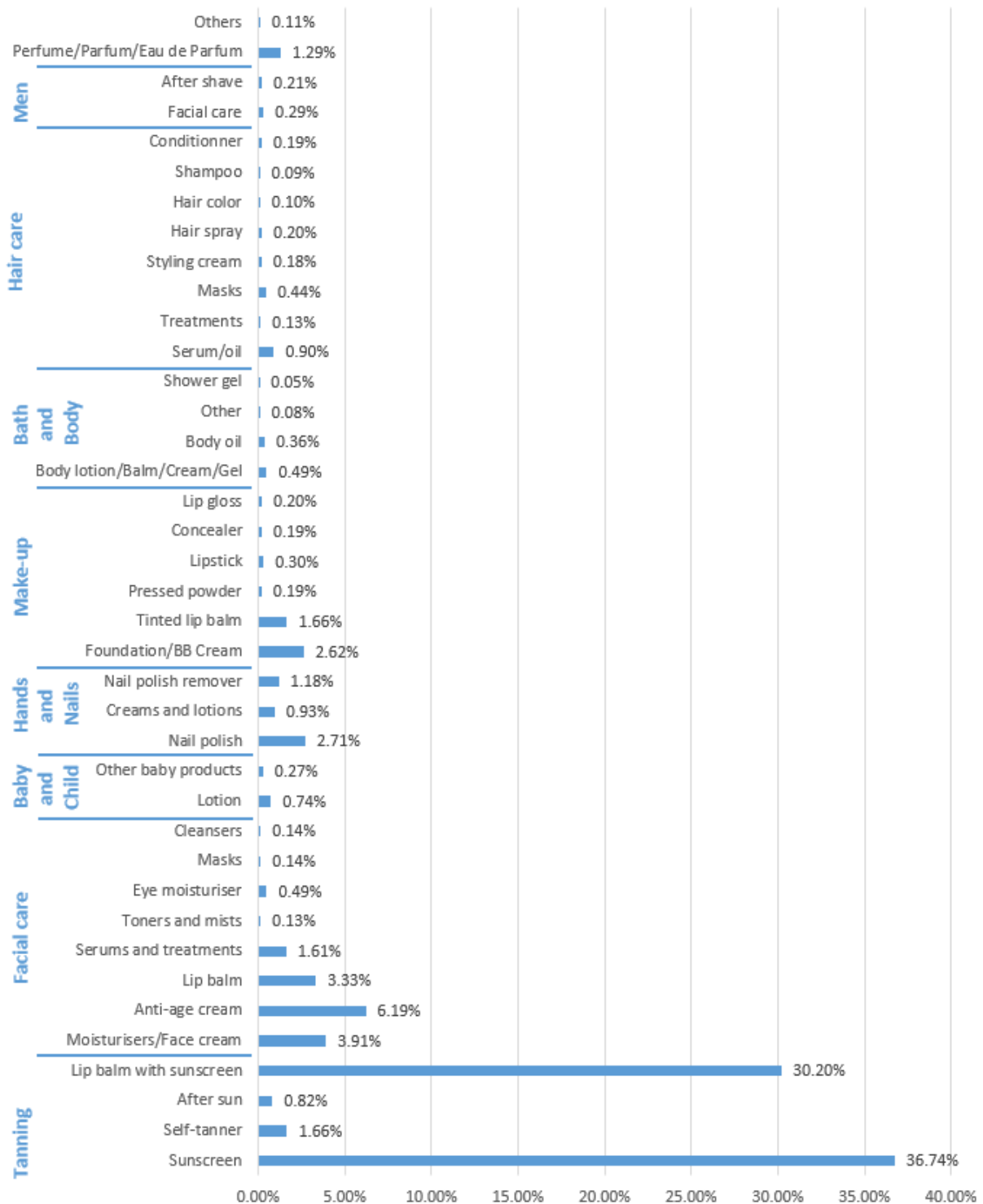
#### **1.3.4. Specific uses of octocrilene in cosmetic products**

Based on these specific properties against UV radiation, Octocrilene is therefore used as a UV (mainly UVB) filter in cosmetic products offering sun protection, i.e., in primary sunscreens as well as in secondary photo-protection products. Importance of sun protection is detailed in Appendix A.2.1.

Octocrilene is listed as an authorised UV filters in the Annex VI of the Regulation (EC) N° 1223/2009 on cosmetic products for its role as a UV filter (mainly UVB), as a general photo-stabilizer and as a solvent for other crystalline and oil soluble UV filters. Full details are available in Appendix E2. As a reminder, the cosmetic regulation solely covers human health endpoint and these are the sole endpoints considered when delivering authorisation of substance to be used as a cosmetic ingredient. Finally, octocrilene can also be used to improve sweat and water resistance of primary sunscreens (CfE#1756; CfE#1759).

Figure 2 provides an overview of the proportion of products per cosmetics category containing octocrilene (detailed figures are available in Appendix A.2.1). This figure is based on the analysis of the dataset obtained from the YUKA application (YUKA, 2024) and is broadly consistent with the results derived from the analysis of other datasets (CosmEthics, 2023; BNPC, 2024; Dossier Submitter's own database) and information provided by other stakeholders during the call for evidence and hearings. Although octocrilene is used in a wide range of cosmetic products, in proportion, octocrilene is mainly used in the formulation of primary sunscreens (36.74%). A non-negligible proportion of lip balms (especially with sunscreen: 30.20%), facial care (facial moisturisers, anti-ageing creams) and make-up products (foundation, BB cream, nail polish) available on the market also contains octocrilene.

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**Figure 2: Proportion of products per cosmetic category containing octocrilene.**

Source: Data from YUKA (YUKA, 2024).

The information presented in Figure 2 is consistent with the results of a survey of Cosmetics Europe members, performed in 2023, on the use of UV filters covering the year 2019 (Cosmetics Europe, personal communication). The results show that in 2019, across all

membership of Cosmetics Europe, more than 80% of the use-volumes of octocrilene (by weight) for the formulation of cosmetic products was used to formulate primary sunscreens. Remaining volumes were mainly used to formulate secondary photo-protection products, and other uses (product protection, others) were minimal.

In addition, Table 5 gives an overview of concentration levels (minimum, median and maximum) of octocrilene found in cosmetic products. Two cases can be distinguished: products where octocrilene is used on product protection purposes contain low concentration levels (2% maximum), while primary sunscreens and secondary photo-protection products show higher concentrations (up to the maximum authorised concentration of 10%).

**Table 5: Concentration of octocrilene in cosmetic products.**

| Product category                                                     | Minimum reported concentration (%w/w) <sup>[1]</sup> | Median reported concentration (%w/w) <sup>[1]</sup> | Maximum reported concentration (%w/w) <sup>[1]</sup> |
|----------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| Primary sunscreen products                                           | 0.5%                                                 | 5%                                                  | 10%                                                  |
| Secondary photo-protection products                                  | 0.2%                                                 | 4%                                                  | 10%                                                  |
| Products in which Octocrilene is used on product protection purposes | No information                                       | No information                                      | 2%                                                   |

Source: Cosmetics Europe (Cosmetics Europe, personal communication). Data are broadly consistent with concentration levels reported in the French National Database of Products and Compositions and information provided by other formulators during the hearings that took place early 2024.

[1]: information provided by Cosmetics Europe (Cosmetics Europe, personal communication) based on a survey to Cosmetics Europe members performed in 2024 as a follow-up of a survey conducted in 2023 on the use of UV filters and covering the year 2019. Seventeen big international companies and 45 SMEs (Small and Medium-sized Enterprises) participated in the initial survey. However, only eight companies participated in the follow-up survey therefore the concentration levels presented here are a rough estimation of concentration values used in the year 2019.

### 1.3.5. Recycling

No specific data were available regarding recycling of octocrilene and cosmetic products containing octocrilene (see previous section). As detailed before, a part of cosmetic products packaging will end up in recycling facility, which will lead to emission to water if the packaging are shredded and washed. During storage, a reaction of retroaldolisation of octocrilene in benzophenone can occur (Downs et al., 2021). To the Dossier Submitter's knowledge, there is no impact of this phenomenon on the potential recycling of cosmetic products containing octocrilene. Nevertheless, benzophenone is classified as carcinogenic 1B by the ATP 18<sup>th</sup><sup>13</sup> and as such the occurrence to a certain concentration of this substance could have an impact on specific waste management systems.

Octocrilene is also used in food contact material and in plastisol for the formulation and production of plastic products. As such a fraction of these products can potentially be recycled into new packaging for food or plastics products. Another fraction of these materials can be directly oriented to incineration or landfill facilities, depending on recycling behaviour or type of materials. Nevertheless, concentration increase for high molecular weight substances (such

<sup>13</sup> <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R0692&from=EN>

as octocrilene) into pellets of plastics for producing new articles such as food contact material is real. It needs to be more investigated as being potentially problematic and overpassing fixed migration limit (Su et al., 2021 a and b).

No specific focus was made regarding recycling and waste management as the use of octocrilene in the production and formulation of plastisol, formulation of preparations containing additives and FCM are not targeted by this restriction proposal. Moreover, as no risks were identified from the preparations containing additives, the production of plastic articles, inclusion into a matrix, the wide dispersive use of plastics and the service life of plastic, these uses are not substantiated anymore in the dossier.

## 1.4. Risk assessment

### 1.4.1. Classification and Labelling

There is no harmonised classification available for octocrilene. The lead registrant for this substance proposed a classification as Aquatic chronic 1 (H410): Very toxic to aquatic life with long lasting effects. Other self-classifications are available in ECHA website as follows:

**Table 6: Summary of available Classification and Labelling.**

| Classification                    |                          | Labelling                |                                 | Specific Concentration limits, M-Factors | Number of Notifiers | Joint Entries |
|-----------------------------------|--------------------------|--------------------------|---------------------------------|------------------------------------------|---------------------|---------------|
| Hazard Class and Category Code(s) | Hazard Statement Code(s) | Hazard Statement Code(s) | Pictograms, Signal Word Code(s) |                                          |                     |               |
| Aquatic Chronic 1                 | H410                     | H410                     |                                 | M(Chronic)=10                            | 13                  | ✓             |
| Aquatic Chronic 4                 | H413                     | H413                     |                                 |                                          | 328                 |               |
| Aquatic Chronic 1                 | H410                     | H410                     | GHS09 Wng                       | M(Chronic)=10                            | 230                 |               |
| Aquatic Chronic 1                 | H410                     | H410                     | GHS09 Wng                       |                                          | 30                  |               |
| Aquatic Chronic 3                 | H412                     | H412                     |                                 |                                          | 24                  |               |
| Not Classified                    |                          |                          |                                 |                                          | 16                  |               |
|                                   |                          | H413                     | GHS09                           |                                          | 1                   |               |
| Aquatic Chronic 4                 | H413                     | H413                     | Wng                             |                                          | 1                   |               |

Consulted in (May 2024)<sup>14</sup>.

<sup>14</sup> <https://echa.europa.eu/fr/information-on-chemicals/cl-inventory-database/-/discli/details/25958>

## 1.4.2. Hazard assessment

***Full details are available in Appendix B.***

### 1.4.2.1. Environmental fate and Hazard assessment

#### 1.4.2.1.1. Degradation

##### 1.4.2.1.1.1. Abiotic degradation

###### 1.4.2.1.1.1.1. Hydrolysis

According to the results from Hydrowin v2.00, the substance has an estimated half-life at pH 7 and 25 °C of about 206 years. Therefore, the substance is considered as slowly hydrolysing in water.

###### 1.4.2.1.1.1.2. Phototransformation/photolysis

According to the AOPWIN v1.92 model, octocrilene is considered to have a half life of 6.19 hours in the atmosphere (within the following conditions:  $1.5\text{E}+06 \text{ OH}\cdot\text{cm}^{-3}$ ). It is therefore not considered to be persistent in air based on this estimated rapid photodegradation potential.

##### 1.4.2.1.1.2. Biotic degradation

###### 1.4.2.1.1.2.1. Biodegradation in water

Two guideline studies on the ready biodegradability of octocrilene are available. They both indicate that octocrilene is not readily biodegradable according to OECD criteria.

Two new simulation tests studies (***Full details are available in Appendix B.4.1.2.1***) were provided during the preparation of this restriction proposal. They investigated the rate and route of degradation of radiolabeled octocrilene in a freshwater sediment system and in an estuarine water-sediment system under aerobic conditions.

In each study, the water-sediment systems were treated with  $54 \mu\text{g}\cdot\text{l}^{-1}$   $^{14}\text{C}$ -labelled octocrilene solution applied into the water phase. Two samples of treated water and sediment were taken for extraction and analysis immediately after treatment (day 0) and after 7, 14, 30, 64 and 100 days of incubation. The degradation rates of octocrilene were calculated using a step-wise approach according to the FOCUS Guidance (2014) and evaluations were performed with the software tool CAKE (version 3.7). Both studies were performed under test conditions that deviated with the OECD TG 308 (Full details are available in Appendix B.4.1.):

- One sediment type was used in the freshwater-sediment system (fine texture/high organic carbon content) and another sediment type was used in the estuarine-sediment system (coarse texture/low organic carbon content sediment, while at least two sediment types and their associated waters are required for each of the two water-sediment systems;
- It is not clear if duplicates samples are true, independent replicates or rather replicate analytical results from a single sample;

High variability between duplicates analysis was noted in both systems. Some deviations from the trend of the values were noted without being significant outliers (Grubb's statistical test) or experimentally justified. Consequently, it was not possible to exclude them from the dataset in order to improve the model-fits and determine most relevant  $\text{DT}_{50}$ .

In both freshwater and estuarine-sediment tested systems, the kinetics model considered to best fitted the experimental data for octocrilene degradation was the bi-phasic Hockey-Stick (HS) model (DT<sub>50</sub> of 7030 days and >1000 days, respectively), rather than the single first-order (SFO) model as proposed by the registrant (DT<sub>50</sub> of 45.5 days and 47.2 days, respectively). It should be noted that the absence of degradation at the test end lowered the confidence of the estimated parameter then, this (HS) DT<sub>50</sub> should be interpreted with caution. Finally, mineralization of octocrilene has also been used and resulted in **DT<sub>50</sub>, mineralization of 1810 days<sup>15</sup> and 1870 days**, respectively.

Four and three relevant metabolites (out of 9 and 14) were detected in the freshwater and estuarine water-sediment systems, respectively. It should be noted that the DT<sub>50</sub> determination was possible only for the metabolite M1 in both systems (DT<sub>50</sub> = 14.8 days and 24.7 days, respectively). The DT<sub>50</sub> for the metabolites M2, M3 and M6 were undefined caused by their accumulation and slow degradation until the end of the test (DT<sub>50</sub> > 1000 days, Default value).

The DT<sub>50</sub> based on octocrilene mineralization in the single water-sediment system tested in each simulation test were considered as the most conservative values covering the potential P/vP concern of its degradation products.

#### 1.4.2.1.1.2.2. Biodegradation in soil

The substance is poorly biodegradable in water. Comparable degradation rates are expected for biodegradation in soil.

#### 1.4.2.1.2. Environmental distribution

##### 1.4.2.1.2.1. Adsorption/desorption

In an OECD 106 adsorption study, an average K<sub>oc</sub> of 48 898 L/kg (range: 29 934 -79 018) in various soil types was determined. In conclusion, the substance has a **high adsorption potential on organic matrices and did not fulfil classification criteria for being identified as a mobile substance in the environment**.

##### 1.4.2.1.2.2. Volatilisation

With a Henry's law constant (H) of 0.00031 Pa m<sup>3</sup>.mol<sup>-1</sup> at 25°C, octocrilene will not evaporate from water into the atmosphere.

The calculated Henry's law constant and log K<sub>oc</sub> of octocrilene suggest that the substance will not evaporate from the water surface into the atmosphere but probably adsorbs to soil and suspended particles. Thus, **octocrilene will mainly distribute into soil and sediment**.

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<sup>15</sup> Calculation: Parent after mineralization (%AR) = 100% - %AR <sup>14</sup>CO<sub>2</sub>

### 1.4.2.1.3. Bioaccumulation

#### 1.4.2.1.3.1. Aquatic bioaccumulation

Two studies are available in the registration dossiers on the bioaccumulation of octocrilene in fish (*Unpublished study report (2008)* and *Unpublished study report (2018d)*). In the first study, the bioaccumulation of octocrilene in *Danio rerio* was investigated according to OECD TG 305. Fish were exposed *via* the water phase to the  $^{14}\text{C}$ -radiolabeled test substance for 28 days. After a depuration period of 15 days the test was terminated and a mean growth and lipid corrected BCF of 858 was determined based on the whole-body wet weight. **Full details are available in Appendix B.4.3.**

The second fish bioaccumulation study using *Oncorhynchus mykiss* and dietary exposure was carried out with  $^{14}\text{C}$ -radiolabeled test substance ( $1\ \mu\text{g}\cdot\text{g}^{-1}$  feed) for 14 days. The resulting lipid and growth corrected kinetic BMF was 0.034.

In another study from the literature, the bioaccumulation kinetics of octocrilene in wild *Mytilus galloprovincialis* mussels was investigated (*Vidal-Liñán et al., 2017*). The kinetics of bioaccumulation of octocrilene significantly fitted to an asymptotic model with BCF values of  $2\ 210\ \text{l}\cdot\text{kg}_{\text{dw}}^{-1}$ , even if no plateau was reached. However, the authors underlined that actual octocrilene BCF in wild mussels subjected to longer exposure periods may be even higher than the value they obtained using a first order kinetics model, which assumes saturation of the tissular concentration (see also monitoring data). It is common for mussels' data to be expressed in dry weight but the B criterion is based on a BCF value expressed in wet weight. Consequently, the BCF would be far below the B criterion.

Octocrilene is not classified as bioaccumulative and very bioaccumulative according to the REACH PBT/vPvB criteria.

#### 1.4.2.1.3.2. Environmental biomonitoring in the biota

Data on the occurrence of octocrilene in biota sampled in European countries are available mainly in aquatic organisms and throughout the food chain (**Full details are available in Appendix B.8.5.3**).

Octocrilene was measured to a maximum value of  $19\ 369\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in macrophyte (*Cadena-Aizaga et al., 2022a*),  $7\ 112\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in mussel (*Bachelot et al., 2012*),  $166\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in echinoderms (*Cadena-Aizaga et al., 2022a*),  $23.1\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in crustacean (*Langford et al., 2015*),  $2\ 698\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in fish and  $108\ \text{ng}\cdot\text{g}_{\text{dw}}^{-1}$  in dolphin.

Octocrilene was found systematically, in winter and summer, in almost all European areas studied (from Spain, to Norway, in sea, lake, river, in protected area).

Field bioaccumulation factor in marine clam *Ruditapes philippinarum* caged in Cadiz Bay (Spain) was estimated to be  $\text{Log BAF} = 2.9$  (mean) and 4.89 (max) (*Pintado-Herrera et al., 2020*). A dynamic BCF value of  $2\ 204\ \text{l}\cdot\text{kg}^{-1}$  was observed in laboratory study on marine clams (*Pintado-Herrera et al., 2024*) and confirmed that octocrilene may be considered bioaccumulative. The  $\text{Log BAF}$  calculated in the primary marine consumers ranged from 5.4 to 8.4, whereas the BMF was lower than 0.3 (*Cadena-Aizaga et al., 2022a*). The  $\text{Log BAF}$  values were also calculated at 3.8 and 4.3 in the two species of cold-water corals sampled in Spain (*Tovar-Salvador et al., 2025*). Those data suggest that octocrilene could be considered as a 'bioaccumulative' compound even if not classified according to criteria set out in annex XIII on PBT, vPvB.

It has to be underlined the tendency of octocrilene to form octocrilene-fatty acids conjugated in some organisms as demonstrated in marine Coral (*Stien et al., 2018*) and marine sediment

worm species (*Clergeaud et al., 2022*). The authors showed that octocrilene was transformed into fatty acid conjugates via oxidation of the ethylhexyl chain, yielding very lipophilic octocrilene analogues that accumulate in tissues. The formation of octocrilene analogues suggests that octocrilene concentrations measured in the environment, and organisms may have been largely underestimated as those conjugates were not measured in these studies and their toxicity not studied.

#### **1.4.2.1.4. Summary of environmental fate and behaviour properties**

##### **1.4.2.1.4.1. Abiotic degradation**

The substance will slowly hydrolyse in water.

##### **1.4.2.1.4.2. Biotic degradation**

According to the two studies (*Unpublished study report (1995a)*; *Unpublished study report (1991a)*) performed in compliance with EU method C.4 -D and OECD guideline TG 301F, octocrilene is not readily biodegradable.

According to the two simulation tests (*Unpublished study report (2024a)*; *Unpublished study report (2024b)*) in water-sediment systems under aerobic conditions (OECD TG 308), the degradation kinetics of octocrilene and its metabolites in the whole system resulted in the following estimated DT<sub>50</sub>:

- Freshwater environment (Freshwater):  
Octocrilene: DT<sub>50</sub>, mineralization (Fw) = 1 810 days;  
Metabolite M1: DT<sub>50</sub>, whole system (Fw) = 14.8 days; Metabolites M2, M3 and M6: DT<sub>50</sub> >1000 days (Default value).
- Estuarine environment (Estuarine water):  
Octocrilene: DT<sub>50</sub>, mineralization (Ew) = 1 870 days;  
Metabolite M1: DT<sub>50</sub>, whole system (Ew) = 24.7 days; Metabolites M2, M3: DT<sub>50</sub> >1000 days (Default value).

##### **1.4.2.1.4.3. Environmental distribution**

The calculated Henry's law constant of 0.00031 Pa m<sup>3</sup>.mol<sup>-1</sup> and the K<sub>oc</sub> of 48 898 l.kg<sup>-1</sup> of octocrilene suggest that the substance will not evaporate from the water surface into the atmosphere but adsorbs to soil and suspended particles. Thus, the substance will mainly distribute into soil and sediment.

##### **1.4.2.1.4.4. Aquatic bioaccumulation**

Octocrilene is not classified as bioaccumulative and very bioaccumulative according to the REACH PBT/vPvB criteria.

Nevertheless, based on environmental biomonitoring in the biota, octocrilene demonstrated the ability to bioaccumulate in environmental species.

##### **1.4.2.1.5. Aquatic compartment toxicity (including sediments)**

***Full details are available in Appendix B.7.1)***

#### 1.4.2.1.5.1. Fish

##### 1.4.2.1.5.1.1. Short-term toxicity to fish

The acute toxicity of octocrilene to fish was tested in two guideline studies (*Unpublished study report (1990)*; *Unpublished study report (1995c)*) and several publications from the scientific literature (*Kaiser et al., 2012a*; *Blüthgen et al., 2014*; *Lucas et al., 2022*; *Gayathri et al., 2023*; *Cahova et al., 2021*). Considering the methodological limitations in those studies, **acute fish toxicity endpoints are considered as supportive data.**

##### 1.4.2.1.5.1.2. Long-term toxicity to fish

No long-term toxicity studies with octocrilene were provided by registrants.

##### Read-across with etocrilene

A fish early life-stage toxicity test (OECD TG 210) (*Unpublished study report (2024c)*) has been provided for another substance, **etocrilene** (CAS 5232-99-5). This substance was used by the registrant in a read across (no justification provided) and as a potential alternative to octocrilene (not approved as a cosmetic ingredient). Nevertheless, and based on the registration dossier of etocrilene and aquatic toxicity data available, it is not possible to conclude on a similarity level between octocrilene and etocrilene for:

- physicochemical, environmental fate and behaviour properties;
- aquatic toxicity.

In conclusion, **a read-across with etocrilene has not been taken into account in the weight of evidence.**

##### Others long term fish studies available

Several studies provided by the registrant or from the literature (*Unpublished study report (2018b)*; *Blüthgen et al., 2014*; *Zhang et al., 2016*; *Li et al., 2018*; *Yan et al., 2020*) in which fish were exposed to octocrilene over a long exposure period are available. These studies have been designed to observe the effects potentially related to an endocrine disruption (EAS modalities). They provide some indications of the systemic toxicity and about parameters considered as 'sensitive to, but not diagnostic of, EATS' in an ED assessment. These studies give indications relevant to ecotoxicological hazard and risk assessment of octocrilene (*i.e.* to determine the most sensitive species and to establish the safety factor for the PNEC<sub>aquatic</sub> calculation), but they cannot be used to provide the Lowest Observed Effect Concentration (LOEC) and NOEC which are necessary for risk assessment as they were not designed to investigate these adverse effects. **These studies indicate a low systemic toxicity of octocrilene on adult fish and effects on the early life stage of fish (larval stage) from 2.7 µg.l<sup>-1</sup> (Yan et al., 2020), without being able to distinguish at this stage any EAS-mediated effect (full details in Appendix B.7.).**

**No OECD study on the long-term toxicity of octocrilene on fish is provided to derive a chronic endpoint value in the context of an environmental risk assessment. Long-term toxicity studies from literature are considered as supportive data and indicate that octocrilene can affect fish species at doses as low as 2,7 µg/L.**

#### 1.4.2.1.5.2. Aquatic invertebrates

##### 1.4.2.1.5.2.1. Short-term toxicity to aquatic invertebrates

The acute toxicity of octocrilene to invertebrates was tested in three guideline studies

(Unpublished study report 1995d; Unpublished study report 2000; Unpublished study report 2001b). Several data are also available in the scientific literature (Pablos et al., 2015; Park et al., 2016b; Thorel et al., 2020; De C S de Paula et al., 2022; Marcin et Aleksander., 2023; Boyd et al., 2020). Considering the methodological limitations in these three GLP guideline studies and in the available literature, **acute invertebrates toxicity endpoints are considered as supportive data.**

#### 1.4.2.1.5.2.2. Long-term toxicity to aquatic invertebrates

In an Unpublished study report (2018c), a 21d *Daphnia magna* reproduction GLP study (OECD TG 211) using  $^{14}\text{C}$  octocrilene at 3.1, 6.3, 12.5, 25 and 50  $\mu\text{g.l}^{-1}$  (nominal) was conducted. The measured concentrations deviated markedly from the nominal concentrations (56-85%), then the mean time-weighted measured concentrations 2.66, 5.25, 9.62, 19.2, 40.6  $\mu\text{g.l}^{-1}$  (77 – 86% of the nominal) were used for the evaluation of the test substance. At the end of the study, a clear concentration response on daphnid reproduction was observed: a significant 16% to 100% decreases of the number of living young per surviving adult were observed from 5.25 to 40.6  $\mu\text{g.l}^{-1}$  respectively. At the highest concentration, results showed 100% parent mortality. A significant 15.2% increase of mean days to first brood was shown at 19.2  $\mu\text{g.l}^{-1}$ . At this concentration, no significant effect on growth, number of immobile young and aborted eggs was shown. In the water and solvent controls, the parent mortality was <20% and mean number of living young per parent  $\geq 60$ , then this test was compliant with the validity criteria. A NOEC of 2.66  $\mu\text{g.l}^{-1}$  (TWM measured concentration) was derived based on reproduction (living offspring per surviving adult).

In Pablos et al., 2015, a semi-static reproduction test with juveniles *D. magna* (less than 24h old) was performed (OECD TG 211). Daphnids were exposed to 3.12, 6.25, 12.5, 25, 50  $\mu\text{g.l}^{-1}$  (nominal). Analytical determinations of the test compounds were not conducted. This paper did not give any information on the exposure conditions and the results were not detailed. The proposed NOEC was >50  $\mu\text{g.l}^{-1}$ .

In Boyd et al., 2020, 21-day chronic exposure to 7.5  $\mu\text{g.l}^{-1}$  (nominal) octocrilene yielded complete mortality within 7 days whereas 0.5  $\mu\text{g.l}^{-1}$  did not affect the mortality rate throughout the 21 days exposure. No effect on physiological and reproductive outputs were observed until the test end. However, this study followed the OECD TG 211 with modifications and exposure conditions were not detailed. This study is considered as supportive data.

De C S de Paula et al., 2022 investigated the chronic effects of octocrilene in two consecutive generations of *D. magna* exposed 21 days at 4.45  $\mu\text{g.l}^{-1}$  nominal. The longevity, time to produce the first offspring, reproduction rate, and oxidative stress (activities of the catalase and glutathione-S-transferase enzymes) were evaluated at day 21. No significant effect was observed in both generations for all tested parameters.

**The NOEC (21d) of 2.66  $\mu\text{g.l}^{-1}$ , has been considered as the relevant endpoint for long-term toxicity to aquatic invertebrates.**

#### 1.4.2.1.5.3. Algae and aquatic plants

A guideline study (OECD TG 201, Unpublished study report (2010)) showed a slight toxicity of the substance to *Desmodesmus subspicatus* with a 72h-EC<sub>50</sub> > 220  $\text{mg.l}^{-1}$  based on algal growth rate.

In a test modified from OECD TG 201 for low volumes, Pablos et al., 2015, showed a low toxicity of octocrilene for the freshwater algae *C. vulgaris* with a 72h-IC<sub>50</sub> of 3.6  $\text{mg.l}^{-1}$  (nominal). In another freshwater species *D. subspicatus*, the EC<sub>50</sub> was 1.54  $\text{mg.l}^{-1}$  (nominal) (De C S de Paula et al., 2022) confirming its low toxicity.

*Thorel et al., 2020*, studied octocrilene toxicity on marine organisms including autotrophs (*Tetraselmis sp.*) and heterotrophs (*Artemia salina*). After 7 days of exposure, a significant decrease of metabolic activity of the marine microalgae *Tetraselmis sp.* at 0.1 mg.l<sup>-1</sup> (nominal) and above was observed.

Octocrilene induced a significant decrease in frond number and frond area of *Lemna minor* after 7 days exposition to octocrilene. The calculated IC<sub>50</sub> was 1.95 mg.l<sup>-1</sup> (nominal) (*Marcin et Aleksander, 2023*).

**The available OECD TG 201 study on Algae is not valid to determine an endpoint value (nominal value far from the solubility limit). Nevertheless, it can be concluded that algae are not the most sensitive trophic level.**

#### 1.4.2.1.5.4. Conclusion for surface water

**Invertebrate *Daphnia magna* has been considered as the most sensitive species with the NOEC determined at 2.66 µg.l<sup>-1</sup>. Even if a fish study from the literature (*Yan et al., 2020*) showed effects on the early life stage of fish (larval stage) at similar concentration levels (from 2.7 µg.l<sup>-1</sup>). However, this fish study cannot be used to provide the Lowest Observed Effect Concentration (LOEC) and NOEC which are necessary for a risk assessment as it was not designed to investigate adverse effects and did not follow any relevant OECD test guideline. This fish study is therefore only used as supportive information.**

#### 1.4.2.1.5.5. Sediment organisms

In an *Unpublished study report (2023b)*, the toxic effects of 62.5, 125, 250, 500 and 1000 mg.kg<sub>dw</sub><sup>-1</sup> octocrilene (nominal) on the submerged aquatic plant *Myriophyllum spicatum* were determined in a 14-day static water-sediment growth inhibition test (OECD TG 239). After 14 days of exposure, no toxic effects were found. The NOEC was determined to be at least the nominal tested concentration of 1000 mg.kg<sub>dw</sub><sup>-1</sup>.

In an *Unpublished study report (2023a)*, first-instar larvae of *Chironomus riparius* were exposed to the octocrilene spiked into the sediment in a static sediment-water system (OECD TG 233) with nominal test concentrations of 62.5, 125, 250, 500 and 1000 mg.kg<sub>dw</sub><sup>-1</sup>. After 28 days exposure of parental generation, no effects were observed. However, it should be noted that the toxicity was probably underestimated since for strongly adsorbing substances as octocrilene (K<sub>oc</sub>=48898 l.kg<sup>-1</sup>), ingestion of contaminated food is known to be a more significant route of exposure.

In an *Unpublished study report (2023c)*, the toxic effects of octocrilene were evaluated on the reproduction and biomass of the freshwater oligochaete *Lumbriculus variegatus*. This study fulfilled the OECD TG 225 validity criteria. A statistically significant reduction in worm reproduction and biomass was found at day 28 in the three higher treatments 103, 227 and 500 mg.kg<sub>dw</sub><sup>-1</sup>. The NOEC was determined to be at 47 mg.kg<sub>dw</sub><sup>-1</sup>. The EC<sub>10</sub> was determined to be at 62.8 and 46.3 mg.kg<sub>dw</sub><sup>-1</sup> for reproduction and dry biomass, respectively.

Additional data available from the scientific literature show significant reduction in the size of *C. riparius* larval at the highest tested concentration 2.33 mg.kg<sub>dw</sub><sup>-1</sup> (*Campos et al., 2017*), or an increase of *hsp70* gene expression in embryos after 24h exposure of egg mass to 1 mg.l<sup>-1</sup> octocrilene. A significant decreased in the percentage of eclosed eggs was also shown at 1 mg.l<sup>-1</sup> suggesting a lethal impairment in embryo development (*Ozaez et al., 2016*).

**Test results show that *L. variegatus* is the most sensitive species in the 28-day sediment tests performed, with the EC<sub>10</sub> for biomass as the most sensitive endpoint determined at 46.3 mg.kg<sub>dw</sub><sup>-1</sup>.**

#### 1.4.2.1.5.6. Other aquatic organisms

In the scientific literature, the toxicity of octocrilene was studied in several species such as on the marine bacteria *Aliivibrio fischeri* (Marcin et Aleksander, 2023) and the freshwater protist *T. thermophila* (Gao et al., 2013), showing no effects. In *Isochrysis galbana*, *Mytilus galloprovincialis* and *Paracentrotus lividus* (Giraldo et al., 2017), NOEC values range from 20 to 40 µg.l<sup>-1</sup>.

Several studies were also available in coral reef species such as *Pocillopora damicornis*, *Seriatopora caliendrum* and *Stylophora pistillata* and showed that octocrilene exposure induced impairment of polyps leading to their closure (300 µg.l<sup>-1</sup>), accumulation due to its transformation into fatty acid conjugates via oxidation of the ethylhexyl chain, effects on biomarkers of cell toxicity (higher acylcarnitines levels) (Stien et al., 2018; He et al., 2018b; Thorel et al., 2022).

#### 1.4.2.1.6. Terrestrial compartment

Several studies are available in the registration dossier focusing on the effects of octocrilene on the terrestrial compartment. **Full details are available in Appendix B.7.2.**

Octocrilene did not induce significant toxicity on the soil microbial activity (OECD TG 216; unpublished study report, 2023c).

In an *Unpublished study report (2017)*, the effects of octocrilene on mortality, biomass and the reproductive potential of the earthworm species *Eisenia fetida* were determined according to OECD TG 222. After 28 days of exposure in soil, no adverse effects were observed. After a further four weeks, the earthworm reproduction was significantly reduced when exposed to 250 and 500 mg.kg<sub>dw</sub><sup>-1</sup>. Then, the NOEC (reproduction) was determined to be 125 mg.kg<sub>dw</sub><sup>-1</sup>.

In an *Unpublished study report (2024d)*, the effects of octocrilene on seedling emergence and early growth of six terrestrial plants (2 monocotyledonous and 4 dicotyledonous plants) were tested according to OECD TG 208. No effect was shown on emergence and mortality of emerged plants for all species. Statistically significant differences between the solvent control and test item treatments were observed for the following species and endpoints: barley (shoot length), mung bean (phytotoxic symptoms, shoot length, shoot weight), buckwheat (phytotoxic symptoms, shoot length, shoot weight) and tomato (shoot length, shoot weight).

The lowest significant effect was observed at the lowest concentration tested of 12.3 mg.kg<sub>dw</sub><sup>-1</sup> (mung bean shoot weight). Even if slightly above 20% effect compared to control, and considering the study limitations described in Appendix B.7.2., this LOEC should be considered in the overall dataset as relevant to be used for the NOEC calculation (LOEC/2; ECHA, 2008c), then, the NOEC=6.15 mg.kg<sub>dw</sub><sup>-1</sup> was determined. This value corresponds to a NOEC=19.5 mg.kg<sub>dw</sub><sup>-1</sup><sup>16</sup> after normalization to a standard soil at 2% organic carbon (OC) (Table R16-8, Echa 2016).

Additional data are available from the scientific literature showing that after 48 h exposure, the concentrations 0.1 and 1 mg.l<sup>-1</sup> caused lipid peroxidation in *Eisenia fetida* and *Allium cepa*.

<sup>16</sup> Tested soil with 0.63% OC, NOEC<sub>soil, 2%(OC)</sub> = 6.15\*(2/0.63) mg.kg<sub>dw</sub><sup>-1</sup>

In *A. cepa*, an induction of genotoxic cellular changes, a reduced cell proliferation in root meristems, a delayed mitosis and seeds germination was observed at 0.1 and 1 mg.l<sup>-1</sup>. After 48h exposure of *E. fetida*, there was no death of earthworms, however, the avoidance rate of earthworms from the contaminated soil increased to 20% and 80% at 0.1 and 1 mg.l<sup>-1</sup> respectively (Santo *et al.*, 2022). Reduction of body weight of *E. fetida* at high dose when exposed to concentration ranging from 500 to 2000 mg.kg<sup>-1</sup> was also highlighted by Gautam *et al.*, 2023, resulting in a LC<sub>50</sub> value >2000 mg.kg<sup>-1</sup>.

**Experimental data on the toxicity of octocrilene are available for three different terrestrial species: soil microorganisms, terrestrial worm and terrestrial plants. Test results show that terrestrial plants are the most sensitive species with a NOEC of 19.52 mg.kg<sub>dw</sub><sup>-1</sup> (After conversion to a standard soil defined as a soil with organic carbon content of 2%).**

#### 1.4.2.1.7. Atmospheric compartment

Not investigated

#### 1.4.2.1.8. Microbiological activity in sewage treatment systems

The toxicity of octocrilene to microorganisms was tested in three studies investigating the respiration inhibition by the substance in activated sludge and *Pseudomonas putida*. Two tests on respiration inhibition in activated sludge were performed. In a study following guideline ISO 8192 (*Unpublished study report (1995b)*), an IC<sub>50</sub>>10000 mg.l<sup>-1</sup> was determined. A guideline study (*Unpublished study report (1991a)*) according to OECD TG 209 detected no respiration inhibition up to a substance concentration of 1000 mg.l<sup>-1</sup>. The results are comparable to a test on *P. putida* conducted in compliance with DIN 38412, part 27 and reporting an EC<sub>50</sub>>10000 mg.l<sup>-1</sup>. However, a solubiliser was used in this study. Moreover, toxicity control of the manometric respirometry test to determine the “ready” biodegradability revealed that octocrilene is not toxic for bacteria. It should be noted that the initial biomass concentration recommended in the OECD 301F guideline is 1.6 g.l<sup>-1</sup> instead of 800 mg.l<sup>-1</sup>. This argument justified that this data is reliable with restriction.

**The inhibition of the degradation activity of activated sludge is not anticipated when introduced in appropriate low concentrations. An IC<sub>50</sub> for microorganisms > 1000 mg.l<sup>-1</sup> is considered as the most sensitive endpoint value.**

#### 1.4.2.1.9. Non compartment specific effects relevant for the food chain (secondary poisoning)

Not investigated. However, biomagnification of octocrilene was suggested as shown in the scientific literature in marine species of high trophic level such as tuna (Pintado-Herrera *et al.*, 2024) and dolphin (Gonzalez-Bareiros *et al.*, 2023).

#### 1.4.2.1.10. Environmental Hazard conclusion

##### 1.4.2.1.10.1. Reference values

##### Assessment Factor (AF) to derive the PNEC<sub>aquatic</sub>

No OECD TG study on the long-term toxicity of octocrilene on fish is available to derive a chronic endpoint value (NOEC/LOEC) in the context of an environmental risk assessment. Nevertheless, in a Weight of Evidence approach, long term exposure data from several fish tests (from literature) with octocrilene are available to define the assessment factor (AF) for

the PNEC value. These studies indicate a low systemic toxicity of octocrilene on adults and effects on the early life stage of fish (larval stage) from  $2.7 \mu\text{g.l}^{-1}$ , but without following a testing protocol similar to OECD test guidelines for testing chronic or reproductive effects or fish life-cycle test (Yan *et al.*, 2020).

An OECD TG 211 test is considered as the only reliable long term aquatic toxicity test for octocrilene. These results show that *Daphnia magna* is the most sensitive species in a 21-day test with a  $\text{NOEC}_{\text{Reproduction}}$  of  $2.66 \mu\text{g.l}^{-1}$  (living offspring per surviving adult).

The available OECD TG 201 study on algae is not valid to determine an endpoint value (nominal value far from the solubility limit). However, based on the overall available data including literature studies, it can be concluded that algae is clearly not the most sensitive trophic level and this endpoint can be used to reduce the assessment factor in the determination of a  $\text{PNEC}_{\text{aquatic}}$ .

**In a weight of evidence approach, an AF of 10 for freshwater and 100 for marine water were considered. In this case, the  $\text{PNEC}_{\text{aquatic}}$  is determined to be  $0.27 \mu\text{g.l}^{-1}$  for freshwater based on the  $\text{NOEC}$  of  $2.66 \mu\text{g.l}^{-1}$  in *Daphnia* and  $0.027 \mu\text{g.l}^{-1}$  for marine water. Nevertheless, and given uncertainties about the toxicity of octocrilene on fish at the larval stage, an AF of 50 for freshwater and 500 for marine water could not be proposed as an acceptable surrogate. However, risks are already identified with PNEC based on *Daphnia* and an AF of 10, the question of increasing the AF does not need to be elaborated further.**

#### **Assessment Factor (AF) to derive the $\text{PNEC}_{\text{sediment}}$ .**

Regarding sediments, three chronic tests with species representing different living and feeding condition (*Chironomus*, *Lumbriculus*, *Myriophyllum*) are available to derive the PNEC value. However, the study with *Chironomus* was judged reliable with restrictions since the toxicity was probably underestimated. Indeed, the exposure route was not modified from spiked-sediment to contaminated food, as required for strongly adsorbing substances as octocrilene ( $K_{oc}=48898 \text{ l.kg}^{-1}$ ).

**Chronic data are considered available for two trophic levels leading to an AF of 50 for the freshwater environment and 500 for the marine environment. The  $\text{PNEC}_{\text{sediment}}$  is determined to be  $0.926$  and  $0.0926 \text{ mg.kg}_{dw}^{-1}$  sediment (for freshwater and marine water respectively) based on the  $\text{EC}_{10}$  of  $46.3 \text{ mg.kg}_{dw}^{-1}$  in *Lumbriculus variegatus*.**

#### **Assessment Factor (AF) to derive the $\text{PNEC}_{\text{soil}}$**

Regarding soil, long-term toxicity tests are available for three trophic levels (microorganisms, earthworm, and plants) to derive the  $\text{PNEC}_{\text{soil}}$  value. The most sensitive species was demonstrated to be the mung bean, with a  $\text{NOEC}$  of  $6.15 \text{ mg.kg}_{dw}^{-1}$  for the emergence of seeds.

**Chronic data are considered available for three trophic levels leading to an AF of 10. In this case, the  $\text{PNEC}_{\text{soil}}$  is determined to be  $1.95 \text{ mg.kg}_{dw}^{-1}$  considering the organic carbon normalisation (at 2%) (Table R16-8, Echa 2016).**

#### **Assessment Factor (AF) to derive the $\text{PNEC}_{\text{STP}}$**

Regarding microorganisms in STP, three studies are available investigating the respiration inhibition by the substance in activated sludge and *Pseudomonas putida*. According to OECD TG 209, an  $\text{IC}_{50}>10000 \text{ mg.l}^{-1}$  was determined.

**Considering an AF of 100 and the IC<sub>50</sub> for microorganisms of 1000 mg.l<sup>-1</sup>, the PNEC<sub>STP</sub> is determined to be > 10 mg.l<sup>-1</sup>.**

Based on the data presented above and in details in Appendix B.7, Table 7 summarizes the PNEC available for each compartment.

**Table 7: PNEC derivation and other hazard conclusions.**

| <b>PNEC DERIVATION AND OTHER HAZARD CONCLUSIONS</b>                 |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazard assessment conclusion for the environment compartment</b> | <b>Hazard conclusion as proposed by the eMS</b>                                                                                     | <b>Remarks/Justification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Freshwater                                                          | PNEC <sub>aquatic</sub> (freshwater):<br><br>0.27 µg.l <sup>-1</sup>                                                                | Assessment factor: 10<br><br>(Three chronic aquatic toxicity tests representing three trophic levels (algae, invertebrates and fish))<br><br>Extrapolation method: assessment factor<br><br>OECD TG 211 (Daphnia reproduction study; NOEC of 2.66 µg.l <sup>-1</sup> , mean measured)                                                                                                                                                                                                                |
| Marine water                                                        | PNEC <sub>aquatic</sub> (marine water):<br><br>0.027 µg.l <sup>-1</sup>                                                             | Assessment factor: 100<br><br>(Three chronic aquatic toxicity tests representing three trophic levels (algae, invertebrates and fish) but no marine water data following OECD guidelines available.)<br><br>Extrapolation method: assessment factor<br><br>OECD TG 211 (Daphnia reproduction study; NOEC of 2.66 µg.l <sup>-1</sup> , mean measured)                                                                                                                                                 |
| Sediments (freshwater)                                              | PNEC <sub>sediment</sub> (freshwater):<br><br>0.926 mg.kg <sub>dw</sub> <sup>-1</sup><br><br>0.20 mg.kg <sub>ww</sub> <sup>-1</sup> | Assessment factor: 50<br><br>(Three chronic tests with species representing different living and feeding condition ( <i>Chironomus</i> , <i>Lombriculus</i> , <i>Myriophyllum</i> ) but <i>Chironomus</i> mostly non exposed to the substance (unrelevant exposure route))<br><br>Extrapolation method: assessment factor<br>Conversion dw to ww: 4.6 according to R.16<br><br>OECD TG 225 (spiked-sediment dwelling worm study; EC <sub>10</sub> of 46.3 mg.kg <sub>dw</sub> <sup>-1</sup> nominal) |

| PNEC DERIVATION AND OTHER HAZARD CONCLUSIONS                 |                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hazard assessment conclusion for the environment compartment | Hazard conclusion as proposed by the eMS                                                                                                                                                   | Remarks/Justification                                                                                                                                                                                                                                                                                            |
| Sediments (marine water)                                     | PNEC <sub>sediment</sub> (marine water):<br><br>$0.0926 \text{ mg.kg}_{dw}^{-1}$<br><br>$0.020 \text{ mg.kg}_{ww}^{-1}$                                                                    | Assessment factor: 500<br><br>Extrapolation method: assessment factor<br>Conversion dw to ww: 4.6 according to the R.16<br><br>OECD TG 225 (spiked-sediment dwelling worm study; EC <sub>10</sub> of $46.3 \text{ mg.kg}_{dw}^{-1}$ nominal)                                                                     |
| Sewage treatment plant                                       | PNEC STP:<br><br>$>10 \text{ mg.l}^{-1}$                                                                                                                                                   | Assessment factor: 100<br><br>Extrapolation method: assessment factor<br><br>OECD TG 209 (Activated sludge); EC <sub>50</sub> $>1\,000 \text{ mg.l}^{-1}$                                                                                                                                                        |
| Soil                                                         | PNEC soil:<br><br>$1.95 \text{ mg.kg}_{dw}^{-1}$ considering the organic carbon normalisation (2%)<br><br>$1.72 \text{ mg.kg}_{ww}^{-1}$ considering the organic carbon normalisation (2%) | Assessment factor: 10<br><br>(Long-term toxicity tests of three trophic levels microorganisms, worm, plants)<br><br>Extrapolation method: assessment factor<br>Conversion dw to ww: 1.13 according to the R.16<br><br>OECD 208 (Seedling emergence test); NOEC of $19.52 \text{ mg.kg}_{dw}^{-1}$ nominal, 2% OC |
| Air                                                          | No hazard identified                                                                                                                                                                       | -                                                                                                                                                                                                                                                                                                                |

#### 1.4.2.1.10.2. PBT assessment

##### Evidence of P or vP properties

Two degradation studies (freshwater- and estuarine-sediment simulation tests) were provided.

High variability in the degradation data of octocrilene and uncertainties in degradation behaviour of several metabolites were observed then, the DT<sub>50</sub> for octocrilene was estimated based on mineralisation data: DT<sub>50</sub>=1810 days (freshwater); DT<sub>50</sub>=1870 days (estuarine water).

Octocrilene appears to be a persistent substance (P) based on the half-life of degradation products. No further testing is required for the time being.

**Conclusion on P/vP properties: Octocrilene is P and vP based on metabolites degradation half-life.**

**Evidence of non-B / non-vB properties**

Based on the results of two OECD 305 fish bioaccumulation studies (exposure *via* water resp. diet), the substance is not classified as bioaccumulative (B) nor very bioaccumulative (vB).

**Conclusion on B/vB properties: not B/vB.** Nevertheless, data from literature showed the bioaccumulative properties of octocrilene in the biota. Therefore, the Dossier Submitter considers octocrilene as a potential bioaccumulative substance.

**Evidence of T properties**

Based on the results from a chronic daphnia toxicity (OECD TG 211, NOEC of 2.66 µg.l<sup>-1</sup>), the substance is considered as toxic (T) for the environment.

**Conclusion on T properties: Octocrilene is T**

**Overall conclusion:** Because it does not fulfil the B criteria, octocrilene cannot be classified as PBT.

**1.4.2.2. Human health hazard assessment**

As the restriction proposal covers solely risks arising from the wide dispersive uses of cosmetic products for the environment, this section will not be further developed at the exception of the section on endocrine disruption data in human health, as these data are considered in addition to environmental data for the whole ED assessment of octocrilene (one health concept and conservation of the modalities considered). Full details of human health hazard assessment will be available in the conclusion document of the substance evaluation under REACH when it is completed.

**1.4.2.3. Assessment of endocrine disrupting properties for human health****1.4.2.3.1. Endocrine disruption - Human health**

The ED assessment has been performed according to the ECHA/EFSA Guidance.

**1.4.2.3.2. ED assessment for T-modality****Lines of evidence for adverse effects related to T-modality**

Thyroid follicular hypertrophy/ hyperplasia and pale staining colloid were consistently observed in the available repeated dose oral toxicity studies in rats (*Unpublished study report, 1993, 2018, 2019*) even at dose level where no general or liver toxicity was noted. While it is acknowledged that thyroid mediated adverse effects may be driven by liver enzymatic induction (further discussed in the below reported MoA analysis for T-modality), **they can therefore not be considered as secondary to overt general or liver toxicity.**

The LOAEL for a thyroid-based adverse effect was mainly based on the observed patterns of histopathological changes, in line with EFSA/ECHA ED GD. The lowest effective dose for the thyroid effects was 2100 ppm (127 mg/kg bw per day) observed in F0 males (statistically significant) and in F1A males (non-statistically significant) of the Extended One Generation Reproductive Toxicity Study (EOGRTS). At this dose level a 7% and 10% increase of relative liver weight was observed in F0 and F1A males respectively without any correlated liver histopathological or clinical changes (*Unpublished study report, 2019*).

A 7 to 10% increase in relative liver weight is not considered adverse in the absence of correlated histopathological or clinical findings (WHO, 2015).

### Parameters sensitive to but not diagnostic of T-modality

- Pre-implantation loss:

Lower mean number of implantation sites, and consequently, a lower number of pups delivered in high-dose group females (7000 ppm) were observed in F0 generation (Unpublished study report, 2019). It was also observed in the dose-range finding study (15 000 ppm) in the presence of general toxicity (*Unpublished study report, 2018*).

- Pituitary histopathology:

In the 90-d rat study, the number of hypertrophic cells in the pituitary gland of high dose males (15000 ppm) was increased compared to controls (*Unpublished study report, 2019*).

- Developmental neurotoxicity:

As regard thyroid related developmental neurotoxicity (DNT) concerns, no final conclusion can be drawn due to the numerous limitations of the DNT part of the EOGRTS. Poor reporting of the methods and lack of raw data for the auditory startle response) **hamper to draw any final conclusion on developmental neurotoxicity.**

### Lines of evidence for endocrine activity related to T-modality

#### In vitro data:

Octocrilene was positive in TOX21\_CAR\_Agonist assay with an AC<sub>50</sub> of 2.72 µM but also positive in TOX21\_CAR\_Antagonist assay with an AC<sub>50</sub> of 8.34 µM (borderline activity). The substance was positive in TOX21\_PXR\_Agonist assay with an AC<sub>50</sub> of 12.9 µM and was negative in TOX21\_AhR\_LUC\_Agonist assay<sup>17</sup>.

No information is available on other relevant molecular initiating events such as inhibition of sodium-iodide symporter (NIS) or thyroid peroxidase (TPO).

#### In vivo data:

There is indication of octocrilene ability to induce hormonal changes, such as an increase in TSH levels in males (not significant) and females (significant) and to decrease T4 level in pups at high doses (*Unpublished study report, 2019*). Octocrilene also induces effect on thyroid enzymes, such as T4-UDPGT, Benzyloxyresorufin-O-debenzylase (BROD), Pentoxeresorufin-O-depentylase (PROD) to different extent depending on the doses administered (*Unpublished study report, 2019*). Liver type III deiodinase (D3, induction) and liver type I deiodinase (D1, inhibition) were impacted following octocrilene oral exposure in a rat mechanistic study (*Unpublished study report, 2019*).

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<sup>17</sup><https://comptox.epa.gov/dashboard/dsstoxdb/results?search=DTXSID9025299#bioactivity> (2022-02).

The available information in rats shows that the substance has endocrine disruption activity (decreased T4 and increased TSH) indicating a chemical capable of perturbing TH-signalling.

### **MoA analysis for T-modality**

The registrant proposed that the T-modality mode of action (MoA) is via liver enzyme induction without performing a WoE analysis in line with ECHA/EFSA ED guidance.

Based on the lines of evidence available, nuclear receptor activation via liver enzyme induction can be identified as a plausible Molecular Initiating Event (MIE) as postulated MoAs.

As regard adverse outcomes, follicular cell hypertrophy/hyperplasia is consistently observed in all rat studies.

Additional MIEs (interference with TSHR or TR) were investigated *in vitro* within TOX21 program (namely TSHR and TR were investigated within TOX21 program, they were considered negative and consequently a MoA was not postulated for these MIEs). Based on current knowledge, evidence for the investigation of further MIEs is lacking (e.g. TPO or NIS inhibition), and this is reported later as uncertainties.

### **Deiodinase I inhibition/ deiodinase III induction as an alternative MIE.**

Regarding effects on neurodevelopment, the numerous limitations of the DNT part of the EOGRTS hamper to draw any final conclusion on developmental neurotoxicity.

### **Conclusion for T-modality**

Based on the MoA analysis performed, while some uncertainties remain, the available data seem to support the postulated MoA.

As regard the human relevance of the proposed MoA leading to thyroid hyperplasia, no substance-specific data have been submitted that provide proof of the non-relevance of this MoA for human. Elevated TSH may not be a reliable key event (KE) biomarker for thyroid cancer in humans; however, increased TSH in rodents does indicate a chemical capable of perturbing TH signalling in other species including human and potentially inducing relevant other adverse outcomes than thyroid cancers.

Impaired pre- and postnatal- neurological development, a crucial adverse outcome potentially linked to a drop in T4 (due to liver induction or other MIEs) during critical periods has not been adequately investigated. Indeed, due to severe limitations, the DNT part of the EOGRTS is considered inconclusive.

**Considering the remaining uncertainties, no final conclusion could be set on ED-potential of octocrilene regarding T-modality for Human health.**

#### **1.4.2.3.3. ED assessment for EAS-modality**

While EAS-mediated parameters have been sufficiently investigated according to ECHA/EFSA guidance (2018), no EAS-mediated parameters were affected by treatment across all available studies including an EOGRTS (level 5 of OECD Conceptual Framework for Testing and Assessment of Endocrine Disrupting Chemicals).

Based on the available data, some EAS activity (mainly antiandrogenic activity) has been

observed, but there is no indication of adversity based on EATS-mediated parameters. In the absence of EAS-mediated pattern of effects, the first condition of the ED criteria is not met; therefore, based on the current knowledge and according to ECHA/EFSA ED guidance, the substance does not meet the ED criteria for Human health (scenario 1a of EFSA/ECHA ED GD).

#### 1.4.2.4. Endocrine disruption - Environment

The available information in human health section and especially in rats show that octocrilene led to adverse effects related to T-modality. Briefly, studies *in vitro* and *in vivo* in rats describe T-mediated effects as well as changes of thyroid hormone levels (see Appendix B.5.). Octocrilene was noted to induce hormonal changes in the 14-d/28-d mechanistic study in rat (*Unpublished study report, 2019*), a 2-fold increase of TSH levels was observed both in high dose (10000 ppm) males (non-statistically significant) and females (statistically significant) at different time points. Non-statistically significant decrease of T4 (10-25%) was observed at the same dose levels and considered biologically relevant. In this study, induction of enzyme involved in TH catabolism was noted (increased T4-UDPGT activity in mid-dose females (3000 ppm), in high-dose males and females (10000 ppm) along with other CAR and PXR-regulated enzymes (BROD and PROD activities) from 3000 ppm onwards in both sexes. Furthermore, liver type III deiodinase, D3, was induced (2-fold increased activity) in mid and high-dose males while liver type I deiodinase, D1, was inhibited (1.5-fold reduced activity) in high-dose males. It should be highlighted that D3 induction decreases the biological activity of thyroid hormones via inner ring deiodination of T4 (thyroxine) and T3 (triiodothyronine) leading to rT3 (reverse T3, an inactive form of T3) and T2 (diiodothyronine) respectively.

These adverse effects may also be relevant for the environmental species. Indeed, in the absence of substance-specific data that provide proof of the contrary, other taxonomic groups should be considered as sensitive to thyroid-disruption such as rat. Indeed, it is well acknowledged that thyroid signalling, regulation and development are highly conserved in vertebrates and are comparable in mammals, hatching birds and amphibians (Buchholz, 2017).

Thyroid hormones are essential for vertebrate development; hence it is important to consider the environmental impact of octocrilene on the actions of the TH nuclear receptor system during development. Deiodinases are involved in metamorphosis in amphibian and impairment of these molecular initiating events will impact metamorphosis as reflected in adverse outcome pathways under development (AOP 189, 190 and 191).

However, none of the available ecotoxicity studies provide information on apical and adverse effects specific to T-modality in other taxonomic groups. The available and current information is therefore not sufficient to draw a final conclusion on the potential hazard for the environment. According to the ECHA/EFSA guidance for the identification of endocrine disruptors in the context of Regulations (EU) No 528/2012 and (EC) No 1107/2009, section 3.3.1.4., no data on apical endpoints related to disruption of thyroid function have been identified in available studies that would lead to environmental adversity of octocrilene at the population level.

It was concluded by the MSC in June 2022<sup>18</sup> that further data need to be generated to

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<sup>18</sup> <https://echa.europa.eu/documents/10162/a9fe9bc3-1dc3-03f2-b307-c25de6f6e6fb>

investigate T-mediated parameters in other taxonomic groups. To do so, a Larval Amphibian Growth and Development Assay (LAGDA, test method: OECD TG 241, 2015) has been requested. Results of this assay are awaited at the time of preparation of this restriction proposal due to delays related to methodology and analytical analyses. Results are expected during finalisation of the restriction proposal. They are not expected to modify the analysis of the present restriction dossier. If Octocrilene is an ED for the environment, it will emphasize the need to restrict its use into cosmetics.

#### **1.4.2.5. Conclusion on hazard assessment**

As detailed in previous section, octocrilene is a hazardous substance for aquatic and sediment species. As such a risk assessment need to be performed to demonstrate the safe use of the substance.

As additional consideration, octocrilene is persistent and very persistent, is toxic for the environment and has the ability to bioaccumulate to some extent when considering the monitoring data but without meeting the criteria set out in annex XIII of REACH for being classified as a PBT and/or vPvB substance. Moreover, octocrilene is a suspected endocrine disruptor for environment under assessment. However, due to a pending data generation, final conclusions on the endocrine effect cannot be drawn yet.

#### **1.4.3. Releases, exposure and risks assessment**

Physicochemical and environmental fate and distribution properties used for exposure estimation are available in Appendix B.8.

##### **1.4.3.1. Substance life cycle**

From all registered CSRs, a mapping of the octocrilene's life cycle is proposed in Figure 3. In Appendix B.8.2, a mapping with the tonnages declared by each registrant and for each stage in the substance's life cycle is available.

Uses have been described by sector associations taking place at "industrial sites" for the following life-cycle stages:

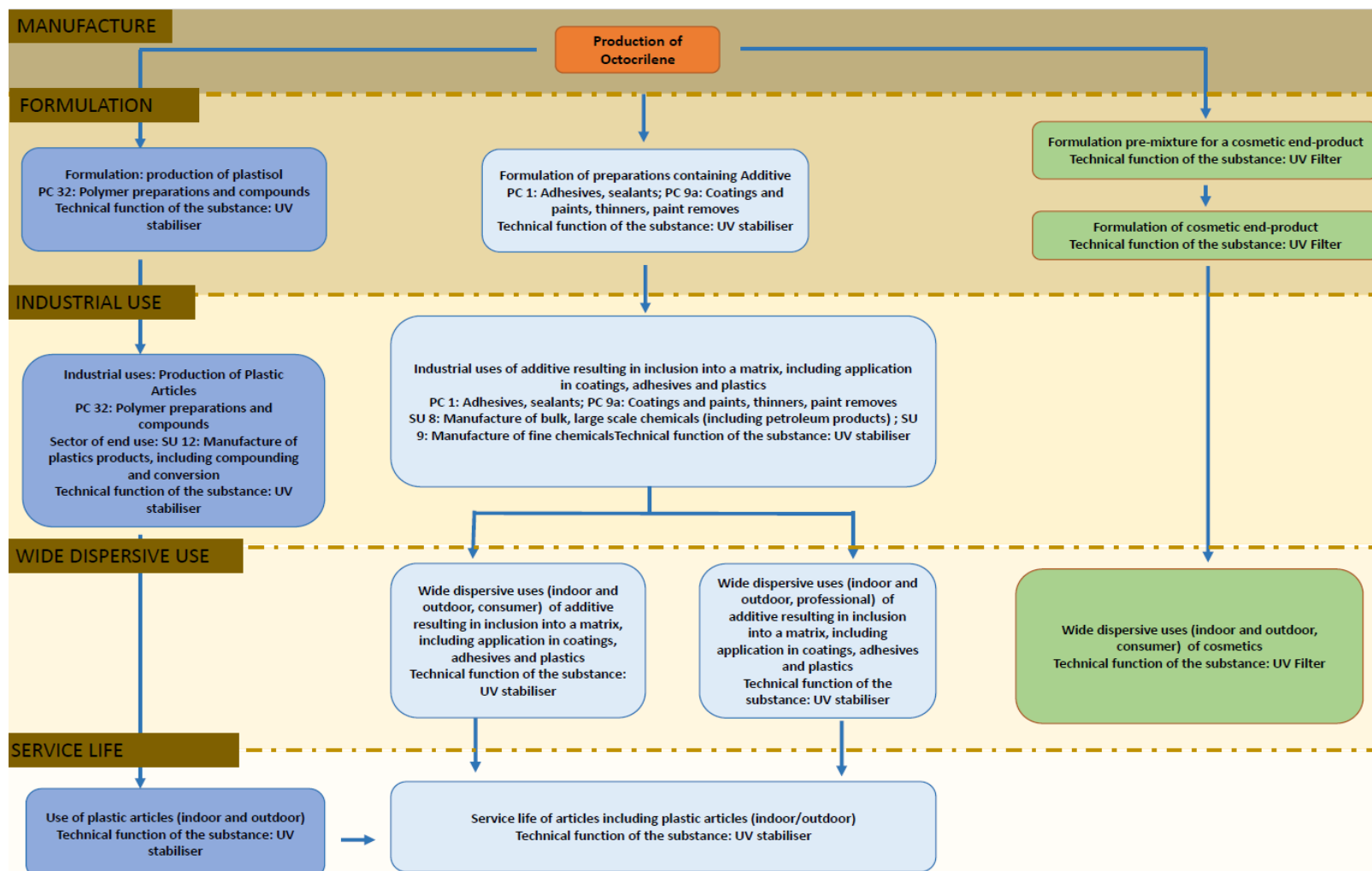
- ✓ Manufacture;
- ✓ Formulation;
- ✓ Use at industrial site;

Or to be "widespread", i.e., used by consumers or by professionals:

- ✓ Widespread use by professional workers;
- ✓ Use by consumers;
- ✓ Service life.

Octocrilene can be released to the environment at all stages of its life cycle: manufacture, formulation of plastisol, formulation of preparations containing additive and formulation of finished cosmetic products, industrial use for the production of plastic articles, industrial use of preparation containing additives for inclusion into a matrix, wide dispersive use by professionals and consumers indoor and outdoor of octocrilene included into a matrix, wide dispersive uses of cosmetics, service life articles containing octocrilene, as well as from disposal as waste and waste treatment. The large diversity of products containing octocrilene that are placed on the market and used, leads to wide dispersive releases to the environment during their life cycle.

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**Figure 3: Mapping of the octocrilene's life cycle**

The Dossier Submitter estimated that the annual use of octocrilene from finished cosmetic products is of **1752 t/y**, estimation confirmed with the data from Euromonitor (2025)<sup>19</sup> (1721 t/y).

All risk assessments are performed following the guidance document of ECHA on the "Guidance on Information Requirements and Chemical Safety Assessment Chapter R.16: Environmental exposure assessment" (ECHA, 2016).

#### **1.4.3.2. Tonnage approach - Environmental exposure assessment and associated risk**

The release of a substance and subsequent exposure of the environment are in principle assessed on two spatial scales: **locally** in the vicinity of a representative source of the release to the environment, and **regionally** for a larger area which includes all release sources in that area. The regional concentrations are used as background concentrations. At last, and for most of the compartments (except for the biological STP), the **PEC regional must be integrated in the calculations of the PEC at the local scale**, according to the R16 guidance, to calculate environmental risk ratios.

##### **1.4.3.2.1. Tonnage approach - Local exposure assessment**

The concentrations of substances released from a single point source (industrial site or standard municipal biological STP) are to be assessed after release to the environment.

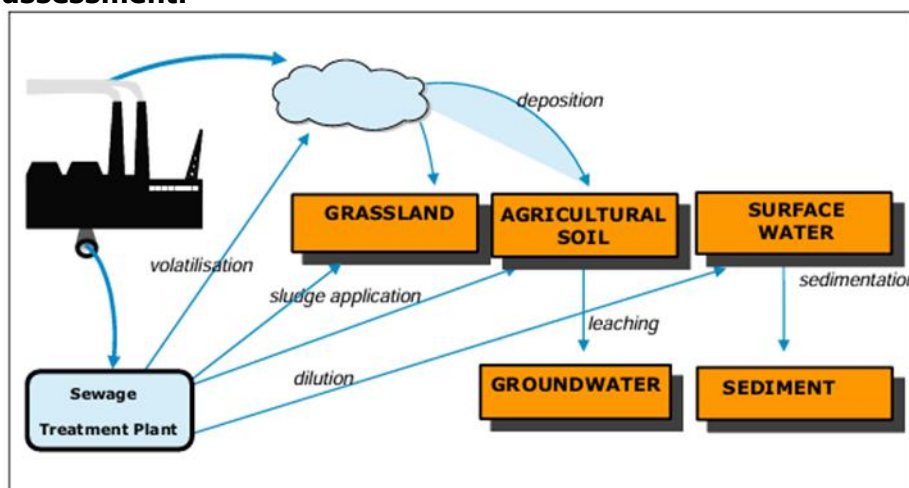
Figure 4 below<sup>20</sup> shows the relationship between the local release routes and the subsequent distribution process modelled for the environmental compartments.

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<sup>19</sup> Euromonitor database (Passport) is a database that contains information on the consumption of different categories of sun care products in terms of value and volume between 2009 and 2028 in different countries of the EEA. It should be noted, first, that it does not provide any information on the other suncare products available (but not purchased) to consumers during their shopping trips. Thus, the choice set of each consumer is not available. Second, the data accessed and used in this section do not cover the whole EEA and cover some countries outside the EEA (Switzerland, Turkey and UK).

<sup>20</sup> Source: ECHA Guidance on information requirements and chemical safety assessment, chapter R.16: Environmental exposure assessment, figure R.16-5

**Figure 4: Local distribution processes considered for environmental risk assessment.**



The set of parameters used in a tonnage-based approach to estimate local exposure for each life-cycle stage is summarized in Appendix B.8.3.1. **These parameters values have been proposed by each registrant** (except for some values described below), **as recorded in their respective CSR.**

Differences in the determination of “release factors” in water/air/soil are to be noted between registrants for a same emission scenario (see Table 8 below, FORMULATION: production of plastisol, SERVICE LIFE: plastic articles or articles including plastic articles).

The only deviation in the parameters from the registration dossiers supplied by registrants concerns the input parameter “application to the STP sludge on agricultural soil” when releases are made in municipal sewage treatment plant. Several registrants have indicated the absence of sludge application to agricultural soil when the releases to water are treated in a standard biological STP (e.g. by incineration of sewage sludge). However, it is not possible to regulate the sludge treatment from municipal wastewater treatment plants at a national or EU level. In fact, this type of practice is regulated at the local level for each sewage treatment plant at least in certain countries as France. In consequence, the value “yes” for the input parameter “application to the STP sludge on agricultural soil” is used by default.

Finally, the tonnage values applied in the emission scenarios for each specific life-cycle stage and by each registrant do not always refers to the real used tonnage. Indeed, and according to R16 guidance, the assessment for the industrial stage can be based on a tonnage used by a large site to cover all situations (large and small industrial sites), and not at the actual tonnage for this specific use (sum of tonnage used by small and large industrial sites).

It worth noting that the local assessment must not take into account the degradation/dissipation of the substance in the estimation of local aquatic PECs (including sediment) considering a continuous release over a short distance from the emission source.

The parameters proposed by the registrants are not discussed, but used as a first approach to perform an environmental exposure assessment specific to the registration dossiers based on the data declared by the registrants. **If risks occur for the environment with these scenarios, it shows that the measures put in place by the registrants do not guarantee the safe use of octocrilene and that the proposed restriction is necessary to tackle the demonstrated risks (see RMOA describing why this is the**

**best risk management option).**

The predicted environmental local concentrations (PECs local) values have been estimated and summarized in the

Table 8. Note that, the PECs local values for soil are given only for agricultural soils and not for grassland in a realistic worst-case approach (no possibility to limit application of sludge to grassland by RMMs). Moreover, the industrial soil is not considered relevant at the local level according to the R.16 guidance.

**Table 8: Local PECs values.**

| MANUFACTURE                                                             |                   |     |              |              |      |
|-------------------------------------------------------------------------|-------------------|-----|--------------|--------------|------|
|                                                                         | ES1 <sup>21</sup> |     | ES2          |              |      |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                | 2.43E-01          |     | 8.53E-03     |              |      |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                         | 2.18E-02          |     | 2.49E-05     |              |      |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> ) | 2.32E+01          |     | 2.65E-02     |              |      |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )   | NR                |     | NR           |              |      |
| FORMULATION: production of plastisol                                    |                   |     |              |              |      |
|                                                                         | ES3               |     | ES4          |              |      |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                | 4.11E-03          |     | No emission* |              |      |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                         | 3.82E-04          |     |              |              |      |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> ) | 4.07E-01          |     |              |              |      |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )   | 6.63E-01          |     |              |              |      |
| FORMULATION: preparations containing additives                          |                   |     |              |              |      |
|                                                                         | ES5               |     |              |              |      |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                | No emission*      |     |              |              |      |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                         |                   |     |              |              |      |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> ) |                   |     |              |              |      |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )   |                   |     |              |              |      |
| FORMULATION: pre-mixture for finished cosmetic products                 |                   |     |              |              |      |
|                                                                         | ES6               |     |              |              |      |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                | 2.48E-03          |     |              |              |      |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                         | 7.25E-06          |     |              |              |      |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> ) | 7.71E-03          |     |              |              |      |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )   | NR                |     |              |              |      |
| FORMULATION: finished cosmetic products                                 |                   |     |              |              |      |
|                                                                         | ES7               | ES8 | ES9          | ES10         | ES11 |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                | No emission*      |     | 1.84E-03     | No emission* |      |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                         |                   |     | 1.72E-04     |              |      |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> ) |                   |     | 1.83E-01     |              |      |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )   |                   |     | 2.98E-01     |              |      |

<sup>21</sup> ES: emission scenario. An ES is defined for each use declared by a registrant. This is done for all registrants

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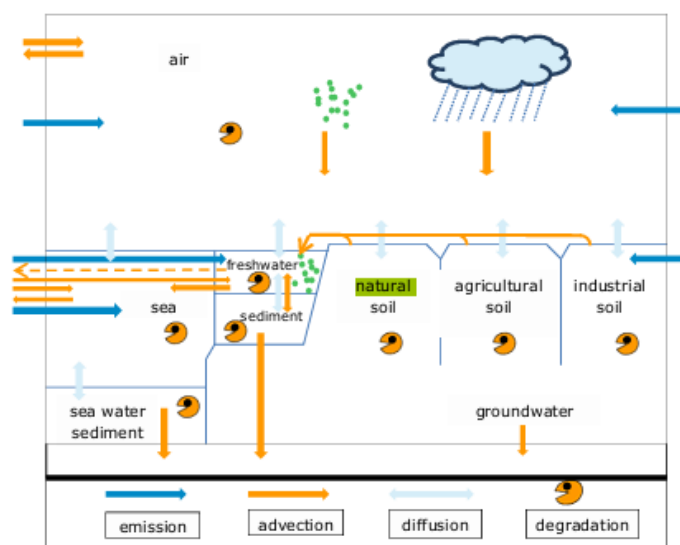
| INDUSTRIAL USE: production of Plastic Articles                                                                                   |                     |          |              |          |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|--------------|----------|
|                                                                                                                                  | ES12                |          | ES13         |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | 3.73E-04            |          | No emission* |          |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  | 3.48E-05            |          |              |          |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          | 3.70E-02            |          |              |          |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            | 6.03E-02            |          |              |          |
| INDUSTRIAL USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics        |                     |          |              |          |
|                                                                                                                                  | ES14                |          |              |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | No emission*        |          |              |          |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  |                     |          |              |          |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          |                     |          |              |          |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            |                     |          |              |          |
| WIDE DISPERSIVE USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics** |                     |          |              |          |
|                                                                                                                                  | Sum of ES15 to ES18 |          |              |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | No emission*        |          |              |          |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  |                     |          |              |          |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          |                     |          |              |          |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            |                     |          |              |          |
| WIDE DISPERSIVE USE: finished cosmetic products**                                                                                |                     |          |              |          |
|                                                                                                                                  | Sum of ES19 to ES23 |          |              |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | 1.78E-02            |          |              |          |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  | 1.66E-03            |          |              |          |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          | 1.76E+00            |          |              |          |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            | 2.87E+00            |          |              |          |
| SERVICE LIFE: plastic articles**                                                                                                 |                     |          |              |          |
|                                                                                                                                  | Sum of ES24 to ES27 |          |              |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | 2.56E-06            | 2.56E-06 | 1.28E-06     | 8.18E-05 |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  | 2.38E-07            | 2.38E-07 | 1.19E-07     | 7.62E-06 |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          | 2.53E-04            | 2.53E-04 | 1.27E-04     | 8.11E-03 |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            | 4.13E-04            | 4.13E-04 | 2.07E-04     | 1.32E-02 |
| SERVICE LIFE: articles including plastic articles**                                                                              |                     |          |              |          |
|                                                                                                                                  | Sum of ES28 to ES31 |          |              |          |
| PEC <sub>STP</sub> (mg.l <sup>-1</sup> )                                                                                         | 5.11E-11            | 2.56E-06 | 5.11E-11     | 1.64E-04 |
| PEC <sub>freshwater</sub> (mg.l <sup>-1</sup> )                                                                                  | 4.76E-12            | 2.38E-07 | 4.76E-12     | 1.52E-05 |
| PEC <sub>freshwater sediment</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                          | 5.07E-09            | 2.53E-04 | 5.07E-09     | 1.62E-02 |
| PEC <sub>agricultural soil</sub> (mg.kg <sub>ww</sub> <sup>-1</sup> )                                                            | 8.26E-09            | 4.13E-04 | 8.26E-09     | 2.64E-02 |

\* No emission: no release to sewage treatment plant (STP)

\*\* According to R16 guidance, the wide dispersive use and service-life have to be summed to estimate  $PEC_{local}$

#### 1.4.3.2.2. Tonnage approach - Regional exposure assessment

The concentrations of octocrilene released from all sources in a larger area are assessed for a generic regional environment. The distribution and fate of the substance further than in the vicinity of the release are taken into account. The regional concentrations are used as background concentrations for the local scale. Regional computations are done by means of multimedia fate models based on the fugacity concept (EUSES, v.2.2.0, based on Mackay concepts). The results from the model are steady-state concentrations, which can be regarded as estimates of long-term average exposure levels.



**Figure 5: regional calculations<sup>22</sup>.**

Details of how regional PECs are calculated are given in Appendix B.8.3.2.

The regional predicted environmental concentrations ( $PEC_{regional}$ ) are presented in the Table 9. The Regional PEC per compartment is based on the sum of the regional total release of all environmental emission scenarios and all CSRs.

**Table 9:  $PEC_{regional}$**

| Protection target        | Regional PEC                |
|--------------------------|-----------------------------|
| Surface water            |                             |
| Freshwater (dissolved)   | 1.15E-04 mg.l <sup>-1</sup> |
| Marine water (dissolved) | 6.86E-06 mg.l <sup>-1</sup> |

<sup>22</sup> Source: ECHA Guidance on information requirements and chemical safety assessment, chapter R.16: Environmental exposure assessment, figure R.16-5

|                                                                          |                                                                                            |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Sediment<br>Freshwater sediment (total)<br>Marine water sediment (total) | 2.11E-01 mg.kg <sub>wwt</sub> <sup>-1</sup><br>7.91E-03 mg.kg <sub>wwt</sub> <sup>-1</sup> |
| Soil<br>Natural soil                                                     | 1.80E-02 mg.kg <sub>wwt</sub> <sup>-1</sup>                                                |

The PEC values are particularly high for the background concentrations on a regional scale.

However, it seems unrealistic to consider all industrial sites (manufacture, formulation, industrial use) from all registrants as sources in a unique region.

Therefore, a more realistic regional scenario is developed with a focus on the main release pathway (i.e., via the wastewater) and the most impacting use, the wide-dispersive use of cosmetic products. In fact, based on the tonnage data, **82% of the total regional release to wastewater and surface water comes from the wide dispersive use of cosmetic products**. This is also highlighted below by the higher risks identified for the use of cosmetics than any other. As a refinement, all regional releases associated with the different identified uses of cosmetic products, for both industrial and wide disperse sources, are cumulated to estimate the total regional release (kg/day) to surface water, wastewater, air and soil.

The PEC regional values (considering only the identified uses of cosmetic products) are given in the Table 10.

**Table 10: PEC regional associated for uses of cosmetic products (industrial and wide disperse sources)**

| Protection target                                        | Regional PEC<br>Dispersive uses of cosmetic products                                       |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Surface water<br>Freshwater<br>Marine water              | 9.39E-05 mg.l <sup>-1</sup><br>5.61E-06 mg.l <sup>-1</sup>                                 |
| Sediment<br>Freshwater sediment<br>Marine water sediment | 1.73E-01 mg.kg <sub>wwt</sub> <sup>-1</sup><br>6.47E-03 mg.kg <sub>wwt</sub> <sup>-1</sup> |
| Soil<br>Natural soil                                     | 1.40E-02 mg.kg <sub>wwt</sub> <sup>-1</sup>                                                |

Results show PECs values very close to the values calculated for a region including all emission scenarios – all industrial sites and widespread sources. **This demonstrates the major significance of emissions related to the use of cosmetics in regional overall emissions into the environment.**

#### 1.4.3.2.3. Tonnage approach – Environmental risk assessment

##### 1.4.3.2.3.1. Local scale assessment

The risk characterisation ratios (RCR) can be derived in order to decide if risks are adequately controlled for each environmental sphere. RCRs are derived by comparing predicted environmental concentrations (PECs) to suitable predicted no-effect concentrations (PNECs). RCRs at the local scale for each use and each emission scenario

(ES<sup>23</sup>) are summarised in the Table 11.

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<sup>23</sup> ES: emission scenario. An ES is defined for each use declared by a registrant. This is done for all registrants

**Table 11: Local RCR values**

| MANUFACTURE                                                                                                                      |                     |     |              |              |      |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------|-----|--------------|--------------|------|
|                                                                                                                                  | ES1                 |     | ES2          |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | 2.43E-02            |     | 8.53E-04     |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    | 8.19E+01            |     | 9.38E-02     |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           | 1.16E+02            |     | 1.33E-01     |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             | NR                  |     | NR           |              |      |
| FORMULATION: production of plastisol                                                                                             |                     |     |              |              |      |
|                                                                                                                                  | ES3                 |     | ES4          |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | 4.11E-04            |     | No emission* |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    | 1.44E+00            |     |              |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           | 2.03E+00            |     |              |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             | 3.85E-01            |     |              |              |      |
| FORMULATION: preparations containing additives                                                                                   |                     |     |              |              |      |
|                                                                                                                                  | ES5                 |     |              |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | No emission*        |     |              |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    |                     |     |              |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           |                     |     |              |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             |                     |     |              |              |      |
| FORMULATION: pre-mixture for finished cosmetic products                                                                          |                     |     |              |              |      |
|                                                                                                                                  | ES6                 |     |              |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | 2.48E-04            |     |              |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    | 2.72E-02            |     |              |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           | 3.85E-02            |     |              |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             | NR                  |     |              |              |      |
| FORMULATION: finished cosmetic products                                                                                          |                     |     |              |              |      |
|                                                                                                                                  | ES7                 | ES8 | ES9          | ES10         | ES11 |
| RCR <sub>STP</sub> (-)                                                                                                           | No emission*        |     | 1.84E-04     | No emission* |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    |                     |     | 6.46E-01     |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           |                     |     | 9.14E-01     |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             |                     |     | 1.73E-01     |              |      |
| INDUSTRIAL USE: production of Plastic Articles                                                                                   |                     |     |              |              |      |
|                                                                                                                                  | ES12                |     | ES13         |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | 3.73E-05            |     | No emission* |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    | 1.31E-01            |     |              |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           | 1.85E-01            |     |              |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             | 3.51E-02            |     |              |              |      |
| INDUSTRIAL USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics        |                     |     |              |              |      |
|                                                                                                                                  | ES14                |     |              |              |      |
| RCR <sub>STP</sub> (-)                                                                                                           | No emission*        |     |              |              |      |
| RCR <sub>freshwater</sub> (-)                                                                                                    |                     |     |              |              |      |
| RCR <sub>freshwater sediment</sub> (-)                                                                                           |                     |     |              |              |      |
| RCR <sub>agricultural soil</sub> (-)                                                                                             |                     |     |              |              |      |
| WIDE DISPERSIVE USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics** |                     |     |              |              |      |
|                                                                                                                                  | Sum of ES15 to ES18 |     |              |              |      |

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|                                                     |                      |
|-----------------------------------------------------|----------------------|
| RCR <sub>STP</sub> (-)                              | No emission*         |
| RCR <sub>freshwater</sub> (-)                       |                      |
| RCR <sub>freshwater sediment</sub> (-)              |                      |
| RCR <sub>agricultural soil</sub> (-)                |                      |
| WIDE DISPERSIVE USE: finished cosmetic products**   |                      |
|                                                     | Sum of ES19 to ES 23 |
| RCR <sub>STP</sub> (-)                              | 1.78E-03             |
| RCR <sub>freshwater</sub> (-)                       | 6.14E+00             |
| RCR <sub>freshwater sediment</sub> (-)              | 8.83E+00             |
| RCR <sub>agricultural soil</sub> (-)                | 1.67E+00             |
| SERVICE LIFE: plastic articles**                    |                      |
|                                                     | Sum of ES24 to ES 27 |
| RCR <sub>STP</sub> (-)                              | 8.82E-06             |
| RCR <sub>freshwater</sub> (-)                       | 3.08E-02             |
| RCR <sub>freshwater sediment</sub> (-)              | 4.37E-02             |
| RCR <sub>agricultural soil</sub> (-)                | 8.28E-03             |
| SERVICE LIFE: articles including plastic articles** |                      |
|                                                     | Sum of ES28 to ES31  |
| RCR <sub>STP</sub> (-)                              | 1.67E-05             |
| RCR <sub>freshwater</sub> (-)                       | 5.82E-02             |
| RCR <sub>freshwater sediment</sub> (-)              | 8.24E-02             |
| RCR <sub>agricultural soil</sub> (-)                | 1.56E-02             |

\* No emission: no release to sewage treatment plant (STP)

\*\* According to R16 guidance, the wide dispersive use and service-life have to be summed to estimate PEC<sub>local</sub>

From results given in the table above, environmental risks for the aquatic compartment (including sediment) are identified at the local scale for the following steps:

- ✓ **Manufacture of octocrilene** for one registrant. No risks are observed for the second registrant. The main difference lies in the choice of the fraction of release to water for the manufacture plant. The first registrant takes into account the value (2%) provided in the Technical Guidance Document on Risk Assessment, Part II (2003) (section IC 2: Chemical Industry: Basic Chemicals (Production), table A1.1 (All MCs, Tonnage <1000 tonnes/year). The second registrant considers a refinement of this value at 0.01% from site-specific information based on the very low solubility of the octocrilene.
- ✓ **Formulation: production of plastisol** for one registrant. No risks are observed for the second registrant due to the application of risk management measure (RMM), *i.e.*, incineration of the washing liquid and exclusion of any release to wastewater.
- ✓ **Wide dispersive use: use of finished cosmetic products.** Concerning the use of cosmetic products by consumers and considering that it's a wide dispersive use, all the tonnage values must be summed up. The results of the risk assessment using the aggregated tonnage for the cosmetic use (presented in the Table 11) show unacceptable risks for the aquatic and also for soil compartments.

While risks for manufacture and formulation of plastisol can be managed (RMM and OC as defined elsewhere), those arising from the cosmetic uses cannot as they are wide dispersive. Additionally, at the local level, the risks from the use of cosmetics treated by octocrilene are much higher than the other releases. As a reminder, 82% of the total regional release to wastewater and surface water comes from the wide dispersive use of

cosmetic products.

#### 1.4.3.2.3.2. Regional scale assessment

The risk characterisation ratios (RCR) at the regional scale are summarised in the Table 12.

**Table 12: Regional RCRs calculated from the sum of the regional total release of all environmental emission scenarios and all CSRs.**

| Protection target                                                        | Risk characterisation |
|--------------------------------------------------------------------------|-----------------------|
| Surface water<br>Freshwater (dissolved)<br>Marine water (dissolved)      | 0.43<br>0.25          |
| Sediment<br>Freshwater sediment (total)<br>Marine water sediment (total) | <b>1.06</b><br>0.40   |
| Soil<br>Natural soil                                                     | 1.05E-02              |

For most of the compartments (except for the biological STP, according to the R16 guidance), the PEC regional must be integrated in the calculations of the PEC at the local scale. However, if we consider regional risk characterisation ratios related to sum of the regional total release of all environmental emission scenarios and all CSRs (table above), **unacceptable risks are already identified for the sedimentary compartment**. It is therefore not necessary to sum the PEC<sub>regional</sub> (associated to all environmental emission scenarios and all CSRs) to the local PECs to carry out additional risks' characterisations.

Considering as explained above that the wide-dispersive use of cosmetic products is the most impacting use, a focus to cosmetic uses has been carried out.

The Regional RCRs for the use of cosmetic products (industrial and wide disperse sources) are given in the Table 13.

**Table 13: RCR associated for widespread use of cosmetic products (industrial and wide disperse sources).**

| Protection target                                       | Risk characterisation |
|---------------------------------------------------------|-----------------------|
| Surface water<br>Freshwater<br>Marine water             | 0.35<br>0.21          |
| Sediment<br>Freshwater sediment<br>Marinewater sediment | 0.86<br>0.32          |
| Soil<br>Natural soil                                    | 8.14E-03              |

#### 1.4.3.2.3.3. Focus on Cosmetic uses

Considering, the major significance of emissions related to the use of cosmetics in the regional overall emissions into the environment, the regional RCRs for the use of cosmetic products (industrial and wide disperse sources) are given in the table below.

Focusing on consumer use of cosmetic products, the RCRs regional (Table 13) have been integrated in the calculations of the RCR at the local scale for use of cosmetic products (Table 11, **Sum of ES19 to ES 23**) by consumers in the Table 14:

**Table 14: RCR values from the tonnage approach for wide dispersive use of cosmetic**

**products by consumers.**

| Protection target     | Risk characterisation |
|-----------------------|-----------------------|
| Surface water         |                       |
| Freshwater            | <b>6.57</b>           |
| Marine water          | <b>6.35</b>           |
| Sediment              |                       |
| Freshwater sediment   | <b>9.69</b>           |
| Marine water sediment | <b>9.15</b>           |
| Agricultural soil     | 1.68                  |

**Wide dispersive use of octocrilene in finished cosmetic products leads to risks for all environmental compartments.****1.4.3.3. Refined approach for the dispersive use of cosmetics end-products - Environmental exposure assessment and associated risk**

As risks are demonstrated from the wide dispersive use of finished cosmetic products in the generic approach for risks assessment, the Dossier submitter also performed a specific and refined approach based on the methodology applied to determine the risks for other cosmetic products (D4, D5, D6 restriction) and on specific consumer rate for sunscreen.

In view of the unacceptable risks for the environment related to the use of **cosmetics end-products** as described in above section, it seems essential to tend towards a more realistic approach. To achieve this, two strategies will be developed here:

- ✓ A second-tier tonnage approach based on the evaluation model applied in the annex XV restriction report dossier for D4, D5, D6 substances;
- ✓ An approach based on consumer rates.

The two approaches are described in the Figure 6.

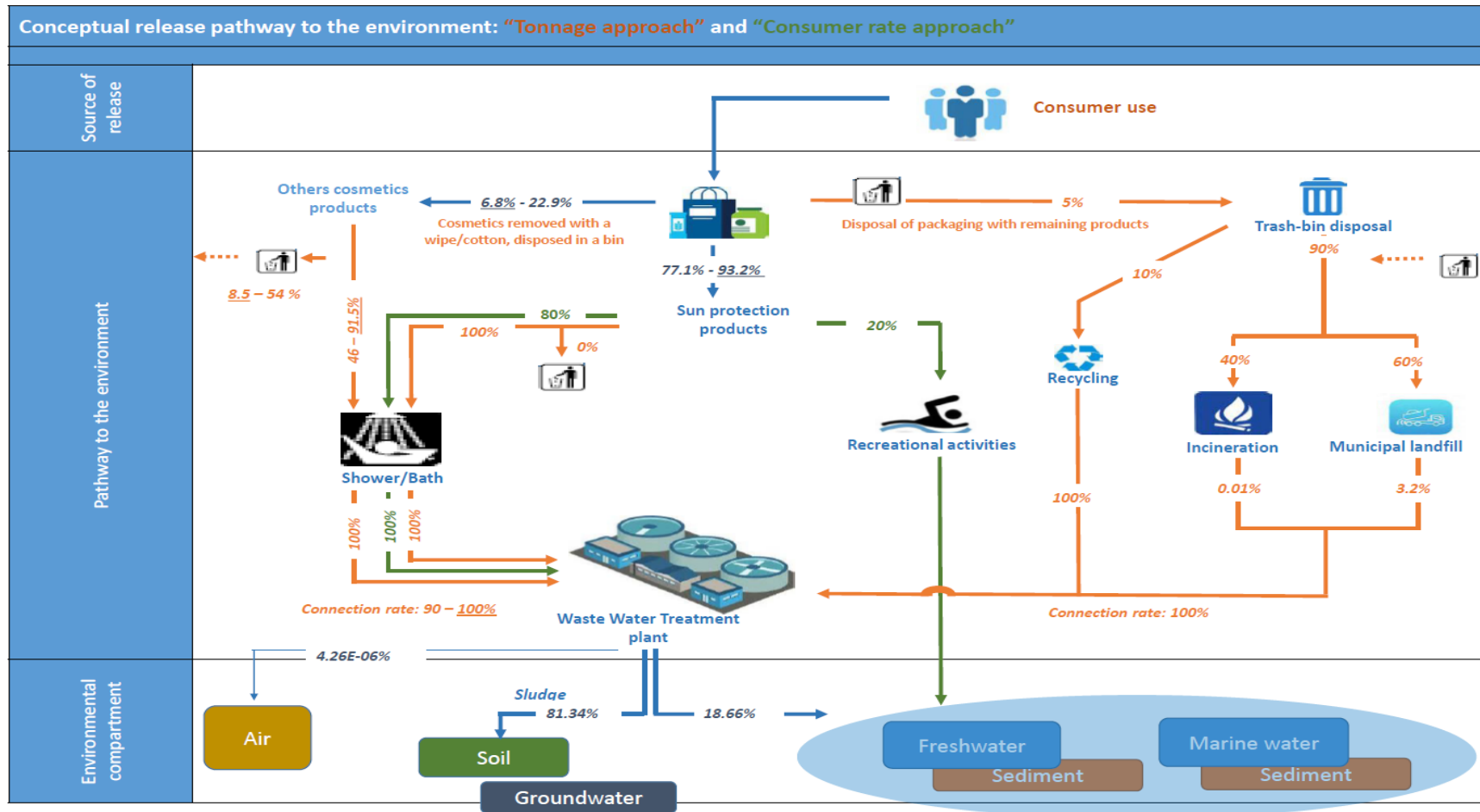
**1.4.3.3.1. Refined tonnage approach**

The approach proposed here is based on the evaluation model applied in the annex XV restriction report dossier for D4, D5, D6 substances. Full description is available in Appendix B.8.4.

The calculation of releases to the aquatic compartment for the various consumer uses are based on the following elements:

- ✓ The quantity of octocrilene used;
- ✓ The consumer behavior in terms of product used (that defines how products are removed from skin and disposed after);
- ✓ The consumer behavior in terms of disposal after use of unused/out of date products, or product packaging, that contains residual octocrilene;
- ✓ Waste water treatment plant connection rate;
- ✓ The waste disposal route for articles and product packaging containing residual octocrilene (e.g., municipal landfill, municipal incineration or recycling)

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**Figure 6: Conceptual release pathways for wide dispersive cosmetics uses.**

Removal of cosmetics and other products applied on skin/body

Leave-on products (7-23% of products) are intended to stay for a certain period of time in contact with the skin, the hair or the mucous membranes. Leave-on products include a broad range of different products such as lipstick, face-care products, body-cream, nail-care products, etc. Depending on the type of products, consumers use different techniques to remove the products from their body, skin, or mucous membranes. The removal methods impact directly the releases of octocrilene to the environment (directly into water, on a wipe, on cotton). Octocrilene in water (46-91%) from other cosmetic products than sunscreens will end up in drain and directed to wastewater treatment plan (when connection is available) and solid waste will be directed to recycling facility and/or in municipal land-fill and/or incineration facility.

In the case of sunscreen products (77 to 93% percent of octocrilene containing products), it is estimated that 20% of the octocrilene quantity will be directly released through wash off of the cosmetic product, or light abrasion of the skin, to water or surrounding area (beaches, shores, river bank). Another fraction will be removed from the skin after shower/bath, and water effluent will be directed to WWTP depending on the connection rate of the considered area (80%).

Most of the time cosmetics are not fully used before their expiry date and the empty packaging, which is disposed of, includes a residual amount of the product. In the CSRs, it was estimated that around 5% of the remaining quantity of cosmetic products containing octocrilene are disposed of in a trash-bin. It is assumed that 10% of the cosmetic packaging are recycled. During the first steps of the recycling of the uncleaned packaging (*i.e.*, shredding/washing of plastic materials), the product is discharge to water. Discarded cosmetic product packaging that is not recycled will be either incinerated (40%) or disposed in landfill (60%).

Tonnage value of octocrilene used by consumer of finished cosmetic products

Due to the discrepancies between CSRs and the data gathered during the CfE (also provided by non registrants), the Dossier Submitter has decided to focus their calculations on the amount of octocrilene declared to be used for cosmetics products from CSRs and compared this value to the data retrieved from Euromonitor<sup>24</sup> to ensure the robustness of the estimates. Euromonitor database allows to have the amount of octocrilene in cosmetic products really put on the market and sold in the European Union (the data available for the considered year of 2023 do not cover Cyprus, Iceland, Luxembourg, Malta, Norway and Liechtenstein). This approach is considered as being the most conservative and close to the reality of the market.

In conclusion for the quantities used in cosmetics products, the Dossier Submitter estimated that the annual amount of octocrilene for cosmetic uses is of **1752 t/y**, estimation confirmed with the Euromonitor data (1721 t/y).

If we relate this value of 1752 t/y to the various percentages proposed in the figure 10 for each pathway to the environment, depending on the type of product (sun protection products and other cosmetic products), two values (**low value: 1473 t/y and high value: 1665 t/y**) are defined for the tonnage entering in the waste water treatment plant.

The Figure 7 below describes the tonnage values for each release pathway of octocrilene to the environment through the use of cosmetic products.

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<sup>24</sup> [Euromonitor, 2025](#) consulted on 15/01/25

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For this scenario, the parameters used to estimate local PEC values after release to a municipal WWTP were detailed in Appendix B.8.4.1.

The following estimates were retrieved from the performed calculations (no regional background concentrations are taking into account here):

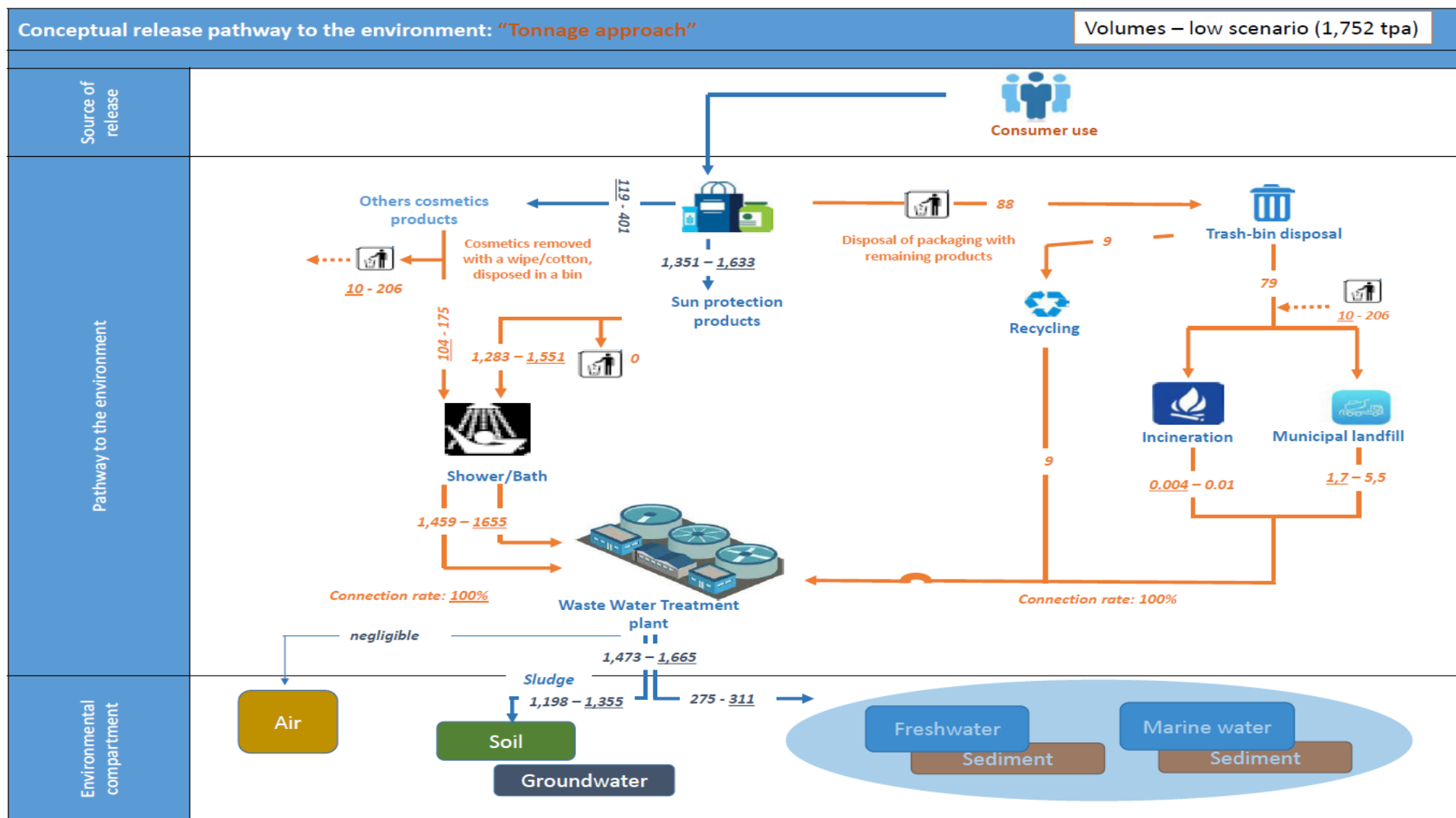
**Table 15: PEC<sub>local</sub> and associated RCR for widespread use of cosmetic products.**

| Protection target                                | Local PEC                               | Risk characterisation |
|--------------------------------------------------|-----------------------------------------|-----------------------|
| STP                                              |                                         |                       |
| Low value                                        | 1.50E-02                                | 1.50E-03              |
| High value                                       | 1.69E-02                                | 1.69E-03              |
| Freshwater (covering marine water)*              |                                         |                       |
| Low value                                        | 1.39E-03 mg.l <sup>-1</sup>             | <b>5.24</b>           |
| High value                                       | 1.58E-03 mg.l <sup>-1</sup>             | <b>5.93</b>           |
| Freshwater sediment (covering marine sediment) * |                                         |                       |
| Low value                                        | 1.48 mg.kg <sub>wwt</sub> <sup>-1</sup> | <b>7.42</b>           |
| High value                                       | 1.68 mg.kg <sub>wwt</sub> <sup>-1</sup> | <b>8.38</b>           |
| Agricultural soil                                |                                         |                       |
| Low value                                        | 2.42 mg.kg <sub>wwt</sub> <sup>-1</sup> | <b>1.41</b>           |
| High value                                       | 2.73 mg.kg <sub>wwt</sub> <sup>-1</sup> | <b>1.59</b>           |

\* No specific PNECs are available for the marine ecosystem. Because an additional dilution factor of ten is applied to both PEC and PNEC in case of releases *via* the WWTP, the risk ratios for the marine compartment are always equal to those for fresh water and fresh water sediment. Considering that the risk assessment for marine water is covered by the assessment for fresh water, PECs and PEC:PNECs ratios were not calculated for the marine ecosystem.

**Results show unacceptable risks for the aquatic compartment including sediment and for soil whatever the range of tonnage value of finished cosmetic products reaching WWTP.**

The tonnage approach used above does not integrate direct emissions in the environment, for instance during bathing activities. This will be assessed below in the consumer approach.



**Figure 7: Conceptual release pathway of octocrilene to the environment through the use of cosmetic products (releases in tonnes per year determined through a tonnage approach).**

#### 1.4.3.3.2. Consumer approach for sunscreens

The second strategy developed by the Dossier Submitter consists in a consumer approach, based on the ESD PT19: Emission scenarios for repellents and attractants (ECHA, 2015). It is worth noting that the consumer approach is only relevant for sunscreens and not for all cosmetic products (due to a lack of input data).

Octocrilene is used in various consumer products in particular finished cosmetic products. Its use results in several potential routes of entry into the environment:

- ✓ *via* rinse-off from bathers in both fresh surfacewaters and marine coastal waters;
- ✓ *via* municipal wastewater from homes, hotels, and other buildings (sunscreens rinse-off from shower/bathing after use of sunscreen for daily activities or outdoor recreation).

81% of the EU population was connected to wastewater treatment (at least secondary wastewater treatment) in 2022<sup>25</sup>.

Other entries into the environment exist but cannot be estimated in the dossier by lack of scenario and data:

- ✓ *via* stormwater from combined sewer overflows;
- ✓ *via* runoff from beach areas;
- ✓ *via* leaching into surface waters from on-site wastewater systems.

UV filters may also be released into the environment during sunscreen application, particularly if aerosol/spray formulations are used. Application near water (i.e., on the beach) may result in aerosol dispersion with direct deposition onto water, deposition and binding to sand where it may run off into the water during flushing events (i.e., rainfall or tides). For example, one study reports that 50 to 60 percent of the sunscreen used during application as spray on skin may be wasted directly to the close environment (*Broussard et al., 2019*).

An analysis of input parameters is carried out and fully available in Appendix B.9.4.2.

##### 1.4.3.3.2.1. Swimming scenario - Emissions via the rinse-off from swimmers in both fresh surfacewaters and marine coastal waters

Two scenarios have been used considering two distinct pairs (number of swimmer/volume of water body). The scenario 1 is a realistic worst-case scenario for standing inland water bodies (ponds, shallow lakes, lakes). The scenario 2 cover marine coastal waters but cannot be considered as a realistic worst-case scenario in the absence of more robust data on the volume of water bodies related to a number of swimmers.

Considering that wastewater effluents are released in rivers, it was considered that the scenario “*via* municipal wastewater” covered releases into rivers.

From the input values determined by the registrants, the scenarios for direct release of UV filters during recreational activities is detailed in Table 16. From this table, the parameters driving the PECs are the number of swimmers and the volume of water body. The

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<sup>25</sup> [Population connected to at least secondary wastewater treatment \(Indicator\) | European zero pollution dashboards](#)

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concentrations in water were calculated for one day and after a bathing season of 3 months considering the dissipation of the substance in the environment.

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**Table 16: Direct Release of UV Filters during Recreational activities.**

| Input                                                                        |                             | Scenario1<br>Inland water bodies<br>(ponds, shallow lakes,<br>lakes) | Scenario2<br>Swimming area of<br>the Marine coastal<br>waters |                       |                                                                                                                                           |
|------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Variable                                                                     | Symbol                      | Value                                                                | Value                                                         | Unit                  |                                                                                                                                           |
| Concentration of UV-filter in sun screen product <sup>2</sup>                | C <sub>product</sub>        | 10                                                                   | 10                                                            | %                     |                                                                                                                                           |
| Quantity of sunscreen lotion applied per day by a consumer<br>(Median value) | Q <sub>sunscreen_appl</sub> | 18                                                                   | 18                                                            | [g.d <sup>-1</sup> ]  |                                                                                                                                           |
| Daily number of swimmers                                                     | N <sub>swimmer</sub>        | 1500                                                                 | 3000                                                          | [-]                   |                                                                                                                                           |
| Volume of water body <sup>1</sup>                                            | V <sub>waterbody</sub>      | 435000                                                               | 4530                                                          | [m <sup>3</sup> ]     |                                                                                                                                           |
| Fraction of swimmers using sunscreen product <sup>2</sup>                    | F <sub>swim_sunscreen</sub> | 0.7                                                                  | 0.7                                                           | [-]                   |                                                                                                                                           |
| Fraction of sunscreen products with OC                                       | F <sub>sunscreen_OC</sub>   | 0.36                                                                 | 0.36                                                          | [-]                   |                                                                                                                                           |
| Fraction of OC released into the environment                                 | F <sub>after_swimming</sub> | 0.2                                                                  | 0.2                                                           | [-]                   |                                                                                                                                           |
| First order rate constant for biodegradation in surface water                | K <sub>degwater</sub>       | 1.54E-02                                                             | 1.47E-02                                                      | [d <sup>-1</sup> ]    |                                                                                                                                           |
| Number of emission days (emission period of 1 day) <sup>2</sup>              | T <sub>emission,1d</sub>    | 1                                                                    | 1                                                             | [d]                   |                                                                                                                                           |
| Number of emission days (emission period of 91 days) <sup>2</sup>            | T <sub>emission,91d</sub>   | 91                                                                   | 91                                                            | [d]                   |                                                                                                                                           |
| Output                                                                       |                             | Scenario1                                                            | Scenario2                                                     |                       |                                                                                                                                           |
| Variable                                                                     | Symbol                      | Value                                                                | Value                                                         | Unit                  | Calculation formula                                                                                                                       |
| Local emission rate to water                                                 | E <sub>localwater</sub>     | <b>1.36E-01</b>                                                      | <b>2.72E-01</b>                                               | [kg.d <sup>-1</sup> ] | $E_{localwater} = C_{product} * Q_{sunscreen\_appl} * N_{swimmer} * F_{swim\_sunscreen} * F_{sunscreen\_OC} * F_{after\_swimming} / 10^3$ |
| Local concentration in water body after one day                              | C <sub>localwater,1d</sub>  | <b>2.91E-04</b>                                                      | <b>5.59E-02</b>                                               | [mg.l <sup>-1</sup> ] | $C_{localwater,1d} = E_{localwater} * 10^3 * T_{emission,1d} / ((1 + K_{psusp} * SUSP_{water} * 10^{-6}) * V_{waterbody})$                |
| Local concentration in water body over 91 days                               | C <sub>localwater,91d</sub> | <b>2.65E-02</b>                                                      | <b>5.09E+00</b>                                               | [mg.l <sup>-1</sup> ] | $C_{localwater,1d} = E_{localwater} * 10^3 * T_{emission,91d} / ((1 + K_{psusp} * SUSP_{water} * 10^{-6}) * V_{waterbody})$               |

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| Refined local concentration in water body over 91 days (including degradation) | $C_{local,water,91d-ref}$ | <b>1.44E-02</b> | <b>2.82E+00</b> | [mg.l <sup>-1</sup> ] | $C_{local,water,91d-ref} = C_{local,water,1d} * \frac{[1-(e^{-k_{degwater} * T_{emission,1d}})^{91}]}{[1-e^{-k_{degwater} * T_{emission,1d}}]}$ |
|--------------------------------------------------------------------------------|---------------------------|-----------------|-----------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Risk characterisation water                                                    | RCR 1 day                 | <b>1.10</b>     | <b>2072</b>     | [-]                   |                                                                                                                                                 |
| Risk characterisation water (including degradation)                            | RCR 91 days               | <b>54</b>       | <b>104544</b>   | [-]                   |                                                                                                                                                 |
| Risk characterisation sediment                                                 | RCR 1 day                 | <b>1.55</b>     | <b>2975</b>     | [-]                   |                                                                                                                                                 |
| Risk characterisation sediment (including degradation)                         | RCR 91 days               | <b>76.4</b>     | <b>150137</b>   | [-]                   |                                                                                                                                                 |

1: the volume water bodies are defined as follow: for Inland, total water volume is considered while for marine costal water, this is approximately limited to the swimming area.

2: realistic worst case

**Results show unacceptable risks for the aquatic compartment including sediment whatever the scenario considered.** In order to ensure scenario as realistic as possible and covering the maximum of situations encountered in Europe, the Dossier Submitter also performed a refinement of these scenarios by excluding water bodies with minimal water exchange (e.g., small lakes or shallow, enclosed coastal bays, lagoon<sup>26</sup>...).

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<sup>26</sup> For lagoons, inflow of oceanic water is sometimes very low or discontinues, so that outflows are negligible and less than 10-20% of the incoming volume.

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**Table 17: Refinement of direct release of UV filters during swimming activities.**

| Input refinement                                                 |                                    | Scenario1<br>Inland water<br>bodies<br>(ponds,<br>shallow lakes,<br>lakes) | Scenario2<br>Marine coastal<br>waters |                       |                                                                                                                                                      |
|------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------|---------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable                                                         | Symbol                             | Value                                                                      | Value                                 | Unit                  |                                                                                                                                                      |
| Tidal period (h)                                                 | Tp                                 | 12.41                                                                      | 12.41                                 | [h]                   |                                                                                                                                                      |
| Dilution per tide (%)                                            | DIL <sub>tide</sub>                | 50                                                                         | 50                                    | [%]                   |                                                                                                                                                      |
| Rate constant for dilution per tide                              | k <sub>dilution tide</sub>         | 1.34E+00                                                                   | 1.34E+00                              | [d <sup>-1</sup> ]    |                                                                                                                                                      |
| Output refinement                                                |                                    | Scenario1                                                                  | Scenario2                             |                       |                                                                                                                                                      |
| Variable                                                         | Symbol                             | Value                                                                      | Value                                 | Unit                  | Calculation formula                                                                                                                                  |
| Local concentration in water body after one day and Tidal period | C <sub>localwater,1d, Tidal</sub>  | 7.62E-05                                                                   | 1.46E-02                              | [mg.l <sup>-1</sup> ] | $C_{localwater,1d, Tidal} = C_{localwater,1d} * EXP(-k_{dilution tide} * 1)$                                                                         |
| Local concentration in water body over 91 days and tidal period  | C <sub>localwater,91d, Tidal</sub> | 3.95E-04                                                                   | 7.58E-02                              | [mg.l <sup>-1</sup> ] | $C_{localwater,91d, Tidal} = C_{localwater,91d} * ((1-(EXP(-k_{dilution tide} * T_{emission,1d}))^9)/(1-EXP(-k_{dilution tide} * T_{emission,1d})))$ |
| Risk characterisation water                                      | RCR 1 day                          | <b>0.29</b>                                                                | <b>542</b>                            | [-]                   |                                                                                                                                                      |
| Risk characterisation water                                      | RCR 91 days                        | <b>1.48</b>                                                                | <b>2806</b>                           | [-]                   |                                                                                                                                                      |
| Risk characterisation sediment                                   | RCR 1 day                          | <b>0.41</b>                                                                | <b>779</b>                            | [-]                   |                                                                                                                                                      |
| Risk characterisation sediment                                   | RCR 91 days                        | <b>2.10</b>                                                                | <b>4030</b>                           | [-]                   |                                                                                                                                                      |

**Even with a refinement excluding water bodies with minimal water exchange, results show unacceptable risks for inland water bodies and marine coastal environment for the water and/or the sediment whatever the scenario.**

#### 1.4.3.3.2.2. Showering scenario – Emissions via the sunscreen rinse-off from showering/bathing (released to the municipal wastewaters)

The Dossier Submitter also performed an estimation of risks for the environment arising through the emissions *via* the sunscreen rinse-off from showering/bathing (released to the municipal wastewaters) using specific inputs values based on sunscreen consumption.

**Table 18: Removal through showering and bathing and releases to WWTP**

| Input                                                      |                         |                 |                           |
|------------------------------------------------------------|-------------------------|-----------------|---------------------------|
| <i>Variable</i>                                            | <i>Symbol</i>           | <i>Value</i>    | <i>Unit</i>               |
| Concentration of UV-filter in sun screen product           | Cproduct                | 10              | [%]                       |
| Quantity of sunscreen lotion applied per day by a consumer | Qsunscreen_appl         | 18              | [g.d <sup>-1</sup> ]      |
| Number of inhabitants feeding one sewage treatment plant   | Nlocal                  | 10000           | [cap]                     |
| Fraction of inhabitants with sunscreen products            | Finhab_sunscreen        | 0.7             | [-]                       |
| Fraction of sunscreen products with OC                     | Fsunscreen_OC           | 0.36            | [-]                       |
| Fraction of OC released into the wastewater                | Fwastewater             | 0.2             | [-]                       |
| Output                                                     |                         |                 |                           |
| <i>Variable/parameter</i>                                  | <i>Symbol</i>           |                 | <i>Unit</i>               |
| Local emission rate to water                               | Elocal <sub>water</sub> | <b>9.07E-01</b> | [kg.d <sup>-1</sup> ]     |
| PEC <sub>STP</sub>                                         |                         | <b>8.46E-02</b> | [mg.l <sup>-1</sup> ]     |
| RCR <sub>STP</sub>                                         |                         | <b>8.46E-03</b> | [-]                       |
| PEC freshwater                                             |                         | <b>7.88E-03</b> | [mg.l <sup>-1</sup> ]     |
| RCR freshwater                                             |                         | <b>2.96E+01</b> | [-]                       |
| PEC freshwater sediment                                    |                         | <b>8.39E+00</b> | [mg.kgwwt <sup>-1</sup> ] |
| RCR freshwater sediment                                    |                         | <b>4.19E+01</b> | [-]                       |
| PEC agricultural soil                                      |                         | <b>1.37E+01</b> | [mg.kgwwt <sup>-1</sup> ] |
| RCR agricultural soil                                      |                         | <b>7.97E+00</b> | [-]                       |

**Results show unacceptable risks for freshwater and associated sediments and for soil arising after removal of sunscreen through showering and bathing.**

#### 1.4.3.4. Overall conclusion on the risk assessment

Environmental risks for the aquatic compartment (including sediment) are identified based on the classical tonnage approach at the local scale of the following steps:

- ✓ **Manufacture of octocrilene** only for one registrant. No risks are observed for the second registrant due to a refinement of the release factor to water considering the very low solubility of the octocrilene.
- ✓ **Formulation: production of plastisol** but only for one registrant. No risks are observed for the second registrant due to the application of risk management measure (RMM: incinerate the washing liquid and exclude any release to wastewater);
- ✓ **Wide dispersive use: finished cosmetic products** once all the scenarios have been summed up. Risks for soil's compartment were also demonstrated.

A refinement of the tonnage approach for the use of cosmetic products was performed and unacceptable risks for the aquatic compartment (including sediment) and soil were shown, taking into account the annual amount of octocrilene declared by the registrant's of 1752 t/y (estimation confirmed with the data from Euromonitor (1721 t/y)).

Moreover, consumer approaches covering the direct releases *via* the swimming activities and only relevant for sunscreens products, were developed and clearly showed unacceptable risks for inland water bodies (ponds, shallow lakes and lakes) and for marine coastal waters. When the water exchanges are taken into account, the risks remain unacceptable for aquatic and/or sedimentary compartments for water bodies with open water exchange (e.g., large lakes, marine coastal waters: estuaries, open-coast shorelines).

Finally, consumer approach covering the indirect releases of sunscreen products during bathing or showering (and the subsequent emission to the WWTP) confirms unacceptable risks for the aquatic compartment (including sediment) and the terrestrial compartment (after STP sludge spreading on land). These unacceptable risks were already observed by the tonnage approach. It should be noted that this consumer approach was considered to cover the risk assessment for direct releases *via* the swimming activities in rivers.

**The different risk assessments performed by the Dossier Submitters clearly show that the wide dispersive use of finished cosmetic products containing octocrilene, and especially the use of sunscreens, led to unacceptable risks for the environment. The Dossier Submitter conclude that the restriction of octocrilene in finished cosmetic products is deemed necessary to tackle the demonstrated risks.**

#### 1.4.4. Environmental monitoring data

As indicated in the previous section, the use of octocrilene as a UV-filter in cosmetic products demonstrates clearly unacceptable risks for the aquatic organisms (including sedimentary organisms). It should be noted that, although no environmental risks have been demonstrated for the use of octocrilene as a stabiliser in plastics, it remains a source of environmental contamination.

Octocrilene can enter aquatic ecosystems *via* two primary pathways after use of sunscreen for daily activities or outdoor recreation:

- ✓ Direct release to both fresh surfacewaters and marine coastal waters *via* the sunscreen rinse-off from swimmers;
- ✓ Indirect release *via* municipal wastewater from homes, hotels, and other buildings (sunscreen rinse-off from shower/bathing).

Soil is contaminated mainly through the spreading of sewage sludge containing octocrilene. Moreover, the accumulation of octocrilene in biota should not be ignored.

In order to support these results, the section below aims to resume the review of the environmental occurrence of octocrilene in the European areas. This review of the scientific literature was conducted over the last 25 years. A summary of the studies by environmental compartment (wastewater, aquatic ecosystems, solid matrices, biota) is given below, while details, in table form, are provided in Appendix B.8.5.

#### 1.4.4.1.1. Wastewater

*The measured data obtained by monitoring can be compared to the toxicity threshold for this compartment (PNEC STP: 10 000 000 ng.l<sup>-1</sup>) in an environmental risk assessment approach.*

The figures on the wastewater matrix come from fifteen scientific literature studies, where the studied waste water treatment plants (WWTPs) are very diverse in nature, due to the treatment process used, the volume of treated water, as well as the number of connected inhabitants.

Six studies were carried out with the goal to validate the determination methods of UV sunscreen chemicals present in waste water samples. These studies included a validation step by analysis of real waste water samples (Rodil et al., 2008; Gomez et al., 2009; Negreira et al., 2011; Cunha et al., 2015; Ramos et al., 2019b; Magi et al., 2012a). The nine other studies (Biel-maeso et al., 2019; Cadena-Aizaga et al., 2022a; Amankwah et al., 2024; Blum et al., 2016; Langford et al., 2015; Krzeminski et al., 2017; Balmer et al., 2005; Kupper et al., 2006; Magi et al., 2012b) were primarily aimed at determining the occurrence of octocrilene in wastewater samples and/or quantifying the removal rate of UV filters by wastewater treatment processes from the water phase. Firstly, **high concentrations of octocrilene in influent were observed** in 60% of the studies, with concentration values exceeding 500 ng.l<sup>-1</sup> found in Ramos et al., 2019b, Cunha et al., 2015, Negreira et al., 2011 and exceeding 1 000 ng.l<sup>-1</sup> in Kupper et al., 2006; Balmer et al., 2005; Blum et al., 2016;. Finally, two studies (Biel-maeso et al., 2019 and Cadena-Aizaga et al., 2022a) revealed extremely high octocrilene concentrations (in excess of 100 000 ng.l<sup>-1</sup>). Moreover, these papers suggest **time variability of UV filter inputs to WWTPs with higher loads during the warmer season (Balmer et al., 2005; Biel-maeso et al., 2019; Magi et al., 2012b; Negreira et al., 2011), most likely as a result of a more frequent use of sunscreens during that time. Higher loads of octocrilene are also noted in WWTPs located near bathing areas and near the most densely populated regions (Cunha et al., 2015).**

An important element that emerges from these results is the high removal efficiencies of octocrilene by WWTPs from the water phase. A **clear reduction (most of the time >80%)** in the octocrilene concentration in the **water effluent** compared with the influent was observed in 9 studies (Rodil et al., 2008, Cadena-Aizaga et al., 2022a, Amankwah et al., 2024, Blum et al., 2016, Balmer et al., 2005, Ramos et al., 2019b, Kupper et al., 2006, Magi et al., 2012a, Magi et al., 2012b). It should be noted that even if an average annual elimination rate of more than 80% is found, variations over the year were observed (e.g. in Cadena-Aizaga et al., 2022a with a range of 54% in August to 99.8% in May). The main reason for the high removal of octocrilene in the water phase of WWTP is related to the **hydrophobic nature of octocrilene enhancing its sorption onto the activated sludge of the waste water treatment plant when set up**. This hypothesis is verified in Langford et al., 2015, where only a mechanical treatment is available and where high effluent concentrations of octocrilene were found (max= 6 969 ng.l<sup>-1</sup>). Moreover, in Biel-maeso et al., 2019, octocrilene was measured over one year in both dissolved and particulate fractions of the influent and effluent and the majority of octocrilene was measured in the particulate fraction (from 77 to 100%) compared to the dissolved fraction indicating the very high affinity of octocrilene towards the organic matter of the particulate phase.

High concentrations were found in the effluents (Negreira et al., 2011; Biel-maeso et al.,

2019; Cadena-Aizaga et al., 2022a; Blum et al., 2016) despite a significant removal rate from the water phase. This may be explained by very high octocrilene levels in the influents of WWTPs related to a high consumption of products containing octocrilene in some regions of Europe where the WWTPs are undersized during the high tourist season to absorb the influx of tourists.

### Sludge

From the literature studies, octocrilene is measured in sludge from different urban sewage plants at high concentrations. Results confirm the hydrophobic nature of octocrilene enhancing its sorption onto sludge when an activated sludge treatment is set up.

High concentrations of octocrilene in WWTP sludge reflect the effective elimination of such compounds from the water phase through the treatment by activated sludge. However, in Biel-maeso et al., 2019, sludge (from the anaerobic digester) samples in Spain at the outlet (after digestion) presented similar concentrations of octocrilene compared to those at the inlet. Results in Kupper et al., 2006 shows comparable concentrations between raw sludge and sludge after digestion treatment. These results suggest that octocrilene has only a limited capacity to degrade in activated sludge.

Finally, as explained above for the water phase of the WWTP, higher concentrations levels of octocrilene were found in sludge of WWTPs in summer than in winter, indicating a seasonality effect (Biel-maeso et al., 2019; Ramos et al., 2018; Langford et al., 2015).

## **1.4.4.1.2. Aquatic ecosystems**

### **1.4.4.1.2.1. Freshwater environment**

The measured data obtained by monitoring can be compared to the toxicity threshold for this compartment ( $PNEC_{aquatic}$  (freshwater): 266 ng.l<sup>-1</sup>) in an environmental risk assessment approach. (In red, value >  $PNEC_{aquatic}$ ).

### Lake

Only five studies are available to characterise the occurrence of octocrilene in European lakes.

Three studies were carried out with the goal to validate or optimize the determination method of UV sunscreen chemicals in water samples. Rodil et al., 2009 and Rodil et al., 2007 included a validation step by analysis of water samples. In Lukić et al., 2022, a second objective was to determine the occurrence of octocrilene in lakes, like in Balmer et al., 2005 and Poiger et al., 2004.

All studies covered emissions from WWTPs in urban area and from recreational activities (swimming/bathing). Moreover, the sampling methods were different between the different studies. In Lukić et al., 2022 water samples were taken at 10 cm below the water surface. In Poiger et al., 2004 and Balmer et al., 2005, samples were generally taken at different depths, 1 to 2.5 m depth (in the middle of the lake).

Octocrilene were detected only in some of the samples and at low concentrations. The choice of sampling locations and method seems to be one of the main reasons for these low values. The depth of sampling seems to have a strong impact on the results (1 to 2.5 m depth in Poiger et al., 2004 and Balmer et al., 2005 compared to 10 cm depth in Lukić et al., 2022). In Poiger et al., 2004, a **seasonal trend** was observed with the highest concentrations measured during summer.

However, in Rodil et al., 2007 and in Rodil et al., 2009, high levels of octocrilene were found in the lake Cospuden (mean: 4 381 ng.l<sup>-1</sup> during hot summer days, 250 ng.l<sup>-1</sup> in June, 10 ng.l<sup>-1</sup>

<sup>1</sup> in May). Authors indicate that these results could be **related to the sunny bathing season** at sampling time. However, the values obtained in *Lukić et al., 2022* of 2 order magnitude less (max= 58 ng.l<sup>-1</sup>) were measured in August 2021, at the peak of the recreational season.

### **River**

Only six studies are available to characterise the occurrence of octocrilene in European rivers.

The values measured in rivers appear to be higher than those found in lakes. Available data, although limited to these 6 studies show an exposure of aquatic organisms over the aquatic PNEC value.

Only one study (*Rodil et al., 2008*) show very low concentration, below the LOD (18 ng.l<sup>-1</sup>). Samples were collected in a river in Galicia (North-western Spain), near the waterworks station, where water is collected for treatment and drinking water supply.

Two studies in Spain (*Gomez et al., 2009* and *García-Pimentel et al., 2023*) show values higher than the PNEC value for the aquatic organisms (266 ng.l<sup>-1</sup>). These two studies were carried out in **areas of high anthropic pressure**.

In *Gomez et al., 2009*, samples (max of 283 ng.l<sup>-1</sup>) were collected from three different zones (north, centre and south) between January and July, all located in the centre of Spain (Madrid), the most developed and densely populated region of Spain. The streams run through several residential and commercial areas; therefore, urban and industrial wastewaters represent a significant input into the rivers, but agricultural areas can also contribute to the contamination.

In *García-Pimentel et al., 2023*, samples (max>2000 ng.l<sup>-1</sup>) were taken in Ebro River, located in the Ebro Delta area (North-eastern Spain). The Delta is considered as an ecosystem of crucial importance for the environment with also a high economic activity (agriculture (rice), farming, shellfish culture, and fisheries), chemical industries, a nuclear powerplant and several urban wastewater treatment plants (WWTPs). The high concentrations detected could be explained by the discharges from **irrigation canals of the Deltebre agricultural area flow, which could also be carried into the bays along with the pollutants discharged from the Deltebre WWTP**.

High octocrilene concentrations are found **during the summer season confirming the seasonal trend with a more intensive use of octocrilene (sunscreens, among others)**. In *García-Pimentel et al., 2023*, octocrilene was detected in all samples. The median and maximum concentrations in summer were significantly higher than in winter (max >2000 ng.l<sup>-1</sup> and = 788 ng.l<sup>-1</sup> respectively). In *Homem et al., 2022*, the highest values were found in summer (Leça river: max= 118.7 ng.l<sup>-1</sup>, Certima river: max= 128.6 ng.l<sup>-1</sup>). The authors indicate that higher concentrations in summer in Leca River may be also explained from **the lower river flow rates due to the weather**.

In *Magi et al., 2012a* the highest concentration was measured in May (112 ng.l<sup>-1</sup>), **an unusual hot and dry climate was observed**.

#### **1.4.4.1.2.2. Marine environment**

*The measured data obtained by monitoring can be compared to the toxicity threshold for this compartment (PNEC<sub>aquatic marine water</sub>: 27 ng.l<sup>-1</sup>) in an environmental risk assessment approach. (In red, value > PNEC<sub>aquatic</sub>).*

Fourteen studies investigating the presence of octocrilene in seawater between 2007 and 2022 are presented below. These studies cover nine European countries (including the United Kingdom).

Concentrations above the PNEC were found in twelve out of fourteen studies (*Aminot et al., 2019; García-Pimentel et al., 2023; Mozas-Blanco et al., 2023; Rodríguez et al., 2015b; Cadena-Aizaga et al., 2022a; Agawin et al., 2024; Langford et al. 2008; Picot-Groz et al., 2018; Labille et al., 2019d; Thallinger et al., 2023; Magi et al., 2012a; Schaap et al., 2018*) and in 8 out of nine countries (Spain, Portugal, Norway, Italy, Greece, Albania, France and Netherlands).

**Literature highlights the ubiquity of octocrilene in the environment at high concentrations, particularly in coastal waters, presenting risks for the aquatic organisms, in almost all the studied locations and throughout the year.**

**The high concentrations of octocrilene are likely a consequence of its common use as a UV-filter and stabiliser in many products commonly used in urban areas, including not only cosmetic products but also plastic packaging. A link between environmental risks and proximity with treated waste water outfall and also high tourism and recreational use of seaside places is shown. Risks are also demonstrated in some beaches close to protected natural areas or close to fragile ecosystems.**

**Seasonnality effect on the risk level are shown by studies (effect which will surely be accentuated with the expected multiplication of extreme meteorological events). The highest concentrations were obtained in summer, compared to winter, when the weather was warmer and tourist number was higher. The highest concentrations in winter were found in areas where lower hydrodynamics take place (semi-closed beaches and closed bays with less water exchange with the open sea). A diurnal variation was also observed in octocrilene levels in seawater correlated with the bathing activity. In fact, the maximum measured concentrations generally coincide with the peaks of attendance on the beaches, i.e. when sampled in the middle to end of the afternoon. Regarding the vertical distribution, octocrilene is detected in the top surface layer of water and clearly also in the water column of the bathing zone. Results suggests that octocrilene can be found in the water column despite its hydrophobic character and its high tendency to adsorb to sediment.**

#### 1.4.4.1.2.3. Sediment

Octocrilene is a highly lipophilic compound, therefore, **tending to adsorb upon sediment organic matter** as demonstrated by the concentrations measured in the environment.

**The variability in the reported concentrations appears to be highly dependent on the objective of the study, the location and the sampling protocols (location, date, depth, etc.), as explained below.**

##### 1.4.4.1.2.3.1. Freshwater environment

*The measured data obtained by monitoring can be compared to the toxicity threshold for this compartment (PNEC for freshwater sediment: 926 ng.g<sup>-1</sup><sub>dw</sub>) in an environmental risk assessment approach. (In red, value > PNEC for freshwater sediment).*

Five studies are available to characterise the occurrence of octocrilene in sediment in freshwater.

One study was carried out to develop an analytical method of determination of UV sunscreen chemicals in sediment (*Gago-Ferrero et al., 2011a*). The aim of the other studies was to investigate the occurrence of UV filter in sediment (*Kaiser et al., 2012; Langford et al., 2015; Mandaric et al., 2017; Aminot et al., 2019*).

*Langford et al., 2015*, carried out in the Norwegian aquatic environment, showed all results

below the 7 ng.g<sup>-1</sup>. All sediment samples (of the top 2 cm of the sediment) were collected in at the outlet of a WWTP discharge point a water depth of 25–35 m.

Another singular study, *Mandaric et al., 2017*, revealed low concentrations with 10/12 samples < LOD in summer and 5/12 < LOD in winter. Samples were collected (from the uppermost 10 cm sediment layer) in the Adige River (Italy), at the Alpine River basin level. Nevertheless, a maximum concentration of **633 ng.g<sup>-1</sup> in the Alpine ecosystems**, was obtained in location impacted by the release of a WWTP (282 residents – 50 000 tourists). This study shows the **octocrilene contamination (due to tourism activity) of the water quality of the Alpine aquatic ecosystems, considered as fragile area**.

In *Gago-Ferrero et al., 2011a* and *Kaiser et al., 2012*, the **presence of octocrilene** in sediment was demonstrated with a frequency of detection of more than 95%. The relatively high values observed in sediment samples (0-2 cm top layer, in the middle river bed), must be underlined in *Gago-Ferrero et al., 2011a*. A maximum of **2 400 ng.g<sup>-1</sup><sub>dw</sub> (above the PNEC value of 926 ng.g<sup>-1</sup><sub>dw</sub>)** was determined in the sediment gathered at a sampling site situated downstream Zaragoza (680 000 inhab.), in the main Ebro River, which receives the effluents from the wastewater treatment plant of the city. Results are confirmed by *Kaiser et al., 2012*, where the main emission sources into the aquatic environment are **via indirect emission (from showering, washing, etc. through STPs) and via direct emission from recreational activities (swimming)**.

A **seasonal trend** was observed only in the sediments gathered from lakes (not in river) in *Kaiser et al., 2012*, with medium level in spring, peaked in summer and dropped to a minimum in autumn.

#### 1.4.4.1.2.3.2. Marine water environment

*The measured data obtained by monitoring can be compared to the toxicity threshold for this compartment (PNEC<sub>sediment marine</sub>: 92.6 ng.g<sup>-1</sup><sub>dw</sub>.) in an environmental risk assessment approach. (In red, value > PNEC for marine sediment).*

Nine studies are available to characterise the occurrence of octocrilene in marine sediment.

One study, *Langford et al., 2015*, carried out in the Norwegian aquatic environment, showed low concentrations (median < 7 ng.g<sup>-1</sup>). All sediment samples (of the top 2 cm of the sediment) were collected at the outlet of a WWTP discharge point a water depth of 98–140 m.

The **presence of octocrilene** in sediment was found with a frequency of detection 93% in *Apel et al., 2018*, 94% in *Tovar-Salvador et al., 2025*, 100% in *Pintado-Herrera et al., 2017* and *Agawin et al., 2024*.

**High influence of WWTPs** as source of octocrilene was highlighted in *Tovar-Salvador et al., 2025* (high concentrations were measured at the submarine outfall discharge of WWTP), in *Pintado-Herrera et al. 2017* and in *Agawin et al., 2024*. This last study, carried out in Formentera in Spain, had the highest concentrations found in scientific literature. Octocrilene concentrations (**up to 2575 ng g<sup>-1</sup><sub>dw</sub>**) in rhizosphere sediments near sewage discharge points exceeded reported maxima, highlighting wastewater treatment plants as significant sources of octocrilene.

*Tovar-Salvador et al., 2025* also demonstrated that the most contaminated area in La Herradura Bay in Spain was located in a site with multiple **uncontrolled untreated wastewater discharges**. In *Pintado-Herrera et al., 2017*, **direct inputs through recreational activities** as source of octocrilene was highlighted.

A **seasonal trend** was also demonstrated in *Tovar-Salvador et al., 2025*.

Finally, two studies showed the **transfer of octocrilene from coastal waters to the open sea, indicating the persistence of octocrilene and its accumulation in the sediment**. The transfer of contaminants from coastal waters to the open sea in the Adriatic was highlighted by Combi et al. 2016. In *Azaroff et al., 2019*, transfer of octocrilene from coastal release to continental shelf and Capbreton Canyon was demonstrated.

#### 1.4.4.1.3. Terrestrial compartment

To date, only few data are available on the occurrence of octocrilene in terrestrial compartment in the European area.

Among entry routes of hydrophobic compounds as octocrilene into soils, there is releases from conventional WWTPs, *via* application of sewage sludge as fertilizer and/or re-use of reclaimed water for irrigation of agricultural soils. *Biel-Maeso et al., 2018* studied soil samples from 20 cm to 150 cm depth collected at Jerez de la Frontera WWTP gardens (Spain). Those soils are irrigated roughly five times per week during summer months with wastewater effluents whereas during the rest of the year the irrigation frequency decreases to approximately twice per week. In this study, octocrilene was measured in a range of 20.9-69.7 ng.g<sup>-1</sup>dw in winter 2015) and a range of 11.9-35.1 ng.g<sup>-1</sup>dw in summer 2015. No trend was observed in the vertical distribution profile of octocrilene in soil and concentrations of octocrilene remained relatively constant between the seasons despite the higher inputs from reclaimed wastewater during the warmer months.

*Ramos et al., 2021a* collected different types of soil in Portugal from the top layer (0–30 cm). All the sampled soil contained octocrilene (near a beach: 6 ng.g<sup>-1</sup>dw; on agricultural lands: 5.2 ng.g<sup>-1</sup>dw, in gardens: 3.3 ng.g<sup>-1</sup>dw, on industrial areas: 3 ng.g<sup>-1</sup>dw, and in schoolyards: 137 ng.g<sup>-1</sup>dw). The highest octocrilene concentration was measured in the schoolyards loamy sand soil and in agricultural land silty loam soil with respectively the highest percentage of organic carbon (14.7%) and clay (4.8%), giving them high adsorption capacity compared to the others soil type. Higher octocrilene concentration was detected in the amended agricultural soil (84 ng.g<sup>-1</sup>dw) than in the initial soil (5.1 ng.g<sup>-1</sup>dw) after an application of a sewage-sludge-based fertilizer which contains 1519 ng.g<sup>-1</sup>dw of octocrilene. Higher concentration of octocrilene in tomato (147 ng.g<sup>-1</sup>dw in fruit) that grew up in this amended agricultural soil than in the initial soil (33 ng.g<sup>-1</sup>dw) was shown in *Ramos et al., 2021b* and underline the capacity of octocrilene to be transferred into edible plants.

### 1.5. Justification for Union-wide action

One of the primary reasons to act on a Union-wide basis is the cross-boundary human health or environmental problem: a risk from exposure exists in all Member States and trans-boundary trade between Member States exists.

Octocrilene is a chemical widely used on the EU market in a large variety of mixture and articles that can be produced inside and outside EU and put on the EU market. Octocrilene and products containing octocrilene in their formulation produced in one Member State may be transported to and used in other Member States. Even if specific restricting measures were introduced at Member State level, there is potential for discrepancies in the definitions and scope of any national restrictions (e.g. uses covered, concentration thresholds, and transition periods). This has implications not only for the degree to which the environment and human health is protected, but also in terms of ensuring the functioning of the internal market. Union wide action avoids trade and competition distortions within the EU/EEA, thereby ensuring a level playing field in the internal EU/EEA market as compared to action undertaken by individual Member States. Additionally, the emitted substances are transported across global borders via water. Hence, EU-wide measures for risk reduction are the only way to implement controls efficiently and uniformly within the EU/EEA. An EU/EEA restriction also allows an adequate regulation of imported goods containing these substances. Moreover, a restriction within the EU/EEA may be the first step for global action, which is needed as octocrilene and

## ANNEX XV RESTRICTION REPORT – OCTOCRILENE

octocrilene-containing formulation and products are manufactured and distributed in the global market. Product containing octocrilene as UV filter, light stabiliser or UV absorber, especially cosmetic product is imported from outside the EU/EEA. Exposure of octocrilene to humans, wildlife and the environment occurs in all EU/EEA Member States. A Union-wide restriction is needed as the persistence and hazards of octocrilene lead to cross-border pollution that cannot be adequately managed by several national regulatory actions.

An EU/EEA wide restriction will prevent and reduce the manufacture, placing on the market (including imports), use and release of octocrilene within the EU/EEA in a harmonised manner.

A Union-wide regulatory measure would also ensure a harmonised high level of protection for human health and/or the environment across the Union.

The Dossier Submitter has published a RMOA showing that several environmental exposure scenarios to octocrilene show unacceptable risks.

*From tonnage approach:*

- *Production of plastisol for freshwater and freshwater sediments;*
- *Wide dispersive use of cosmetic products;*

*From consumer approach:*

- *Exposure scenario for UV-filters used in sunscreen products applied on human skin. Emissions during swimming on both freshwater lakes and marine coastal waters*

Unacceptable risks were demonstrated in section 1.4.3 for the aquatic compartments and associated sediments. Moreover, risks were also demonstrated for soil when WWTP resulting sludge were applied onto them.

Considering all the elements described above, **the Dossier Submitter considers that there is a need for risk management.**

Section 1.4 illustrates that the hazards of octocrilene associated with its uses leads to concerns. Due to the ongoing releases from wide dispersive uses of cosmetic products, the risks are currently not adequately controlled. The Dossier Submitter considers that the risk management measures adopted so far are not sufficient as quantitative threshold for risks are overpassed. Based on these unacceptable risks for the environment identified from the wide dispersive uses of octocrilene, measures are needed to minimize the releases to the environment as much as possible to reduce the risks for the environment and in the meantime to reduce human exposure to a minimum (especially as being a potential ED, which is still under investigation at the time of preparing this restriction proposal).

Octocrilene is manufactured in several European countries. The annual tonnage manufactured and imported in the EEA is estimated to be comprised between ca. 2500 and 4500 t/y of octocrilene with at least 70 t/y being directly exported. According to the data provided during the CfE, the estimated imported amount of octocrilene is even greater, around 5500 t/y (CfE#1758), on which a substantial amount is used for cosmetics. Moreover, the Dossier Submitter highlights that the real imported tonnage may in reality be much higher and not accounted for in CSRs, as tonnages imported below 1 tonne/year of substance per company are not registered.

Due to the discrepancies between individual CSRs, as well as between CSRs and CfE, the Dossier Submitter has decided to focus their estimates on the amount of octocrilene declared to be used for cosmetics products from CSRs. It can be estimated from the different CSRs that the annual amount of octocrilene for formulation of finished cosmetic products is of 1752 t/y. Despite the fact that a part of octocrilene volume as UV filter in sunscreen are currently

replaced by alternatives, the increasing demand for UV-filters, the performance of the substance and its relatively affordable cost led to the fact that substitution to less hazardous UV filters is not naturally happening. Emission sources contribute to the exposure of the environment and human with marine water, surface water, sediment and soil being the key receiving environmental compartments.

Based on the information presented in section 1.4.2, octocrilene is a substance likely to persist in the environment for a very long period. Consequently, if releases are not minimised, humans and other organisms will be exposed to progressively increasing amounts of octocrilene until such levels are reached where effects are likely. In such an event the exposures are practically irreversible. Even if further releases of octocrilene were immediately prevented, existing environmental stocks would continue to be a source of exposure for several years. Removal of contamination is technically challenging, energy intensive, and thus costly. Additionally, loss of biodiversity and loss of ecosystem services need to be taken into account. In this Annex XV Dossier, the Dossier Submitter therefore supports a preventive approach, seen as more protective. Nevertheless, the potential implementation of a curative approach as well as the associated performances in terms of implementability and cost is discussed in the Appendix (see section E.1.1.2).

Monitoring studies demonstrate the large presence of octocrilene in all bathing area, in sediment and in soil after application of sludge from WWTP. Widespread occurrence of this persistent substance in the environment, e.g. via aqueous compartments potentially lead to global distribution. Additionally, soil and sediment can serve as long-term reservoir for octocrilene and potentially be a long-term source of contamination for groundwater. Thus, effects will not only occur at the point of release but also at some distance from that point. Therefore, it may affect a very large number of ecosystems and subsequent population. Human biomonitoring demonstrates that, for example, German population exposure to octocrilene increase along years since 1996 (*Bury et al., 2023*).

Moreover, the proposed restriction will contribute to the objectives set out in the Water Framework Directive (2000/60/EC)<sup>27</sup>, to submit proposals for control measures for the cessation or phasing-out of discharges, emissions and losses of the concerned substances to surface waters.

Based on the above reasons, a Union-wide action to address the risks associated with EU manufactured or imported finished cosmetic products containing octocrilene is needed.

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<sup>27</sup> [https://eur-lex.europa.eu/resource.html?uri=cellar:5c835afb-2ec6-4577-bdf8-756d3d694eeb.0004.02/DOC\\_1&format=PDF](https://eur-lex.europa.eu/resource.html?uri=cellar:5c835afb-2ec6-4577-bdf8-756d3d694eeb.0004.02/DOC_1&format=PDF)

## 2. Identification of REACH restriction option(s)

### 2.1. Justification that a restriction is the most appropriate Union-wide measure

Several options have been considered by the dossier submitter on how to best manage the risks arising from the use of octocrilene before preparing this restriction report.

One particularity of octocrilene is its large use as UV filter and preservative in cosmetics products. Octocrilene led to risks for the environment through these specific uses. As being largely used in sunscreen, release of octocrilene can occur in specific areas such as beaches or during mountain activities for inland releases. The European countries present a large part of their surfaces in contact to the ocean or seas. It is particularly the case for the overseas countries and territories which are part of the European countries but with specific relationship with EU as detailed below together with their specific legislations.

#### 2.1.1. Overseas Countries and Territories, applicable regulatory systems

France, like a certain number of other member states in the European Union, has overseas territories, one of the particularities of which is that they have coral reefs, especially sensitive to global warming and anthropogenic pollution. The overseas territories that could benefit from this proposed restriction are of different juridical natures regarding their relationship to EU and this leads to different applications of community law:

- all Community law applies in the European Outermost Regions (OR);
- the other territories are grouped in the category of Overseas Countries and Territories (OCT) where community law does not apply.

##### 2.1.1.1. Regulations applicable to Overseas Countries and Territories

The overseas countries and territories (OCTs), whose status<sup>28</sup> was created under the Treaty of Rome in 1957, are not an integral part of the European Union and are not members of the Schengen area but are dependencies and overseas territories of the Member States of the European Union. These different territories have special statuses and relationships with the States to which they belong, just as with the EU (reinforced association statuses defined by Articles 198 to 204 of the Treaty on the Functioning of the European Union (TFEU)<sup>29</sup>). The OCTs have wide-ranging autonomy, covering areas such as economic affairs, employment market, public health, home affairs and customs, while defence and foreign affairs usually remain within the remit of the Member States. Thus, for example, **in matters of the environment, community law does not apply in Overseas Countries and Territories which have dedicated environmental skills**. The regency of these territories is defined on a case-by-case basis in the Treaty on European Union (Annex II of the TEU and the TFEU). The status of overseas country and territory concerns 13 countries and territories belonging to three of the Member States of the European Union which as of 1 February 2020 are as follows:

linked to Denmark: Greenland

linked to Netherlands: Aruba, Bonaire, Curaçao, Saba, Sint Eustatius Sint Maarten

<sup>28</sup> <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32021D1764>

<sup>29</sup> <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:12016ME/TXT>

linked to France: French Polynesia, French Southern and Antarctic Territories, New Caledonia, Saint Barthélemy, St. Pierre et Miquelon, Wallis et Futuna Islands.

As for the French OCTs, local governance manifested by competent local territorial assemblies takes decisions. An Environmental Code specific to each community defines the different regulations that apply to the protection of nature. As such, **restriction does not apply in these territories, unless local government decide to follow what was decided in the EU.**

#### 2.1.1.2. Regulations applicable to the European Outermost Regions (OR)

Different member states are concerned by the fact of having territories with the community status of OR (defined by article 349 of the Treaty on the Functioning of the European Union (TFEU)). This special status, officially recognized for the first time in 1992 by the Treaty of Maastricht, was formally defined in 2009 with the entry into force of the Treaty of Lisbon. This status concerns 9 island regions and, or, geographically distant from the European continent, namely:

- France with the islands of Reunion, Guadeloupe, Martinique, Mayotte and Saint-Martin and the territory of Guyana;
- Spain with the Canary Islands;
- Portugal with the Azores and Madeira Islands.



**Figure 8: Geographical location of outermost region.**

[https://ec.europa.eu/regional\\_policy/policy/themes/outermost-regions\\_en](https://ec.europa.eu/regional_policy/policy/themes/outermost-regions_en)

Article 349 of the TFEU defines European policy with regard to the ORs and specifies the way in which European law applies to them, taking into account their specificities. In that sense, restriction applies to these territories.

## **2.1.2. Specific legal text and regulation applicable to marine, coastal and overseas territory**

### **2.1.2.1. Council Directive 92/43/EEC of 21 May 1992 on the conservation of natural habitats and of wild fauna and flora<sup>30</sup>**

The “Habitats – Fauna – Flora” directive, adopted on May 21, 1992, applies to all member countries of the European Union and establishes a framework for the conservation of natural habitats, wild fauna and flora. It aims to ensure the maintenance or recovery of species and habitat types significant for European biodiversity in a favourable conservation state.

It allows the Member States of the European Union, under a common legal framework, to protect the most vulnerable species and habitat types by designating special protection zones. It constitutes the foundation of the European network of Natura 2000 protected areas called Special Areas of Conservation. Each Member State identifies and proposes to the European Commission important protection sites for the species and habitats listed in the directive. Management measures must then be implemented in these areas in order to maintain or restore the species and habitats to a good conservation state.

The directive also provides for the establishment of a strict protection system for certain species listed in Annex IV, including when they are recorded outside Natura 2000 sites. Member States are in particular required to take the necessary measures to prevent their intentional or accidental capture, the deterioration of their breeding or resting sites, or even the intentional disturbance of individuals.

To follow progress in the implementation of the defined objectives, the Member States of the European Union submit every six years to the European Commission a report on the conservation status of species and habitats of Community importance present on their territory.

The Dossier Submitter considers that the provisions of the “Habitats – Fauna – Flora” directive will not achieve the goal of minimising all environmental releases from octocrilene as focusing on specific endangered species or threatened environmental area.

### **2.1.2.2. Directive 2008/56/EC of the European Parliament and of the Council of 17 June 2008 establishing a framework for community action in the field of marine environmental policy (Marine Strategy Framework Directive)<sup>31</sup>**

In order to make the development of human activities compatible with the preservation of marine ecosystems and their functionalities, integrated management is provided into the Marine Strategy Framework Directive (MSFD).

Adopted on June 17, 2008, this directive applies to all European countries with a coastline. It establishes a framework for Community action in the field of policy for the marine environment. Member States are required to implement strategies aimed at (i) the protection and preservation of the marine environment, the prevention of its deterioration or, where practicable, the restoration of marine ecosystems in areas where they have been adversely affected; (ii) the prevention and reduction of inputs in the marine environment, in order to ensure that there are no significant impacts on or risks to marine biodiversity, marine

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<sup>30</sup> <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:31992L0043>

<sup>31</sup> <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32008L0056>

ecosystems, human health or legitimate uses of the sea. This legislative framework takes into account the following marine regions: (a) the Baltic Sea; (b) the North-east Atlantic Ocean; (c) the Mediterranean Sea; and (d) the Black Sea.

The directive indicates that a list of chemicals giving rise to concerns such as sediment contamination, hotspots, health issues and contamination of biota (especially biota meant for human consumption) should be listed and monitored. No other specific action regarding chemicals is provided. Moreover it solely concerns marine environment.

The Dossier Submitter considers that the provisions of the marine environmental policy directive will not achieve the goal of minimising all environmental releases from octocrilene as solely focusing on monitoring chemicals in the marine environment only.

### **2.1.2.3. Regional seas conventions**

The Mediterranean Action Plan (MAP) of the United Nations Environment Programme (UNEP/MAP) is a regional cooperation platform established in 1975 as the first regional action plan under the UNEP Regional Seas Programme.

The Regional Seas Programme launched in 1974 to coordinate activities aimed at the protection of the marine environment through a regional approach. UNEP/MAP became a model for other UNEP-administered Regional Seas Action Plans around the world.

The MAP was the first initiative developed within the framework of this program. The adoption (2008, 2012) of the vision, goals and 11 Ecological Objectives represents a decisive milestone and development of the MAP system in its support of national and regional efforts aimed at achieving Good Ecological Status in the Mediterranean.

The Convention for the Protection of the Marine Environment of the North-East Atlantic or OSPAR<sup>32</sup> Convention is the mechanism by which fifteen governments and the European Union cooperate to protect the marine environment of the North-East Atlantic. Its scope was extended to pollution of land-based origin and the oil industry by the Paris Convention in 1974 and in 1998 to cover non-polluting human activities that can have a negative effect on the marine environment.

The Baltic Marine Environment Protection Commission or “Helsinki Commission” or “HELCOM”<sup>33</sup>, established in 1974, updated in 1992 with Denmark, Estonia, the European Union, Finland, Germany, Latvia, Lithuania, Poland, Russia and Sweden adopts recommendations related to the protection of the marine environment and sustainable maritime activities, decides on the budget and makes other key decisions by consensus. HELCOM objective is to address the increasing environmental challenges from industrialisation and other human activities that were having a severe impact on the marine environment. It includes the protection of the Baltic Sea from all sources of pollution from land, air and sea and ensure the involvement of all committing countries to take measures to conserve habitats and biological diversity and to ensure the sustainable use of marine resources.

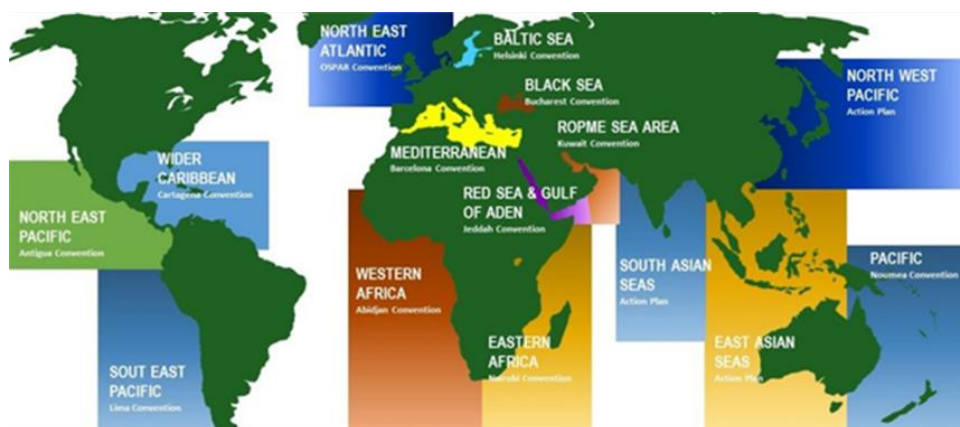
The UNEP is the only effective legal framework which aims for coordinated management of coastal and marine environments between States at the “regional” level (e.g. North East

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<sup>32</sup> <https://helcom.fi/>

<sup>33</sup> <https://helcom.fi/>

Atlantic, Wider Caribbean, Pacific and Eastern Africa) (Figure 9).



**Figure 9: UNEP coastal and marine environment at the “regional” level.**  
(Source : [UNEP](#))

#### 2.1.2.4. Regarding coral reef of ORs

- Caribbean – Cartagena Convention<sup>34</sup> :

The Cartagena Convention for the protection and development of the marine environment in the Caribbean region applies to Saint-Martin, Saint-Barthélemy, Martinique and Guadeloupe. The convention aims to promote: the fight against pollution, the protection of the environment, cooperation in the event of a major environmental risk, the assessment of impacts on the environment and scientific and technical cooperation between the parties.

- Indian Ocean – Nairobi Convention<sup>35</sup> :

The Nairobi Convention concerns the protection, management and development of the marine and coastal environment of the East African region. It does not apply to discharges from swimming activities or everyday discharges (cosmetic products in general).

Dossier Submitter considers that the provisions of the UNEP/MAP will not achieve the goal of minimising all environmental releases from octocrilene due to a lack of data, risk assessment, measurement and enforcement.

#### 2.1.3. Within the European Economic Area

##### 2.1.3.1. Identification as SVHC/Candidate list for inclusion in Annex XIV

Octocrilene could be identified as SVHC, according to REACH article 57 and put on the candidate list only if meeting the criteria for:

- classification as carcinogenic, mutagenic or toxic for reproduction (CMR) category 1A or 1B in accordance with the CLP Regulation.

<sup>34</sup> <https://www.unep.org/cep/who-we-are/cartagena-convention>

<sup>35</sup> <https://www.milieuamfrance.fr/Nos-rubriques/Cadre-reglementaire/Conventions-des-mers-regionales/Convention-de-Nairobi>

- for being persistent, bioaccumulative and toxic (PBT) or very persistent and very bioaccumulative (vPvB) according to REACH Annex XIII.
- causing an equivalent level of concern as CMR or PBT/vPvB substances.

Once listed on the Annex XIV, the substances may not be used or placed on the market without authorisation. The prioritisation for inclusion in Annex XIV from the candidate list is driven by several criteria that are set by Article 58 of REACH and implemented by ECHA following a methodology that has been agreed by the Member States Committee (MSC) that includes consideration related to hazards and to exposure parameters.

Octocrilene is persistent (P) and toxic (T) in the environment but not bioaccumulative (B). The M criterion (mobility) has not been evaluated but its log K<sub>oc</sub> is greater than 3, meaning that octocrilene is very unlikely to be mobile. Since octocrilene is only identified as persistent and toxic, it cannot meet the SVHC criteria regarding PBT issues. Octocrilene was presented to the PBT expert group of ECHA in the course of the SEv and it was concluded that the substance does not fulfil the classification criteria for being identified as a B substance according to annex XIII of REACH. It would have eventually been possible to present the substance again to the PBT expert group of ECHA and then to the Member State Committee with the new data updated from the initial dossier to potentially identify the substance as a PBT (especially in regard to potential new data on the bioaccumulation of octocrilene) but the RMOA performed for the octocrilene showed that, even with new data, restriction is the most efficient management measure. Moreover, an identification as a PBT will have also led to a probable restriction for the same identified risks. As quantitative threshold for risks to the aquatic compartment from the use of octocrilene have been overpassed, there is no need to await for a potential PBT identification of octocrilene as a SVHC.

Regarding CMR properties, the classification proposed for the reproductive toxicity criterion is only a category 2, this classification does not meet the SVHC criteria.

Furthermore, the Dossier Submitter notes that the ED identification for the environment is awaiting the results of the LAGDA test currently being carried out. If octocrilene would be identified as an endocrine disruptor for the environment, its SVHC identification could be considered.

It should be noted that, even if octocrilene were to be identified as an SVHC and prioritized for listing in Annex XIV, the authorization requirements only apply to substances and mixtures produced in the European Union. It would not prevent mixtures containing octocrilene to be imported from third countries.

The Dossier Submitter considers that the conditions set up in REACH regarding SVHC identification under article 57 are not met presently and that the SVHC process would not address the identified risk.

#### **2.1.3.2. REACH Authorisation process**

Octocrilene is not classified as CMR category 1a or 1b, nor is classified as PBTs, vPvBs, PMT, ED nor has been identified as substances of equivalent concern. Therefore, authorisation cannot be used as a Risk Management Measure. Moreover, REACH authorisation does not apply to imported products. REACH Authorisation would not be appropriate to address the risks.

For all these reasons, authorisation is thus not considered an appropriate risk management option.

### **2.1.3.3. Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy**

The Water Framework Directive (WFD) applies to inland, coastal, transitional surface waters (including estuaries and lagoons) and groundwaters. Their management is carried out on the geographical scale of hydrographic basins, established under the WFD.

The WFD applies to all outermost regions (ORs) (see section 2.1.2.2). It aims for “good chemical status” and “good ecological status” of water (article 1) towards which Member States act to respect water quality standards. Annex X gives the authorized limits for these 45 substances. Good chemical status is achieved when the concentrations of pollutants or groups of pollutants in water, sediments or biota do not exceed a threshold concentration defined as an “environmental quality standard” (EQS).

Generally speaking, the WFD and other related directives establish a framework for an overall Community policy in the field of water (surface water and groundwater). However, its application still remains limited in the ORs, particularly due to the fact that the Environmental Quality Standards (EQS) are identical to those of mainland and do not take into account the specificities of tropical regions (different sensitivities of species in the marine environment to substances chemicals).

Furthermore, the WFD only applies to coastal waters, even though there are coral reefs located beyond this boundary. However, the WFD also provides that the Water Offices search, in their hydrographic basins, for substances relevant to the local context and establish lists and associated standards.

Since the implementation of the directive, a watch list has been introduced in France and Europe in order to identify substances likely to be included, at the end of a monitoring period, among the priority substances of the WFD. Since the update of the 22 July 2022 of the annex X watch list of substances for Union-wide monitoring referred to in Article 8b of Directive 2008/105/EC, the list was updated with the inclusion of octocrilene<sup>36</sup> with a maximum acceptable method detection or quantification limit of 266 ng/l.

The WFD would make it possible to define an environmental limit value for regulating discharges from industrial installations formulating and producing mixtures containing octocrilene. However, in the case of octocrilene, the risks identified using data from the scientific literature mainly result from its wide dispersive use as cosmetic, in particular due to its role as a UV filter in consumer products, and are not arising at the industrial location. The inclusion of octocrilene in this directive does not affect consumer uses that are the main sources of environmental concentration.

The definition of a guide value within the framework of the WFD would, on the other hand, be a good way of monitoring the application of concentration limits that could be set by means of a restriction.

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<sup>36</sup> [https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32022D1307&qid=1658824912292#ntr12-L\\_2022197EN.01012001-F0012](https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32022D1307&qid=1658824912292#ntr12-L_2022197EN.01012001-F0012)

#### **2.1.3.4. Directive 2006/7/EC of the European Parliament and of the Council of 15 February 2006 concerning the management of bathing water quality and repealing Directive 76/160/EEC**

This directive concerns the management of bathing water quality.

At the end of each bathing season, the classification of bathing water quality is carried out. The quality of bathing water is determined on the basis of analysis results on samples taken at a monitoring point in the zone of maximum bather attendance. The regulated parameters are *Escherichia coli* and intestinal enterococci indicators but does not include any chemical substance.

The provisions of the management of bathing water quality directive will therefore not achieve the goal of minimising all environmental releases from octocrilene.

#### **2.1.3.5. Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006**

In the current state of the assessment and considering the available data, no endocrine disruptor or PBT classification can be proposed within the framework of the CLP regulation.

The harmonized classification of substances according to the CLP regulation makes it possible to provide information on hazards, notably through labeling. It should be noted, however, that the CLP Regulation does not apply to substances or mixtures used as cosmetic products in the finished state, i.e. ready for use for consumers (art.1.5.c of the CLP regulation; unlike cosmetic ingredients as raw materials or cosmetic mixtures - in bulk - to which the CLP regulation applies).

The harmonized classification is not the most appropriate regulatory management measure to address the issues identified for octocrilene. This is not likely to address the risks associated with the cosmetic use of the substance and will not achieve the goal of minimising all environmental releases from octocrilene.

#### **2.1.3.6. Regulation (EC) No 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products.**

In the EEA, substances used in the composition of cosmetic products are regulated by Regulation No. 1223/2009 relating to cosmetic products, complementary to Regulation No. 1907/2006, known as the REACH regulation. Annex VI of the regulation related to cosmetic products establishes the list of UV filters permitted in cosmetic products (known as the positive list) such as sunscreen and Annex V of the same regulation, that of preservative agents. These annexes are established by the European Commission, following the opinion of the Scientific Committee for Consumer Safety (SCCS) on human health and safety risks (chemical, biological, mechanical and other physical risks) for non-food products and certain services. On the other hand, with regard to the environmental concerns from substances present in cosmetic products, these are examined within the framework of the REACH regulation, which implements an assessment of environmental safety in a trans-sectorial manner.

Currently, cosmetic regulations authorize the use of octocrilene up to a concentration of 9 or 10% depending on cosmetic uses. As explained in the French RMOA on octocrilene and summed up in this restriction proposal, a classification for sexual function and fertility based on available data is possible and is justified (Repr. Category 2 (H361f)). According to Article

15 of the Cosmetic Regulation, a substance classified as Category 2 may be used in cosmetic products where the substance has been assessed by the SCCS and found to be safe for use in cosmetic products. However, the SCCS has already evaluated octocrilene, in 2021, and deemed it safe for use in cosmetic products. It is therefore difficult to predict the impact of reprotoxic classification 2 on this use and its subsequent consequence on environmental issues.

Dossier Submitter considers that the provisions of the cosmetic products regulation is not directed to tackle the environmental issues and that putative inflexion of the use for human health issues will not sufficiently minimise environmental releases from octocrilene.

**2.1.3.7. Regulation (EC) No 850/2004 of the European Parliament and of the Council of 29 April 2004 on persistent organic pollutants and amending Directive 79/117/EEC<sup>37</sup>.**

The Stockholm Convention, which aims to control, reduce or eliminate emissions of persistent organic pollutants (POPs). This regulation prohibits (Annex I) the production, placing on the market and use of these substances. It is stipulated to limit exemptions as much as possible, so that they are only possible in cases where a substance fulfils an essential function in a specific application. In general, the identification of a substance as a POP is complex and might not be agreed easily nor reach promptly a consensus as octocrilene is not classified as PBT, vPvB nor PMT.

Dossier Submitter considers that the provisions of the POP regulation will not achieve the goal of minimising all environmental releases from octocrilene.

**2.1.3.8. Council Directive 91/271/EEC of 21 May 1991 concerning urban waste-water treatment**

The objective of this Directive is to protect the environment from the adverse effects of urban waste water discharges and discharges from certain industrial sectors and concerns the collection, treatment and discharge of: Domestic waste water; Mixture of waste water; and Waste water from certain industrial sectors. Increasing the efficiency of waste-water treatment through measures under this Directive could help prevent the release of octocrilene in surface water. However, this would mean more octocrilene end up in the sludge and therefore would be an increase in the burden to the terrestrial environment and would not adequately control the identified risk. Moreover, it will only help to deal with part of the emission not dealing with the direct water exposure due to recreational bathing.

Dossier Submitter considers that the provisions of the urban wastewater treatment directive will not achieve the goal of minimising all environmental releases from octocrilene.

**2.1.3.9. Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste**

A mandatory destruction (incineration) scheme could be considered as a risk management option for the waste lifecycle stage in particular for the industrial uses. However, this option is not currently considered to be feasible because of the implementation challenges associated with harmonising waste management practices across the EU, the (non) availability of incineration facilities in all EU countries and the identification of the articles and/or cosmetic

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<sup>37</sup> <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:32004R0850>

containers containing octocrilene.

Dossier Submitter considers that the provisions of the waste directive will not achieve the goal of minimising significantly all environmental releases from octocrilene.

#### **2.1.3.10. Directive 2010/75/EU of the European Parliament and of the Council of 24 November 2010 on industrial emissions (integrated pollution prevention and control)**

This Directive requires operators of industrial installations to obtain a permit from the national authorities to continue operating. Permits place a requirement for the use of Best Available Techniques (BAT) to reduce emissions and the impact on the environment as a whole.

The Industrial Emission Directive (IED) has the potential to limit emissions from permitted sites. However, many downstream users, such as manufacturers of mixtures containing octocrilene, are unlikely to require a permit and be subject to IED. Additionally, IED only controls parts of the lifecycle and will thus have no effect on the service life emissions or release from the waste stage of mixtures.

Exposure to the environment via emissions occurs mainly when mixtures containing octocrilene are used, not during the production phase. Therefore, measures aiming at limiting sources, including Waste Water Treatment Works (WWTWs), would not effectively address the source of exposure. The derivation and implementation of Environmental Quality Standards (EQS) values for the aquatic environment would not be an effective risk management measure. Moreover, this Directive is unable to manage release from wide dispersive uses of cosmetic products from consumer.

Dossier Submitter considers that the provisions of the IED will not achieve the goal of minimising all environmental releases from octocrilene.

#### **2.1.3.11. REACH Art. 68.2**

REACH Article 68(2) stipulates those substances that are CMR categories 1 or 2 can be subject to a proposal from the Commission to inclusion in Annex XVII for consumer uses without using the procedures in article 69-73 in the REACH Regulation. Octocrilene is not classified as such and this measure is therefore not applicable.

#### **2.1.3.12. Other potential regulations that could be applied to manage risks arising from octocrilene**

**Table 19: Other potential regulations that could be applied to manage risks arising from octocrilene.**

| Legal text                                                                                                                                                  | Domain of concern                                                                                                                                                                                                                                                                                                                                                                                            | Applicability to all identified issues |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Council Directive 86/278/EEC of 12 June 1986 on the protection of the environment, and in particular of the soil, when sewage sludge is used in agriculture | Encourage the use of sewage sludge in agriculture and try to prevent harmful effects on soil, vegetation, animals and man. It prohibits the use of untreated sludge on agricultural land unless it is injected or incorporated into the soil. It also requires that sludge should be used by taking into account the nutrient requirements of plants and that the quality of the soil and of the surface and | No                                     |

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|                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|                                                                                                                                                                                                                                                                                                                                                         | groundwater is not impaired.                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| Regulation (EC) No 648/2004 of the European Parliament and of the Council of 31 March 2004 on detergents                                                                                                                                                                                                                                                | The Regulation harmonises the rules for the placing on the market of detergents and of surfactants for detergents; the biodegradability of surfactants in detergents; restrictions or bans on surfactants on grounds of biodegradability; the additional labelling of detergents, including fragrance allergens; However, it does not cover the degradability of UV filters even if used to protect the formulation                            | No |
| Regulation (EU) No 305/2011 of the European Parliament and of the Council of 9 March 2011 laying down harmonised conditions for the marketing of construction products and repealing Council Directive 89/106/EEC                                                                                                                                       | Information on the content of hazardous substances in the construction products should be included in the declaration of performance to reach all potential users. It is not evident if this legislation would apply, or if it would be an effective tool for reducing releases to the environment. Based on data, octocrilene could be classified as Aquatic Chronic 1, H410, but the impact on all environmental emissions is not guarantee. | No |
| Regulation (EC) No 1935/2004 of the European Parliament and of the Council of 27 October 2004 on materials and articles intended to come into contact with food and repealing Directives 80/590/EEC and 89/109/EEC. Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food. | Octocrilene is listed in Annex I of this regulation; octocrilene can be used as an additive or auxiliary in polymer production. The specific migration limit applicable to the substance is 0.05 mg/kg of foodstuffs. It is not permitted to use octocrilene as a monomer or other starting substance.<br><br>This Directive is unable to manage the wide dispersive uses of finished cosmetic products from consumer uses.                    | No |
| Directive 2001/95/EC of the European Parliament and of the Council of 3 December 2001 on general product safety                                                                                                                                                                                                                                         | Addresses risks to consumers related to specific products and not risks related to a cumulated exposure from different products, or to risks posed to the environment. It can be used to restrict articles but this needs annual renewal.                                                                                                                                                                                                      | No |
| Land and soil Regulations                                                                                                                                                                                                                                                                                                                               | Octocrilene may be released to soils from industrial, professional or consumer activities and the application to soil of sewage sludge.                                                                                                                                                                                                                                                                                                        | No |

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|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>Currently no legal EU framework would be able to address all these releases. There is no dedicated European legislation on soil quality.</p> <p>For large industrial sites, there are provisions in the IED that relate to soil protection and remediation. More generally, the Environmental Liability Directive (Directive 2004/34/EC) establishes a framework for preventing and remedying environmental damage.</p> <p>Under the EU Green Deal strategy, the Zero Pollution Action Plan and the revision of the thematic strategy for soil protection could also provide a framework to address the concerns raised by the contamination of soils.</p> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

The risks for the environment identified from the wide dispersive use of finished cosmetic products make it necessary to reduce emissions of octocrilene from these uses to a minimum. Different regulatory risk management options have been considered (e.g., Industrial Emission Directive, RoHS, Biocidal Products Regulation, Product Safety Directive, Waste Directive, REACH authorisation...). A restriction can cover a wide range of uses and can address the risks arising from the manufacture and use of the substances as such as well as in mixtures and in articles, including imported articles from outside the EU. It was concluded that a REACH restriction would be the most appropriate EU-wide measure to address the risk posed by octocrilene for the environment arising from the wide dispersive use of finished cosmetic products.

### 2.2. Identification of REACH restriction option(s)

Based on the conclusions of the risk assessment, the releases of octocrilene occurring from the wide dispersive uses of cosmetic products (by consumers of end-products) are considered to pose a risk to the environment that is not adequately controlled.

In response to the identification of this risk, diverse risk management options (RMOs) were analysed to identify the most appropriate measure to address the risk. In Section 2.1, it is concluded that a REACH restriction is the preferred risk management option as being the sole regulatory measure to be able to act at the European level on all uses of octocrilene in finished cosmetic products put on the market and ensure the same level playing fields among all professionals and consumers to achieve the objective of minimising the risks for the environment arising from these cosmetic products uses. The assessment of the various risk management options carried out by the Dossier Submitter was underpinned by the information on uses, releases and socio-economic impacts, but also by the effectiveness and practicality criteria outlined in Annex XV of REACH for assessing the appropriateness of a REACH restriction.

On the basis of this assessment, the following restriction options were further considered for the impact, practicality and monitorability assessment:

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- **RO1:** Restriction on placing octocrilene on the market in consumer and professional finished cosmetic products (concentration limit of 0.001% w/w). A two-year transitional period is implemented after the enforcement of the proposed restriction.
- **RO2:** Restriction on placing octocrilene on the market in consumer and professional finished cosmetic products (concentration limit of 0.001% w/w). A five-year transitional period is implemented after the enforcement of the proposed restriction.

Both ROs will be effective in managing the identified risks as they will potentially eliminate almost all release of octocrilene from finished cosmetic products to the environment by forbidding the use of the substance in finished cosmetic products put on the market in the EEA (representing at least 1752 t/y, as estimated by the Dossier Submitter).

Considering that risks were arising from the wide dispersive uses of cosmetic products, derogations for the following uses will be proposed in the restriction entry proposal: manufacturing of octocrilene, formulation and industrial uses for the production of plastisol, and preparation with additives and service life of plastic articles.

The implementation of the proposed restriction would result in the cessation of the following activities within the EEA:

- production and import of octocrilene for the formulation of cosmetic products containing octocrilene and intended for the EEA market;
- formulation of cosmetic products containing octocrilene and intended for the EEA market;
- import and placing on the EEA market of cosmetic products containing octocrilene.

Besides, the proposed restriction would result in the reformulation of existing products intended for the EEA market (formulation based on alternative substances).

However, the following activities would continue to be allowed within the EEA (see uo3 under the Uncertainty Analysis section):

- production and import of octocrilene for the formulation of cosmetic products containing octocrilene and intended for export outside the EEA market;
- formulation of cosmetic products containing octocrilene intended for export outside the EEA market.

While developing the proposed ROs, the Dossier Submitter examined available data on existing alternative substances. Thus, the Dossier Submitter considers that the proposed ROs meet the practicality criterion as both alternative substances and end-products formulated based on the latter are already available on the market (see section 3.2.4.3).

The key difference between RO1 and RO2 lies in the duration of the transitional period: a two-year transitional period is considered under RO1 while a five-year transitional period is considered for RO2. In defining the duration of the transition period, the Dossier Submitter took the following information into consideration:

- Availability of alternative substances;
- Time required to reformulate a cosmetic product in the context of a regulatory change, in particular sunscreen products;
- Availability of octocrilene-free cosmetic products on the market;

## Existence of voluntary reformulation initiatives by cosmetics manufacturers

- Seasonality of production and consumption of sunscreens as well as the time needed to sell existing stocks.<sup>38</sup>

The risk management options indicated below were also initially considered, but were not considered to be as appropriate as the options identified above, and were therefore discarded:

- Restriction on placing octocrilene on the market in consumer and professional finished cosmetic products consisting in the definition of an updated and lower concentration limit (higher than the LoD). This option was discarded as it did not meet the effectiveness criterion;
- Elimination of octocrilene by treatment of sewage plant effluents (degradation by micro-organisms, membrane filtration, chlorination). These options were discarded as they do not meet the practicality criterion.

### 2.3. General considerations for the concentration limit

Octocrilene's wide dispersive uses from cosmetic products lead to unacceptable risks for the environment and especially in fresh and marine waters, associated sediments and agricultural soils after sludge application as demonstrated by the tonnage and consumer approach for determining the risks. The lowest NOEC for various trophic level was determined to be on daphnia with a value of 2.66 µg/L. The resulting PNEC for aquatic environment were determined to be of 0.266 µg/L for freshwaters and 0.0266 µg/L for marine waters.

For the two assessed ROs, the Dossier Submitter is proposing that octocrilene shall not be placed on the market, or used, as substances or in mixtures in a concentration equal to or greater than 0.001% w/w. This concentration was proposed based on the calculated risks demonstrated for the environment, the fact that technical function of octocrilene as preservative in some cosmetic products are as low as 0.1% (nails products for example), the persistence of the substance, the occurrence of octocrilene in every environmental compartment in several EEA countries (e.g. WWTP influent and effluent in Switzerland, Norway, Italy; Seawater in Spain, France; Sediments in Spain; Sludge in Norway) as described in Appendix B.9.6 on monitoring and in various biota as described in hazards section 1.4.2. Moreover, the available analytical technique limit of quantification (LoQ) and limit of detection (LoD) retrieved in the literature for cosmetic products analysis in different matrix was also used for setting this concentration limit.

By setting a concentration limit of 0.001% w/w, and knowing that octocrilene is authorised to a maximum concentration of 10 % in sunscreen, the corresponding maximum value under the restriction proposal will allow a maximum concentration in mixture of 0.001% of octocrilene. With this threshold in finished cosmetic products, the function of octocrilene as UV filter, UV absorber and light stabiliser will be lost and as such the interest to use the substance.

The proposed concentration limit of 0.001% w/w correspond to a value of 10 ppm,

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<sup>38</sup> The reformulation and production of sun protection products must be completed early enough to allow them to be placed on the market. If this is not the case, the launch of new products is delayed by one year. In addition, sunscreen producers are subject to the "take-back rule": unsold products must be taken back by the producer. If the transition period were too short, producers would have to bear the cost of destroying unsold products in addition to the logistical costs.

representing 10 µg/g or 10 mg/L depending on the matrix considered (liquid or solid).

### 3. Impact assessment

#### 3.1. Scope and approach

As outlined in section 1.4.3, a risk has been demonstrated for water compartments (fresh- and marine waters), sediments and soils in relation to the use of octocrilene in cosmetic products (wide dispersive use). No risk not avoidable by other RMMs have been demonstrated in relation to other uses of the substance. Consequently, the restriction proposal addresses the use of octocrilene in cosmetic products only and so does the impact assessment.

As good quality and detailed economic information was available regarding cosmetic products (albeit with some uncertainties), the Dossier Submitter has been able to perform a quantitative impact assessment.

Firstly, the Dossier Submitter has undertaken a detailed quantitative analysis of the **costs** expected from the restriction for the manufacturers of cosmetic products (see section 3.4.1 to 3.4.5. for details on the methodology). Sensitivity analysis has been undertaken on key uncertainties. For this analysis, the Dossier Submitter considered a 10 years assessment period to account for the cosmetics industry's baseline major reformulations cycle, while assuming the restriction proposal would be enforced in 2027. As the use of octocrilene for the manufacture of cosmetic products is significantly driven by the use in sun protection products<sup>39</sup>, the Dossier Submitter has quantified the expected cost as the sum of the extra-cost associated to the manufacture of sun protection products and the extra-cost associated to the manufacture of other cosmetic products. Based on the results of the cost assessment, the Dossier Submitter discussed the impact of the restriction on the consumers through an increase in the market price of cosmetic products as a result of the extra-cost borne by manufacturers based on a quantitative analysis.

Moreover, the Dossier Submitter has also undertaken a qualitative analysis of the impact of the restriction on the manufacturers of cosmetic ingredients (octocrilene and its alternatives) and discussed the impacts for importers and retailers of cosmetic products. The Dossier Submitter also proposed a quantification of enforcement and control costs.

Secondly, the **benefits** of the restriction were estimated in monetary terms based on a specific study eliciting the willingness to pay of European households (commissioned by the Dossier Submitter for the specific purpose of this restriction proposal) to avoid the negative impacts of octocrilene used in cosmetic products on marine and freshwater ecosystems (Sotis et al., unpublished and Appendix E.4.4).

Thus, the Dossier Submitter was able to discuss the proportionality of the restriction based on a cost-effectiveness and a cost-benefit analysis, as well as a break-even analysis. The scope of the cost and benefit assessments as well as main assumptions considered are presented in Box 1 and Box 2 below. Details on the methodology are provided in section 3.4.

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<sup>39</sup> As a reminder, throughout this document, the Dossier Submitter uses the term 'sun protection products' to refer to both primary sunscreens and secondary photo-protection products. Although the proportion of products containing octocrilene within these product categories is different (see Appendix A.2.1), the claim of sun protection (and therefore the functional importance of Octocrilene) is nevertheless a central common element between these two types of products. Moreover, according to the data provided by Cosmetics Europe, these types of products have similar levels of concentration of octocrilene (see Table 5).

**Box 1: Assessment of the benefits of the restriction - Main methodological points**

The evaluation of the benefits induced by the restriction is based on a stated preference study – combining discrete choice experiment and contingent valuation methods – conducted in September 2024 among a representative sample of 7,200 Europeans (1,200 people surveyed in 6 EEA 'case-study' countries). The case study countries were chosen so that the survey results would reflect the heterogeneity of preferences within the EEA. The Dossier Submitter therefore selected countries with a diversity of economic conditions (purchasing power, income level), geographical location, access to the sea and environmental preferences. From this survey, the Dossier Submitter determined an average and median willingness to pay at household level for improving the states of aquatic species in coastal and freshwaters, which it then extrapolated to all households in the EEA.

**Table 20: Scope of the stated preference study**

|                                      |                                                                                                                                                          |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sample size</b>                   | 7,200 adults (>18 years old): 1,200 respondents per case-study country                                                                                   |
| <b>Case-study countries</b>          | Bulgaria, France, Germany, Hungary, Spain, Sweden                                                                                                        |
| <b>Representativeness</b>            | Gender, age, occupation and region of residence. <sup>a</sup>                                                                                            |
| <b>Product categories considered</b> | All cosmetic products                                                                                                                                    |
| <b>Impact on the environment</b>     | Improvement in the state of marine and freshwater aquatic environments in terms of biodiversity in the event of a restriction on the use of octocrilene. |

a: The Dossier Submitter also aimed at the representativeness of the sample for the level of education. However, representativeness for this variable could not be fully attained.

**Box 2: Assessment of the costs of the restriction - Main methodological points****Table 21: Scope of the cost assessment**

| Affected stakeholders                               | Impacts considered                                                                                                                | Type of analysis |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------|
| Manufacturers and importers of cosmetic ingredients | Loss or gain in terms of sales opportunities                                                                                      | Qualitative      |
| Manufacturers of cosmetic products                  | Reformulation (extra) cost<br>Raw material (extra) cost                                                                           | Quantitative     |
| Importers of cosmetic products and retailers        | Control (extra) cost                                                                                                              | Qualitative      |
| Public authorities                                  | Control and enforcement cost                                                                                                      | Quantitative     |
| Consumers                                           | Increase in market price of cosmetic products<br><br>(Loss of quality/performance of end products: investigated but not relevant) | Quantitative     |

**Table 22: Main assumptions considered**

| Parameter                                              | Assumptions considered in the present Annex XV Dossier         | Assumptions considered in ECHA (2020) | Sensitivity analysis performed? |
|--------------------------------------------------------|----------------------------------------------------------------|---------------------------------------|---------------------------------|
| Reference year (PV and adjustments of monetary values) | 2024                                                           | Not relevant                          | Not relevant                    |
| Entry into force of the restriction                    | 2027: this is when the transitional period starts to apply     | Not relevant                          | Not relevant                    |
| Length of the transitional period                      | <b>2 years (proposed restriction RO1)</b><br><br>5 years (RO2) | 2 years<br><br>5 years                | Yes                             |
| Useful life of a cosmetic formulation (=study period)  | 10 years                                                       | 20 years                              | Yes                             |
| Rate of reformulation in the event of a restriction    | Partial reformulation <sup>a</sup>                             | Partial reformulation                 | Yes                             |
| Discount rate                                          | 3% <sup>b</sup>                                                | 4%                                    | No                              |

a: Under the partial reformulation hypothesis the Dossier Submitter assumes that only a fraction of cosmetic formulations that contain octocrilene would actually be reformulated in the event of a restriction.

b: The Dossier Submitter followed the EU Better regulation guidelines (European Commission, 2023) and considered a 3% discount rate.

## 3.2. Baseline and response to the proposed restriction options

### 3.2.1. Baseline

In this section, the Dossier Submitter defines the Baseline scenario regarding the future trends of the use of octocrilene in the cosmetic industry and associated releases of this substance to the environment. This work is based on information on current use volumes as well as on other mechanisms at work likely to influence the use of this substance. The baseline scenario is compared to the proposed restriction scenario in the impact assessment, in terms of both expected costs and benefits.

The evolution of use-volumes of octocrilene in the formulation and production of cosmetic products under the baseline scenario is defined by extrapolation from the data obtained for sun protection products. The Dossier Submitter was not able to collect specific information on use-trends for other categories of cosmetic products. However, given that the use-volumes of the substance are significantly driven by its use in sun protection products<sup>40</sup>, considering this product category as a basis for extrapolation seems acceptable.

On the one hand, several mechanisms at work could lead to an increase in the volumes of octocrilene used in the production of sun protection and cosmetic products. Indeed, regarding the market dynamic, both the market for sun protection products and cosmetic products is expected to grow within the next five years: a 13% (Statista, 2024b) to 31.5% (Euromonitor International, 2024b) increase in sales is expected for the sun care market in Western Europe between 2023 and 2028. As mentioned in section 1.3.4 and as underlined by industrial stakeholders during the initial consultation phases, the use of octocrilene in sun protection products ensures good sensory performances<sup>41</sup> at lower production cost. This sensory performance influences the purchase and use of skincare products by consumers (see Appendix A.3). Several alternative UV filters are available that can be used to formulate sunscreen products offering equivalent performances in terms of effectiveness and sensory characteristics (see section 3.2.4), however these substances have a higher market price (see Appendix E.2.3.). In this context, in the absence of a restriction on octocrilene and any other regulatory constraint, it can be expected that producers of sun protection products will continue to use this substance.

On the other hand, other mechanisms linked to consumer preferences as well as the ban of octocrilene in other jurisdictions could lead to a reduction in the use-volumes of the substance in the future. Indeed, while consumers grant importance to the sensory performance of sun protection products, they also value the presence of natural ingredients, the absence of chemical filters and the sustainability of these products (see Appendix A.3). Besides, in the context of the ban of several substances worldwide that was recently enforced in Hawaii (US Senate, 2018) and the Republic of Palau<sup>42</sup>, several chemical filters - including octocrilene - are currently suffering from "bad press" (stakeholder consultations). In the case of octocrilene, publications mention its suspected effects on coral reefs, as well as its potential endocrine disrupting properties (see for example Wecf France and Agir pour l'environnement,

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<sup>40</sup> Sun protection products account for 84% of use-volumes of Octocrilene in 2023 across all cosmetic products in Western Europe (Euromonitor, 2025). Approx. 75% of cosmetic products available on the market that contain octocrilene are sun protection products (CosmEthics, 2023)

<sup>41</sup> According to industrial stakeholders, consumers attach great importance to the sensory performance of sun protection products. The 'ideal' sunscreen product is transparent, non-greasy and applies easily (see Appendix A.3) for more information on consumer preferences regarding sun protection products.

<sup>42</sup> <https://www.palau.gov.pw/the-republic-of-palau-adopts-the-worlds-strictest-national-sunscreen-standard/>

2020; BUND, 2023). During its data collection efforts, the Dossier Submitter was able to observe the presence of "coral/reef friendly" labels on sunscreens and sometimes even an explicit mention of the absence of octocrilene<sup>43</sup>. According to recent Euromonitor data, 'Reef/Ocean Safe' claims recorded a 36% growth between 2020 and 2022 in Western Europe (De Carlo Coleman, 2023). In this context, some industry stakeholders have indicated that they intend to abandon the use of octocrilene in the near future (stakeholder consultation). Some producers of sun protection products have already started reformulating existing formulations with alternative chemical filters and/or not considering octocrilene for the development of new products (only alternative UV filters)<sup>44</sup>. However, it is difficult to appraise the definitive nature/trend of this strategic choice on the use of octocrilene in the absence of a restriction. Besides, stakeholders have also stressed the difficulty of actually formulating and placing sun protection products on the market in the short term due to the limited availability of alternative filters (see section 3.2.4).

The data available to the Dossier Submitter does not allow to determine which one of these two mechanisms prevails over the other. In this context, the Dossier Submitter has calculated the impact of the proposed restriction assuming a 'business as usual' baseline scenario, where octocrilene would continue to be used in the same quantities as reported today. This amounts to assuming that new formulations and growth in the cosmetics market are driven by octocrilene-free products, without however assuming voluntary reformulations. Given the uncertainty regarding the impact of the two effects mentioned above ('bad press' versus 'market dynamic') and which one would prevail, assuming constant volumes of use for octocrilene could result in an over- or underestimation of the costs of the restriction. In this scenario, all of the costs of replacing octocrilene would be assumed to be the result of the proposed restriction and not of a possible strategic choice by the sector. Thus, the Dossier Submitter also presents a sensitivity analysis measuring the impact of different levels of potential voluntary reformulation on the costs of restriction (see Appendix E.4.3.4).

### 3.2.2. Response to Restriction option 1

The concentration limit set by the proposed restriction for octocrilene in RO1 being very low and far lower than the minimum concentration observed in cosmetic products (see section 1.3.4), companies producing products containing octocrilene will have to reformulate them to remove the substance if they want to continue placing them on the market.

**For sun protection products**, the reformulations that would be implemented in the event of a restriction are major<sup>45</sup> reformulations. Based on the analytical framework proposed in the study prepared by RTI for the US Food and Drug Administration (RTI, 2002), this is

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<sup>43</sup> The Dossier Submitter observed this phenomenon in Germany in particular, without being able to identify the proportion of sunscreen products concerned by this type of claim or their impact on consumer purchasing behaviour (see Appendix A.2.1.2. for details regarding the data collection process). The Dossier Submitter was unable to determine the extent of this phenomenon in the rest of the EEA.

<sup>44</sup> The Dossier Submitter also notes that industrial stakeholders reported a high number of reformulations currently taking place in relation to regulations that were recently enforced (e.g., restriction on the use of homosalate in (sun protection) body products, REACH restriction on the use of microplastics, cyclosiloxanes, etc.). Given the reputational issues associated with octocrilene, it is likely that at least part of companies manufacturing sun protection products are able to address multiple regulatory concerns in one single reformulation.

<sup>45</sup> As mentioned in section 3.4.2, the Dossier Submitter relies here on the framework used in recent restriction dossiers concerning the use of cyclo-siloxanes and microplastics in cosmetic products (ECHA, 2015; ECHA, 2020a; ECHA, 2020b). In the latter, the Dossiers Submitters consider two types of reformulations (major or minor) with significantly different levels of costs (with major reformulations being more costly than minor ones). The minor/major nature of a reformulation depends on the role and concentration level of the targeted ingredient in the end-product: minor ingredients are used at low concentration levels and do not necessarily have a functional role while major ingredients are used at high concentration levels and always have a functional role.

explained both by the concentration levels of octocrilene in sunscreen products and its functional role. Reformulations are possible for sun protection products targeted by the restriction as alternatives to octocrilene (alternative UV filters) exist that enable producing and placing on the market end-products offering the same performance in terms of effectiveness (sun protection) and quality (sensory performances in particular). Sun protection products based on such alternatives are already available on the market. However, several factors indicate that a few years would be necessary to complete the reformulations required by a restriction for octocrilene-containing sun protection products currently on the market. The reformulation of such products (particularly when it comes to modifying UV filters) has been described as a complex and lengthy process by industrial stakeholders (two to five years needed for a reformulation), even if it is possible for some major players to carry out several reformulations at the same time (source: stakeholder consultations). Tensions were also highlighted regarding testing<sup>46</sup> capacities as well as regarding the supply of alternative UV filters. Indeed, according to Cosmetics Europe (source: personal communication), 3 to 4 years would be needed for manufacturers of alternative UV filters to adjust production capacities in order to meet an increase in demand.

**Regarding 'other cosmetic products'** and due to the diversity of product categories containing octocrilene, the Dossier Submitter was not able to determine the magnitude of the reformulations (major or minor) induced by a restriction for these products. However, the Dossier Submitter was still able to make assumptions regarding the value of reformulation costs for these products categories based on information gathered during the stakeholder consultation. It is considered that the cost of reformulation is lower for 'other cosmetic products' than for sun protection products (see section 3.4.2.1 for further details). Besides, the Dossier Submitter was not able to identify alternatives to octocrilene for these product categories, with as much precision as for sunscreen products. Nevertheless, 'other cosmetic products' that contain octocrilene belong to a wide range of different product categories, within which they represent a small share of the supply (in number of products). The Dossier Submitter considers that this demonstrates both the feasibility of reformulations and the availability of alternatives for these product categories.

For the first restriction option 'RO1' (the proposed restriction), the Dossier Submitter considers a **transitional period of two years**. Considering the time needed for the examination of the proposed restriction by the ECHA committees (RAC and SEAC) and for the final decision of the European Commission (approx. 2.5 years in total), a two-year transitional period appears sufficient if it can be assumed that concerned industrial stakeholders would anticipate the required changes once the proposed restriction has passed the conformity stage. This is a realistic assumption given that the majority of proposed restrictions that are considered to conform by the ECHA committees are then adopted by the European Commission. Based on this assumption, industrial stakeholders would have 4.5 years in total. Thus, the two-year transitional period seems sufficient in view of the information mentioned previously (time needed for a reformulation and constraints on testing, production capacities, seasonality and 'take-back rule'). Finally, and as mentioned previously (see section 2.2), the availability of alternative substances and end-products based on these alternatives on the market allows considering such a short transitional period.

### 3.2.3. Response to Restriction option 2

The concentration limit set by the restriction for octocrilene is identical for RO1 and RO2. Consequently, the response to RO2 is identical to the response to RO1: companies producing products containing octocrilene will have to reformulate these products to remove the

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<sup>46</sup> Several tests have to be performed before placing a new formulation on the market (stability, SPF, water resistance, brand specific claims). Depending on the company, these tests may be carried out in-house and/or by a sub-contractor.

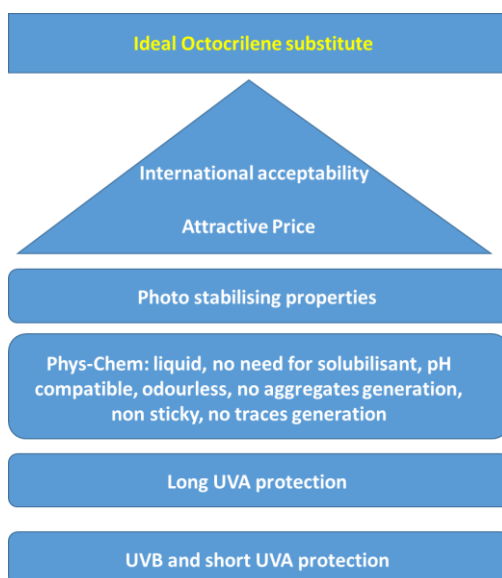
substance if they want to continue placing them on the market. However, for this second restriction option 'RO2', the Dossier Submitter considers a **transitional period of five years**. This alternative duration for the transitional period allows assessing the impact of the proposed restriction when assuming that industrial stakeholders are unlikely to anticipate the changes required by the restriction after it has passed the conformity stage within the ECHA committees (RAC and SEAC). In this case, the Dossier Submitter assumes that concerned industrial stakeholders would start implementing the required changes only after the proposed restriction is enforced. Compared to a two-year transitional period, considering a five-years one also gives the opportunity to producers of cosmetic products to better coordinate the reformulations required in the event of a restriction with the baseline reformulation schedule (see Section 3.4.2.1 for more details).

### 3.2.4. Identification of the most likely alternatives and conclusion on the likelihood of substitution and

As detailed in section 1.3.4, octocrilene absorbs UV radiation efficiently from 290 to 340 nm with a peak absorption at 307 nm offering sun protection, i.e., in primary sunscreens as well as in secondary photo-protection products (see appendix E.2.1). It also acts as a general photo-stabilizer, allowing protecting the formula as a whole and specifically ingredients being less photo-stable, such as other UV filters (such as Butyl Methoxydibenzoylmethane (CfE#1756; CfE#1757; CfE#1759; CfE#1760; Cosmed, 2024; L'Oréal, 2024).

Besides, octocrilene is used as a solvent for other crystalline and oil soluble UV filters (ex: Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine or Butyl Methoxydibenzoylmethane; CfE#1757; CfE#1758; CfE#1759; CfE#1760; CfE#1761; Cosmed, 2024). This allows reducing/avoiding the use of lipophilic solvents, which is also beneficial for the sensoriality of the end-products (as it reduces the 'oily and sticky skin' sensation left by sunscreen products) and for the formula cost (CfE#1759; Cosmed, 2024; L'Oréal, 2024). Finally, octocrilene can also be used to improve sweat and water resistance of primary sunscreens (CfE#1756; CfE#1759).

As illustrated in Figure 10, the ideal alternative to octocrilene should meet all these requirements.



**Figure 10: Ideal alternative to octocrilene.**

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The considered alternatives must comply with the obligation set up in the Regulation (EC) No 1223/2009 on cosmetic products and, in particular with the fact that any alternative to be used as an UV filter in cosmetic product must be authorised and listed in the annex VI of this regulation also known as the positive list. These UV filters are listed in Table 23 which details the product type, the potential regulatory ongoing activity under REACH, the role as UV absorber and light stabilizer.

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**Table 23: Authorised UV filter in annex VI in Regulation (EC) No 1223/2009 and available information under REACH regulation.**

| Reference number | Name of Common Ingredients Glossary | CAS number | EC number | Harmonised Classification | Self-Classification                                                                                                            | Product type, body parts                                      | Maximum concentration in ready for use preparation | Information requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Suspected Hazard                                                               |
|------------------|-------------------------------------|------------|-----------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 2                | Camphor Benzalkonium Methosulfate   | 52793-97-2 | 258-190-8 |                           | Skin Irrit. 2 H315<br>Eye Irrit. 2 H319<br>Aquatic Chronic 2 H411                                                              |                                                               | 6 %                                                | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                |
| 3                | Homosalate                          | 118-56-9   | 204-260-8 |                           | Repr. 2 H361d<br>STOT SE 3 H335<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 4 H416 | Face products with the exception of propellant spray products | 7,34 %                                             | <p>Data generation under Compliance Check requesting:</p> <ol style="list-style-type: none"> <li>1. Sub-chronic toxicity study (90-day), oral route (OECD TG 408) in rats;</li> <li>2. Pre-natal developmental toxicity study (OECD TG 414) in a first species (rat or rabbit), oral route;</li> <li>3. Extended one-generation reproductive toxicity study (OECD TG 443) in rats, oral route as follows: <ul style="list-style-type: none"> <li>- Ten weeks premating exposure duration for the parental (PO) generation;</li> <li>- Dose level setting shall aim to induce some toxicity at the highest dose level;</li> <li>- Cohort 1A (Reproductive toxicity);</li> <li>- Cohort 1B (Reproductive toxicity) without extension to mate the Cohort 1B animals to produce the F2 generation;</li> <li>- Cohorts 2A and 28 (Developmental neurotoxicity); and</li> <li>- Cohort 3 (Developmental immunotoxicity),</li> </ul> </li> <li>4. Identification of degradation products using an appropriate test method.</li> </ol> <p>Deadline: 25 February 2024</p> <p>Data generation for point 1, 2 and 3 were suspended due to an appeal from industry in the EU Court of Justice of ECHA. This appeal was dismissed (22 November 2023)<sup>47</sup> and data generation should be ongoing.</p> | Suspected ED, Suspected PBT, Reproductive toxicity Exposure of the Environment |

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<https://curia.europa.eu/juris/document/document.jsf?jsessionid=5EEF21635DFFA5CBBAF06F32427304E5?text=&docid=244530&pageIndex=0&doclang=EN&mode=lst&dir=&occ=first&part=1&cid=1590412>

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|   |                                           |                        |           |  |                                                                                                                                                                          |                                                                                                                                                                                        |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |
|---|-------------------------------------------|------------------------|-----------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 4 | Benzophenone-3                            | 131-57-7               | 205-031-5 |  | Repr. 2 H361d<br>STOT RE 2 H373<br>STOT SE 3 H335<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319<br>Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 2 H411 | a) Face products, hand products, and lip products, excluding propellant and pump spray products<br>b) Body products, including propellant and pump spray products<br>c) Other products | a) 6 %<br>b) 2,2 %<br>c) 0,5 % | Substance under evaluation by Denmark.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Suspected ED                                      |
| 6 | Phenylbenzimidazole Sulfonic Acid         | 27503-81-7             | 248-502-0 |  |                                                                                                                                                                          |                                                                                                                                                                                        | 8 % (as acid)                  | A Testing Proposal Evaluation (TPE) and a Compliance Check (CCH) in case of Dossier Evaluation are ongoing. No details available on the subject covered by this TPE and CCH.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |
| 7 | Terephthalylidene Dicamphor Sulfonic Acid | 92761-26-7, 90457-82-2 | 410-960-6 |  |                                                                                                                                                                          |                                                                                                                                                                                        | 10 % (as acid)                 | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |
| 8 | Butyl Methoxydibenzoylmethane             | 70356-09-1             | 274-581-6 |  | Skin Irrit. 2 H315<br>Eye Irrit. 2 H319                                                                                                                                  |                                                                                                                                                                                        | 5 %                            | A substance evaluation decision number SEV-D-2114357832-44-01/F ongoing requesting:<br>1. Aerobic mineralisation in surface water – simulation biodegradation test, (OECD TG 309).<br>2. Aerobic and anaerobic transformation in aquatic sediment systems, (OECD TG 308),<br>3. Long-term toxicity testing on aquatic invertebrates (OECD TG 211),<br>4. Long-term toxicity testing on fish (OECD TG 210).<br>Based on the available information, the evaluating MSCA considers that the Substance is potentially PBT as the substance is considered to fulfil the P and B criteria; information on T is currently considered inconclusive despite the availability of standard long-term studies on aquatic organisms. | Suspected ED and PBT, Exposure of the environment |

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|    |                                          |             |           |                |                                                                                                                                                                                   |  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                  |
|----|------------------------------------------|-------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|    |                                          |             |           |                |                                                                                                                                                                                   |  |                   | An amphibian metamorphosis assay (OECD TG 231)<br>Deadline: 16 February 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                  |
| 9  | Benzylidene Camphor Sulfonic Acid        | 56039-58-8  |           | Eye Dam 1 H318 |                                                                                                                                                                                   |  | 6 % (as acid)     | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                  |
| 10 | Octocrylene                              | 6197-30-4   | 228-250-8 |                | Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 2 H411<br>Aquatic Chronic 3 H412<br>Aquatic Chronic 4 H413<br>a) Propellant spray products<br>b) Other products |  | a) 9 %<br>b) 10 % | A substance evaluation decision number SEV-D-2114287467-34-01/F ongoing requesting:<br>The Larval Amphibian Growth and Development Assay (LAGDA, OECD TG 241).<br><br>Deadline: 04 June 2024 (expected 31 October 2024)<br><br>A Testing Proposal was also ongoing:<br>A. Information required from the Registrants subject to Annex IX of REACH<br>1. Sediment simulation testing (OECD TG 308),<br>2. Identification of degradation products (OECD TG 308),<br>3. Effects on soil micro-organisms (OECD TG 216),<br>4. Long-term toxicity to terrestrial plants (OECD TG 208 with at least six species).<br><br>B. Information required from the Registrants subject to Annex X of REACH 1. Long-term toxicity testing to sediment organisms (OECD TG 225 and OECD TG 233 and OECD TG 239).<br><br>Deadline: 03 December 2023, submitted in June 2024. | Potential PBT, Suspected ED for human health and the environment |
| 11 | Polyacrylamidomethyl Benzylidene Camphor | 113783-61-2 |           |                | STOT SE 3 H335 (Lungs)<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319                                                                                                                 |  | 6 %               | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                  |
| 12 | Ethylhexyl Methoxycinnamate              | 5466-77-3   | 226-775-7 |                | H361f<br>Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 3 H412<br>Aquatic Chronic 4 H413                                                                       |  | 10 %              | No data requested                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Uncertainties related to genotoxicity, Potential ED              |
| 13 | PEG-25 PABA                              | 116242-27-4 |           |                |                                                                                                                                                                                   |  | 10 %              | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                  |

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|    |                                |                       |               |  |                                                                                                                              |  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |
|----|--------------------------------|-----------------------|---------------|--|------------------------------------------------------------------------------------------------------------------------------|--|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 14 | Isoamyl p-Methoxycinnamate     | 71617-10-2            | 275-702-5     |  | No data                                                                                                                      |  | 10 % | <p>1. Either an Amphibian Metamorphosis Assay, (OECD 231), or a Larval Amphibian Growth and Development Assay, (OECD 241);</p> <p>2. Fish Sexual Development Test, test method (OECD 234);</p> <p>3. Provide information and justification for parameters in the environmental exposure assessment within the Chemical Safety Report.</p> <p>Combined assessment of OMC and IPMC.</p> <p>ECHA considers the results obtained in the FSDT and AMA with OMC as unsuitable to conclude on the ED properties for IPMC as well.</p> <p>Combination of Fish short term reproduction assay (OECD TG 229) and Fish Sexual Development Test (OECD TG 234)</p> <p>Deadline: 19 December 2025</p> | Suspected ED, Exposure of the environment.           |
| 15 | Ethylhexyl Triazone            | 88122-99-0            | 402-070-1     |  | Acute Tox. 4 H302, H312<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319, H335<br>Aquatic Chronic 2 H411<br>Aquatic Chronic 4 H413 |  | 5 %  | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |
| 16 | Drometrizole Trisiloxane       | 155633-54-8           | 423-660-5     |  | No data                                                                                                                      |  | 15 % | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |
| 17 | Diethylhexyl Butamido Triazone | 154702-15-5           | 421-450-8     |  | Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 4 H413                                                     |  | 10 % | <p>Initial ground for potential PBT.</p> <p>Substance is P, vP</p> <p>Inconclusive results for B vB.</p> <p>Classification: Not PBT, vPvB</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                      |
| 18 | 4-Methylbenzylidene Camphor    | 38102-62-4/36861-47-9 | - / 253-242-6 |  | Aquatic Chronic 4 H413                                                                                                       |  | 4 %  | <p>SVHC dossier according to REACH Art. 57 (f) owing to its endocrine disrupting properties was prepared by Germany and later withdrawn due to data generation awaited. A follow up action for SVHC dossier by Denmark was done. Identified as a SVHC according to art. 57f in the 76<sup>th</sup> meeting of the MSC.</p> <p>Listed for eventual inclusion in Annex XIV of REACH.</p> <p>Indication of potential PBT properties.</p>                                                                                                                                                                                                                                                  | <p><b>SVHC ED art. 57f</b></p> <p>Potential PBT.</p> |

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|----|-----------------------------------|------------------------|----------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 20 | Ethylhexyl Salicylate             | 118-60-5               | 204-263-4      |  | No data                                                                                                                                         |  | 5 %           | <p>1. Pre-natal developmental toxicity study (OECD TG 414) in a first species (rat or rabbit), oral route;</p> <p>2. Extended one-generation reproductive toxicity study (OECD TG 443) in rats, oral route;</p> <p>3. Long-term toxicity testing on aquatic invertebrates (OECD TG 211);</p> <p>4. Long-term toxicity testing on fish (OECD TG 234))</p> <p>Deadline: 25 February 2024</p> <p>Data generation for point 1, 2 and 4 were suspended due to an appeal from industry in the EU Court of Justice of ECHA. This appeal was dismissed (22 November 2023)<sup>48</sup> and data generation should be ongoing.</p> | Suspected ED<br>Aquatic toxicity<br>Reproductive toxicity                  |
| 21 | Ethylhexyl Dimethyl PABA          | 21245-02-3             | 244-289-3      |  | Aquatic Chronic 3 H412<br>Aquatic Chronic 4 H413                                                                                                |  | 8 %           | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Indication of potential classification as Repro 1B from data               |
| 22 | Benzophenone-4,<br>Benzophenone-5 | 4065-45-6<br>6628-37-1 | 223-772-2<br>- |  | Repr. 2 H361d<br>STOT RE 2 H373<br>(Other: organs)<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319<br>Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410 |  | 5 % (as acid) | <p><b>Benzophenone-4</b><br/><u>CCH 1:</u></p> <p>1. In vitro gene mutation study in bacteria (OECD TG 471);</p> <p>2. Short-term toxicity testing on aquatic invertebrates (OECD TG 202);</p> <p>3. Growth inhibition study aquatic plants (OECD TG 201).</p> <p>4. In vitro cytogenicity study in mammalian cells (OECD TG 473) or In vitro micronucleus study (OECD TG 487);</p> <p>5. If negative results are obtained in tests performed, then: In vitro gene mutation study in mammalian cells (OECD TG 476 or TG 490).</p> <p>Deadline: 20 August 2024<sup>49</sup>.</p>                                           | Suspected ED<br><br>Considered safe by the SCCS (April 2024) <sup>51</sup> |

<sup>48</sup>

<https://curia.europa.eu/juris/document/document.jsf?jsessionid=81FF4CB4A599CB25D8A22F9F8F19271D?text=&docid=279983&pageIndex=0&doclang=en&mode=req&di r=&occ=first&part=1&cid=1802221>

<sup>49</sup> <https://echa.europa.eu/documents/10162/36364f70-c144-8623-7023-bfcb7250447d>

<sup>51</sup> [https://health.ec.europa.eu/publications/benzophenone-4-cas-no-4065-45-6-ec-no-223-772-2\\_en](https://health.ec.europa.eu/publications/benzophenone-4-cas-no-4065-45-6-ec-no-223-772-2_en)

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|     |                                                            |             |           |  |                                                                                                                                                                                                 |  |                |                                                                                                                                                                                                                                                                                                                                                        |  |
|-----|------------------------------------------------------------|-------------|-----------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|     |                                                            |             |           |  |                                                                                                                                                                                                 |  |                | <u>CCH 2:</u><br>1. Short-term toxicity testing on aquatic invertebrates (OECD TG 202);<br>2. Growth inhibition study aquatic plants (OECD TG 201 or OECD TG 221).<br>3. In vitro gene mutation study in mammalian cells (OECD TG 476 or TG 490).<br><br>Deadline: 20 August 2024 <sup>50</sup> .<br><br><b>Benzophenone-5</b><br><br>No relevant data |  |
| 23  | Methylene Bis-Benzotriazolyl Tetramethylbutylphenol        | 103597-45-1 | 403-800-1 |  | Repr. 2 H361d<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319<br>Aquatic Acute 1 H400<br>Aquatic Chronic 1 H410<br>Aquatic Chronic 4 H413                                                            |  | 10 % (5)       | No relevant data                                                                                                                                                                                                                                                                                                                                       |  |
| 23a | Methylene Bis-Benzotriazolyl Tetramethylbutylphenol (nano) | 103597-45-1 | 403-800-1 |  | Repr. 1B H360, H360f<br>Repr. 2 H361d<br>STOT SE 3 H335<br>Skin Irrit. 2 H315<br>Eye Irrit. 2 H319                                                                                              |  | 10 % (5)       | No relevant data                                                                                                                                                                                                                                                                                                                                       |  |
| 24  | Disodium Phenyl Dibenzenimidazole Tetrasulfonate           | 180898-37-7 | 429-750-0 |  | Repr. 2 H361<br>STOT SE 3 H335 (not provided or lungs)<br>Skin Sens. 1 H317<br>Skin Irrit. 2 H315<br>Skin Irrit. 3 H316<br>Eye Irrit. 2 H319<br>Aquatic Chronic 3 H412<br><br>Skin sens. 1 H317 |  | 10 % (as acid) | Ongoing Dossier Evaluation for completeness and comprehension                                                                                                                                                                                                                                                                                          |  |

<sup>50</sup> <https://echa.europa.eu/documents/10162/c6ac4ed0-2f4e-b974-5b02-84aacb7b4a8d>

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|-----|------------------------------------------------|--------------------------------|-------------------------------|------------------------|----------------|--|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 25  | Bis-Ethylhexyloxyphenyl Methoxyphenyl Triazine | 187393-00-6                    | 425-950-7                     | Aquatic Chronic 4 H413 |                |  | 10 %     | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | P, vP                                                                   |
| 26  | Polysilicone-15                                | 207574-74-1                    | 426-000-4                     | Aquatic Chronic 4 H413 |                |  | 10 %     | No active registration under REACH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                         |
| 27  | Titanium Dioxide                               | 13463-67-7/1317-70-0/1317-80-2 | 236-675-5/215-280-1/215-282-2 |                        | Not classified |  | 25 % (4) | <p>A substance evaluation decision requesting:</p> <p>Intratracheal instillation study combined with comet assay in rats on 11 forms of Titanium dioxide (TiO<sub>2</sub>) on 5 male rat. Each instillation (at 0 and 24 h) must comprise 3 different concentrations, followed by 3 post-exposure observation times (2-6 h, 1 day and 28 day post exposure);</p> <p>. The lowest concentration must produce little or no evidence of toxicity and the highest concentration must result in a clear level of toxicity but not inducing overload where feasible;</p> <p>Observations:</p> <p>1. bronchoalveolar lavage fluid (BALF) and histopathology of the lung according to GD39.</p> <p>2' Oxidative stress by measuring reactive oxygen species (ROS) and reactive nitrogen species (RNS) by fluorimetric probes, malondialdehyde (MDA) by HPLC and Haeme oxygenase 1 activity.</p> <p>3. Histopathology of liver, kidney, testis and brain.</p> <p>4' Cardiovascular function by measuring at least nitrogen oxide (NO) content, endothelial nitrogen oxide synthase (eNOS) activity and high sensitivity C reactive protein (hs-CRP) content in the serum.</p> <p>5. Comet assay on lung tissues must be performed according to the OECD TG 489 with and without specific modification to detect oxidative damage.</p> | Carcinogenic<br>Suspected<br>Mutagenic<br>Other hazard<br>based concern |
| 27a | Titanium Dioxide (nano)                        | 13463-67-7/1317-70-0/1317-80-2 | 236-675-5/215-280-1/215-282-2 |                        | Not classified |  | 25 % (4) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |

# ANNEX XV RESTRICTION REPORT – OCTOCRILENE

|     |                                                                         |              |           |               |                                                      |  |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                   |
|-----|-------------------------------------------------------------------------|--------------|-----------|---------------|------------------------------------------------------|--|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 28  | Diethylamino<br>Hydroxy benzoyl<br>Hexyl Benzoate                       | 302776-68-7  | 443-860-6 |               | Acute Tox. 2 H330<br>Aquatic Chronic 4<br>H412, H413 |  | 10 %     | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                   |
| 29  | Tris-biphenyl<br>triazine<br><br>Tris-biphenyl<br>triazine (nano)       | 31274-51-8   | 479-950-7 | Carc. 2 H351i |                                                      |  | 10 %     | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                   |
| 30  | Zinc Oxide                                                              | 1314-13-2    | 215-222-5 | Carc. 2 H351i |                                                      |  | 25 % (3) | Subchronic Inhalation Toxicity: 90-Day Study (OECD 413) combined with the Reproduction/ Developmental Toxicity Screening Test (OECD 427) in rat (nose-only)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Other hazard based concern<br>Consumer use<br>Exposure of environment<br>Other exposure/risk based concern<br>Wide dispersive use |
| 30a | Zinc Oxide (nano)                                                       | 1314-13-2    | 215-222-5 |               |                                                      |  | 25 % (3) | 2. In Vivo Mammalian Alkaline Comet Assay (OECD TG 489) in rat, inhalation route (nose-only),<br><br>3. Information on transformation, dissolution and dispersion stability of manufactured and imported nanoforms of zinc oxide that are covered by the registration dossier(s) submitted for zinc oxide<br><br>a. Screening Test (24 h) from OECD Guidance Document on Transformation/Dissoluti n of Metals and Metal Compounds in Aqueous Media<br><br>b. OECD TG 318 on dispersion stability of nanomaterials<br><br>4. Freshwater Algae and Cyanobacteria, Growth Inhibition Test (OECD TG 201,)<br><br>5. Long-term toxicity on Daphnia magna (OECD TG 211): Daphnia magna Reproduction Test,<br><br>6. Information on uses and operational conditions of zinc oxide in nanoforms. |                                                                                                                                   |
| 31  | Phenylene Bis-<br>Diphenyltriazine                                      | 55514-22-2   | 700-823-1 |               |                                                      |  | 5 %      | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                   |
| 32  | Methoxypropylamin<br>o Cyclohexenylidene<br>Ethoxyethylcyanoac<br>etate | 1419401-88-9 | 700-860-3 |               |                                                      |  | 3%       | No relevant data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                   |

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|    |                                                                   |             |           |  |  |  |          |                  |
|----|-------------------------------------------------------------------|-------------|-----------|--|--|--|----------|------------------|
| 33 | Bis-<br>(Diethylaminohydroxybenzoyl Benzoyl)<br>Piperazine        | 919803-06-8 | 485-100-6 |  |  |  | 10 % (9) | No relevant data |
| 34 | Bis-<br>(Diethylaminohydroxybenzoyl Benzoyl)<br>Piperazine (nano) | 919803-06-8 | 485-100-6 |  |  |  | 10 % (9) | No relevant data |

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Among the UV filters authorised in annex VI of the cosmetic regulation, 6 do not have UV absorber properties (Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine nano and non nano, Methoxypropylamino Cyclohexenylidene Ethoxyethylcyanoacetate, Phenylene Bis-Diphenyltriazine, Methylene Bis-Benzotriazolyl Tetramethylbutylphenol nano and non-nano) and 6 do not have light stabilizing properties (Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine nano and non nano, Titanium dioxide nano and non-nano, homosalate and Camphor Benzalkonium Methosulfate). In total, there are 161 substances used as light stabilizer and 171 substances used as UV absorber which are listed as cosmetic ingredients. Among them, 28 (27 without octocrilene) are also used as UV filters and can potentially be used as alternatives to octocrilene as presenting several of octocrilene properties against light radiation. These substances are detailed in the table 71 of Appendix E.2.2.1.

As detailed in Table 23, some salicylate (homosalate, ethylhexyl salicylate,...) used as UV filters are under investigation in the EU for reproductive toxicity, suspected as being endocrine disruptor and aquatic and environmental toxicity. There is an on-going call for data to remove these uncertainties. Due to these uncertainties, industrial stakeholders indicated during the audition that there is ongoing work to reformulate products containing homosalate (as it recently has been restrained from being used for body applications).

During the stakeholder consultation, manufacturers pointed out that 6 of these 34 chemical filters are no longer used for photo-protection. The following 4 filters in particular are no longer marketed: Camphor Benzalkonium Methosulfate, Benzylidene Camphor Sulfonic Acid, Polyacrylamidomethyl Benzylidene Camphor, Peg-25 PABA. Specific data on these filters are detailed in annex E2.2.2.

Moreover, during these consultations, stakeholders indicated that the best identified alternatives to octocrilene are the following substances (Table 24):

**Table 24: The most suitable alternatives to octocrilene according to stakeholder consultation.**

| Reference number | Substances information                         |             |           |                                                                     | PNEC Freshwater/Marine water (mg/L) |
|------------------|------------------------------------------------|-------------|-----------|---------------------------------------------------------------------|-------------------------------------|
|                  | Name of Common Ingredients Glossary            | CAS number  | EC number | Suspected Hazard                                                    |                                     |
| 15               | Ethylhexyl Triazone                            | 88122-99-0  | 402-070-1 | No relevant data                                                    | 0.08/0.008                          |
| 17               | Diethylhexyl Butamido Triazone                 | 154702-15-5 | 421-450-8 | Substance is P, vP<br>Inconclusive results for B vB <sup>52</sup> . | 0.002/0.0002                        |
| 25               | Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine | 187393-00-6 | 425-950-7 | Substance is P, vP. Not B, vB<br>No other relevant data.            | No relevant data                    |
| 28               | Diethylamino Hydroxy benzoyl Hexyl Benzoate    | 302776-68-7 | 443-860-6 | No relevant data                                                    | No relevant data                    |

These substances show several physico-chemical properties similar to octocrilene, allowing to use them quite easily in regard to the other authorised UV filters. Specifically, they are also used in cosmetic products as UV absorber and light stabilizer. Moreover, although they have

<sup>52</sup> <https://echa.europa.eu/documents/10162/ea01b414-c7e8-8f3a-0f37-c15f8ae26b83>

some hazardous properties, they do not present hazards in regards to regulatory obligations that would prevent to use them as alternatives; several sunscreens were already formulated with these substances, showing that they are technically-applicable candidates.

In regards to PBT assessment, Diethylhexyl Butamido Triazone is P, vP but the assessment of the B, vB properties of the substance were inconclusive based on predicted and measured Log Kow values, read-across to a structurally related substance and data for sediment organisms and toxicokinetic data. While the substance fulfills the screening criteria for terrestrial bioaccumulation, there is no indication of toxicity or bioaccumulation from the available mammalian data. The high hydrophobicity of the substance raises a concern for slow bioaccumulation which however cannot be investigated further with existing validated methods. It may evolve in the future but for now, no undisputable conclusion is reach on the PBT properties of the substance. As such, the substance is not classified as PBT, nor vPvB.

Regarding toxicity to biota in water systems, Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine and Diethylamino Hydroxy benzoyl Hexyl Benzoate are not classified and do not lead to regulatory obligation of determining PNEC values so far. Contrarily, Ethylhexyl Triazone and Diethylhexyl Butamido Triazone are toxic for biota and especially aquatic invertebrates and led to proposing PNEC values as low as those for octocrilene. Depending on several factors (PBT properties, RMMs, OCs, tonnages, releases ...) diethylhexyl butamido triazone may potentially be a regrettable alternative (it was not investigated by the Dossier Summitter). Depending on the requested physico-chemical properties and the necessity to protect both human health (good UV filter) and the environment (not PBT), several alternatives are available to industry, some being potentially “safer” than other.

**Table 25: Physico-chemical properties of the most suitable alternatives to octocrilene according to stakeholder consultations (see appendix E.2.2.1 for further details).**

|                  | Ethylhexyl Triazone | Diethylhexyl Butamido Triazone              | Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine | Diethylamino Hydroxy benzoyl Hexyl Benzoate |
|------------------|---------------------|---------------------------------------------|------------------------------------------------|---------------------------------------------|
| Physical form    | Solid               | White powder, Solid, odourless              | Pale yellow Solid                              | White powder, Solid                         |
| Water solubility | 0.007 mg/L          | < 0.00075 mg/L at 20.0±0.5°C, pH= 5.4 – 7.1 | < 0.014 mg/L at 20°C                           | 16 µg/l at 20° C                            |
| Oil solubility   | High                | High                                        | High                                           | High                                        |
| Photostability   | High                | High                                        | High                                           | High                                        |
| UV absorber      | Yes                 | Yes                                         | Yes                                            | Yes                                         |
| Light stabilizer | Yes                 | Yes                                         | Yes                                            | Yes                                         |
| λ max            | 314 nm              | 310 nm                                      | 310 and 343 nm                                 | 354 nm                                      |
| Light Spectrum   | UVB, UVA2           | UVB, UVA 1 and UVA2                         | UVB, UVA 1 and UVA2                            | UVA1, UVA2                                  |

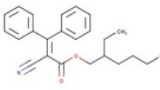
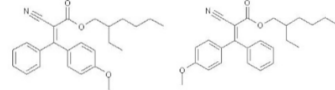
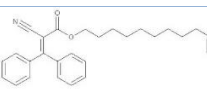
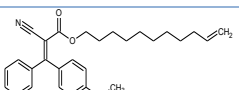
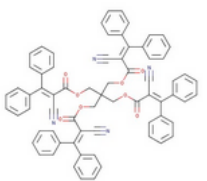
(full data in annex E2).

Industry and researchers are also currently working on new UV filters based on derivatives from in use and authorised UV filters, such as structure-related benzophenone and lactone (SPF = 16, but only ability to absorb UVB radiation), new PABA derivatives, PABA methyl ester and PABA methyl stearate (SPF values of 20.60 and 26.17). Ring-benzimidazole derivatives are also under development. New 1,3,5-triazine derivatives were synthesised that displayed the most promising antioxidant activity and revealed the highest SPF and UVA protection factor.

Derivates of 3-benzylidenecamphor modified at the level of the camphor moiety demonstrated the ability to absorb both UVA and UVB radiations. This ongoing work was also noted for sinapic acid analogues of EPMC. At least 140 potential alternatives are under scrutiny and development (Jesus et al., 2022).

### 3.2.4.1. Potential alternatives not authorised in annex VI of cosmetic regulation from the group of esters of cyanodiphenyl propenoic acid

In March 2020, ECHA initiated an Assessment of Regulatory Needs (ARN) on the group of esters of cyanodiphenyl propenoic acid in which octocrilene belongs. It consisted of a screening assessment of available data and the need for future regulatory action based on warnings arising from this screening. Two group of substances were gathered depending on their molecular weight and properties. The one of the subgroup 2 (EC No 429-160-3, a pentaerythritol ester of 2-cyano-3,3-diphenyl-prop-2-enoate) will not be considered here because, a priori, too different. The considered subgroup 1 is composed of five substances which are either ethyl-, 2-ethylhexyl-, or undec-10-ene- esters of either 2-cyano-3,3-diphenyl-prop-2-enoate or 2-cyano-3-(4-methoxy-phenyl)-3-phenyl-prop-2-enoate and are illustrated in the Figure 11 below.

| Ester group             | 2-Cyano-3,3-diphenyl-prop-2-enoate                                                                                  | 2-Cyano-3-(4-methoxy-phenyl)-3-phenyl-prop-2-enoate                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subgroup 1</b>       |                                                                                                                     |                                                                                                                                     |
| - ethyl ester           | <br>Etocrilene: EC No 226-029-0   | n/a                                                                                                                                 |
| - 2-ethylhexyl ester    | <br>Octocrilene: EC No 228-250-8 | <br>List No 700-213-5<br>(RaX to EC 700-284-2) |
| - undec-10-ene ester    | <br>List No 700-604-0            | <br>List No 700-284-2                          |
| <b>Subgroup 2</b>       |                                                                                                                     |                                                                                                                                     |
| - pentaerythritol ester | <br>EC No 429-160-3              | n/a                                                                                                                                 |

**Figure 11: Potential alternative to octocrilene from the group of esters of cyanodiphenyl propenoic acid.**

The extended conjugation of the acrylate portion of the molecules absorbs UVB and short-wave UVA radiation. Therefore, the substances of this group function mainly as UV stabilisers for plastics and/or as active ingredient of cosmetic products and some members of the group can be envisaged as future alternatives for octocrilene as UV filters, despite the fact that **none are authorised in the annex VI of the cosmetic regulation** for now. When looking at structural similarities and UV absorption properties, two substances of the group appeared to be potential alternatives to octocrilene, namely etocrilene (EC No 226-029-0, CAS: 5232-99-5) and ethylhexyl methoxycrilene (EC no 700-213-5, CAS 947753-66-4) (highlighted in

green). The two substances are listed in CosIng as cosmetic ingredients used for their function as skin conditioning and UV absorber.

#### **3.2.4.1.1. Human Health effect**

Briefly, and based on the ARN published by ECHA, all substances in subgroups 1 and 2 are unlikely to have carcinogenic (C) or mutagenic (M) hazards. This is based on available data on some substances and extrapolation to the others due to structural similarity for subgroup 1 substances which are either ethyl-, 2-ethyhexyl-, or undec-10-ene- esters of cyanodiphenyl propenoic acid. The substances are expected to be hydrolysed by esterases and further oxidized in the liver by the CYP enzymes.

Skin sensitisation and photosensitisation cannot be excluded based on octocrilene data. In conclusion, uncertainties remain for the group as being skin sensitisers.

#### **3.2.4.1.2. Environmental health effect**

The substances are non-volatile liquids or low melting solids with negligible vapour pressure (VP). They are poorly soluble in water (WS <0.1 mg/l) with logKow values >6 (indicating a bioaccumulation potential) and logKoc values >ca.4 (indicating a high potential for adsorption).

Based on the ARN published by ECHA, no substances in the group are readily biodegradable and therefore they screen as potentially persistent (P). Sediment simulation studies are available for octocrilene (see section 1.4.2.1.1.2), and a testing proposal was submitted for etocrilene.

Octocrilene has been assessed as not bioaccumulative and consequently not classified as PBT by the PBT expert group despite presenting some bioaccumulation in several species (see section 1.4.2.1.10.2). Etocrilene and the substances List Nos 700-213-5 and 700-604-0 have a potential for bioaccumulation based on the high Kow values (above 6). In a weight of evidence approach using log Kow, QSAR and toxicological data, the bioaccumulation potential is below 500 for etocrilene. No indication for bioaccumulation could be found in the available toxicological studies.

The substances are not T from the available data.

In conclusion, octocrilene does not fulfil the criteria for being classified as a PBT or (v)P(v)B, whereas there is uncertainty for etocrilene and the substances List Nos 700-213-5 and 700-604-0.

#### *ENV hazards identified for subgroup 1 substances*

To assess effects on aquatic species for poorly-water soluble substances, long-term studies are necessary.

There are chronic effects in Daphnia for octocrilene and List No 700-213-5 (which are read-across) but no effects in fish at the water solubility limit. Both List Nos 700-213-5 and 700-604-0 are incorrectly self-classified, they should be classified as aquatic chronic 1.

There are no chronic effects in Daphnia for etocrilene. Following a recent CCH<sup>53</sup>, etocrilene present a NOEC of 0.09 mg/L for fish based on mortality (OECD TG 210) meaning that the substance should be classified as aquatic chronic 1 (see annex B.7.).

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<sup>53</sup> <https://echa.europa.eu/documents/10162/c28b58e2-00f3-da49-6f21-ef8520833842>

### 3.2.4.2. Other potential alternatives to be used as UV filters

#### 3.2.4.2.1. Nature derived UV-filters

Various natural molecules are available and can possibly be used in sunscreens formulation as UV-filters (Chavda et al., 2023, He et al., 2021). The main investigated natural molecules or components to be used in sunscreen formulations are detailed below:

- |                           |                                                    |                    |
|---------------------------|----------------------------------------------------|--------------------|
| - Rosmarinic acid         | - Flavonoid compounds                              | - Silymarin        |
| - Lignin                  | - Phenolic compounds                               | - Quercetin        |
| - Carotenoids             | - Microsporin-like                                 | - Rutin            |
| - Raspberry extract       | - Amino Acids conjugated with chitosan (MAAs)      | - Chlorogenic acid |
| - Blueberry extract       | - Buddleja Officinalis Flower Extract (Soliberine) | - Ellagic acid     |
| - Peels of Benitaka grape | - Coral powder                                     | - Ferulic acid     |
| - Naringenin              |                                                    | - Marin extracts   |
| - Mangiferin              |                                                    |                    |
| - Epigallocatechin        |                                                    |                    |
| - Hyaluronic acid         |                                                    |                    |

These substances or mixtures cannot be considered directly as potential alternatives as some exert UV protection (e.g. carotenoids, rosmarinic acid with absorption peak in both UVA (330 nm) and UVB (229 nm) regions, marin extract) and others act more as antioxidant (e.g. phenolic compounds, rutin), anti-inflammatory or photo-protective enhancers of the formula (e.g. lignin extract, MAAs, ferulic acid) (He et al., 2021, Chavda et al., 2023). Moreover, some of these substances are used in combination to already authorised UV filters to improve the photostability of the formulation and increase the SPF of the sunscreen (e.g. mangiferin, Kawakami and Gaspar, 2015, Patent EP0793476A1, WO2019150087A1). They also are used when photoprotection is not the main purpose of the cosmetic products considered (Hegde et al., 2024).

Recent work by Tsuno Group in association with Matsumoto Trading on ferulic acid, well known for acting as a melanin production blocker by stopping tyrosinase activity, while expressing anti-oxidant, anti-inflammatory and antimicrobial properties, demonstrated that it can be used efficiently as a UV filter. Indeed, a patent was recently filed for using high concentration of ferulic acid derivated from rice bran as UV filter (SPF 50+, ferulic acid as sole UV filter in the formulation). The activity of ferulic acid as a UV filter was stated as being in the same range as those of octyl methoxycinnamate<sup>54</sup>.

These data demonstrated that new alternatives to classical UV filters are under investigation and can become available in the future to increase the number of authorised UV filter in annex VI of the cosmetic regulation.

#### 3.2.4.2.2. Advanced nanotechnology-based sunscreen formulations

A possible way to improve sunscreen protection in the future, without awaiting for newly authorised UV filter, is the possible use of nanotechnology for formulating sunscreen with already authorised UV filters. It is already the case as few of them are already authorised and

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<sup>54</sup> <https://www.cosmetic-business.com/en/news/new-natural-sunscreen-made-from-rice-bran>

used, either for mineral filters such as TiO<sub>2</sub> and ZnO or for organic UV Filter such as Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine or Tris-biphenyl triazine.

The use of nanomaterial is authorised in cosmetic regulation, which defines them as “an insoluble or biopersistent and intentionally manufactured material with one or more external dimensions, or an internal structure, on the scale from 1 to 100 nm”. These structures are illustrated in annex E.2.

The preparation of these nanocarriers required specific technologies and can possibly be tricky. They all present advantages and disadvantages depending on the required function in the sunscreen formulation or more generally, in any of the cosmetics. They are detailed in figure 12 of annex E.2.

There is large research and development in that fields that are promising and might result in alternative products in the future. These alternatives, when using already authorised UV filter can be available very soon. At the opposite, the development of new nanosized UV filter can take much more time until the final approval as UV filter, indicating that these types of alternatives could only be available in several years, if not decades.

### 3.2.4.3. Conclusion on alternatives

In conclusion, no drop-in alternatives of octocrilene are available in the market as octocrilene plays multiple roles when used in cosmetics. Nevertheless, alternatives to octocrilene as a UV filter, light stabiliser and UV absorber are available on the market and are already used on the market. Some present better environmental, ecotoxicological and toxicological profiles depending on the choice of the alternatives for the specific requested function. However, it is important to note that some of the alternatives are currently suspected to be substances with ED, PBT/vPvB properties, or might become subject to regulatory actions under REACH in the future (see section 3.2.4).

As formulation of finished cosmetic products based on alternative substances – for each of the abovementioned technical functions – are already available on the EEA market, it demonstrates the technical feasibility and affordability of the proposed restriction (see also Appendix E.2.3).

## 3.3. Risk reduction and environmental impacts

As mentioned in section 1.4.3.3.1, the estimation of environmental releases of octocrilene is based on a tonnage approach from the annual use volumes of octocrilene in cosmetic products as determined in section 1.3.2. Thus, approximately 1,700 tonnes of octocrilene are released from consumer use of cosmetic products each year under the baseline scenario. The proposed restriction options (RO1 and RO2) share the same scope regarding the product categories that would be targeted by the restriction (i.e., sun protection products and ‘other cosmetic products’) but differ in terms of the duration of the transitional period: a transitional period of 2 years is considered for RO1 while a transitional period of five years is considered for RO2. Consequently, the reduction in emissions permitted by RO1 is greater than RO2. **Compared to the baseline, RO1 and RO2 are expected to reduce releases of octocrilene into the environment by about 13,300 tonnes (10,800 tonnes to soil and 2,500 tonnes to water) and 8,300 tonnes (6,800 tonnes to soil and 1,600 tonnes to water)<sup>55</sup> respectively over the 10-year period considered for the impact assessment** (see also Figure 12 displaying the conceptual release pathways of octocrilene to the environment through consumer use of

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<sup>55</sup> Main scenario, values rounded to the nearest hundreds.

cosmetic products).

Octocrilene is P and T, and could also bioaccumulate to some extent despite not being classified as B according to criteria set out in REACH annex XIII. Consequently, the use of octocrilene and associated releases give rise to specific concerns based on the substance's potential to accumulate in the environment, associated effects that are unpredictable in the long-term and are difficult to reverse even when releases cease. The environmental risk assessment highlighted risks for the following environmental compartments arising from the use of cosmetic products (see section 1.4.3.4):

- agricultural soil
- fresh and marine water
- sediments

Therefore, the proposed restriction and the associated significant abatement of releases will prevent the identified risks to occur for the abovementioned compartments. However, the Dossier Submitter notes that current releases will not be entirely abated as the use and disposal of mixtures containing octocrilene will continue after the restriction enters into force, and after the transition periods have elapsed.

It should be noted that the risk reduction resulting from the proposed restriction for agricultural soils depends on the use of wastewater treatment plant sludge and in particular its use for fertilizing agricultural soils (i.e., spreading). The elements presented above regarding avoided releases in the event of a restriction are based on the assumption that 100% of the sludge is spread (in the baseline, RO1 and RO2 scenarios). Considering that part of the sludge is incinerated and not spread leads to a decrease in the risk reduction (i.e., avoided releases) resulting from the proposed restriction for agricultural soil.

The assumption regarding the use of WWTP-sludge does not affect the risk reduction for freshwater and marine water resulting from the proposed restriction all else equal. Besides, it should be noted that the quantification of avoided releases through a tonnage approach leads to an underestimation of releases to water. Indeed, under this approach it is assumed that releases associated with the consumption of sun protection products entirely take place through showering and bathing while, in reality, a significant share of these releases corresponds to direct emissions in the environment through swimming/spending time in water. Also, as explained in section **Error! Reference source not found.**, when quantifying avoided releases, the Dossier Submitter assumed that 100% of sewage networks are connected to a wastewater treatment plant. In reality, this rate is lower (approx. 81% in 2022 according to Eurostat data<sup>56</sup>) and a proportion of wastewater is discharged untreated directly into water.

It is difficult to quantify further the risk posed by octocrilene for the environment and the Dossier Submitter could not go beyond quantifying the avoided releases for agricultural soil to characterise the risk reduction therefore minimizing this reduction. However, the Dossier Submitter sought to go further in taking into account the impacts on aquatic environments of the proposed restriction in the impact assessment and to quantify associated benefits.

To do so, the Dossier Submitter commissioned a stated preference study (see Appendix E.4.4. for details) that assessed the economic value of a reduction in potential risks to fresh- and coastal water ecosystems resulting from the proposed restriction. The economic value of such a risk reduction is quantified based on Citizens' Willingness to Pay (WTP). This study, carried out by a team of consultants (Sotis et al., unpublished), was based on a qualitative description

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<sup>56</sup> [Data set](#) "Population connected to at least secondary wastewater treatment" consulted on 20/01/2025.

of the impacts of current octocrilene releases caused by the use of cosmetic products on freshwater and coastal water ecosystems. This qualitative description was established based on expert judgement by the team of consultants in collaboration with the Dossier Submitter and in particular its experts in eco-toxicology. The main elements used in this description are listed below:

- Release pathways: freshwater and marine ecosystems can be exposed to octocrilene via two routes. On the one hand, there are direct emissions during bathing/swimming as part of the recreational use of aquatic environments. On the other hand, indirect emissions can occur and correspond to the elimination of cosmetic products containing octocrilene during showering/bathing.
- Frequency and duration of exposure: Indirect emissions can occur throughout the year. The same applies to direct emissions, although the latter are particularly problematic during the tourist season. Indeed, during the tourist season, the aquatic environment is continuously exposed to high concentrations of octocrilene for several weeks or even months. For instance, continuous exposure lasts around 7 months a year in Mediterranean areas, and 3 months a year in mountain lakes. This continuous exposure is particularly problematic because it undermines the resilience of the environment.
- Magnitude of the impact on freshwater and marine ecosystems: octocrilene is toxic to aquatic species, and particularly highly toxic to invertebrates. For aquatic invertebrates, the concentration levels observed in the environment each year can lead to the death of individuals. Octocrilene is P and T, and therefore posing a long-term risk for aquatic, and especially sediment, organisms. Despite not meeting the B criteria, bioaccumulation studies and monitoring data demonstrate that octocrilene might anyway bioaccumulate to some extent. Thus, other aquatic species are affected indirectly through the food chain, but it is not possible to precisely determine the impact on them. Exposure and impact are greater in areas frequented by tourists as emissions from tourists are added to those from locals (e.g., the Mediterranean). This applies to emissions due to beach going/swimming and via wastewater treatment plants.
- Risk reduction allowed by the restriction: The Dossier Submitter considered that the impact of the proposed restriction (i.e., the expected environmental improvement) depends on the exposure of the different water bodies to octocrilene (duration of exposure, concentration level) but also on the relative weight of octocrilene in relation to the pressure, particularly chemical pressure, to which the ecosystems are subject. Given the possible heterogeneity of the water bodies impacted by the restriction in terms of exposure and environmental pressure and the associated uncertainties, the Dossier Submitter has chosen to consider three qualitative improvement levels in the abovementioned study (low, medium and high). The improvement levels were defined in collaboration between the Dossier Submitter and the consultants on the basis of the results of regulatory tests studying the effect of octocrilene on indicator species. They correspond to a recovery of different group of species according to their level of sensitivity to octocrilene and the relationships between species within trophic chains (see Appendix E.4.4. for details). Finally, given the persistent nature of the substance, there is uncertainty as to the timeframe within which the environmental improvement enabled by the restriction will be realised. This uncertainty was accounted for in the benefits assessment (see section 0).

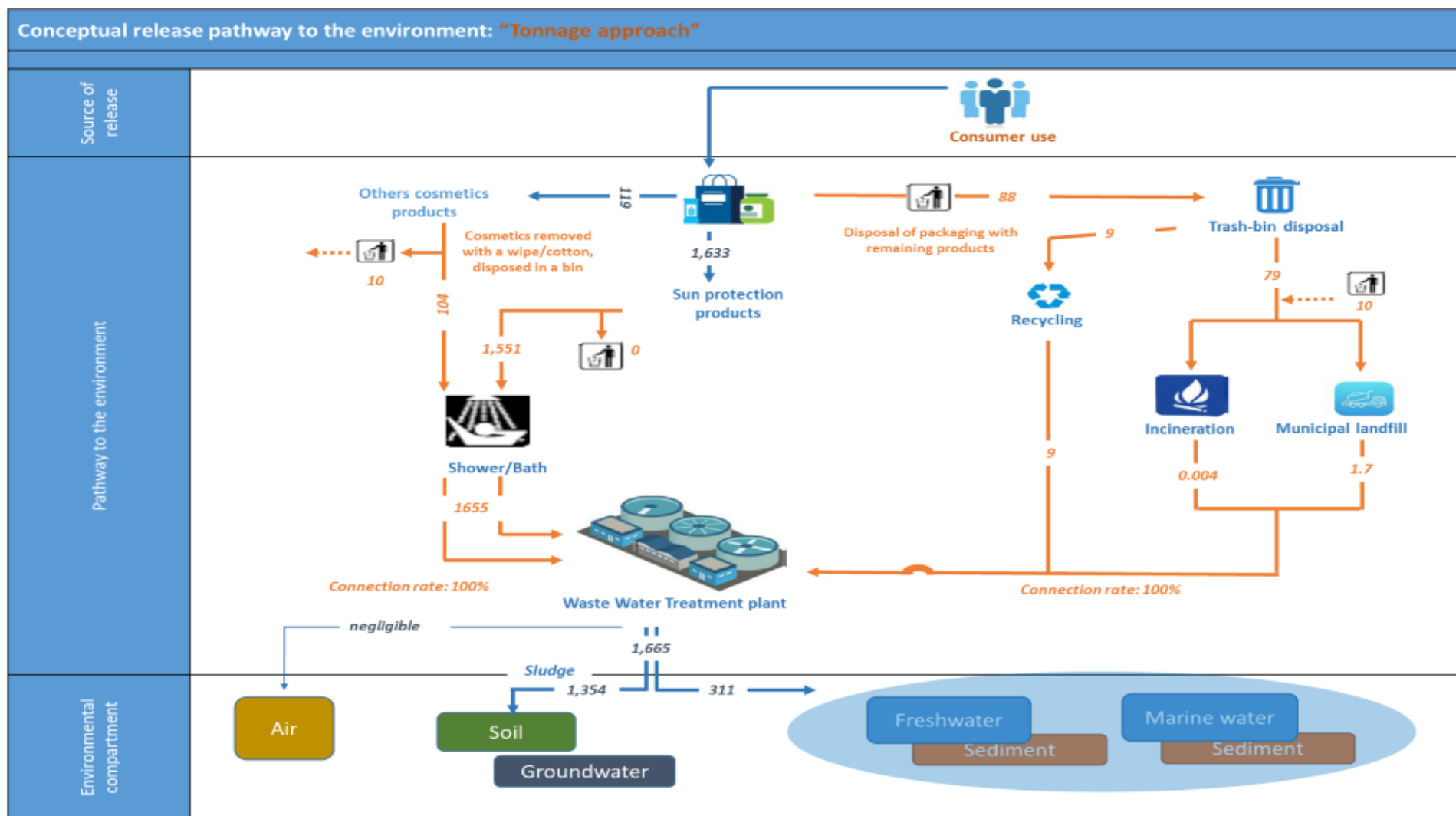
The methodology of the stated preference study and the survey questionnaire are available in Appendix E.4.4. The results of this study, their use in the economic assessment of the benefits of the restriction and elements for discussion are presented in section 0.

Finally, multiple alternatives to octocrilene were identified, some currently suspected to be substances with ED, PBT/vPvB properties. If industry decided to replace octocrilene in cosmetic products by one or a multiple of these substances (likely to result to regrettable substitution), and if the hazard of these substances was confirmed, then the risk reduction capacity of the proposed restriction would therefore be lowered compared to the net risk

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reduction capacity of the proposed restriction if industry selected a technically feasible and available alternative with no identified hazard or concern. Currently, it is not possible to pre-empt the hazard conclusions for these alternative substances as assessments and evaluation are currently on-going.

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**Figure 12: Conceptual release pathway of octocrilene to the environment through the use of cosmetic products (releases in tonnes per year determined through a tonnage approach).**

### 3.4. Economic impacts

#### 3.4.1. Impacts of the restriction on the manufacturers and importers of cosmetic ingredients

In this section the Dossier Submitter investigated the impact of the proposed restriction on manufacturers/importers of cosmetic ingredients in the EEA. Identifying these impacts requires looking at two types of manufacturers/importers of cosmetic ingredients:

- Manufacturers/importers of octocrilene in the EEA whose sales opportunities would be reduced (at least in part) in the event of a restriction;
- Manufacturers/importers of alternative substances to octocrilene in the EEA that would benefit from new opportunities.

More specifically, the Dossier Submitter used the public registration data available on the ECHA Chem platform<sup>57</sup> to characterize the production/import activity of octocrilene and alternative substances within the EEA according to the following aspects: global tonnage per year, number and names of active registrants for each substance. Based on these data, the Dossier Submitter determined that there is octocrilene production/import activity in the EEA likely to be impacted by the proposed restriction. Indeed, according to the data extracted from the ECHA Chem platform, there are 8 registrants reporting this activity for a total production/import volume of 1,000 to 10,000 tonnes per year at the EEA level (Table 26). The sales opportunities – use volumes of octocrilene for the production of cosmetic products – for these stakeholders would be reduced<sup>58</sup>.

Furthermore, the Dossier Submitter determined that presently there is also production/import activity for alternative substances to octocrilene in the EEA. As Table 26 shows, there are active registrants for 20 of the 29 chemical UV filters authorised under the Cosmetics Regulation<sup>59,60</sup>. Thus, although there is significant uncertainty regarding the reported (wide ranged) tonnages and the type of registrant activity (import or production, the registration data not allowing to differentiate them), this shows that associated stakeholders already have the technical knowledge and/or are already integrated into commercial networks for these substances within the cosmetics market.

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<sup>57</sup> Echa Chem platform (<https://chem.echa.europa.eu/>): last access on 24/09/2024. Under the REACH regulation, companies must register substances they import or manufacture in the EEA at 1 tonnes per year and above. As part of the registration, companies submit a so-called registration dossier to ECHA, with information on the identity, properties, classification and uses of the substance.

<sup>58</sup> The Dossier Submitter considers that in the event of a restriction, part of the current production/import volumes would be maintained in order to supply the production of cosmetic products for export as the use of octocrilene remains authorised abroad.

<sup>59</sup> Here the Dossier Submitter has considered the nano and non-nano forms of a UV-filter as a single substance, since both share a single identifier ('EC-number'). In addition, as the industrial stakeholders stressed that the substitution of octocrilene would be based on chemical UV filters only and not on mineral filters, the Dossier Submitter only looked at the production/import activity of the other authorised chemical filters.

<sup>60</sup> During the stakeholder consultation, manufacturers pointed out that 6 of these 29 chemical filters are no longer used for photo-protection. The following 4 filters in particular are no longer marketed: Camphor Benzalkonium Methosulfate, Benzylidene Camphor Sulfonic Acid, Polyacrylamidomethyl Benzylidene Camphor, Peg-25 Paba.

**Table 26: Number of active registrants and tonnage for chemical UV-filters authorized under the Cosmetic Products Regulation.**

| Reference number in annex VI - Cosmetic regulation | Name of Common Ingredients Glossary                          | EC number | Number of active registrants <sup>a</sup> | Tonnage per year (tpa) |
|----------------------------------------------------|--------------------------------------------------------------|-----------|-------------------------------------------|------------------------|
| 3                                                  | Homosalate                                                   | 204-260-8 | 9                                         | 1000-10000             |
| 4                                                  | Benzophenone-3                                               | 205-031-5 | 6                                         | 100-1000               |
| 6                                                  | Phenylbenzimidazole Sulfonic Acid                            | 248-502-0 | 11                                        | 1000-10000             |
| 7                                                  | Terephthalylidene Dicamphor Sulfonic Acid                    | 410-960-6 | 6                                         | 100-1000               |
| 8                                                  | Butyl Methoxydibenzoylmethane                                | 274-581-6 | 9                                         | 1000-10000             |
| 10                                                 | Octocrilene                                                  | 228-250-8 | 8                                         | 1000-10000             |
| 14                                                 | Isoamyl p-Methoxycinnamate                                   | 275-702-5 | 1                                         | 10-100                 |
| 15                                                 | Ethylhexyl Triazone                                          | 402-070-1 | 17                                        | 1000-10000             |
| 17                                                 | Diethylhexyl Butamido Triazone                               | 421-450-8 | 13                                        | 100-1000               |
| 20                                                 | Ethylhexyl Salicylate                                        | 204-263-4 | 14                                        | 1000-10000             |
| 21                                                 | Ethylhexyl Dimethyl PABA                                     | 244-289-3 | 4                                         | 100-1000               |
| 22                                                 | Benzophenone-4,<br>Benzophenone-5                            | 223-772-2 | 5                                         | 100-1000               |
| 23                                                 | Methylene Bis-Benzotriazolyl Tetramethylbutylphenol          | 403-800-1 | 12                                        | 1000-10000             |
| 24                                                 | Disodium Phenyl Dibenzimidazole Tetrasulfonate               | 429-750-0 | 2                                         | 10-100                 |
| 25                                                 | Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine               | 425-950-7 | 16                                        | 1000-10000             |
| 28                                                 | Diethylamino Hydroxybenzoyl Hexyl Benzoate                   | 443-860-6 | 10                                        | 1000-10000             |
| 29                                                 | Tris-biphenyl triazine, Tris-biphenyl triazine               | 479-950-7 | 1                                         | 10-100                 |
| 31                                                 | Phenylene Bis-Diphenyltriazine                               | 700-823-1 | 1                                         | 10-100                 |
| 32                                                 | Methoxypropylamino Cyclohexenylidene Ethoxyethylcyanoacetate | 700-860-3 | 1                                         | 1-10                   |
| 33                                                 | Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine          | 485-100-6 | 1                                         | 1-10                   |

a: Companies with several subsidiaries have been considered as a single entity. For example, the entities 'Symrise AG', 'Symrise H17' and 'Symrise Ibérica, S.L.' have all been grouped together as a single entity called 'Symrise'

Source: ECHA Chem accessed on 27/09/2024

In addition, the Dossier Submitter studied the concordance between the active registrants for octocrilene and the active registrants for the alternative substances. As shown in Table 27, the results of this analysis show that among the 8 active registrants for octocrilene, 6 of them also have production/import activity:

- for 4 or more other UV filters – if it is considered that all the chemical filters listed in Annex VI of the Cosmetic Products Regulation qualify as an alternative;
- for 3 or more other UV filters - if it is considered that among the chemical filters listed in Annex VI only filters absorbing at least partially in the UVB range qualify as an alternative.

Thus, there is already significant production/import activity of alternative substances to octocrilene within the EEA, and these are already incorporated into the production of cosmetic products. In addition, there is significant overlap between the stakeholders involved in the production/import of octocrilene and these alternatives. In addition, the Dossier Submitter collected qualitative information from a few producers of cosmetic products concerning the origin of the use volumes of octocrilene and its alternatives (production within the EEA or

import). The consulted stakeholders did not indicate any major difference regarding their supply regime between octocrilene and its alternatives: they rely on both EEA and non-EEA producers. **Based on these elements, the Dossier Submitter therefore considers that the impact of the proposed restriction - even if uncertainty remains as to the nature of the net effect (positive or negative) - is likely to be moderate for producers/importers of cosmetic ingredients in the EEA** both in terms of sales volumes and in terms of employment in the sector.

**Table 27: Number of registrations for alternative UV filters per company identified as an active registrant for octocrilene**

| Registrant <sup>a</sup>   | Number of registrations for<br>alternative UV filters | Number of registrations for<br>alternative UV filters |
|---------------------------|-------------------------------------------------------|-------------------------------------------------------|
|                           | -<br>All chemical UV filters                          | -<br>'UVB' and 'UVA + UVB' filters only               |
| BASF                      | 8                                                     | 5                                                     |
| DSM Nutritional Products  | 7                                                     | 5                                                     |
| L'Oréal                   | 0                                                     | 0                                                     |
| Merck                     | 5                                                     | 4                                                     |
| Reach24h Consulting Group | 11                                                    | 8                                                     |
| Symrise                   | 9                                                     | 6                                                     |
| Wilkinson Sword GmbH      | 4                                                     | 3                                                     |
| Confidential registrant   | No information                                        | No information                                        |

a : Companies with several subsidiaries have been considered as a single entity. For example, the entities 'Symrise AG', 'Symrise H17' and 'Symrise Ibérica, S.L.' have all been grouped together as a single entity called 'Symrise'  
Source: ECHA Chem accessed on 27/09/2024

### 3.4.2. Impacts of the restriction on the manufacturers of cosmetic products

In general terms, this analysis follows the approach used by the UK Annex XV restriction proposal for the use of D4 and D5 in wash-off cosmetics (ECHA, 2015) and the ECHA Annex XV restriction proposal for the use of D4, D5 and D6 in leave-on personal care products and other consumer/professional products (as well as the use of D6 in wash-off cosmetics) (ECHA, 2020a). However, the Dossier Submitter has adapted some of the key assumptions used in that assessment to the specific case at hand<sup>61</sup>.

The Dossier Submitter considers the following broad categories of impacts for producers of cosmetics/sun protection products that may arise from the proposed restriction:

- extra reformulation costs: one-off (fixed) costs associated to R&D and regulatory tests and adaptation of industrial processes (including acquisition of new production equipment);
- increase in raw material costs (variable costs): increase in the formulation cost<sup>62</sup> due to the use of alternative (and more expensive) ingredients in new formulations as well as additional ingredients to ensure similar product performance (e.g., solvents).

<sup>61</sup> In assessing the reformulation costs, the Dossier Submitter considered a shorter useful life of a formulation (10 years compared to 20 years for ECHA, 2020a) as well as reformulation costs specific to the product categories targeted by the restriction. Concerning the evaluation of the raw material costs, the Dossier Submitter determined the evolution of the formulation cost instead of focusing on the difference in market price between octocrilene and its alternatives for constant use-volumes.

<sup>62</sup> Formulation cost is the total ingredients cost of producing one kilogram of a cosmetic formulation. It can be derived from the cost per kg of ingredients and their concentration in the end-product.

The Dossier Submitter did not consider potential costs associated with performance losses for sun protection products, as products based on alternatives are already available on the market and offer equivalent performances in terms of effectiveness of sun protection and sensory performance (source: stakeholder consultation)<sup>63</sup>. Regarding performance losses for cosmetic products other than sun protection products, the Dossier Submitter relied on YUKA/CosmEthics data (YUKA, 2024; CosmEthics, 2023, see Appendix A.2.1.1) and concludes from the small share of octocrilene-containing products for these products categories that products based on alternatives are likely to offer similar performances.

Finally, the following general assumptions have been applied to the calculation of the costs:

- Entry into force of a restriction is the year 2027. This is when the transitional period starts to apply;
- Two- and five-years transitional periods have been considered (2-years proposed);
- The period of analysis is 10 years, to coincide with the cosmetics industry's baseline major reformulations cycle;
- A discount rate of 3% has been used;
- Costs are in 2024 prices, and so the Present Value (PV) is presented to 2024;
- The reformulations undertaken as a result of the restriction take place throughout the transitional period, in equal numbers each year. The Dossier Submitter considers this is a more realistic assumption than all the reformulations taking place at the end of the transitional period, due to specialists who can undertake these reformulations being scarce resources.

### **3.4.2.1. Reformulation costs**

If a restriction prevents/limits the use of octocrilene in cosmetic products, companies producing them will have to reformulate their products to remove octocrilene if they want to continue placing them on the market. The annualised net cost of product reformulation is estimated by considering:

- the number of products facing formulation;
- the reformulation cost per formulation;
- the useful life of a formulation and baseline reformulation cycle (annualisation period).

#### **3.4.2.1.1. Number of products facing formulation**

The Dossier Submitter determined the number of products facing formulation (to estimate the expected number of reformulations) by considering the total number of cosmetic product formulations available on the EEA market, the share of these products containing octocrilene

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<sup>63</sup> The Dossier Submitter received confirmation from Cosmetics Europe (personal communication) that it is technically possible to formulate sunscreen products without octocrilene. Cosmetics Europe also confirmed that products that are 'cosmetically appealing' and octocrilene-free (based on alternative chemical filters) are already available on the market. Other consulted industrial stakeholders have also confirmed the availability of sun protection products based on alternatives to octocrilene and offering equivalent performances.

and assumptions on the actual reformulation rate<sup>64</sup>.

The Dossier Submitter was not able to access updated data regarding the total number of cosmetic product formulations available on the EEA market. Therefore, the Dossier Submitter relies on the data used in the Annex XV restriction proposal for the use of PFASs, which focused in particular on the use of these substances in cosmetics (ECHA, 2023a). In the latter, the authors considered the European Commission impact assessment on simplification of the Cosmetics Directive, which indicated that 300,000 cosmetic product formulations were available on the market in 2008. This estimate was updated to 520,000 formulations available on the EEA market in 2023 based on information provided by Cosmetics Europe (100,000 formulations from 'larger companies' and 420,000 from SMEs).

Only part of these formulations contains octocrilene and would be affected by the proposed restriction. The Dossier Submitter opted for a similar method to that used in the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) and determined the number of formulations containing octocrilene at the EEA level by extrapolation, using the databases generated by the CosmEthics (CosmEthics, 2023) and YUKA (YUKA, 2024) applications. Both these applications gather data on the ingredients contained in cosmetics products across Europe and provides a mobile application that allows consumers to scan products and inform themselves about their composition. Data within these two databases stem from both the cosmetic industry - directly transmitting data on its products and their composition - and consumers transmitting the references and composition of products not yet listed. These two databases offer broad coverage in terms of both product types and countries within the EEA, as well as quality processes guaranteeing data reliability. However, it should be noted that both applications list products and not formulations. Using these data biases, the distribution of the number of formulations in favour of those that exist in several package sizes. The Dossier Submitter has worked more closely with the company YUKA and has consequently a more detailed knowledge of the data provided by this app and associated uncertainties. Thus, the central estimate of the cost assessment performed used data from the YUKA database. The Dossier Submitter has also calculated the reformulation costs using CosmEthics data to provide a basis for comparison (see Appendix E.4.3.5).

According to the data received from YUKA, 2.31% of cosmetic formulations contain octocrilene and would need reformulation to be able to be kept on the market under the proposed restriction. As mentioned in section 1.3.4, the share of octocrilene-containing products varies greatly from one product category to another (e.g., approx. 37%, 2.7% and 0.24% in sunscreens, foundation/BB cream and hair care products respectively). Assuming 520,000 formulations on the EEA cosmetic market, this results in approximately 12,000 formulations<sup>65</sup> (approx. 9,300 formulations for sun protection products and 2,700 formulations for 'other' cosmetic products; see Appendix E.4.2 for exact values) containing octocrilene according to YUKA data<sup>66</sup>.

The YUKA and CosmEthics data do not make it possible to identify which/how many facial

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<sup>64</sup> Similarly to the UK Annex XV restriction proposal for the use of D4 and D5 in wash-off cosmetics (ECHA, 2015) and the ECHA Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a), the Dossier Submitter considered that, in the event of a restriction, it is not necessarily optimal for a firm to reformulate. This is also consistent with the results presented in the RTI study (RTI, 2002), which states that ceasing production (i.e., product withdrawal) may be the firm's least cost option depending on the size of the reformulation costs.

<sup>65</sup> In the main text, the Dossier Submitter uses rounded figures signaled by the word 'approximately' but has performed all calculations with exact values given in Appendix E.4.2.

<sup>66</sup> As mentioned above, the Dossier Submitter uses the term 'sun protection products/formulations' to refer to both primary sunscreens and secondary photo-protection products (e.g. facial care and make-up products that also offer protection against UV radiation).

care and make-up products actually correspond to secondary photo-protection products. In the absence of more precise data, the Dossier Submitter assumes that all products belonging to product categories that could contain secondary photoprotection products are secondary photo-protection products<sup>67</sup>. On the one hand, considering the assumptions made for the reformulation cost (see section B below), this assumption could lead to an overestimation of reformulation costs. On the other hand, not making the distinction in these products categories between products that offer sun protection and those that do not (ex: daycreams and daycreams offering sun protection) could lead to an incorrect appraisal of the share of products containing octocrilene, of the reformulation rate (see below) and ultimately lead to an underestimation of reformulation costs. In the absence of more precise data, the Dossier Submitter considers that these effects cancel each other out.

As mentioned also in section 1.3.4, the Dossier Submitter cross-checked the data from the YUKA and CosmEthics applications with the Dossier Submitter's own database on the share of primary sunscreens products containing octocrilene in 5 EU countries in order to determine the quality of the extrapolation process (i.e., based on the proportion of products containing octocrilene within each product category). All these data sources are broadly consistent with each other. In addition, as mentioned above, a sensitivity analysis was carried out and the cost of reformulation was calculated using CosmEthics data.

Regarding the expected number of (actual) reformulations, the Dossier Submitter follows the approach used in the ECHA Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) and considered that only part of formulations containing octocrilene would actually be reformulated in the event of a restriction<sup>68</sup>. As a general approach, the Dossier Submitter relied on YUKA data and assumed that the lower the proportion of products that contain octocrilene within a product category, the lower the proportion of products within this category that would actually be reformulated in the event of a restriction. YUKA data show that for most subcategories, the products using octocrilene are a minority. Out of 39 subcategories where products containing octocrilene are present, in 32 categories (82% of them) products containing octocrilene are less than 2% of the total products identified in their category. Only two product categories have a share of products containing octocrilene above 30% (lip balms with sunscreen: 30.2%; sunscreens: 36.74%; see also Figure 2). The proportion of products with octocrilene for each subcategory as well as the number of expected reformulations is included in Appendix E.4.2.

The reformulation rate is therefore determined for each product category on the basis of the proportion of products containing octocrilene within that category. The specific assumptions used are as follows:

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<sup>67</sup> If we refer to the nomenclature of the CosmEthics database adapted to the YUKA data (see Appendix A.2.1 for details), products categories for which octocrilene-containing products are considered to be secondary photo-protection products are:

- Facial care: Moisturisers/Face cream and Anti-age cream;
- Make-up: Foundation/BB Cream;
- Male grooming: Facial care.

<sup>68</sup> The following rationale is provided in the ECHA Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020 p46): "if there are many products within a category that do not contain D4, D5 and D6 then it is likely that these already offer comparable product performance to products that contain D4, D5 and D6. In this scenario companies (particularly large ones, which are also likely to produce alternative formulations within the same category) will accept that customers will switch to an existing alternative product rather than invest in reformulation. Where there are relatively few products within a subcategory that do not contain D4, D5 and D6 it is assumed that alternative products, not containing D4, D5 and D6, do not currently offer comparable product performance and that companies will, in this instance, prefer to reformulate existing products." This assumption seems relevant in view of the data collected by the Dossier Submitter on primary sunscreens, which show that the majority (64%) of producers of primary sunscreens own a portfolio in which formulations with and without octocrilene coexist.

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- For product categories where products containing octocrilene represent less than 30% of the market (this is the case for anti-age creams for example), the alternatives products are expected to take over their market share and very few of these products are expected to be reformulated in the event of a restriction (assumed 5%);
- For product categories where products containing octocrilene represent between 30% and 70% of all products (this is the case for sunscreens for example), it is assumed that half of these products would be reformulated in the event of a restriction. The remaining 50% of products are expected to be discontinued;
- For product categories where products containing octocrilene represent over 70% of all products, it would be assumed that 95% of those products would be reformulated in the event of a restriction. However, no categories in the YUKA (nor CosmEthics) data show such high prevalence of products containing octocrilene.

Based on YUKA data and the assumptions described above, the Dossier Submitter estimates that approximately **3,600 reformulations are expected in the event of a restriction** (see Appendix E.4.2 for exact values):

- 3,400 sun protection products<sup>69</sup>;
- 200 'other' cosmetic products.

Thus, considering all types of cosmetic products together, approx. 30% of all the formulations containing octocrilene would actually be reformulated (approx. 37% of sun protection products and 8% of 'other' cosmetic products)<sup>70</sup>. The reasoning used to determine the expected number of reformulations and the cost of reformulation as a whole in the event of a restriction is detailed in Figure 13.

In the absence of specific data, the Dossier Submitter considers the same distribution as in the Annex XV restriction proposal for the use of PFASs (ECHA, 2023) and assume that 19% of formulations belong to larger companies and 81% to SMEs.<sup>71</sup>

The Dossier Submitter also carried out a sensitivity analysis for the 'partial reformulation hypothesis', estimating reformulation costs in a scenario where all products containing octocrilene would be reformulated in the event of a restriction (see Appendix E.4.3.2).

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<sup>69</sup> This figure (3,413 for the exact value) is obtained by the following calculation (see also Appendix E.4.2):

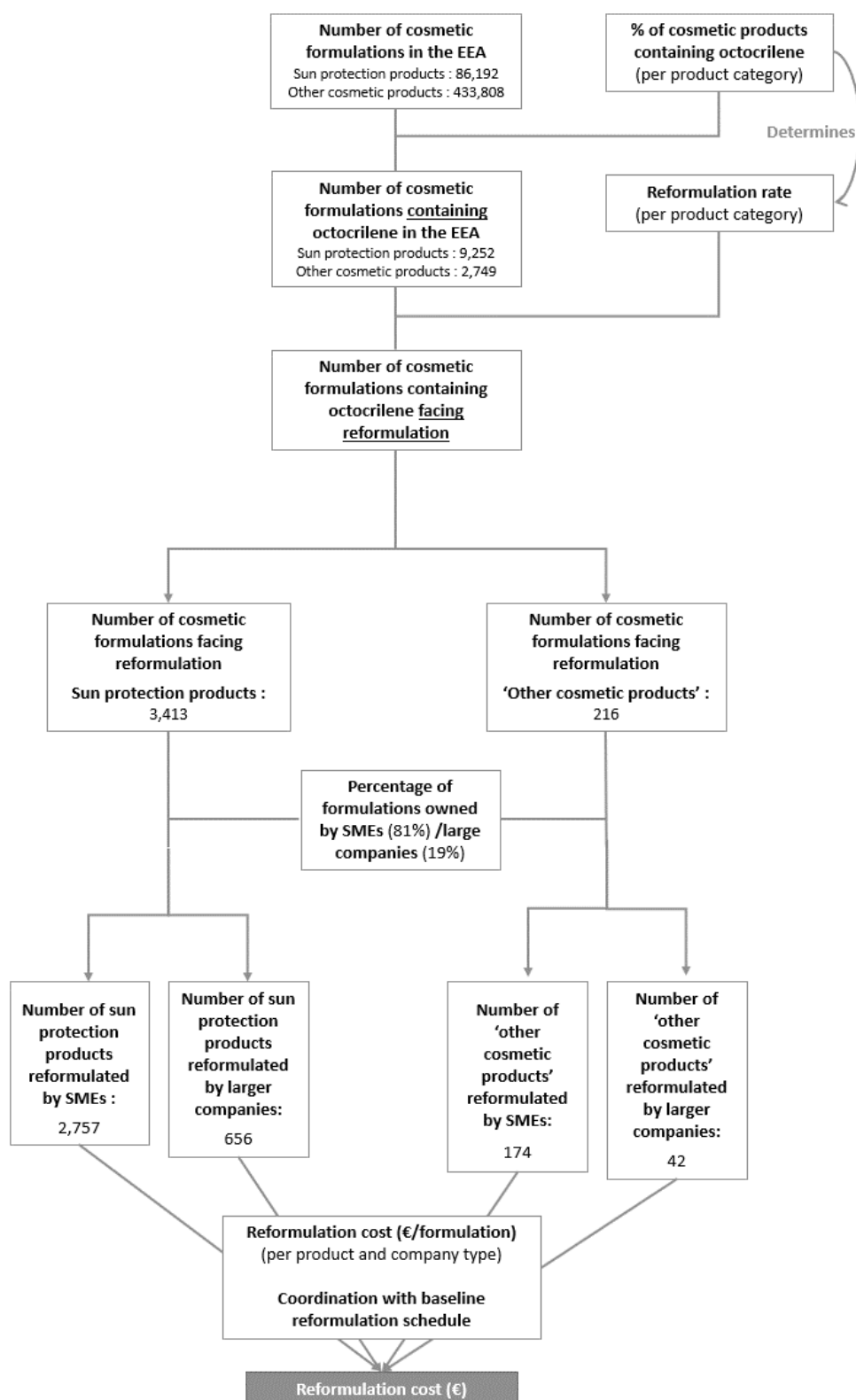
- 50% of the 6,555 formulations of primary sunscreens that contain octocrilene are reformulated in the event of a restriction;
- As well as 5% of the 1,313 face cream formulations, the 880 anti-age cream formulations, the 500 Foundation/BB cream formulations and the 4 face creams formulations for men that contain octocrilene.

<sup>70</sup> These reformulation rates are global rates calculated for an aggregate of several product categories. For example, the 37% reformulation rate calculated for sun protection products assumes a 50% reformulation rate for products belonging to the 'sunscreens' product category, while a 5% reformulation rate applies to all secondary photo-protection products (Moisturisers/Face cream and Anti-age cream, Foundation/BB Cream and Male grooming- Facial care).

<sup>71</sup> The Dossier Submitter uses rounded figures in the main text but performed the calculations considering the following exact values:

- Share of cosmetic formulations owned by SMEs: 420,000/520,000;
- Share of cosmetic formulations owned by larger companies: 100,000/520,000.

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**Figure 13: Calculation method of the reformulation costs induced by the proposed restriction.**

Note: figures indicated in the boxes are figures used in the assessment. Exact figures regarding the share of cosmetic products containing octocrilene and reformulation rate per product category are available in Annex E.4.2

### 3.4.2.1.2. Reformulation cost per formulation

As explained previously (see 3.2.2) and as underlined by industrial stakeholders, substituting for octocrilene in sun protection products (i.e., primary sunscreens and secondary photo-protection products) requires undertaking a major reformulation.

In the Annex XV restriction proposals for the use of D4, D5 and D6 in cosmetic products (ECHA, 2015 and ECHA, 2020a), the Dossier Submitters also considered that the proposed restrictions would require major reformulations of the concerned cosmetic products. The ECHA assessment, consisting of a refinement of the methodology proposed in the UK assessment, considered the costs of a major reformulation to be different depending on the size of the company. A cost of €365,000 was considered for major reformulations implemented by larger companies, while a cost of €42,000 was considered for SMEs. This cost differentiation was based on information provided by Cosmetics Europe at the time the UK assessment was developed. This cost differentiation is also consistent with the results presented in the RTI study prepared for the US FDA (RTI, 2002), which states that certain processes that can be implemented by larger companies as part of major reformulations are usually not implemented by SMEs (e.g. market tests).

As the great majority of reformulations that would be triggered by the restriction would concern sun protection products, the Dossier Submitter has consulted industrial stakeholders to gain a better understanding of reformulation costs for this specific product category. From a qualitative perspective, several industrial stakeholders indicated that sun protection products are particularly complex to (re)formulate and that placing these products on the market requires carrying out numerous (regulatory and non-regulatory) tests (effectiveness of the sun protection but also as many tests as there are claims, e.g. resistance to water, sand, etc.). These stakeholders consequently stressed that sun protection products were among the most expensive products to (re)formulate. From a quantitative perspective, the Dossier Submitter was able to collect data regarding the value of reformulation costs from a few SMEs and 'larger' companies for sun protection products (see Table 28 below as well as Appendix G for details on stakeholder consultation).

**Table 28: Reformulation costs for major reformulation of sun protection products (cost per formulation in €2024).**

|                         | Minimum <sup>a</sup> | Mid-point       | Maximum <sup>a</sup> |
|-------------------------|----------------------|-----------------|----------------------|
| <b>SMEs</b>             | €50,000-€70 000      | <b>€85,000</b>  | €100,000-€120,000    |
| <b>Larger companies</b> | €90,000-€110 000     | <b>€175,000</b> | €240,000-€260,000    |

a: As the cost data provided by stakeholders are confidential, they are expressed in the form of intervals.

Source: stakeholder consultations

Regarding the reformulation of other cosmetic products than 'sun protection products', the Dossier Submitter was not able to determine its magnitude (major or minor) and was only able to collect information on the reformulation cost from a single industrial stakeholder and a single product category (use of octocrilene in nail polish). The associated cost data is confidential but suggests that the reformulation cost for this type of product is significantly lower (around five times lower) than the reformulation cost for a sun protection product. In the absence of better-quality data, the Dossier Submitter considers that the reformulation cost depends on the type of product considered and assumes that sun protection products are more expensive to reformulate than other cosmetic products. In addition, quantitative data collected from industrial stakeholders on sun protection products confirm that the cost of a reformulation varies according to the size of the company: it is higher for 'larger companies' compared to SMEs in absolute terms. The Dossier Submitter assumed that the same holds for 'other cosmetic products' and applied a 1/5 ratio to the reformulation costs presented in Table 28. The reformulation cost estimated based on these assumptions for 'other' cosmetic products is shown in

Table 29 below.

**Table 29: Reformulation costs for reformulation of 'other' cosmetic products (cost per formulation in €2024).**

|                         | Minimum         | Mid-point      | Maximum         |
|-------------------------|-----------------|----------------|-----------------|
| <b>SMEs</b>             | €10,000-€14 000 | <b>€17,000</b> | €20,000-€24,000 |
| <b>Larger companies</b> | €18,000-€22,000 | <b>€35,000</b> | €48,000-€52,000 |

A comparison of the reformulation costs considered by the Dossier Submitter with the values used in the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) leads to the following observations:

- For SMEs and as previously mentioned, the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) considers a unique reformulation cost, 'all cosmetic products combined' (42,000 euros). The assumptions made by the Dossier Submitter in this dossier result in a reformulation cost higher than this single value for sunscreen products (as expected) and lower for 'other cosmetic products'.
- For 'larger companies', the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) also considers a unique reformulation cost (365,000 euros). Contrary to the SMEs and contrary to what was expected based on the qualitative elements shared by the industrial stakeholders, the assumptions considered by the Dossier Submitter in the present dossier result in considering a reformulation cost lower than this unique value not only for 'other cosmetic products' but also for sunscreen products.

In this context, the Dossier Submitter gave priority to the data provided by industrial stakeholders over data provided in ECHA (2020a). Besides, the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a) discussed extensively the fact that cost values considered in the assessment might be overestimates. **Thus, the Dossier Submitter assessed the reformulation cost induced by the proposed restriction by considering the mid-point cost values presented in Table 28 and**

**Table 29 as a best estimate.** A sensitivity analysis is presented in Appendix E.4.3.1 in which the Dossier Submitter assessed the reformulation costs based on the minimum and maximum reformulation cost. This Appendix also provides the results of an additional sensitivity analysis assessing the reformulation costs likely to be triggered by the restriction based on the reformulation cost value considered in the Annex XV restriction proposal for the use of D4, D5 and D6 for 'larger' companies, i.e., €466,130 (€365,000 adjusted to 2024 values based on the harmonised index of consumer prices<sup>72</sup>).

In order to put the reformulation costs communicated by the consulted industrial stakeholders into perspective with the size of companies, the Dossier Submitter calculated an 'effort rate'

<sup>72</sup> [https://ec.europa.eu/eurostat/databrowser/view/prc\\_hicp\\_aind/default/table?lang=en](https://ec.europa.eu/eurostat/databrowser/view/prc_hicp_aind/default/table?lang=en) (last access on 29/01/2025)

for each of them.<sup>73</sup> This effort rate corresponds to the ratio of the reformulation cost to the company's turnover. On the one hand, according to the results of these calculations the effort rate appears significantly higher for an SME compared to a larger company. In other words, the reformulation cost represents a significantly larger proportion of turnover and reformulating is relatively more costly for SMEs compared to larger companies. In addition, the results of the calculations confirmed the heterogeneity of stakeholders within the 'large company' category. This category includes both multinationals and intermediate-sized companies. Ideally, the assessment of the reformulation cost induced by the proposed restriction should consider this heterogeneity by determining the number of formulations belonging to each of these two sub-categories and the corresponding cost of reformulation. However, the Dossier Submitter was unable to carry out such an analysis.

Finally, and as discussed in the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a), the costs mentioned above are per *successful* reformulation of one formulation. As in the latter report, the Dossier Submitter has no information on the proportion of failed reformulations and their impact on the cost of reformulation. The Dossier Submitter considers - as in the aforementioned report also - that these failed reformulations generate knowledge and experience that can be used to improve the efficiency of other subsequent reformulations and therefore have a neutral effect on the cost of reformulation.

#### **3.4.2.1.3. Coordination with baseline reformulations**

The cosmetics industry is highly innovative and R&D/reformulations are undertaken regularly to ensure the product portfolios on the market respond to the latest market demands and advancements in the industry. In the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2020a), it is considered that a formulation undergoes a minor reformulation every 6-7 years and a major reformulation every twenty years.

The Dossier Submitter has consulted industrial stakeholders to gain a better understanding of the baseline reformulation cycle for sun protection products and noted some disparities between information shared by the latter. Indeed, some stakeholders stated that they perform major reformulations regularly (due to regulatory constraints but also on their own initiative). Others stress that the complexity of formulating sun protection products calls for undertaking reformulations only when required/needed, resulting in a very low frequency of reformulations for sunscreen products compared to other types of cosmetics products.

The frequency of reformulations reported by manufacturers for sun protection products - although collected from a limited number of stakeholders - suggests that the useful life of a formulation is shorter than that considered in the Annex XV restriction proposals mentioned above (UK-HSE, 2015 and ECHA, 2020a): a sun protection product is reformulated approximately every five to ten years at most. In this context, the Dossier Submitter considers that sun protection products undergo a major reformulation every ten years. Reformulation costs are also calculated by considering a reformulation every twenty years as part of a sensitivity analysis (see Appendix E.4.3.1). With regard to minor reformulations, consulted stakeholders pointed out that these were uncommon for a sun protection product, given that even the modification of a minor ingredient such as a fragrance can have an impact on the effectiveness of a product (in particular on the sun protection factor - SPF) and can therefore prove complex. The Dossier Submitter therefore considers that sun protection products are only subject to major reformulations.

The Dossier Submitter was unable to gather specific data on the frequency of implementation of the different types of reformulation for other cosmetic products containing octocrilene (outside sun protection products). The Dossier Submitter therefore considered identical

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<sup>73</sup> This effort rate is not disclosed in this report for confidentiality reasons.

assumptions for both product categories (one major reformulation every ten years, no assumption regarding minor reformulations).

Finally, regarding the possibility to coordinate some of the reformulations required by the proposed restriction to remove octocrilene with those that would already have happened in the baseline, the Dossier Submitter adopted the same approach as that taken in the Annex XV restriction proposal for the use of D4, D5 and D6 (ECHA, 2015 and ECHA, 2020a):

- i. For products where the baseline major reformulation would have taken place during the transitional period (periods of 2 and 5 years analysed), it is assumed that they would be coordinated with removal of octocrilene and therefore there would be no additional costs as a result of the restriction;
- ii. For products where the baseline major reformulation would have taken place within the 5 years after the end of the transitional period (be it 2 or 5 years), it is assumed that reformulation would be coordinated with removal of octocrilene and done earlier, during the transition period. So there would be only the costs of bringing those reformulations forward;
- iii. For products where the baseline major reformulation would have taken place six years or more after the end of the transitional period, it is assumed that no coordination would be possible, as it would be impossible to predict market demands that far in advance. For those products, there would be an additional major reformulation during the transition period (and the reformulations that would have taken 6+ years after the end of the transitional period would still take place). Therefore, for these products, the full cost of an additional reformulation would be incurred as a result of the restriction.

As in the Annex XV restriction proposals mentioned above, the Dossier Submitter also assumes that the cost of the coordinated reformulations would not increase by incorporating the removal of octocrilene (beyond the extra cost in (ii) of bringing them forward in time).

**As a result of the costs assessment, the Dossier Submitter's central estimate of the reformulation costs associated<sup>74</sup> with introducing a restriction on octocrilene, assuming a transitional period of 2 years (5 years), is a 10-year PV of €111.22 million (€20.97 million). This represents annualized reformulation costs of €12.29 million (€2.32 million)<sup>75</sup>. For illustrative purposes, the annualized cost of reformulation under RO1 represents 0.013% of total sales in value of cosmetic products in Europe in 2023<sup>76</sup> and 0.35% of total sales in value for the same year if we focus on sun protection products<sup>77</sup>.**

**Table 30 and Table 31 below provide a breakdown of the reformulation cost between the two main product categories considered (sun protection products and 'other cosmetic products') and also between SMEs and larger companies. The vast majority (nearly 99%) of reformulation costs are borne by sun protection products. It is also worth noting that the cost of reformulation is borne mainly by SMEs. This can be explained by the fact that the basis for calculating the reformulation cost is the number of reformulations and by the assumptions made about the distribution of**

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<sup>74</sup> Assuming partial reformulation and mid-point values for reformulation costs per formulation.

<sup>75</sup> Equivalent annuity over a period of 10 years, assuming that reformulation costs would be borne from 2027.

<sup>76</sup> The consumption value of cosmetics and personal care products reached 95.7 billion euros in 2023 in Europe (Statista, 2024a).

<sup>77</sup> The consumption value (retail selling price) of the sun care market reached 3.5 billion euros in 2023 in Western Europe (Euromonitor International, 2024b).

cosmetic product formulations between SMEs (approx. 81%) and larger companies (approx. 19%). Putting the costs of reformulation induced by the proposed restriction into perspective with the turnover of each of these two categories of companies on the sun protection products market shows that annualized reformulation costs account for 1.61 % of sales for SMEs and 0.15 % for larger companies.<sup>78,79</sup>

**Table 30: Breakdown of total reformulation costs expected from a restriction between sun protection products and other cosmetic products (transition periods of 2 and 5 years).**

|                         | Number of reformulations | Reformulation costs <sup>a</sup><br>RO1: 2 years transitional period<br>(€ million) | Reformulation costs <sup>a</sup><br>RO2: 5 years transitional period<br>(€ million) |
|-------------------------|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Sun protection products | 3,413                    | 109.83                                                                              | 20.71                                                                               |
| Other cosmetic products | 216                      | 1.39                                                                                | 0.26                                                                                |
| <b>Total</b>            | <b>3,629</b>             | <b>PV = €111.22 million<br/>(annualized cost<sup>b</sup> = €12.29 million)</b>      | <b>PV = €20.71 million<br/>(annualized cost<sup>b</sup> = €2.32 million)</b>        |

a: PV 2024 at 3%; mid-point reformulation cost; useful life of a formulation = 10 years.

b: Equivalent annuity over a period of 10 years, assuming that reformulation costs would be borne from 2027.

**Table 31: Breakdown of total reformulation costs expected from a restriction between SMEs and larger companies (RO1: transition period of 2 years).**

|                  | Number of reformulations | Reformulation costs <sup>a</sup><br>(€ million) |
|------------------|--------------------------|-------------------------------------------------|
| SMEs             | 2,931                    | 74.64                                           |
| Larger companies | 698                      | 36.59                                           |
| <b>Total</b>     | <b>3,629</b>             | <b>PV = 111.22</b>                              |

a: PV 2024 at 3%; mid-point reformulation cost; useful life of a formulation = 10 years.

<sup>78</sup> The following figures were considered for this calculation: according to Euromonitor data, SMEs own 15.8 % of the adult sun protection market and 19.6 % of the market for child and baby specific sun care. In other words, total sales by SMEs in these markets amount to 512 million euros in 2023, while these of larger companies amount to 2.64 billion euros (Euromonitor, 2024a).

<sup>79</sup> Euromonitor (2024a) provides the name of the 61 manufacturers producing adult sun protection or baby and child specific sun care in Western Europe with a market share higher than 0.1 % in the overmentioned markets. The firms with a market share lower than 0.1 % were grouped together as "others". These firms are assumed to be SMEs, although this is not necessarily the case. The firms that sell private labels are assumed to be larger companies. The sales or similar (net sales, income before taxes, turnover) of 54 firms were collected by the Dossier Submitter through their annual report or financial results that were publicly available on their sites or obtained via sites providing stock market information or public declaration on media. Such data could not be found for 7 firms. The Dossier Submitter not having additional information to further distinguish them, manufacturers with group sales of less than €50 million, or for whom it was impossible to collect information, were then considered as SMEs. The other were considered larger firms. Thus, the Dossier Submitter database includes at least 16 SMEs and 45 larger manufacturers.

### 3.4.2.2. Raw material costs

#### 3.4.2.2.1. Primary sunscreens

Following a substance-by-substance approach, the Dossier Submitter was able to collect data regarding the market prices (in €/kg) of alternative UV-filters<sup>80</sup> from industrial stakeholders (stakeholder consultations) supplemented by price data from online chemical substance sales platforms where data was lacking. Three alternatives (Homosalate, Ethylhexyl Methoxycinnamate and Ethylhexyl Salicylate) were identified with market price below or close to that of octocrilene. However, according to industrial stakeholders these alternatives are characterized by a low feasibility as their use is already restricted and/or they are currently under scrutiny from a regulatory perspective<sup>81</sup>. The majority of alternatives are thus more expensive, some substantially so (see Appendix E.2 for more details on technical, regulatory and economic feasibility of alternative substances).

However, and as mentioned previously, it should be noted that octocrilene plays multiple roles when used in a sun protection product (i.e., UV filter, photostabiliser, solvent, etc. see section 1.3.4). As UV filters vary in terms of UV absorption spectrum, solubility (fat-soluble or water-soluble) and solubilisation properties, there is no exact equivalent to octocrilene and therefore no drop-in alternative (see Appendix E.2 for more details on the properties of alternative UV filters). Consequently, a restriction on the use of this substance would require a complete redesign of the formulation. Octocrilene could be replaced by one or a combination of several UV filters and the use of additional substances that are not UV filters (additional solvents, etc.) could be required, depending on the chosen alternatives. Uncertainty remains regarding which alternative would be used but according to Cosmetics Europe the substitution of octocrilene as a UV filter would be based on other organic/chemical filters (Cosmetics Europe, personal communication).

Based on this information, the formulation cost<sup>82</sup> appears to be the relevant indicator to assess the extra-cost borne by manufacturers of cosmetic products due to the substitution for octocrilene with alternative raw materials. The Dossier Submitter was able to collect data from industrial stakeholders on formulation costs of products containing octocrilene as well as estimates of the potential increase in formulation cost due to the proposed restriction. Consulted industrial stakeholders underlined that the formulation cost of a sun protection product depends both on the galenic form (i.e., spray, cream, milk, etc.) and on the level of SPF, as the concentration of UV filter that rises with the level of SPF has a significant influence on formulation price. In the absence of data on the distribution of formulations and sales volumes across SPF levels and galenic forms, the Dossier Submitter calculated the net annual raw material cost by considering the following assumptions:

- The formulation cost of sun protection products containing octocrilene lies between 10€/kg and 20€/kg (15€/kg considered as the central estimate; source: stakeholder consultation).
- In the event of a restriction, products containing octocrilene that are reformulated and

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<sup>80</sup> In this case, the Dossier Submitter considered that only UV filters whose absorption spectrum covered UVB at least in part could be considered as alternatives to octocrilene.

<sup>81</sup> Industrial stakeholders reported that homosalate and ethylhexyl salicylate are both concerned by the European Commission's call for data on salicylic esters. In addition, ethylhexyl methoxycinnamate is suspected of being an endocrine disruptor and was assessed by the SCCS [https://health.ec.europa.eu/document/download/3166a978-06de-4526-b25e-8d5a66943a2e\\_en?filename=ccs\\_o\\_294.pdf](https://health.ec.europa.eu/document/download/3166a978-06de-4526-b25e-8d5a66943a2e_en?filename=ccs_o_294.pdf)

<sup>82</sup> Formulation cost is the total ingredients cost of producing one kilogram of a cosmetic formulation. It can be derived from the cost per kg of ingredients and their concentration level in the end-product.

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produced based on alternative substances would face a 10% to 50% increase in their formulation cost (30% increase considered as a central estimate; source: Cosmetics Europe, personal communication). The resulting formulation cost of octocrilene-free formulations is consistent with the information provided by industrial stakeholders (manufacturers of octocrilene-free sun protection products) as part of the stakeholder consultation on this subject.

In this section, the Dossier Submitter relies in part on the same assumptions as those considered in the section 3.4.2.1 on reformulation costs. It is assumed that only part of the formulations containing octocrilene would actually undergo reformulation and that it is only the production volumes corresponding to these reformulated formulations that would experience an increase in raw material costs.

The raw material (RMC) extra-cost resulting from the proposed restriction is calculated as follows:

$$RMC = P \times s \times \rho \times \Delta$$

Where:

- $P$  is the volume of sun protection products produced in the EEA in kg ;
- $s$  is the share of sun protection products produced in the EEA that contain octocrilene;
- $\rho$  is the share of these 'octocrilene-containing' production volumes that would actually be reformulated and produced based on alternative substances in the event of a restriction;
- $\Delta$  is the difference in formulation cost in €/kg between a formulation with and a formulation without octocrilene.

The Dossier Submitter did not have access to production data within the EEA either for all sun protection products or for sun protection products containing octocrilene but only to sales volumes of primary sunscreens for adults<sup>83</sup> (27,848,000 litres sold in 2023), children and babies<sup>84</sup> (4,813,000 litres sold in 2023). These data are extracted from the Euromonitor database (Euromonitor, 2025) and cover EU-27. Thus, the Dossier Submitter determined the additional raw material (RMC) cost resulting from the proposed restriction for primary sunscreens only and the variables in the equation above were approximated by the parameters listed in Table 32 below. In addition, the method for calculating the raw material extra-cost induced by the proposed is illustrated in Figure 14.

**Table 32: Variables used in the calculation of the raw material cost.**

| Original variable                                                | Variable used                                                                                                                                           |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P$ : Volume of sun protection products produced in the EEA (kg) | Sales volumes of primary sunscreens for adults, babies and children in the EU-27 in litres (L) combined to density of primary sunscreen in kg/L (proxy) |
| $s$ : Share of sun protection products produced in the           | Share of primary sunscreens (proportion of product                                                                                                      |

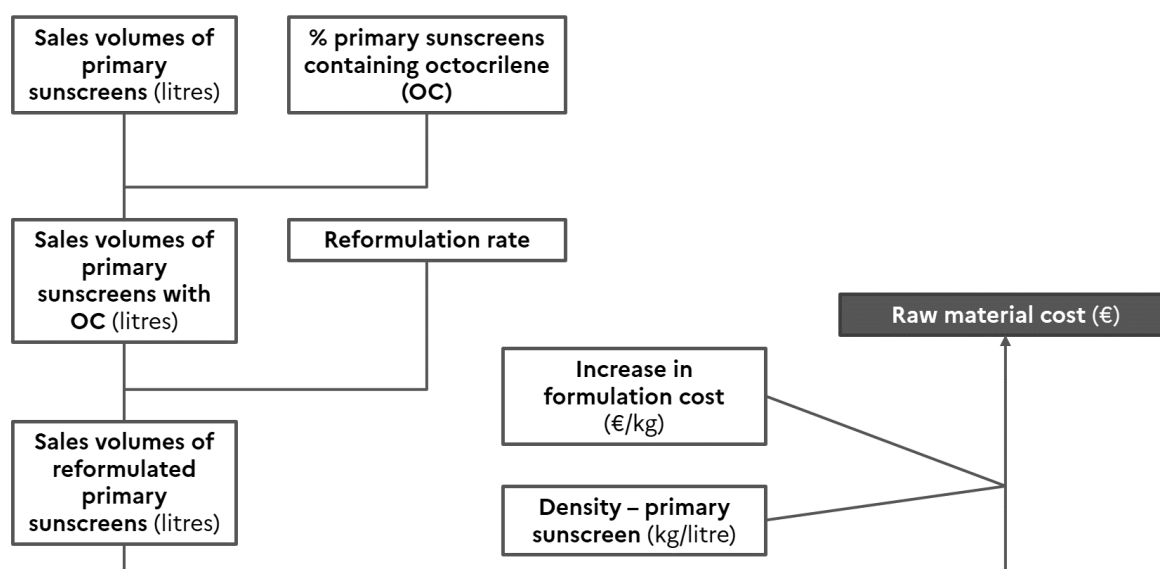
<sup>83</sup> The Dossier Submitter considered the sales volumes for the 'Adult sun protection' category within the Euromonitor database. This product category is defined as follows: 'Includes all premium and mass-market milks, lotions, gels, cushion, mousse and oil formulated with sun protection factors as a sun screen/block. Also includes filter-free tan accelerators and products designed specifically for lip and hair protection. Also included are brands marketed at both adults and children. Excluded are baby/children's brands, which are included in baby sun care.'

<sup>84</sup> The Dossier Submitter considered the sales volumes for the 'Baby and Child-specific Sun Care' category within the Euromonitor database. This product category is defined as follows: 'Includes all sun protection and after-sun products specifically marketed for babies and young children. Excluded are products marketed as 'ideal for adults and children' or those adult products marketed as also 'suitable for children and sensitive skin'.'

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| EEA that contain octocrilene                                                                                                                                             | references) that contain octocrilene (proxy)                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $\rho$ : Share of 'octocrilene-containing' production volumes that would actually be reformulated and produced from alternative substances in the event of a restriction | Reformulation rate of primary sunscreens as considered in section 3.4.2.1 (proxy)                       |
| $\Delta$ : Difference in formulation cost in €/kg between a formulation with and a formulation without octocrilene                                                       | Difference in formulation cost in €/kg between a formulation with and a formulation without octocrilene |

According to YUKA data and considering the assumptions made in section 3.4.2.1, 36.74% of primary sunscreens contain octocrilene and 50% of these formulations would be reformulated in the event of a restriction. Consequently, the Dossier Submitter considered that 18% of primary sunscreens produced in the EEA would face an increase in formulation cost. Net annual raw material costs were calculated by applying the increase in formulation cost (in €/kg) mentioned above to the relevant share of sales volumes after conversion into kilograms<sup>85</sup>. To do so, an average density of 1.44kg/L was considered (8 doses of 36g per 200ml bottle)<sup>86</sup>.



**Figure 14: Calculation method of the raw material extra-cost induced by the proposed restriction.**

Considering mid-point values for both initial formulation cost and cost increase (15€/kg and a 30% increase), The Dossier Submitter estimated **that net annual raw material extra-costs borne by producers of primary sunscreens would amount to 38.87 million euros per year in the event of a restriction. Assuming a transitional period of 2 years**

<sup>85</sup> By applying these shares to sales volumes, we assume that market shares are distributed in the same way as the share of product references with regard to the presence of octocrilene. However, different formulations/product references do not necessarily have the same market share and formulations containing octocrilene may sell better or worse than those based on alternatives substances.

<sup>86</sup> French National Agency for the Safety of Medicines and Health Products (ANSM) quoted in <https://www.tf1info.fr/conso/video-tf1-conseils-creme-solaire-voici-la-bonne-quantite-a-appliquer-pour-etre-vraiment-protge-2263285.html>

**(5 years), this represents a 10-year PV (3% 2024-PV) of €242.47 million (€144.77 million).** Considering minimal and maximal values for the initial formulation cost and the cost increase triggered by the proposed restriction, the following range is obtained for annual raw materials extra-costs: [8.6; 86.39] millions of euros per year.

#### **3.4.2.2.2. Secondary photoprotection products and other cosmetic products**

The Dossier Submitter has no information on the increase in formulation costs triggered by the proposed restriction for other cosmetics other than primary sunscreens (i.e., secondary photo-protection products - see footnote 7 for a definition - and 'other' cosmetics), nor on the associated production or sales volumes for these products. Consequently, raw material costs could not be quantified for these products.

#### **3.4.3. Transmission of additional costs along the value chain and impact on the end consumer**

Considering the welfare implications of the proposed restriction one important question concerns the extent to which the extra-cost borne by the manufacturers of cosmetic products would be passed on to the consumer (pass-through). The additional cost borne by cosmetics manufacturers could be:

- fully absorbed by the latter and/or wholesalers and retailers *via* a reduction in the producer surplus (zero pass-through);
- partially passed on;
- fully passed on (100% pass-through);
- or over-passed.

Thus, the extent of the pass-through depends on the behavior of stakeholders operating at various stages of the value chain: manufacturers of cosmetic products but also wholesalers and retailers. It especially depends on:

- The elasticity of the demand: the more elastic the demand, the lower the pass-through all else equal;
- The level of competition between manufacturers;
- The relationship between manufacturers and retailers and their bargaining power: all else equal the less powerful the retailer, the more the producer would pass on the cost increment. Moreover, when selling the same product to retailers differing in terms of bargaining power, manufacturer of cosmetic products could adopt different levels for the pass-through: a lower pass-through for powerful retailers (e.g., supermarkets working through a central purchasing body) and a higher pass-through for less powerful retailers (e.g., local pharmacies), this is the "waterbed effect" (Inderst and Valletti, 2011).
- The level of competition among retailers/wholesalers: the more intense the competition between retailers, the less likely they are to pass through the cost increment all else equal.

The Dossier Submitter was not able to determine the extent of the pass-through given the lack of empirical data and studies on this topic in the literature<sup>87</sup>. In addition, the following evidence gathered by the Dossier Submitter to characterise the value chain affected by the proposed restriction (see Appendix E.4.1) confirm that it is not feasible to determine the extent of the pass-through based on theoretical reasoning:

- There is significant heterogeneity between manufacturers of cosmetic products: multinationals/large companies operate alongside SMEs, each type of company being able to implement different strategic behaviours due to different financial resources;
- There is significant heterogeneity between retailers of the products concerned by the proposed restriction: these products (in particular sun care products) are distributed in supermarkets (a highly concentrated distribution sector) but also in pharmacies/parapharmacies (a less concentrated distribution sector but with significant heterogeneity between large chains and local shops affected by higher competition).

Furthermore, to the Dossier Submitter's knowledge, there is no empirical evidence of the relationship between cosmetics producers and wholesalers/retailers.

In this context, the Dossier Submitter relied on the formulation cost data provided by the industrial stakeholders (see Appendix E.4.3.5.2) and the market-price data for 'octocrilene-containing' primary sunscreens that it had itself collected (appendix A.2.1.2) to gauge the profit margin for manufacturers of these products. To do so, the Dossier Submitter determined the average share of the formulation cost in the selling price. It is clear to the Dossier Submitter that such an approach does not allow determining the profit margin precisely<sup>88</sup> but provides an order of magnitude for the latter. Based on these results, the Dossier Submitter is able to discuss qualitatively the impact of an increase in the production cost of primary sunscreens for the consumer.

Table 33 shows the share of the formulation cost in the average selling price of primary sunscreens that contain octocrilene for three formulation costs scenarios and five EU countries. According to industrial stakeholders, the formulation cost of primary sunscreens containing octocrilene varies between 10€/kg and 20€/kg. Results show that, depending on the formulation cost scenario and the country considered the formulation cost represents 5.3% to 28.1% of the selling price. Results displayed in this table only cover primary sunscreens for the body, excluding stick formats. The data base contained too few observations for facial primary sunscreens and baby and children's primary sunscreens to carry out the same analysis for these product categories separately in a robust way. Furthermore, it would have been irrelevant to determine the share of formulation cost in the selling price by grouping all the primary sunscreens together, given the difference in margins between the three product categories under consideration. Indeed, the share of the formulation cost seem to be higher for baby and children primary sunscreens and lower for facial primary sunscreens compared to primary sunscreens for the body.

**Table 33: Formulation cost as a percentage of average selling price of primary sunscreens**

<sup>87</sup> According to literature (Harlington et al. 2022, Mahé et al., 2011) and industry representatives, consumers are sensitive to the price of sunscreen but also to their features such as applicability. However, given the difference between the selling price and the cost of production and the fact that the most valued attributes (for women) are the ability to prevent skin cancer, sunburn and sun-related ageing and irritation (Harlington et al., 2023) and the fact that sunscreens with alternative are already on the market, it seems unlikely that the demand for sunscreens would fall.

<sup>88</sup> The Dossier Submitter does not have information on the cost of packaging and logistics.

(in 2024).

| Formulation cost /Countries | France             | Finland | Poland <sup>89</sup> | Ireland | Germany |
|-----------------------------|--------------------|---------|----------------------|---------|---------|
| Formulation cost = 10€/kg   | 7.1%               | 5.3%    | 14.1%                | 8%      | 6.8%    |
| Formulation cost = 15€/kg   | 10.6%              | 8%      | 21.1%                | 12.1%   | 10.2%   |
| Formulation cost = 20€/kg   | 14.2% <sup>a</sup> | 10.6%   | 28.1%                | 16.1%   | 13.6%   |

*Nota bene:* The formula cost varies from one line to another (three formulation-cost scenarios), while the average market price of a primary sunscreen varies from one column to another (from one country to another). The average market price for France, Finland, Poland, Germany and Ireland is 141€/kg, 188€/kg, 304 PLN, 152€/kg and 147€/kg respectively

a: This value can be interpreted as follows: considering a formulation cost of €20/kg, the latter represents on average 14.2% of the market price of a primary sunscreen in France.

Given the costs of alternatives to octocrilene, industrial stakeholders expect a 10% to 50% increase in the formulation cost. In view of the relatively low share of the formulation cost in the selling price, the additional cost induced by the proposed restriction for the manufacturers of primary sunscreen should result in moderate price increases for the consumer, even if the extra-cost is fully passed or over-passed. For example, if we consider a formulation cost of 20€/kg and a 50% increase in the latter due to the proposed restriction, a 100mL bottle sold for €20.31 in France<sup>90</sup> before the restriction may encounter a price increase by approx. €1.5 if the formulation cost increase is fully passed on to the consumer (that is a selling price of €21.75 after the proposed restriction is enforced)<sup>91</sup>.

Based on this analysis, it can be concluded that the share of the formulation cost in the selling price of primary sunscreens is relatively low and suggests significant profit margins. The manufacturers, wholesalers and retailers would therefore be able to absorb at least part of the production cost increase for these products. Besides and as explained above, even if this cost increase was fully passed on to the consumers, this would only lead to a limited increase in the selling price. **It is therefore considered unlikely that the proposed restriction would jeopardise the supply of affordable suncare products.**

#### 3.4.4. Enforcement and control costs

The Dossier Submitter considered whether the proposed restriction would cause additional **costs to retailers/importers of cosmetic products and Member States' enforcement authorities** through control and administrative costs to prove that the products they market do not contain octocrilene.

##### 3.4.4.1. Importers and retailers of cosmetic products

Given that the volume of imports of cosmetic products into the European Union could not be

<sup>89</sup> The cost of formulas with octocrilene is 43 PLN/kg when this cost is low, 64 when it is mid and 86 when it is high.

<sup>90</sup> This corresponds to the average selling price the Dossier Submitter has determined for France based on the collected data (see Appendix A.2.1.2).

<sup>91</sup> We are looking here at a hypothetical product whose price in €/kg corresponds to the average price in France. Thus, considering a formulation cost of €20/kg and a 50% increase in the latter under the proposed restriction amounts to considering a €30/kg formulation cost (€10 increase). If this cost increase is fully passed on to the consumer through the selling price (considering that the average density of primary sunscreen is 1.44kg/L) the latter increases from 141.06€/kg (203.13 €/L) to 151.06 €/kg (217.53€/L). This amounts to a 7.1% increase in the selling price.

obtained, the enforcement and control costs could not be assessed quantitatively for importers of cosmetic products.

Besides, it is considered unlikely that importers and retailers of cosmetic products will incur significant additional control and administrative costs in the event of a restriction, for the following reasons:

- As underlined in previous Annex XV reports (see for instance ECHA, 2020b p 121-124) the EU's Cosmetics Product Regulation has strict requirements for labelling of cosmetics products which mandate that every ingredient must be included on the product label sold to consumers. Therefore, the need to test for the presence of octocrilene in cosmetic products is expected to be minimal for retailers and importers as information on the ingredients is passed on along the supply chain primarily via the information on the label;
- Moreover, the detection of octocrilene in cosmetic products is based on two types of tests: HPLC/UPLC-MS/MS (High or Ultra-performance liquid chromatography coupled with tandem mass spectrometry) and GC-MS/MS (gas chromatography coupled with tandem mass spectrometry) (see section 4.2). These tests require standard technical equipment and can detect the presence of several substances in the same sample. Given that companies operate on a market where the regulatory requirements change regularly and that a number of restrictions on substances used in cosmetic products have recently been adopted (e.g., homosalate in sunscreen products for the body, cyclosiloxanes), the need to detect octocrilene could not result in any additional costs, if octocrilene is tested as a bunch with other restricted substances.

#### 3.4.4.2. Public authorities

Public authorities support administration and enforcements costs that include the costs of publicizing the existence of the new restriction, of testing products in order to ensure that firms comply with it and of implementing systems of regulatory sanctions to respond to non-compliance.

For the purpose of the quantitative analysis, it is assumed that enforcement and administrative costs would amount to €55,600 per year for the entire EEA (as estimated by ECHA, unpublished). These costs would be borne from the entry into effect of the proposed restriction to the end of the study period (after the end of the transitional period) and would amount to 0.35 million euros for the entire study period (PV 2024, 3%). It is likely that this value corresponds to an overestimation of enforcement costs. Indeed, the latter could be reduced when some of these costs are shared with the enforcement costs associated with other interlinked REACH restrictions and other regulations (such as the detection of octocrilene alongside with other restricted substances based on same detection tests).

#### 3.4.5. Overview of the costs of the restriction options

Table 34 and Table 35 below show the overall cost to the cosmetics industry induced by the proposed restriction and its breakdown for the main scenario between:

- The different types of cost considered;
- The different product categories affected.

**Thus, if we consider a transitional period of 2 years (R01), the cost induced by the restriction amounts to 353.69 million euros for the entire study period (2027-2036;**

annualised cost of 39.08 million euros).<sup>92</sup> Most of this cost relates to sun protection products, which would incur an extra-cost of 352.3 million euros for the entire study period (38.93 million euros in annualised cost). To these costs would be added 0.35 million euros in enforcement costs for the entire study period. In terms of market value, the annualized cost of the proposed restriction represents 0.04% of total sales in value of cosmetic products in 2023 in Europe<sup>93</sup> and 1.11% of total sales in value for the same year if we focus on sun protection products in Western Europe<sup>94</sup>.

Considering a transition period of 5 years (RO2) reduces the cost of the restriction by 53%: the cost induced by the restriction amounts to 165.74 million euros for the entire study period (annualised cost of 18.31 million euros). To these costs would be added 0.21 million euros in enforcement costs. This reduction can be explained by the fact that an increase in the length of the transition period allows for greater concordance between the baseline reformulation schedule and the implementation of reformulations during this transition period. It also delays the start of production of the reformulated products and reduces the raw material costs.

It should also be noted that raw material costs represent a significant proportion of the cost of the restriction: 69% (87%) of the annualised cost if a transition period of 2 years (5 years) is considered. This is explained by the fact that the proposed restriction affects a relatively small number of formulations and that the increase in formulation cost<sup>95</sup> induced by the proposed restriction is significant.<sup>96</sup> It should be recalled that the raw material costs presented here represent an underestimate of the raw material costs induced by the proposed restriction over its entire scope in terms of cosmetic products, since they could only be quantified for primary sunscreens.

**Table 34: Total costs of restriction option 1 (2 years transitional period)**

|                                | <b>Reformulation costs<sup>a</sup></b><br>(€ million) | <b>Raw material costs<sup>b</sup></b><br>(€ million) | <b>Total</b><br>(€ million)                           |
|--------------------------------|-------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|
| <b>Sun protection products</b> | 109.83                                                | 242.47                                               | <b>352.3</b>                                          |
| <b>Other cosmetic products</b> | 1.39                                                  | NA                                                   | <b>1.39</b>                                           |
| <b>Total</b>                   | <b>111.22</b>                                         | <b>242.47</b>                                        | <b>353.69</b><br>(354.04 including enforcement costs) |

a: PV 2024 at 3%; mid-point reformulation cost; useful life of a formulation = 10 years; assuming partial reformulation.

b: considering mid-point values for both initial formulation cost and cost increase; assuming partial reformulation.

<sup>92</sup> Equivalent annuity over a period of 10 years, assuming that reformulation costs would be borne from 2027.

<sup>93</sup> The consumption value of cosmetics and personal care products reached 95.7 billion euros in 2023 (Statista, 2024a).

<sup>94</sup> The consumption value (retail selling price) of the sun care market reached 3.5 billion euros in 2023 (Euromonitor, 2024b).

<sup>95</sup> As a reminder, Formulation cost is the total ingredients cost of producing one kilogram of a cosmetic formulation. It can be derived from the cost per kg of ingredients and their concentration in the end-product.

<sup>96</sup> In particular, the UV filters that can be substituted for octocrilene have significantly higher market prices.

**Table 35: Total costs of restriction option 2 (5 years transitional period)**

|                                | <b>Reformulation costs<sup>a</sup></b><br>(€ million) | <b>Raw material costs<sup>b</sup></b><br>(€ million) | <b>Total</b><br>(€ million)                           |
|--------------------------------|-------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|
| <b>Sun products protection</b> | 20.71                                                 | 144.77                                               | <b>165.48</b>                                         |
| <b>Other products cosmetic</b> | 0.26                                                  | NA                                                   | 0.26                                                  |
| <b>Total</b>                   | <b>20.97 Mio €</b>                                    | <b>144.77 Mio €</b>                                  | <b>165.74</b><br>(165.95 including enforcement costs) |

a: PV 2024 at 3%; mid-point reformulation cost; useful life of a formulation = 10 years; assuming partial reformulation.

b: considering mid-point values for both initial formulation cost and cost increase; assuming partial reformulation.

**Table 36: Annualised costs of the restriction (RO1 and RO2)<sup>97</sup>**

|                            | <b>RO1</b><br>(€ million)                           | <b>RO2</b><br>(€ million)                           |
|----------------------------|-----------------------------------------------------|-----------------------------------------------------|
| <b>Reformulation costs</b> | 12.29                                               | 2.32                                                |
| <b>Raw material costs</b>  | 26.79                                               | 15.99                                               |
| <b>Total</b>               | <b>39.08</b><br>(39.12 including enforcement costs) | <b>18.31</b><br>(18.34 including enforcement costs) |

### 3.4.6. Benefits of the restriction

As mentioned in section 3.1, the benefits of the proposed restriction are estimated by considering citizens' willingness to pay (WTP) for improving the state of the aquatic environment (see Appendix E.4.4 for full details). The stated preference study – combining discrete choice experiment (DCE) and contingent valuation (CV) methods – was conducted on representative<sup>98</sup> samples of the general national adult population of six EEA countries: Bulgaria, France, Germany, Hungary, Spain and Sweden (sample size = 7,200 with 1,200 respondents per country). The countries were selected to capture heterogeneity especially regarding preferences for the environment, economic conditions and geographic location in the EEA.

This study was designed and carried out by a team of consultants (Sotis et al., unpublished) in collaboration with the Dossier Submitter. The survey development included several pre-testing steps such as expert reviews and cognitive interviews before conducting two pilot surveys in all study countries (see Appendix E.4.4. for details). The final survey was implemented online in September 2024 by a professional survey company (Ipsos). As mentioned previously (see section 3.3), respondents were asked to express their WTP for three improvement levels (low, medium and high) - induced by the proposed restriction - in

<sup>97</sup> Equivalent annuity over a period of 10 years (2027-2036), assuming that reformulation costs would be borne from 2027.

<sup>98</sup> The Dossier Submitter aimed at representativeness of the general national adult population in terms of gender, age, occupation and geographical location. Occupation serves as a proxy for social position. The Dossier Submitter also aimed at the representativeness of the sample for the level of education. However, representativeness for this variable could not be fully achieved.

the state of freshwater ecosystems on the one hand and coastal ecosystems on the other. These different levels of improvement were defined qualitatively by the team of consultants in collaboration with the Dossier Submitter in order to take into account the impact of several sources of uncertainty.<sup>99</sup> These three improvement levels were presented to respondents prior to the WTP elicitation questions, along with the baseline scenario corresponding to the continuation of the current level of octocrilene emissions into the environment, also translated into the presence and abundance of species in freshwater and coastal ecosystems. The three improvement levels as well as the baseline scenario are presented in Table 37. The payment vehicle was defined as an increase in household expenditure on cosmetic products incurred by respondents due to higher prices of octocrilene-free products. This expenditure-based payment vehicle was described to respondents as an annual payment for 5 years (as 5-years was specified to respondents as the maximum duration of these higher prices).

The WTP was elicited in the following way:

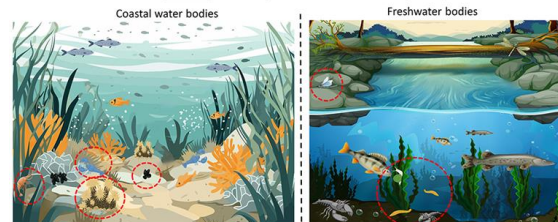
- For the DCE: WTP elicitation was based on choice cards. Respondents were confronted with 6 choice cards. Each card presented three alternatives: a status quo scenario (no improvement = baseline scenario, no cost) and two improvement alternatives affecting the state of coastal and freshwater ecosystems, with associated yearly costs presented in local currencies.<sup>100</sup> The elicitation question was formulated as follows: "Please choose the option you would prefer overall".
- For the CV: first respondents were randomly assigned to one of three scenarios describing either low, medium, or high level of environmental improvement (corresponding to the attribute levels used in the DCE for both coastal and freshwater environments). They were then asked to state their maximum willingness to pay for achieving these improvements using an open-ended elicitation format. The WTP question was formulated as follows: "What would be the most you would be willing to pay more per year for products which do not contain octocrilene?", followed by a standard reminder of the budget constraint.

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<sup>99</sup> As a reminder, the Dossier Submitter considered that the impact of the proposed restriction (i.e., the expected environmental improvement) depends on the exposure of the different water bodies to octocrilene (duration of exposure, concentration level) but also on the relative weight of octocrilene in relation to the pressure, particularly chemical pressure, to which the ecosystems are subject. Given the possible heterogeneity of the water bodies impacted by the restriction in terms of exposure and environmental pressure and the associated uncertainties, the Dossier Submitter has chosen to consider three qualitative improvement levels in the stated preference study (low, medium and high). The improvement levels were defined in collaboration between the Dossier Submitter and the consultants on the basis of the results of regulatory tests studying the effect of octocrilene on indicator species. They correspond to a recovery of different group of species according to their level of sensitivity to octocrilene and the relationships between species within trophic chains (see Appendix E.4.4. for details).

<sup>100</sup> In total 48 choice cards were generated, divided into 8 blocks of 6 cards each, with each respondent completing one block, i.e., 6 choice tasks/cards. Monetary amounts presented on the cards were country specific

**Table 37: Improvement levels considered in the stated preference study**

| Improvement level      | Illustration                                                                                                                                                                    | Explanation                                                                                                                                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO (baseline scenario) | <p>NO octo ban: <b>NO</b> improvement in the state of aquatic species (current situation)</p>  | Many invertebrate species (such as shrimps, lobster, sea anemones and mussels/snails, worms, crayfish and dragonflies) disappear from waterbodies exposed to octocrilene. Fish species also have fewer individuals than if octocrilene were not present in these waterbodies. |
| LOW                    | <p>Octo ban : <b>LOW</b> level of improvement in the state of aquatic species</p>              | Return of some invertebrate species such as sea anemones, lobsters, mussels, crayfish and dragonflies (seen in red circles).                                                                                                                                                  |
| MEDIUM                 | <p>Octo ban : <b>MEDIUM</b> level of improvement in the state of aquatic species</p>          | Return of invertebrate species most sensitive to octocrilene, as well as recovery of other invertebrates such as crayfish/lobster, sea anemone and mussels (seen in red circles).                                                                                             |
| HIGH                   | <p>Octo ban : <b>HIGH</b> level of improvement in the state of aquatic species</p>           | There would be a significant improvement for all invertebrate species. There would also be a recovery of notably some fish species which feed on these invertebrates.                                                                                                         |

The willingness to pay values resulting from this study are presented in Table 38 (results of the DCE) and Table 39 (results of the CV) in Euro PPP<sup>101</sup> per household and per year. When looking at country-specific values, the mean willingness to pay to improve the state of fresh- and coastal water ecosystems by one level (e.g., from baseline to a low level

<sup>101</sup> Purchasing power parity (PPP) is a money conversion rate used to express the purchasing powers of different currencies in common units. This rate expresses the ratio between the quantity of monetary units required in different countries to purchase the same "basket" of goods and services. This conversion rate may be different from the "exchange rate", as the exchange rate of one currency in relation to another reflects their reciprocal values on the international financial markets and not their intrinsic values to a consumer.

of improvement)<sup>102</sup> resulting from the DCE ranges from €74 (Hungary) to €503 (Sweden) per household per year for a period of five years. WTP values derived from the contingent valuation for the improvement from baseline to low level are lower and range from €41 per household per year (Hungary) and €73 (Bulgaria).<sup>103</sup> The difference in these values may be attributed to the different WTP elicitation formats, with the CV being more conservative due to its direct questioning format.

**Table 38: Mean willingness to pay to reduce the risks associated with the use of octocrilene in cosmetic products for freshwater and coastal water ecosystems – Results of the discrete choice experiment (in € PPP per household and per year)**

|                                   | Mean willingness to pay – improvement for coastal water ecosystems |    | Mean willingness to pay – improvement for freshwater ecosystems |    | Total |
|-----------------------------------|--------------------------------------------------------------------|----|-----------------------------------------------------------------|----|-------|
| <b>Bulgaria</b>                   | 167                                                                |    | 15 €                                                            | NS | 182 € |
| <b>Germany</b>                    | 301                                                                |    | 106 €                                                           |    | 407 € |
| <b>Spain</b>                      | 241                                                                |    | 75 €                                                            |    | 316 € |
| <b>France</b>                     | 219                                                                |    | 128 €                                                           |    | 347 € |
| <b>Hungary</b>                    | 38                                                                 | NS | 36 €                                                            | NS | 74 €  |
| <b>Sweden</b>                     | 366                                                                |    | 138 €                                                           |    | 503 € |
| <b>Simple average<sup>a</sup></b> | <b>222</b>                                                         |    | <b>83 €</b>                                                     |    | 305 € |

Note: for two of the countries surveyed (Hungary for both types of ecosystems and Bulgaria for freshwater ecosystems only), estimated mean coefficients for willingness to pay are not statistically significant (see Appendix E.4.4 for details of the statistical analysis). Thus, these coefficients reflect the population's willingness to pay with a low level of precision. "NS" means "not significant" in the table.

a: Mean willingness to pay over the 6 surveyed countries. Willingness to pay values for the countries for which results were not statistically significant (NS) are included. The Dossier Submitter has chosen to present this single value in the main report, while being aware of the associated limitations: including non-significant coefficients obtained for some of the case-study countries in the calculation of the average WTP reduces the level of precision with which it measures the population's willingness to pay to avoid the impacts of octocrilene on the aquatic environment. However, other methods of calculation and taking into account non-significant coefficients were considered to determine the average WTP at the level of the case study countries. These elements and the associated assessment of the benefits of the restriction are presented in Appendix E.4.5.2.

**Table 39: Mean willingness to pay to reduce the risks associated with the use of octocrilene in cosmetic products for freshwater and coastal water ecosystems – Results of the contingent valuation (in € PPP per household and per year)**

| Country         | Low improvement | Medium improvement | High improvement |
|-----------------|-----------------|--------------------|------------------|
| <b>Bulgaria</b> | 73              | 87                 | 95               |
| <b>Germany</b>  | 65              | 79                 | 85               |
| <b>Spain</b>    | 50              | 64                 | 64               |
| <b>France</b>   | 52              | 67                 | 73               |

<sup>102</sup> The willingness to pay for a two- (baseline to medium) or three-level (baseline to high) improvement can be obtained by simply multiplying the values displayed in Table 38 by two or three.

<sup>103</sup> In contrast to the WTP values derived from the DCE, the WTP values derived from the contingent valuation are associated to a specific level of improvement (low, medium or high) affecting both marine and freshwater ecosystems (see Appendix E.4.4 for details of the method).

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|                       |    |    |    |
|-----------------------|----|----|----|
| <b>Hungary</b>        | 41 | 58 | 62 |
| <b>Sweden</b>         | 62 | 73 | 72 |
| <b>Simple average</b> | 57 | 71 | 75 |

a: Values are expressed in Euro PPP per household and per year

The benefits resulting from the proposed restriction at the EEA level are assessed by transferring these willingness-to-pay values while following the methodological steps described below. In the remainder of this section, **the Dossier Submitter chose to present the results of the benefit assessment of the proposed restriction based on the WTP estimates derived from the contingent valuation** (main assessment). This provides a more conservative assessment of the benefits based on a well-established and well-studied valuation method, that was also used to estimate reference values for health impacts from chemicals (ECHA, 2023b). The results of the benefit assessment based on the DCE results are presented in Appendix E.4.5.

The Dossier Submitter first established an EEA-wide reference value, following the SEAC recommendations for the transfer of WTP values (ECHA, 2023b): this EEA-wide reference value (in €/household/year) corresponds to a simple average over the surveyed countries (country-specific WTP values having already undergone a PPP-conversion). **When considering WTP values derived from the CV, the average WTP for improving the state of coastal and freshwater ecosystems** (EEA-wide reference value) **amounts to approx. €57, €71 and €75 per household per year for a low, medium and high level of environmental improvement respectively.**

As mentioned above, for methodological reasons households' willingness to pay has been elicited for the next five years. However, this 5-year timeframe does not correspond to the study period considered to assess the impacts of this restriction (i.e., 10 or 20 years). It is therefore necessary to 'convert' the aggregate willingness to pay established at the EEA level for this 5-year period to the study period under consideration. To do so, the Dossier Submitter determined the corresponding annuity while assuming that the benefits would only start to materialize after the transitional period (i.e for a 10-year study period: after 2 years and for the 8 remaining years for RO1; after 5 years and for the 5 remaining years for RO2)<sup>104</sup>. In addition, the Dossier Submitter wished to take into account two main sources of uncertainty in its assessment. As discussed in section 3.3, the water bodies affected by the proposed restriction are likely to be heterogeneous in terms of the level of exposure to octocrilene but also in terms of the overall environmental pressure faced. This heterogeneity, as well as the uncertainty about the initial state of the water bodies, results in uncertainty about the level of improvement that would result from the proposed restriction. The Dossier Submitter therefore assessed the benefits of the restriction considering that water bodies would only benefit from a low level of improvement<sup>105</sup>. Furthermore, given the persistent nature of the substance, there is uncertainty as to when the benefits will start to accrue. To account for this uncertainty, the Dossier Submitter considered the following three scenarios<sup>106</sup>:

- 1. Immediate materialisation of benefits: the recovery of water ecosystems is complete**

<sup>104</sup> WTP values for the future years were discounted to present values using a 3% discount rate.

<sup>105</sup> As the WTP values resulting from the DCE are derived from the estimation of a linear model, this implies considering one level of improvement. The use of a statistical model assuming linearity in the level of improvement induced by the proposed restriction for the environment does not require to define the initial state.

<sup>106</sup> In the absence of data to characterise more precisely the time required for beneficial environmental impacts to materialise, the Dossier Submitter has defined ad hoc scenarios to account for the impact of a lag on the benefits assessment of the restriction.

at the end of the transition period;

2. Moderate lag in materialisation of benefits: the recovery of water ecosystems is gradual and complete four years after the end of the transition period;
3. Significant lag in the materialisation of benefits: the recovery of water ecosystems is gradual and complete eight years after the end of the transition period<sup>107</sup>.

**For scenarios #2 and #3, the annuity (benefits for the overall EEA) is prorated in relation to the level of recovery of the environment.**

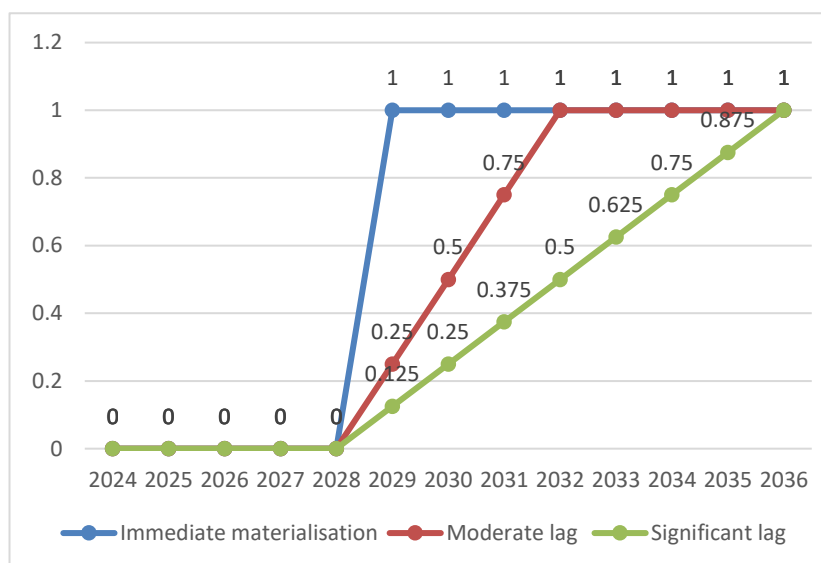
Figure 15 shows the improvement levels (i.e., ratios) considered to do so for each of the abovementioned scenarios. For instance, when considering a 'moderate lag' scenario under RO1, only 50% and 75% of the annuity (yearly benefit) has been accounted for for the years 2030 and 2031 respectively when assessing the benefits over the whole study period (i.e., computing the 2024 PV). **Table 40 shows the benefits of the restriction** calculated on that basis while transferring WTP values derived from the CV to the whole EEA population (considering 203 102 020 households in the EEA, Eurostat 2023 population data): **they amount to €49 billion, €39 billion and €26 billion for an 'immediate realisation', a 'moderate' and a 'significant lag' respectively (PV 2024 at 3%).**

In addition to these three scenarios, the Dossier Submitter assessed the benefits of the restriction for a worst-case scenario under RO1 for comparison with the worst-case scenario established for costs. Under this scenario, the Dossier Submitter considered a 20-year study period and assuming that the recovery of water ecosystems is gradual and complete only at the end of the study period (i.e., after 18 years). Under this scenario, the benefits of the restriction amount to 24 billion euros (PV 2024, 3%).

**Table 40: Benefits of the restriction based on the contingent valuation results (PV 2024 at 3%).**

|                              | RO1<br>(€ billion) | RO2<br>(€ billion) |
|------------------------------|--------------------|--------------------|
| <b>Immediate realisation</b> | 49                 | 44                 |
| <b>Moderate lag</b>          | 39                 | 31                 |
| <b>Significant lag</b>       | 26                 | Not considered     |
| <b>Worst case scenario</b>   | 24                 | Not considered     |

<sup>107</sup> This scenario was only considered for RO1 as for RO2 there are only five years between the end of the transitional period and the end of the study period (10 years).



**Figure 15: Share of improvement achieved under the three benefits assessment scenarios (under RO1).**

The Dossier Submitter assessed the proportionality of the proposed restriction based on the results of the benefit assessment (see section 3.6).

### 3.5. Other impacts

With regard to social and wider economic impacts, and considering that the phasing out of octocrilene in cosmetic products appears to be technically and economically feasible, the Dossier Submitter foresees no major effects on employment. This said, some job losses could occur among the producers of some cosmetics products with a specific portfolio. However, considering that technically and economically feasible alternatives are available, it is likely that the producers of alternatives might need to hire new employees to meet the growing demand and increase their production capacity. Moreover, as cosmetics products containing octocrilene could be manufactured in EEA and sold outside the EEA, the potential job losses are expected to be minimal. The restriction proposal also only covers uses of octocrilene in cosmetic products, meaning that not all jobs and economic activities linked to plastisols will be affected by the proposal.

### 3.6. Proportionality assessment

#### 3.6.1. Cost-effectiveness analysis

First, the Dossier Submitter performed a cost-effectiveness analysis to assess the proportionality of the proposed restriction. Conducting a cost-effectiveness analysis is recommended by SEAC for restrictions on PBT/vPvB substances and when benefits of the restriction cannot be quantified (ECHA, 2014). In such cases, avoided releases are used as a proxy for the benefits of the restriction. The Dossier Submitter chose to perform such an analysis despite the fact that octocrilene is not currently classified as a PBT and/or vPvB under the CLP Regulation. Indeed, as mentioned in section 1.4.2, the substance is persistent and toxic. In addition, although *in vivo* studies do not demonstrate the bioaccumulative nature of octocrilene (octocrilene does not meet the criteria set out in annex XIII of REACH for the B modality), the results of monitoring studies carried out *in situ* show that it does bioaccumulate.

According to Table 41, compared to the baseline RO1 and RO2 are expected to reduce releases of octocrilene to the environment by about 13,300 tonnes (10,800 tonnes to soil and 2,500 tonnes to water) and 8,300 tonnes (6,800 tonnes to soil and 1,500 tonnes to water) respectively over the 10-year period considered for the impact assessment (see also **Error! Reference source not found.** displaying the conceptual release pathways of octocrilene to the environment through consumer use of cosmetic products, see section 3.3). This is equivalent to approx. 10,400 tonnes and 6,200 tonnes when discounted at 3%. The cost-effectiveness (C/E) ratio was calculated as the ratio of the net present value of costs and avoided releases. Both costs and avoided releases were discounted over 10 years using a 3% discount rate. Thus, if the Dossier Submitter considers all releases to water and soil, **the resulting C/E ratios amount to approx. €34 and €27 per kg of releases abated for RO1 and RO2 respectively.** The Dossier Submitter also determined the incremental C/E ratio of RO1 compared to RO2. Indeed, RO1 allows avoiding the release of approx. 4,200 additional tonnes of octocrilene into the environment over the whole study period (3% PV) compared with RO2, while generating an additional cost of 188 million euros (3% PV). The resulting incremental C/E ratio amounts to around €45 per kg of additional releases abated for RO1. Thus, increasing the transitional period from two years to five years makes the cost-effectiveness estimates slightly more favourable. However, the overall impact of changing the duration of the transitional period is not considered to be significant.

**Table 41: Cost-effectiveness of RO1 and RO2.**

|            | <b>Total cost</b><br><br>(PV over 10 years, 3% in € million) | <b>Total releases reduction</b><br><br>(Sum over 10 years, no discounting, tonnes) | <b>Total releases reduction</b><br><br>(PV over 10 years, tonnes) | <b>C/E ratio</b><br><br>(€/kg of avoided releases) | <b>Incremental C/E ratio</b><br><br>(€/kg of additional avoided releases for RO1 compared to RO2) |
|------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>RO1</b> | 354.04                                                       | 13,320                                                                             | 10,384                                                            | <b>34.09</b>                                       | 44.92                                                                                             |
| <b>RO2</b> | 165.95                                                       | 8,325                                                                              | 6,200                                                             | <b>26.77</b>                                       | irrelevant                                                                                        |

To assess the proportionality of the various restriction options with regard to the risk identified, cost-effectiveness ratios were compared to those of previous REACH restrictions adopted to avoid releases of PBT/vPvB substances. As can be seen from Table 42, the cost-effectiveness ratios of the examined ROs are on the lower end compared to other recent REACH restrictions. The Dossier Submitter notes that a recent study by Oosterhuis et al. (2017) concluded that, although cost estimates of previously adopted actions do not allow the derivation of a value for society's willingness-to-pay for reductions in PBT presence, use, and emissions, the available evidence suggests that measures costing less than €1,000 per kg of PBT use or emission reduction would usually not be rejected for reasons of disproportionate costs by the legislator, whereas measures with costs above €50,000 per kg of PBT are likely to be rejected by the legislator. As mentioned previously, the Dossier Submitter is aware of the fact that octocrilene is not currently classified as a PBT and/or vPvB under the CLP Regulation, however these results regarding acceptable costs per kg of PBT use/emission are still presented for additional information as no other data is available.

**Table 42: Cost-effectiveness of recent REACH restrictions.**

| Restriction under REACH               | C/E ratio: €/kg – central value | Status                                     |
|---------------------------------------|---------------------------------|--------------------------------------------|
| Lead in shot in wetlands              | 9                               | Adopted by the Commission (Jan. 2021)      |
| Terphenyl, hydrogenated               | 16                              | SEAC and RAC opinions adopted (June 2023)  |
| PAHs in clay targets                  | 20                              | SEAC and RAC opinions adopted (Dec. 2022)  |
| Micro-plastics in rinse-off cosmetics | 22                              | Adopted by the Commission (Sept. 2023)     |
| MCCP                                  | 55-66                           | SEAC and RAC opinions adopted (Sept. 2023) |
| D4, D5, D6 in cosmetics               | 85                              | Adopted by the Commission (May 2024)       |
| Lead in PVC                           | 308                             | Adopted by the Commission (May 2023)       |
| D4, D5 in wash-off cosmetics          | 415                             | Adopted by the Commission (Jan. 2018)      |
| DecaBDE                               | 464                             | Adopted by the Commission (Feb. 2017)      |
| Phenylmercury compounds               | 649                             | Adopted by the Commission (Sept. 2012)     |
| PFOA-related substances               | 734                             | Adopted by the Commission (June 2017)      |
| Micro-plastics in leave-on cosmetics  | 820                             | Adopted by the Commission (Sept. 2023)     |
| PFOA                                  | 1649                            | Adopted by the Commission (June 2017)      |

**To this respect, the Dossier Submitter considers both RO1 and RO2 proportionate; the C/E analysis being more favourable to RO2 in terms of C/E ratio but being more favourable to RO1 in terms of avoided releases.**

The Dossier Submitter has also analysed whether cost-effectiveness varies between different product groups. As raw material costs could not be quantified for the 'other cosmetic products' category, the Dossier Submitter was only able to compute a comprehensive C/E ratio for sun protection products and compared C/E ratios between sun protection products and other cosmetic products when considering reformulation costs only. Nevertheless, the Dossier Submitter was able to qualitatively appraise the cost-effectiveness of the proposed restriction for the 'other cosmetic products' category based on the information provided by the industrial stakeholders on the function of octocrilene and the use volume in these products as well as the existence of alternative substances. The Dossier Submitter determined that the consumer use of sun protection products is responsible for the release of approx. 1,560 tonnes of octocrilene (93.7%) per year (290 tonnes to water and 1,270 tonnes to the soil) while consumer use of other cosmetic products is responsible for the release of approx. 100 tonnes of octocrilene (6.3%) per year (nearly 20 tonnes to water and approx. 80 tonnes to the soil). The **corresponding C/E ratio for sun protection products** – calculated as the ratio of the present value of costs and avoided releases for this specific product category over the whole study period – amounts to approx. **€36 and €28** per kg of releases abated for RO1 and RO2 respectively. On that basis, the Dossier Submitter considers that the proposed restriction is proportionate for the latter product category. When considering **reformulation cost only**, under RO1 C/E ratios amount to approx **€11 and €2** per kg of releases abated for sun protection products and other cosmetic products respectively. Under RO2, the C/E ratios amount to approx **€4 and €0.7** per kg of releases abated. Thus, the proposed restriction shows the same degree of proportionality for sun protection products and other cosmetic

products. As mentioned above, the Dossier Submitter was unable to quantify the costs of the restriction for the 'other cosmetic products' category concerning raw materials. Nevertheless, the Dossier Submitter considers that **the proposed restriction is at least as proportionate as for sunscreen products**. This conclusion is based on the fact that the category 'other cosmetic products' is associated with lower use-volumes of octocrilene at lower concentration levels (see section 1.3.4).

**Table 43 : Specific cost-effectiveness ratios for the main cosmetic product categories.**

|            | Sun protection products | Sun protection products | Other cosmetic products |
|------------|-------------------------|-------------------------|-------------------------|
|            | Reformulation cost only | Total cost              | Reformulation cost      |
| <b>RO1</b> | 11.28 €/kg              | 36.20 €/kg              | 2.13 €/kg               |
| <b>RO2</b> | 3.56 €/kg               | 28.48 €/kg              | 0.67 €/kg               |

Note: C/E ratios calculated as the ratio of the net present value of costs (reformulation and raw material cost) and avoided releases (PV 2024, 3%) for each product category. For example, the 'total cost' column the C/E ratio for the sun protection products category for RO1 was calculated as follows: 36.20 €/kg = 352.3 Mio€ (PV 2024 of reformulation and raw material costs, see Table 34) /9,733.10<sup>3</sup>kg (PV 2024 of avoided releases).

In addition to the calculation of C/E ratios under the main scenario, the Dossier Submitter has performed several sensitivity analyses to account for the impact of uncertainties on the costs triggered by the proposed restriction as well as on avoided releases on the cost-effectiveness of examined restriction options. With regard to avoided releases, there is uncertainty about the use of sludge from wastewater treatment plants, and in particular whether or not it is spread on agricultural land. The cost-effectiveness ratios shown in Table 41 are based on the assumption that 100% of this sludge is spread (in the baseline, RO1 and RO2 scenarios). The Dossier Submitter determined an alternative cost-effectiveness ratio for RO1 and RO2 considering that none of the sludge is spread on agricultural land (i.e., all of it is incinerated).<sup>108</sup> In this case, emissions to soil are negligible and the proposed restriction only avoids emissions to water. The cost-effectiveness ratio is then approx €183 and €143 per kg of avoided releases for RO1 and RO2 respectively. Thus, the uncertainty surrounding the spreading of wastewater treatment plant sludge does not significantly affect the proportionality of the ROs examined. Besides, it should be noted that the quantification of avoided releases through a tonnage approach leads to an underestimation of releases to water. Indeed, under this approach it is assumed that 100% of releases associated with the consumption of sun protection products take place through showering and bathing while, in reality, a significant share of these releases corresponds to direct emissions in the environment through swimming/spending time in water. Also, as explained in section 1.4.3, when quantifying avoided releases, the Dossier Submitter assumed that 100% of wastewater networks are connected to a wastewater treatment plant. In reality, this rate is lower (approx. 81% in 2022 according to Eurostat data<sup>109</sup>) and a proportion of wastewater is discharged directly into water. The Dossier Submitter has also determined the cost-effectiveness ratio of RO1 under a worst-case scenario regarding the costs of the restriction<sup>110</sup>. Under such a

<sup>108</sup> This is a very strong assumption considered for the sole purpose of discussing the proportionality of the restriction. Indeed, the data consulted by the Dossier Submitter shows that sewage sludge is spread in several EEA countries. See for instance the 'Sewage sludge production and disposal from urban wastewater' [Data set](#).

<sup>109</sup> [Data set](#) "Population connected to at least secondary wastewater treatment" consulted on 20/01/2025.

<sup>110</sup> As a reminder, in this case, the Dossier Submitter calculated the cost of the restriction while considering the most pessimistic assumptions for all the considered parameters (see Appendix E.4.3.6). Also, costs and avoided releases are discounted over 20 years to account for alternative assumptions regarding the useful life of a cosmetic formulation.

scenario, **the proposed restriction still appears proportionate** as the C/E ratio amounts to approx. €251 per kg of avoided releases (discounting of costs and releases at 3%).

Finally, based on the recommendation of SEAC and the extensive discussion included in the Annex XV report regarding the substances containing polycyclic aromatic hydrocarbons (PAHs) in clay targets for shooting (ECHA, 2022 p65) about the impact of the discount rate on the cost-effectiveness of restriction options – especially for persistent substances – the Dossier Submitter computed the C/E ratios while using dual discounting. A 3% discount rate was used for the costs of the restriction and a 0% discount rate was used for avoided releases. **Dual discounting makes the cost-effectiveness estimates even more favourable but does not alter the relative cost-effectiveness of RO1 and RO2: the corresponding C/E ratios amount to approx. €27 and €20 per kg of avoided releases for RO1 and RO2 respectively.**

### 3.6.2. Cost-benefit analysis

In this section, the Dossier Submitter discusses the proportionality of the identified restriction options by comparing the present value of the costs expected by the latter with the associated benefits for the freshwater and coastal water ecosystems.

As presented above, **the total costs triggered by RO1 and RO2 under the main scenario amount to 354.04 Mio€ and 165.95 Mio€ respectively** (PV 2024 at 3%, 10-year study period). The majority of these costs correspond to the costs of reformulating and producing sun protection products using alternative substances.

The environmental benefits expected from the restriction arise from the reduction in potential risks associated with the use of octocrilene in cosmetic products for the water (freshwater, marine water, groundwater and sediments) and soil compartments. However, the Dossier Submitter was only able to monetise the benefits of the restriction for fresh- and coastal water ecosystems based on a stated preference valuation study (see sections 3.3, 0 and Annex E.4.4 for details). As mentioned previously, the Dossier Submitter chose to base the assessment of benefits on the results of the contingent valuation (CV). Thus, **aggregate benefits to the EEA population estimated based on CV results amount to approx. €39 billion and €31 billion for RO1 and RO2** (PV 2024 at 3%, 10-year study period) **respectively when considering a moderate lag in the materialisation of benefits** (due to the persistence of the substance in the environment). **The aggregate benefits amount to €24 billion when considering a worst-case scenario** (considered for RO1 only and assuming a significant lag in the materialisation of the benefits).

Thus, comparing benefits and costs, it is clear that the benefits of the restriction significantly outweigh the costs and this under the main as well as the worst-case scenario. Beyond the absolute value of the benefits estimated on the basis of the results of the CV, this stated preference study shows that Europeans are willing to pay to avoid the environmental impacts of octocrilene on aquatic environment. **WTP values derived from the CV at the household (and aggregate) level are well above the minimum WTP that would be required to obtain benefits equal to the costs generated by the restriction (i.e., break even).** Under RO1, a WTP equal or greater than €0.37 per household per year for the next five years would result in benefits greater than the cost for the proposed restriction. Under RO2 and the worst-case scenario, this minimum willingness to pay is €0.17 and €5.33 per year per household for the next five years respectively.

**Based on these results, both restriction options are considered proportionate.**<sup>111</sup> In contrast with the results of the cost-effectiveness analysis, it should be noted that RO1 performs better than RO2 when considering the results of the CBA ( $NPV_{RO1} > NPV_{RO2}$ ): although RO1 is associated with higher costs, these are largely compensated by an earlier materialisation of benefits (allowed by a shorter transition period than RO2).

## 4. Practicality and monitorability

### 4.1. Implementability and manageability

The restriction options are considered implementable and manageable for the different actors in the supply chain: manufacturers, importers, formulators and downstream users.

As demonstrated in previous section, there is no drop-in one-for-one alternatives to octocrilene for sunscreens and other cosmetic products. Nevertheless, several alternatives to octocrilene are available for each use and technical function, already in use and economically feasible (see section 3.2.4). In addition, the **reformulation of products containing octocrilene is already occurring**, transition to alternatives is feasible considering normal temporality to formulate and/or reformulate sunscreens or cosmetic products. The alternatives appeared to be more costly than octocrilene, but still affordable. In terms of hazards and risks associated with alternatives, the analysis of alternatives looked at both human health and environmental hazard. The identified alternatives, principally those authorised under annex VI of the cosmetic regulation to be used as UV filter in primary and secondary sunscreen products, varied in regard to their relative environmental and human health hazards: for some of them there are indeed concerns about ED, PBT/vPvB properties or environmental hazards. Nevertheless, many alternatives appeared less hazardous and risky than octocrilene. In addition, the reformulation, substitution, and transition to alternatives are feasible if sufficient transition time is given to Industry.

The scope of the restriction is quite narrow and focuses on cosmetics uses of octocrilene, meaning that the manageability is practicable. The fact that it covers specific uses in cosmetics products make it also easily manageable and practical for importation in the EEA. Manufacturing of the substance and uses in plastisol is not covered by this restriction as operational conditions and RMMs allow to manage the risks.

Therefore, both RO1 and RO2 are concluded to be implementable and manageable.

### 4.2. Enforceability

The scope of the proposed restriction is clear and unambiguous, as such is easily understandable for effected parties: it covers the uses of octocrilene as a substance or in mixtures used by consumers and professionals in cosmetics products. Manufacturing, formulation of plastisol, of preparation containing additives and finished cosmetic products, industrial uses for the production of plastic articles, inclusion into a matrix and uses in plastisols by consumers and/or professionals indoor and outdoor are out of the scope of the proposed restriction.

Enforceability of both RO1 and RO2 is considered attainable. To be enforceable, a restriction needs to have a clear scope so that it is obvious to enforcement authorities which products

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<sup>111</sup> Basing the benefit assessment on the WTP values from the DCE would lead to the same conclusion regarding the proportionality of the proposed restriction as the DCE leads to higher WTP estimates (both at the household and the EEA-level) compared to the CV.

are within the scope of the restriction and which ones are not. Moreover, the restriction needs a concentration limit value that can be subject to supervision mechanism. The proposed ROs provides these prerequisites.

The monitoring of the proposed restriction is expected to be done through enforcement. Competent authorities of EU Member States responsible for REACH enforcement activities have experience with REACH restrictions, including restrictions dealing with specific cosmetic products (e.g. restriction on D4, D5, D6). Activities relating to RO1 and RO2 of this proposal can be integrated in current enforcement activities in the Member States. The enforceability is partly dependent on the availability of sufficiently efficient and effective analytical methods for monitoring. Furthermore, it is implementable as companies can test for concentration limits in concerned mixtures or make it a condition of sourcing contracts. In addition, the proposed restriction provides sufficient transition time to the impacted supply chains.

For cosmetic products, a simple preliminary check can already be done by reading the INCI ingredients list on cosmetics packaging: even if this information is (i) not 100% reliable (especially if the brand owner does not respect the EU Regulation in this matter), and (ii) does not indicate the concentration in the product; this quick check can nevertheless give an indication of the presence of octocrilene in the product.

To the Dossier Submitter knowledge, there is no specific standardised laboratory methods for measuring octocrilene in cosmetic products and environmental samples. Standard analytical methods for octocrilene would be very helpful and facilitate the enforcement of the proposed restriction. Development of such methods is therefore highly encouraged by the Dossier Submitters. However, in the absence of European (or international) standard analytical methods for octocrilene, there are different opportunities for checking compliance with the proposed restriction.

Methods can be organised as accredited, standard, validated and research methods, where the former has the most stringent classification. It is advised to use an accredited method in an accredited laboratory when this is available. These methods have been (1) extensively developed and tested, (2) have an inherent quality control guarantee, (3) are cross checked regularly between accredited laboratories and regulatory organs and (4) follow a fixed protocol that cannot be deviated from. This leads to results that can be compared between different laboratories, regions, time points, etc. When an accredited method is not available, it is advised to use a standard or at least a validated method. This validation should be as extensive as possible covering accuracy, precision, linearity and application range, limit of detection (LoD), limit of quantification (LoQ), selectivity/specificity, recovery and robustness/ruggedness. Extensive validation leads mostly to results with a sufficient confidence to be used for reporting or as with accredited methods to compare between different laboratories, regions and time points.

Nevertheless, the substance is under surveillance and is analysed for several years, especially since the ban of the use of octocrilene as UV filter in the Republic of Palau and the state of Hawaii in the US. Moreover, since the inscription of octocrilene to the 4<sup>th</sup> watch list under the Water Framework Directive (Cortes et al., 2022), methodologies for analysing octocrilene in different matrix were recommended. The applied methodologies and technical equipment involved an extraction technique depending on the sample considered, such as solvent extraction, ultrasound assisted extraction (UAE), solid phase extraction (SPE), solid-phase microextraction (SPME), pressurized liquid extraction (PLE) and QuEChERS (Quick, Easy, Cheap, Effective, Rugged and Safe). The techniques applied are High-Pressure Liquid Chromatography (HPLC) and derivated techniques, such as liquid chromatography (LC), Ultra Performance Liquid Chromatography (UPLC), Reversed-Phase High-Pressure Liquid Chromatography (RP-HPLC) and Gaseous Chromatography (GC), which are classical laboratories techniques for analysing substances in different matrices. These equipments are associated to different type of detector, depending on the matrix used for the analysis and the resolution and accuracy needed. It corresponds to mass spectrometry (MS) that can be associated in tandem, coupled with Electrospray Ionisation (ESI) or Quadrupole analyser

(QpQ).

With these technologies, the LoD are as low as 0.3 ng/L and 0.1 ng/g (Kameda et al., 2011) when applying GC-MS analysis on river water and sediment and the LoQ start as low as 5 ng/L and ng/g in water and sediment with LC-MS/MS or GC-MS/MS (Magi et al., 2013, Langford et al., 2015). Depending on variation in apparatus and samples, these threshold can vary but low LoQ and LoD are oftenly reported (Aminot et al., 2019; Balmer et al., 2005; Bratkovics and Sapozhnikova, 2011; Downs et al., 2022; Filho et al., 2024; Gago-Ferrero et al., 2013; Mozas-Blanco et al., 2023; Pafili et al., 2021; Pan et al., 2023, Tsui et al., 2017, <https://sciex.com/tech-notes/environmental-industrial/industrial/sensitive-and-robust-quantification-of-15-common-uv-filters-in-c>). Recently, the FDA also proposed a specific methodology for the identification and quantification of several UV filter, among which is octocrilene. It uses HPLC with a Diode Array Detection (DAD) to detect and quantify the UV filters (Morgan et al., 2021, CARTS No. IR01925). Full protocol is detailed in annex E.7. Moreover, all environmental occurrence studies detailed in Annex B 9.2 provide several reference applying these methodologies to analyse and quantify octocrilene in several matrices, allowing to conclude that monitoring techniques are available.

The proposed concentration limit of 0.001% w/w correspond to a value of 10 ppm, representing 10 µg/g or 10 mg/L depending on the matrix considered (liquid or solid).

In view of the LoD and LoQ of the methods described above, the proposed concentration limit is enforceable and practical.

### **Cost considerations**

As detailed in the previous paragraph, there is a large variety in analytical approaches to analyse octocrilene in various matrices. Therefore, it is very difficult to set a specific price for a typical analysis. A number of different parameters with an analytical project will influence the price per sample, such as: number of samples, matrix, technique(s) used, and post analysis work like modelling or data visualization. However, a rough cost estimate is €100 per sample for a standardized targeted LC-MS/MS analysis in a commercial lab. These prices increase with increasing level of complexity.

Considering the availability of analytical methods on the market to measure the content of various UV filters and especially octocrilene, the affordable prices, the recent inscription of octocrilene in the 4<sup>th</sup> watch list of the Water Framework Directive, the ROs are concluded to be practicable and enforceable with regards to analytical methods and concentration limits.

### **4.3. Monitorability**

The restriction options RO1, RO2, are considered monitorable. The monitoring of the proposed restriction is expected to be done through enforcement. The effectiveness of the proposed restriction could indeed be monitored:

- Via the monitoring (tonnages and uses) of the registration dossiers;
- Via a market survey similar to the one undertaken by the Dossier Submitter;
- Via documentation checking from the supply chain confirming that mixtures do not contain octocrilene (INCI ingredients list on cosmetics packaging);
- Via EU or national monitoring campaign in the environment;
- Via action under the WFD as octocrilene is registered under the watch list of this regulation;
- Via monitoring emissions in wastewater and waste streams;

The monitoring of samples from the biota and the environment is possible using the advanced

analytical techniques available and detailed in enforceability section and in annex E.5.

Nevertheless, the effect of the restriction may be difficult to measure via environmental monitoring campaigns after the entry into force of the restriction proposal. The concentrations of octocrilene measured in the environment may come from previous sources of releases and uses (e.g. landfill disposal prior to the restriction for waste), as well as from uses and releases outside the EU due to the mixing of sediment from water flow. Moreover, octocrilene will still be placed in the market from the other uses not covered by this restriction proposal. However, as these uses are less dispersive and that emissions were mainly occurring from the wide dispersive uses of cosmetic products as demonstrated in section 1.4. and in annex B, environmental concentration should drop quite drastically in the near future of the adoption of this proposal.

An alternative approach to monitoring the effectiveness of the proposed restriction is the monitoring of octocrilene in emissions like wastewater and waste streams. If octocrilene is phased out of cosmetic products, a reduction of octocrilene in these media could be expected within shorter timelines than reduction in environmental and human samples.

A combination of all methods described above will allow a proper monitorability of the restriction.

## 5. Conclusion on the most appropriate restriction option(s)

Based on the impact assessment, the Dossier Submitter considers that both restriction options RO1 and RO2 are:

- Effective, i.e. targeted to the risks posed by octocrilene;
- Effective in reducing the risks (release reduction of almost 100% for finished cosmetic products);
- Limiting the risk for “regrettable substitution” within a reasonable period of time as suitable alternative are already available and in use;
- Proportionate to the risks demonstrated for the environment (costs of the ROs are in line with the previous restrictions for substances of similar concern);
- Proportionate to the risks demonstrated for the environment as unknown (but potentially significant) damage resulting from octocrilene’s emissions are expected to cause as they continue to accumulate in the environment;
- Practicable, i.e. implementable, manageable and enforceable;
- Monitorable, i.e. the result of the implementation of the proposed restrictions can be duly monitored.

Finally, the analysed restriction options ensure a level playing field<sup>112</sup> for EU manufacturers and companies to compete with foreign produced cosmetic mixtures and finished cosmetic products. **Besides, in light of the performance of RO1 - earlier risk reduction and good performance in terms of proportionality – the DS recommends implementing OR1 rather than OR2.**

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<sup>112</sup> In trade and commerce, level playing field is a concept about fairness, not that each player has an equal chance to succeed, but that they all play by the same set of rules.

## 6. Uncertainties and sensitivity analysis

**Table 44: Identified key uncertainties.**

| #   | Section (s) | Short description of the uncertainty                                                                                                | Impact conclusions on the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Probability of impact on the conclusions                                                                                                                                                                                                                                                                                            |
|-----|-------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U01 | 1.3.2       | Exact amount of octocrilene placed on the European market.                                                                          | <p>The quantities available on the EU market impact the total cost of substitution and total release estimates. The quantity used for the estimation performed are underestimated according to CSRs data and CfE.</p> <p>Impossibility to performe a range estimation of the amount really available and put in the market.</p> <p>Cross checking of this estimates from different sources (consumer data on goods put on the market...) ascertain the estimates made by the Dossier Submitter as these estimates clearly underestimates the amount of octocrilene put on the market and released into the environment.</p> | <p>As the restriction focuses on cosmetic products and that tonnages put in the market for these uses were cross checked with consumer database, the estimates sound pretty close to what is present. Cost and benefits would be impacted: Proportionnality should not be impacted.</p> <p>No negative impacts foreseen.</p>        |
| U02 | 1.4.3       | Uncertainties related to the releases                                                                                               | As no study on cosmetic products release pathways are available, estimates were made on release routes and pathways for determining amount of octocrilene reaching the different environmental compartments                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Different approaches of environmental exposure were used, either classical or coming from existing restriction dossiers from other cosmetic products. All the approaches lead to risk for the environment. Proportionnality should rather increase when considering increasing release.</p> <p>No negative impacts foreseen.</p> |
| U03 | 1.4.3       | Uncertainty related to exact amount of octocrilene contained in products coming from outside <sup>113</sup> EEA market (travellers, | Despite the large number of travellers and tourists coming to the EEA, the amount of octocrilene released from foreign                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | If the EU measure is not followed by third countries, environmental exposure could occur from foreign travellers unless communicated                                                                                                                                                                                                |

<sup>113</sup> No risk has been demonstrated for the formulation of cosmetics products containing octocrilene. However, the Dossier Submitter would like to underline that some of the exported products (/substance) could re-enter the EEA and cause a risk for the environment. In addition, it should be noted that this risk could also occur outside the EEA.

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| #   | Section (s)                 | Short description of the uncertainty                                       | Impact conclusions on the                                                                                                                                                                                                                                                                          | Probability of impact on the conclusions                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                             | tourists, ...).                                                            | travellers are not known.                                                                                                                                                                                                                                                                          | that sunscreen produced outside EU cannot be used in EU.<br><br>No impact foreseen as this tonnage will not be impacted by the proposal.                                                                                                                                                                                                                                                                                     |
| U04 | 1.4.3                       | WWTP connection rate, sludge treatment and RMMs                            | Depending on WWTP connection rate and sludge application on soil from urban or industrial WWTP, underestimation of risk may be retrieved for freshwater, freshwater sediment or soil, depending on the % of connection chosen.                                                                     | Some assumption could lead to decreased risk reduction (if some sludge are incinerated), others to increased risk reduction (direct release of octocrilene to water, some water not treated by WWTP). Proportionality can be considered either increased or decreased.<br><br>No negative impacts foreseen as risks were identified for all compartments. The compartment at risk may be modified but risk will still occur. |
| U05 | 1.3.2;<br>1.4.3;<br>3.4.2;7 | Regulatory uncertainty related to EU Chemicals Strategy for Sustainability | Best management of substance with demonstrated risks in the EU according to better policy willingness.                                                                                                                                                                                             | High impact on scope, C/E and proportionality.                                                                                                                                                                                                                                                                                                                                                                               |
| U06 | 1.4.3                       | Industrial waste at management site                                        | Uncertainty on the treatment of the industrial waste. Industrial waste may not all be incinerated/destroyed as demonstrated. As such, risk arise for one registrant at manufacture site. RMM as those described by the lead registrant <b>must</b> be put in place.                                | Potential impact on the entry of the restriction that may covers the manufacture of the substance.<br><br>High impact on cost and proportionality of the restriction if manufactured of the substance is not derogated                                                                                                                                                                                                       |
| U07 | 1.4.                        | Parameters considered in the risk assessment                               | Several parameters considered in the risk assessment determination were best case estimates (tonnage, degradation half-life and no consideration of the toxicity of conjugates, assessment factor for aquatic compartment, normalised PNEC value...) and as such influence the outcome of the risk | As the risk were mainly determined with parameters allowing a best approach, risk may be underestimated. Nevertheless, the conclusions will not be impacted, as risk will remain and even be potentially higher.                                                                                                                                                                                                             |

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| #   | Section (s)    | Short description of the uncertainty                                                                                                                                  | Impact conclusions on the assessment.                                                                                                                                                                                                                                                                                                                                                                                 | Probability of impact on the conclusions                                                                                                                                                                                                                                                            |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U08 | 1.4.3          | Release and monitoring into underground water                                                                                                                         | Monitoring data demonstrated that octocrilene is present in underground water to relatively high concentration. Potential additional hazards for HH (drinking water) and secondary poisoning.                                                                                                                                                                                                                         | No negative impact foreseen. Potentially bringing additional hazards reinforcing the risk assessment and the need of a restriction to tackle the risk.                                                                                                                                              |
| U09 | 1.4.4; 7       | Potential identification of the substance as an ED                                                                                                                    | No changes of the conclusions as these hazards will solely reinforce the need for a restriction.<br><br>Will have widened the scope of the restriction. Impact on SEA and on risk assessment.                                                                                                                                                                                                                         | Impact on the scope of the restriction, will have covered manufacture, formulation and uses of the substance, for professionals and consumers.<br><br>The current restriction would already highly limit the wide-dispersive uses of the substance. Additionnal RMM could be envisaged in due time. |
| U10 | 3.4.2<br>3.4.2 | Parameters impacting the assessment of the costs of the proposed restriction bought by the manufacturers of cosmetic products (reformulation and raw material costs). | To ensure the robustness of the conclusions regarding the proportionality of the restriction, the Dossier Submitter has assessed quantitatively the impact of these uncertainties both individually and in a combined manner via a worst-case scenario. The proposed restriction still appears proportionate when considering the costs for this worst-case scenario.                                                 | No negative impacts foreseen in view of the results of the proportionality assessment under a worst-case scenario.                                                                                                                                                                                  |
| U11 | 0              | Level of improvement enabled by the proposed restriction for fresh- and coastal water ecosystems                                                                      | Uncertainties regarding the safety of the alternatives used. To ensure the robustness of the conclusions regarding the proportionality of the restriction, the Dossier Submitter chose to perform a conservative assessment a considered a low improvement across the EEA. In addition, the Dossier Submitter performed a cost-effectiveness (CEA) assessment to support the conclusion of the cost-benefit analysis. | Proportionality and efficiency of the restriction will depend on the safety of the alternatives used.<br><br>No negative impacts foreseen in view of the results of the CEA.                                                                                                                        |

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| #   | Section (s) | Short description of the uncertainty                                                                                                 | Impact conclusions on the Probability of impact on the conclusions                                                                                                                                                                                                                |
|-----|-------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U12 | 0           | Lag in the materialisation of environmental improvements enabled by the proposed restriction for fresh- and coastal water ecosystems | The Dossier Submitter assessed quantitatively the impact of this uncertainty through the development of alternative scenarios (including a worst-case scenario). The proposed restriction still appears proportionate when considering the benefits for this worst-case scenario. |

### U01 - Potential underestimation of the tonnage of octocrilene put on the market

Based on REACH Article 6(1), Article 7(1) and Guidance on registration v4.0 § 2.2.6, tonnage used in the EU (tonnage manufactured plus imported minus exported) represents the tonnage used at all stages of the life cycle (i.e. including tonnage in imported mixtures and tonnage in imported articles with intended releases, above 1 tonne per year per importer). However, the Dossier Submitter highlights that the real imported tonnage may in reality be higher and not accounted for, as tonnages imported below 1t/y of substance per company, or, for articles, above 1t/y per company with no intended releases, are not registered. Moreover, as detailed in section 1.3.2., discrepancies between CSRs and CfE data were observed, hampering the Dossier Submitter to use these data to determine a sound and plausible range of octocrilene tonnage to be considered in the exposure and risk assessment. As such, the Dossier Submitter confronted the declared tonnage for finished cosmetic products wide dispersive uses with consumer data on finished cosmetic products containing octocrilene sold in the EEA and conclude that the estimated aggregated tonnages were a sound estimates, as both tonnages were substantially identical (1752 t/y vs 1721 t/y). As Euromonitor data for few countries are lacking and that registration dossier is mandatory for manufacturer or importer of octocrilene > 1t/y, this data is a potential underestimation of the tonnage available in the EEA and put on the market.

In addition, the registered tonnages considered for the baseline calculation (and release reduction potential of the different ROs) are still high considering that substitution is already ongoing following the regulatory work on octocrilene and the “bad press” actually surrounding the substance.

Similarly, if substitution would already be considered ongoing without a restriction this would affect the costs as well because then there is no substitution cost, ie both numerator and denominator of the C/E would be affected. Therefore, while this uncertainty on the registered tonnage may lead to an underestimation of the releases, this similarly would not have any implications on the overall conclusions on the assessed restriction options in terms of proportionality.

### U02 - Potential underestimation of the tonnage released to the environment

Due to a lack of consolidated data available to the Dossier Submitter on the total tonnages, specific tonnage data on manufacture, import and export of octocrilene are judged roughly estimated by the dossier submitter. As risk arises from the wide dispersive use of finished cosmetic products and hazards of octocrilene for aquatic species and sediments, focuses was made on this specific use. By focusing on this use, the Dossier Submitter was able to better estimate the tonnage of octocrilene used for finished cosmetic products from exposure scenario of each registrant. This data was confirmed by the crosschecking with Euromonitor

database. As Euromonitor miss data for few countries and that registration dossier is mandatory for manufacturer or importer of octocrilene > 1t/y, this data is an underestimation of the tonnage available in the EEA and put on the market. As such, release to the environment from this wide dispersive use may be underestimated. Moreover, no specific and dedicated study on release pathways to the environment from finished cosmetic products containing octocrilene are available to the Dossier Submitter knowledge. Estimates were made based on the work performed and agreed in the restriction on D4, D5 and D6. It allowed the Dossier Submitter to estimate the release of octocrilene from sunscreen into the different environmental compartment and allowed a dedicated risk assessment from a consumer approach. With no specific release fraction estimates available for octocrilene in finished cosmetic products, release may partly be assigned to the wrong release pathways. As sunscreen is the main driver of the wide dispersive release of finished cosmetic products containing octocrilene, and considering that the tonnage for this specific use is underestimated, the Dossier Submitter considers that the impact of this repartition uncertainty on the risk assessment performed and the restriction conclusion are negligible.

Several parameters not taken into account by the Dossier Submitter may also affect the fraction of octocrilene released to the environment in the different simulations performed. For example, octocrilene can reach the aquatic environment *via* stormwater from combined sewer overflows in particular climatic event; finished cosmetic products containing octocrilene may be subjected to runoff from beaches area where it is deposited onto the beaches after dispersive application of sunscreen (aerosol, sprays, ...); octocrilene may be subject to leaching into surface waters from on-site wastewater systems. Thus, release of octocrilene to the environment may be underestimated, such as the corresponding risks ratio.

Moreover, the other uses of octocrilene in plastisol and as additives to be included in matrix for consumer and professional also led to potential release to the environment. **As such the risk assessment is underestimated as solely performed with the determined tonnage from consumer uses of finished cosmetic products.** Nevertheless, this underestimation is not affecting the outcome of the proposed restriction as risks were demonstrated from tonnage approach and from consumer approach, at a local and regional scale, specific to these uses. This may have an impact on the cost-effectiveness of the restriction. Nevertheless, the concerned additional octocrilene tonnage will not be tremendous, and the Dossier Submitter considers that C/E would remain in the same order of magnitude if additional tonnage were considered.

#### U03 – Potential additional tonnage from outside EEA market

Amount of octocrilene entering the EEA and released from consumer using products bought outside the EU, is unknown. This shows that the risk assessment performed by the Dossier Submitter is rather a best-case scenario and might underestimate the risks as additional releases can occurs.

It is assumed that the tonnage in imported articles (i.e. FCM, PVC articles, articles containing adhesives, sealants, paints, coatings, and any other articles) is 0. Despite release to the environment could occur during the service life of the article and at the waste stage, the Dossier Submitter assessed the sole emission to the environment of the wide dispersive uses of finished cosmetic products as they represent largely the main sources of octocrilene reaching the environment and are the main driver for the identified risks as demonstrated in section 1.4.4. Therefore, the release from the other uses than finished cosmetic products may be underestimated and risks for the environment may be underestimated.

#### U04 – WWTP connection rate, sludge treatment and RMMs

As detailed in section 1.4.4 and in annex B.9., the Dossier Submitter has considered a connection rate of 100% to municipal WWTP for all the uses and life cycle stages. This connection rate was applied in order to minimise the uncertainties regarding direct release to the environment of untreated wastewater and decrease the potential bias for risk assessment

for freshwaters and associated sediments. The connection rate to WWTP was assumed to be 81 % in EU according to the most recent data on Eurostat, in compliance with the provisions of the Urban Waste Water Treatment Directive<sup>114</sup> (EU Commission, 2020). Therefore, a fraction of the wastewater (19 %) is assumed to be discharged to surface water directly, increasing the risks for the concerned environmental compartments. In addition, with regard to the releases associated to waste, 100 % connection rate to municipal WWTP is assumed: meaning that releases to water from operating landfill (via leachate) and incineration (via scrubbing), will be treated and not go directly to surface water.

The municipal WWTP sludge disposal used by the Dossier Submitter is of 100%, approach classically applied in exposure assessment. The Dossier Submitter considered this fraction of sludge application on soil as a worst-case scenario. The most recent data on Eurostat<sup>115</sup> indicated that landfill application of municipal waste in the EU is of 21%. As such, the Dossier submitter could have applied a 79% rate of sludge application on agricultural soil as a more realistic scenario for this restriction proposal. Even when considering a rate of sludge application of 50% as a scenario, RCR ratio for agricultural soil will be > 1 in the tonnage approach (considering local and regional estimates). For consumer rate approach, considering a 50% rate of sludge application will still led to RCR > 1 for agricultural soil (initial RCR of 4.67).

Regarding releases of octocrilene from industrial sites (manufacture, formulation, industrial uses), it is assumed that RMMs are implemented at least to some extent, due to the hazardous nature of the substance, which aim to reduce releases and enable proper waste management. Thus, release factors used in the estimates represent the releases after on-site risk management measures. However, as explained, the RMMs effectiveness in the EU in the current time period is not known. Despite analysis of registration dossiers, calls for information and evidence, no data was received to substantiate the RMMs effectiveness.

For professional and consumer uses, it is assumed that no particular risk management measures are implemented.

#### Manufacturing of the substance:

Little specific information on risk management measures applied at manufacturers' site is available.

Due to the variability in the scarce available information from CSRs, it is likely that risk management measures are applied and able to reduce releases, but their efficiencies in terms of quantitative abatement of concentration in effluents are uncertain. As demonstrated, RMMs are in place for one registrant, corresponding to the incineration of the washing liquid and exclusion of any release to wastewater.

#### U05 - Regulatory uncertainty

A source of regulatory uncertainty comes from the scope of the restriction. Restriction options are built on to the assumption that restriction is applied for placing octocrilene containing finished cosmetic products on the EU market. However, the recent EU Chemicals Strategy for Sustainability<sup>46</sup> states that the EU will lead by example and ensure that hazardous chemicals banned in the EU are not produced for export either. Thus, the decision-maker may want to opt for a restriction that applies to the production of octocrilene containing cosmetic products

<sup>114</sup> Council Directive 91/271/EEC concerning urban waste-water treatment.

<sup>115</sup>

[https://ec.europa.eu/eurostat/databrowser/view/ten00138/default/table?lang=en&category=t\\_env.t\\_env\\_was.t\\_env\\_wasst](https://ec.europa.eu/eurostat/databrowser/view/ten00138/default/table?lang=en&category=t_env.t_env_was.t_env_wasst)

rather than the placing on the market of the resulting cosmetic products. The amount of cosmetic products produced and formulated in the EEA and exported is not known from the Dossier Submitter to complement its analysis.

#### U06 – Handling of industrial waste at management site

Industrial waste (pre-consumer waste) from manufacture, formulation and industrial uses residues (e.g. lost from application process), off-specifications products, waste from equipment and surfaces cleaning, waste from RMM (e.g. PPE, filters, sludges, particles), etc. from industrial sites are assumed to be handled appropriately. RMM were put in place by one registrant of octocrilene and refined release fraction estimates provided, especially in regards to the manufacturing of the substance. This was not the case for the second registrant that manufacture the substance, leading to risks being demonstrated. Similarly, risks were also identified for the formulation step in the production of plastisol for a second registrant due to the absence of this specific RMM, corresponding to the incineration of the washing liquid and exclusion of any release to wastewater. Uncertainty arise regarding the proper application of RMMs and OCs to ensure that release of octocrilene to wastewater and sludge from industrial WWTP are disposed properly to avoid any release to water directed to local WWTP and that sludge are not intended to be used onto agricultural soil.

As octocrilene is a suspected ED substance, the substance could have been handled as a hazardous waste<sup>116</sup>. Classification of waste as hazardous is rather complex and depend on a risk assessment if not listed as “absolute hazardous” in the European List of Waste. In Norway, the concentration limit to render waste as being hazardous is 2 500 mg/kg (Danish EPA, 2014). The EU legislation stipulates that hazardous waste have to be handled so as to provide protection for the environment and human health, which includes traceability, mixing ban, labelling, and treatment in specially designated facilities that have obtained a special permit<sup>117</sup>. The most common treatment methods are incineration and hazardous landfilling (EU Commission, 2021b).

The inappropriate handling of wastewater or sludge or the inability to put in place specific RMMs and OCs could have led the Dossier Submitter to propose no derogation for the manufacture of the substance, formulation of finished cosmetic products or plastisol. This would have led to much higher cost of the restriction proposal as affecting more type of industrial activities, employment and supply chain. The proportionality of the restriction would also have been negatively impacted.

Given that the proposed restriction covers solely the uses of octocrilene in finished cosmetic products, the Dossier Submitter expects that impacts of the restriction may occur solely on cosmetic industry, and solely to a fraction of manufacturers or formulators already manufacturing or using alternatives.

#### U07 – Choice of parameters in the risk assessment determination

The parameters chosen to perform risk assessment were mainly based on those proposed by industry or “best-case” also more stringent parameters could have been identified. However, as risk were already identified using those parameters, it was deemed unnecessary to justify

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<sup>116</sup> In the Waste Framework Directive, ‘hazardous waste’ is defined as ‘waste which displays one or more of the hazardous properties listed in Annex III’. The classification into hazardous and non-hazardous waste is based on the system for the classification and labelling of dangerous substances and preparations. This ensures that similar principles are applied over the whole life cycle of materials. As CLP was updated with new hazard class, it would have ensure the proper hazardous properties identification to these waste stream, for PBT, vPvB and ED waste.

<sup>117</sup> Directive 2008/98/EC; Commission notice on technical guidance on the classification of waste (2018/C 124/01)

the use of those more stringent ones (degradation half-life, inclusion of the toxicity of the conjugates/metabolites, AF, more global input parameters (tonnage ...).

### U08 – Persistence in underground water and secondary poisoning

Octocrilene has been found in underground water to relatively high concentration. The substance being persistent, the accumulation into underground water may pose a risk for human health through consumption of contaminated water for drinking purpose. Moreover, additional risks may occurred for environmental species through secondary poisoning.

### U09 – Identification of octocrilene as an Endocrine Disruptor

Substance Evaluation of octocrilene is actually on-going as being a suspected ED. Formal identification of the substance as an ED-ENV cat 1 or 2 would lead to consider the ban of the substance for any scenario leading to release of the substance to the environment, impacting all industries manufacturing the substance and formulating articles or mixture for the EEA market. As such, the economic impact would be much higher in term of costs. Cost-Effectiveness ratio would be impacted, depending on the considered tonnage of the substance and the concerned industries, especially for industries producing plastisol and their importance on the market. On the other hand, the species concerned by the identified hazard would be wider and the willingness to pay most probably higher.

### U10 - Parameters impacting the assessment of the costs of the proposed restriction borne by the manufacturers of cosmetic products

Although it is based on data provided by the consulted industrial stakeholders and/or on data published in recent Annex XV reports, some of the parameters used by the Dossier Submitter in the cost assessment of the restriction are subject to uncertainty. More specifically, the Dossier Submitter assessed the impact of uncertainties on the following parameters through individual quantitative sensitivity analyses (see Appendix E.4.3):

- Reformulation cost:
  - Share of octo-containing products in each product category;
  - Useful life of a cosmetic formulation (10 years or 20 years);
  - Reformulation cost per cosmetic formulation;
  - Reformulation rate.
- Raw material cost:
  - Formulation cost of sun protection products;
  - Increase in formulation cost in the event of a restriction.

The results of these sensitivity analyses show that the uncertainties affecting the above-mentioned parameters are likely to have a significant impact on the cost of the proposed restriction. In this context, and to ensure the robustness of the conclusions regarding the proportionality of the restriction, the Dossier Submitter has assessed the impact of all these uncertainties in a combined manner via a worst-case scenario. The results of this sensitivity analysis are available in Appendix E.4.3.6. **The proposed restriction still appears proportionate when considering the costs for this worst-case scenario.**

### U11 and U12 - Level of improvement enabled by the proposed restriction for fresh- and coastal water ecosystems and lag in their materialisation

One main driver of how the restriction will improve water ecosystem is the use of safer UV

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filters for the environment but also for human health. Identifying these chemicals is a major challenge for this restriction to be efficient but in general for our future.

See section 0 and Appendix E.4.5. The Dossier Submitter assessed quantitatively the impact of these two sources of uncertainty and results show no impact on the conclusions drawn by the Dossier Submitter regarding the proportionality of the proposed restriction.

## 7. Proposed restriction entry

### 7.1. Wording of the proposed restriction entry

#### Short title:

Restriction on the placing on the market and uses of finished cosmetic products containing octocrilene.

#### Scope description:

The text of the proposed entry in REACH Annex XVII has been drafted to describe the intention of the Dossier Submitter. The final legal wording (i.e. to update the Annex XVII to REACH) is decided by the European Commission during the decision-making phase.

An explanation of the intention of the Dossier Submitter, and the wording proposed is provided in Section 7.2.

**Table 45: Proposed REACH Annex XVII entry.**

| Designation of the substance, of the group of substances or of the mixture | Conditions of restriction                                                                                                            |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Octocrilene                                                                | 1. Shall not be placed on the market, or used, in finished cosmetic products in a concentration equal to or greater than 0.001% w/w. |
| CAS No 6197-30-4                                                           |                                                                                                                                      |
| EC No 228-250-8                                                            | 2. The restriction shall apply 24 months after its entry into force.                                                                 |
|                                                                            | 3. 'Finished cosmetic product' is a product as defined by Regulation (EC) No 1223/2009.                                              |

### 7.2. Justification of the wording for the proposed restriction entries

#### Wording of the column 1 – Designation of the substance

Column 1 lists the identifiers of the substance subject to the restriction proposal.

#### Wording of the column 2 – Paragraph 1

The concentration limit in paragraph 1 sets the maximum concentration of octocrilene allowed in finished cosmetic products and mixtures to be used or placed on the market.

The term 'placing on the market' is defined in REACH Article 3:

- 'Placing on the market' should be understood as 'placing on the EEA market (i.e. EU + Iceland, Norway and Liechtenstein)' as defined under REACH Article 3(12), i.e. 'supplying or making available, whether in return for payment or free of charge, to a third party. Import is deemed to be placing on the market'.

The 'ban on placing on the market' puts therefore the obligation on all actors in the supply chain who supply or make cosmetic mixtures available, whether in return for

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payment or free of charge, to a third party. It includes also the placing on the market via internet.

Finished cosmetic products are products as defined in the regulation (EC) No 1223/2009 on cosmetic products:

- 'finished cosmetic product' means the cosmetic product in its final formulation, as placed on the market and made available to the end user, or its prototype.
- 'cosmetic product' means any substance or mixture intended to be placed in contact with the external parts of the human body (epidermis, hair system, nails, lips and external genital organs) or with the teeth and the mucous membranes of the oral cavity with a view exclusively or mainly to cleaning them, perfuming them, changing their appearance, protecting them, keeping them in good condition or correcting body odours.
- 'end user' means either a consumer or professional using the cosmetic product.

## References

- Agawin, N. S., García-Márquez, M. G., Espada, D. R., Freemantle, L., Herrera, M. G. P., & Tovar-Sánchez, A. (2024). Distribution and accumulation of UV filters (UVFs) and conservation status of *Posidonia oceanica* seagrass meadows in a prominent Mediterranean coastal tourist hub. *The Science Of The Total Environment*, 948, 174784. <https://doi.org/10.1016/j.scitotenv.2024.174784>
- Amankwah, B. K., Šauer, P., Grabicová, K., Von Der Ohe, P. C., Ayikol, N. S., & Kroupová, H. K. (2024). Organic UV filters: Occurrence, risks and (anti-)progestogenic activities in samples from the Czech aquatic environment and their bioaccumulation in fish. *Journal Of Hazardous Materials*, 471, 134338. <https://doi.org/10.1016/j.jhazmat.2024.134338>
- Aminot, Y., Sayfritz, S. J., Thomas, K. V., Godinho, L., Botteon, E., Ferrari, F., Boti, V., Albanis, T., Köck-Schulmeyer, M., Diaz-Cruz, M. S., Farré, M., Barceló, D., Marques, A., & Readman, J. W. (2019). Environmental risks associated with contaminants of legacy and emerging concern at European aquaculture areas. *Environmental Pollution*, 252, 1301–1310. <https://doi.org/10.1016/j.envpol.2019.05.133>
- Ao, J., Yuan, T., Gu, J., Ma, Y., Shen, Z., Tian, Y., Shi, R., Zhou, W., & Zhang, J. (2018). Organic UV filters in indoor dust and human urine: A study of characteristics, sources, associations and human exposure. *The Science of the Total Environment*, 640–641, 1157–1164. <https://doi.org/10.1016/j.scitotenv.2018.05.367>
- Apel, C., Joerss, H., & Ebinghaus, R. (2018). Environmental occurrence and hazard of organic UV stabilizers and UV filters in the sediment of European North and Baltic Seas. *Chemosphere*, 212, 254–261. <https://doi.org/10.1016/j.chemosphere.2018.08.105>
- Avenel-Audran, M., Dutartre, H., Goossens, A., Jeanmougin, M., Comte, C., Bernier, C., Benkalfate, L., Michel, M., Ferrier-Lebouëdec, M. C., Vigan, M., Bourrain, J. L., Outtas, O., Peyron, J. L., & Martin, L. (2010). Octocrylene, an emerging photoallergen. *Arch Dermatol*, 146(7), 753–757. <https://doi.org/10.1001/archdermatol.2010.132>
- Azaroff, A., Miossec, C., Lanceleur, L., Guyoneaud, R., & Monperrus, M. (2019). Priority and emerging micropollutants distribution from coastal to continental slope sediments: A case study of Capbreton Submarine Canyon (North Atlantic Ocean). *The Science Of The Total Environment*, 703, 135057. <https://doi.org/10.1016/j.scitotenv.2019.135057>
- Bachelot, M., Li, Z., Munaron, D., Gall, P. L., Casellas, C., Fenet, H., & Gomez, E. (2012b). Organic UV filter concentrations in marine mussels from French coastal regions. *The Science of the Total Environment*, 420, 273–279. <https://doi.org/10.1016/j.scitotenv.2011.12.051>
- Balázs, A., Krifaton, C., Orosz, I., Szoboszlai, S., Kovács, R., Csenki, Z., Urbányi, B., & Kriszt, B. (2016). Hormonal activity, cytotoxicity and developmental toxicity of UV filters. *Ecotoxicol Environ Saf*, 131, 45–53. <https://doi.org/10.1016/j.ecoenv.2016.04.037>
- Balmer, M. E., Buser, H., Müller, M. D., & Poiger, T. (2005b). Occurrence of Some Organic UV Filters in Wastewater, in Surface Waters, and in Fish from Swiss Lakes. *Environmental Science & Technology*, 39(4), 953–962. <https://doi.org/10.1021/es040055r>
- Berardesca, E., Zuberbier, T., Viera, M. S., & Marinovich, M. (2019). Review of the safety of octocrylene used as an ultraviolet filter in cosmetics. *Journal of the European Academy of Dermatology and Venereology*, 33(S7), 25–33. <https://doi.org/10.1111/jdv.15945>
- Berkmen, Y. M., & Lande, A. (1975). Chest roentgenography as a window to the diagnosis of

- Takayasu's arteritis. *The American Journal of Roentgenology, Radium Therapy, and Nuclear Medicine*, 125(4), 842-846. <https://doi.org/10.2214/ajr.125.4.842>
- Biel-Maeso, M., Corada-Fernández, C., & Lara-Martín, P. A. (2018). Removal of personal care products (PCPs) in wastewater and sludge treatment and their occurrence in receiving soils. *Water Research*, 150, 129-139. <https://doi.org/10.1016/j.watres.2018.11.045>
- Blum, K. M., Andersson, P. L., Renman, G., Ahrens, L., Gros, M., Wiberg, K., & Haglund, P. (2016b). Non-target screening and prioritization of potentially persistent, bioaccumulating and toxic domestic wastewater contaminants and their removal in on-site and large-scale sewage treatment plants. *The Science Of The Total Environment*, 575, 265-275. <https://doi.org/10.1016/j.scitotenv.2016.09.135>
- Blüthgen, N., Meili, N., Chew, G., Odermatt, A., & Fent, K. (2014). Accumulation and effects of the UV-filter octocrylene in adult and embryonic zebrafish (*Danio rerio*). *The Science of the Total Environment*, 476-477, 207-217. <https://doi.org/10.1016/j.scitotenv.2014.01.015>
- Bolon, B., Garman, R., Jensen, K., Krinke, G., & Stuart, B. (2006). A « best practices » approach to neuropathologic assessment in developmental neurotoxicity testing—For today. *Toxicol Pathol*, 34(3), 296-313. <https://doi.org/10.1080/01926230600713269>
- Boyd, A., Stewart, C. B., Philibert, D. A., How, Z. T., El-Din, M. G., Tierney, K. B., & Blewett, T. A. (2020). A burning issue: The effect of organic ultraviolet filter exposure on the behaviour and physiology of *Daphnia magna*. *The Science Of The Total Environment*, 750, 141707. <https://doi.org/10.1016/j.scitotenv.2020.141707>
- Broussard, L., Hirner, S., Dellavalle, R. P., Dunnick, C. A., & Hugh, J. (2019). Spray sunscreen: Characterizing application area density and implications for sun protection. *Journal of the American Academy of Dermatology*, 82(3), 749-751. <https://doi.org/10.1016/j.jaad.2019.07.094>
- Bury, D., Modick-Biermann, H., Leibold, E., Brüning, T., & Koch, H. M. (2019). Urinary metabolites of the UV filter octocrylene in humans as biomarkers of exposure. *Archives of Toxicology*, 93(5), 1227-1238. <https://doi.org/10.1007/s00204-019-02408-7>
- Bury, D., Weber, T., Ebert, K. E., Zülz, S., Brüning, T., Koch, H. M., & Kolossa-Gehring, M. (2023). Increasing exposure to the UV filters octocrylene and 2-ethylhexyl salicylate in Germany from 1996 to 2020: Human biomonitoring in 24-h urine samples of the German Environmental Specimen Bank (ESB). *Environment International*, 182, 108334. <https://doi.org/10.1016/j.envint.2023.108334>
- Cadena-Aizaga, M. I., Montesdeoca-Esponda, S., Sosa-Ferrera, Z., & Santana-Rodríguez, J. J. (2022a). Occurrence and environmental hazard of organic UV filters in seawater and wastewater from Gran Canaria Island (Canary Islands, Spain). *Environmental Pollution*, 300, 118843. <https://doi.org/10.1016/j.envpol.2022.118843>
- Cahova, J., Blahova, J., Plhalova, L., Svobodova, Z., & Faggio, C. (2021). Do Single-Component and Mixtures Selected Organic UV Filters Induce Embryotoxic Effects in Zebrafish (*Danio rerio*)? *Water*, 13(16), 2203. <https://doi.org/10.3390/w13162203>
- Campos, D., Gravato, C., Quintaneiro, C., Golovko, O., Žlábek, V., Soares, A. M., & Pestana, J. L. (2017). Toxicity of organic UV-filters to the aquatic midge *Chironomus riparius*. *Ecotoxicology And Environmental Safety*, 143, 210-216. <https://doi.org/10.1016/j.ecoenv.2017.05.005>
- Clergeaud, F., Fagervold, S. K., Rodrigues, A. M. S., Thorel, E., Stien, D., & Lebaron, P.

- (2022). Transfer of 7 Organic UV Filters from Sediment to the Ragworm *Hediste diversicolor*: Bioaccumulation of Benzophenone-3 and Further Proof of Octocrylene Metabolism. *Pollutants*, 2(1), 23-31. <https://doi.org/10.3390/pollutants2010004>
- Combi, T., Pintado-Herrera, M. G., Lara-Martin, P. A., Miserocchi, S., Langone, L., & Guerra, R. (2016). Distribution and fate of legacy and emerging contaminants along the Adriatic Sea: A comparative study. *Environmental Pollution*, 218, 1055–1064. <https://doi.org/10.1016/j.envpol.2016.08.057>
- CompTOX/EDSP-21. (Accessed February 2022). <https://comptox.epa.gov/dashboard/>
- Connor, J. M. (1981). Food Product Proliferation: A Market Structure Analysis. *American Journal of Agricultural Economics*, 63(4), 607-617. <https://doi.org/10.2307/1241203>
- CosmEthics. (2023). *Confidential data extract [extracted on 15 November 2023]. Statistics retrieved from the mobile application database CosmEthics owned by CosmEthics Oy.* <http://cosmethics.com/>
- Cunha, S., Pena, A., & Fernandes, J. (2015b). Dispersive liquid-liquid microextraction followed by microwave-assisted silylation and gas chromatography-mass spectrometry analysis for simultaneous trace quantification of bisphenol A and 13 ultraviolet filters in wastewaters. *Journal Of Chromatography A*, 1414, 10-21. <https://doi.org/10.1016/j.chroma.2015.07.099>
- De C S de Paula, V., Gomes, M. F., Martins, L. R. R., Yamamoto, F. Y., & De Freitas, A. M. (2022). Acute toxicity characterization of organic UV-filters and chronic exposure revealing multigenerational effects in *DAPHNIA MAGNA*. *Ecotoxicology*, 31(9), 1413-1425. <https://doi.org/10.1007/s10646-022-02598-4>
- de Groot, A. C., & Roberts, D. W. (2014). Contact and photocontact allergy to octocrylene: A review. *Contact Dermatitis*, 70(4), 193-204. <https://doi.org/10.1111/cod.12205>
- Delplace, D., & Blondeel, A. (2006). Octocrylene : Really non-allergenic? *Contact Dermatitis*, 54(5), 295. <https://doi.org/10.1111/j.0105-1873.2006.0698a.x>
- Downs, C. A., DiNardo, J. C., Stien, D., Rodrigues, A. M. S., & Lebaron, P. (2021). Benzophenone Accumulates over Time from the Degradation of Octocrylene in Commercial Sunscreen Products. *Chemical Research in Toxicology*, 34(4), 1046–1054. <https://doi.org/10.1021/acs.chemrestox.0c00461>
- ECHA. (2008c). *Guidance on Information Requirements and Chemical Safety Assessment Chapter R.10: Characterisation of dose [concentration]-response for environment.* European Chemicals Agency.
- ECHA. (2016). *Guidance on Information Requirements and Chemical Safety Assessment Chapter R.16: Environmental exposure assessment.* European Chemicals Agency.
- ECHA. (2020). *Background Document to the Opinion on restriction proposals on D4, D5, D6.*
- ECHA. (2023). *Reference willingness-to-pay values for monetising chemicals health impacts.* [https://echa.europa.eu/documents/10162/17229/seac\\_reference\\_wtp\\_values\\_en.pdf/403429a1-b45f-4122-ba34-77b71ee9f7c9](https://echa.europa.eu/documents/10162/17229/seac_reference_wtp_values_en.pdf/403429a1-b45f-4122-ba34-77b71ee9f7c9)
- ECHA/EFSA (2018) European Chemicals Agency and European Food Safety Authority with the technical support of the Joint Research Centre (JRC), "Guidance for the identification of endocrine disruptors in the context of Regulations (EU) No 528/2012 and (EC) No

- 1107/2009." EFSA Journal 16 (6):e05311. [https://doi: 10.2903/j.efsa.2018.5311](https://doi.org/10.2903/j.efsa.2018.5311)
- ESD PT19. Emission scenario document for Product Type 19 – Repellents and attractants, 2015. ECHA. Appendix 6.3.
- Emission scenario document for Product Type 21: Antifouling products, 2004
- Euromonitor International, 2025. *Use volumes of octocrilene in western Europe in 2023*.
- Euromonitor International, 2024a. *Company shares in western Europe in 2023*.
- Euromonitor International, 2024b. *Sales of Sun Care - Retail Value RSP (2009-2018)*.
- FOCUS (2014). Generic guidance for Estimating Persistence and Degradation Kinetics from Environmental Fate Studies on Pesticides in EU Registration. Version: 1.1, 18 December 2014, 440 pp.
- Ficheux, A., Chevillotte, G., Wesolek, N., Morisset, T., Dornic, N., Bernard, A., Bertho, A., Romanet, A., Leroy, L., Mercat, A., Creusot, T., Simon, E., & Roudot, A. (2016). Consumption of cosmetic products by the French population second part: Amount data. Food and Chemical Toxicology, 90, 130–141. <https://doi.org/10.1016/j.fct.2016.02.008>
- Gago-Ferrero, P., Díaz-Cruz, M. S., & Barceló, D. (2011a). Fast pressurized liquid extraction with in-cell purification and analysis by liquid chromatography tandem mass spectrometry for the determination of UV filters and their degradation products in sediments. Analytical And Bioanalytical Chemistry, 400(7), 2195-2204. <https://doi.org/10.1007/s00216-011-4951-1>
- Gago-Ferrero, P., Díaz-Cruz, M. S., & Barceló, D. (2011b). Occurrence of multiclass UV filters in treated sewage sludge from wastewater treatment plants. Chemosphere, 84(8), 1158-1165. <https://doi.org/10.1016/j.chemosphere.2011.04.003>
- Gao, L., Yuan, T., Zhou, C., Cheng, P., Bai, Q., Ao, J., Wang, W., & Zhang, H. (2013). Effects of four commonly used UV filters on the growth, cell viability and oxidative stress responses of the Tetrahymena thermophila. Chemosphere, 93(10), 2507-2513. <https://doi.org/10.1016/j.chemosphere.2013.09.041>
- García-Pimentel, M., Campillo, J., Castaño-Ortiz, J., Llorca, M., & León, V. (2023). Occurrence and distribution of contaminants of legacy and emerging concern in surface waters of two Western Mediterranean coastal areas: Mar Menor Lagoon and Ebro Delta. Marine Pollution Bulletin, 187, 114542. <https://doi.org/10.1016/j.marpolbul.2022.114542>
- Gautam, K., Mishra, S., & Anbumani, S. (2023). Environmental toxicity of organic UV filters at predicted no-effect concentrations in soil organism, Eisenia fetida. Environmental Technology & Innovation, 31, 103224. <https://doi.org/10.1016/j.eti.2023.103224>
- Gayathri, M., Sutha, J., Mohanthi, S., Ramesh, M., & Poopal, R. (2023). Ecotoxicological evaluation of the UV-filter octocrylene (OC) in embryonic zebrafish (Danio rerio): Developmental, biochemical and cellular biomarkers. Comparative Biochemistry and Physiology Part C Toxicology & Pharmacology, 271, 109688. <https://doi.org/10.1016/j.cbpc.2023.109688>
- Giraldo, A., Montes, R., Rodil, R., Quintana, J. B., Vidal-Liñán, L., & Beiras, R. (2017). Ecotoxicological Evaluation of the UV Filters Ethylhexyl Dimethyl p-Aminobenzoic Acid and Octocrylene Using Marine Organisms Isochrysis galbana, Mytilus galloprovincialis and Paracentrotus lividus. Archives Of Environmental Contamination And Toxicology, 72(4), 606-611. <https://doi.org/10.1007/s00244-017-0399-4>

- Gómez, M., Gómez-Ramos, M., Agüera, A., Mezcua, M., Herrera, S., & Fernández-Alba, A. (2009). A new gas chromatography/mass spectrometry method for the simultaneous analysis of target and non-target organic contaminants in waters. *Journal Of Chromatography A*, 1216(18), 4071-4082. <https://doi.org/10.1016/j.chroma.2009.02.085>
- Gomez-Berrada, M., Ficheux, A., Rakotomalala, S., Roudot, A., & Ferret, P. (2017). Probabilistic exposure assessment of sun care products. *Food and Chemical Toxicology*, 108, 314–325. <https://doi.org/10.1016/j.fct.2017.07.044>
- Gomez-Berrada, M., Ficheux, A., Rakotomalala, S., Guillou, S., Bellec, M., De Javel, D., Roudot, A., & Ferret, P. (2018). Consumption and exposure assessment to sunscreen products: A key point for safety assessment. *Food and Chemical Toxicology*, 114, 170–179. <https://doi.org/10.1016/j.fct.2018.02.035>
- González-Bareiro, E., Montesdeoca-Esponda, S., De La Fuente, J., Sosa-Ferrera, Z., Arbelo, M., Fernández, A., & Santana-Rodríguez, J. J. (2023). Assessment of the presence of UV filters and UV stabilizers in stranded dolphin blubber. *The Science of the Total Environment*, 895, 165041. <https://doi.org/10.1016/j.scitotenv.2023.165041>
- Guidance on information requirements and chemical safety assessment. Chapter R.16: Environmental Exposure Estimation. Draft version 2.0 25/01/2010. ECHA
- Gu, J., Yuan, T., Ni, N., Ma, Y., Shen, Z., Yu, X., Shi, R., Tian, Y., Zhou, W., & Zhang, J. (2019). Urinary concentration of personal care products and polycystic ovary syndrome: A case-control study. *Environ Res*, 168, 48-53. <https://doi.org/10.1016/j.envres.2018.09.014>
- Harlington, H., Patel, S., & Kundu, R. V. (2022). Cost and quality in consumer sunscreen preferences. *Archives of Dermatological Research*, 315(4), 925-931. <https://doi.org/10.1007/s00403-022-02467-4>
- He, T., Tsui, M. M. P., Tan, C. J., Ying, C., MA, Yiu, S. K. F., Wang, L. H., Chen, T. H., Fan, T. Y., Lam, P. K. S., & Murphy, M. B. (2018b). Toxicological effects of two organic ultraviolet filters and a related commercial sunscreen product in adult corals. *Environmental Pollution*, 245, 462-471. <https://doi.org/10.1016/j.envpol.2018.11.029>
- Heerfordt, I. M., Torsnes, L. R., Philipsen, P. A., & Wulf, H. C. (2018). Sunscreen use optimized by two consecutive applications. *PLoS ONE*, 13(3), e0193916. <https://doi.org/10.1371/journal.pone.0193916>
- Hiller, J., Klotz, K., Meyer, S., Uter, W., Hof, K., Greiner, A., Göen, T., & Drexler, H. (2019). Systemic availability of lipophilic organic UV filters through dermal sunscreen exposure. *Environment International*, 132, 105068. <https://doi.org/10.1016/j.envint.2019.105068>
- Homem, V., Llompart, M., Vila, M., Ribeiro, A. R., Garcia-Jares, C., Ratola, N., & Celeiro, M. (2022). Gone with the flow - Assessment of personal care products in Portuguese rivers. *Chemosphere*, 293, 133552. <https://doi.org/10.1016/j.chemosphere.2022.133552>
- Jungman, E., Laugel, C., Kasselouri, A., & Baillet-Guffroy, A. (2012). Study of the potential of stratum corneum lipids and exogenous molecules interaction by fluorescence spectroscopy for the estimation of percutaneous penetration. *International Journal of Pharmaceutics*, 434(1), 183-190. <https://doi.org/10.1016/j.ijpharm.2012.05.049>

- Kaiser, D., Sieratowicz, A., Zielke, H., Oetken, M., Hollert, H., & Oehlmann, J. (2012a). Ecotoxicological effect characterisation of widely used organic UV filters. *Environmental Pollution*, 163, 84-90. <https://doi.org/10.1016/j.envpol.2011.12.014>
- Kaiser, D., Wappelhorst, O., Oetken, M., & Oehlmann, J. (2012b). Occurrence of widely used organic UV filters in lake and river sediments. *Environmental Chemistry*, 9(2), 139. <https://doi.org/10.1071/en11076>
- Karlsson, I., Vanden Broecke, K., Mårtensson, J., Goossens, A., & Börje, A. (2011). Clinical and experimental studies of octocrylene's allergenic potency. *Contact Dermatitis*, 64(6), 343-352. <https://doi.org/10.1111/j.1600-0536.2011.01899.x>
- Krzeminski, P., Schwermer, C., Wennberg, A., Langford, K., & Vogelsang, C. (2016). Occurrence of UV filters, fragrances and organophosphate flame retardants in municipal WWTP effluents and their removal during membrane post-treatment. *Journal Of Hazardous Materials*, 323, 166-176. <https://doi.org/10.1016/j.jhazmat.2016.08.001>
- Kunz, P. Y., & Fent, K. (2006). Multiple hormonal activities of UV filters and comparison of in vivo and in vitro estrogenic activity of ethyl-4-aminobenzoate in fish. *Aquat Toxicol*, 79(4), 305-324. <https://doi.org/10.1016/j.aquatox.2006.06.016>
- Kupper, T., Plagellat, C., Brändli, R., De Alencastro, L., Grandjean, D., & Tarradellas, J. (2006b). Fate and removal of polycyclic musks, UV filters and biocides during wastewater treatment. *Water Research*, 40(14), 2603-2612. <https://doi.org/10.1016/j.watres.2006.04.012>
- Labille, J., Slomberg, D., Catalano, R., Robert, S., Apers-Tremelo, M., Boudenne, J., Manasfi, T., & Radakovitch, O. (2019d). Assessing UV filter inputs into beach waters during recreational activity: A field study of three French Mediterranean beaches from consumer survey to water analysis. *The Science Of The Total Environment*, 706, 136010. <https://doi.org/10.1016/j.scitotenv.2019.136010>
- Langford, K. H., & Thomas, K. V. (2008). Inputs of chemicals from recreational activities into the Norwegian coastal zone. *Journal Of Environmental Monitoring*, 10(7), 894. <https://doi.org/10.1039/b806198j>
- Langford, K. H., Reid, M. J., Fjeld, E., Øxnevad, S., & Thomas, K. V. (2015). Environmental occurrence and risk of organic UV filters and stabilizers in multiple matrices in Norway. *Environment International*, 80, 1-7. <https://doi.org/10.1016/j.envint.2015.03.012>
- Li, A. J., Law, J. C., Chow, C., Huang, Y., Li, K., & Leung, K. S. (2018). Joint Effects of Multiple UV Filters on Zebrafish Embryo Development. *Environmental Science & Technology*, 52(16), 9460-9467. <https://doi.org/10.1021/acs.est.8b02418>
- Lopardo, L., Adams, D., Cummins, A., & Kasprzyk-Hordern, B. (2018). Verifying community-wide exposure to endocrine disruptors in personal care products—In quest for metabolic biomarkers of exposure via in vitro studies and wastewater-based epidemiology. *Water Res*, 143, 117-126. <https://doi.org/10.1016/j.watres.2018.06.028>
- Louviere, J. J., & Hensher, D. A. (1982). Design and Analysis of Simulated Choice or Allocation Experiments in Travel Choice Modeling. *Transportation Research Record*, 890, 11-17.
- Louviere, J. J., & Woodworth, G. (1983). Design and Analysis of Simulated Consumer Choice or Allocation Experiments: An Approach Based on Aggregate Data. *Journal of Marketing Research*, 20(4), 350-367.

- Lucas, J., Logeux, V., Rodrigues, A., Stien, D., & Lebaron, P. (2022). Impact of egg exposure to UV Filter-Spiked sediment on the survival, hatching success, cardiac frequency, and metabolic scope of zebrafish embryos. *Oceans*, 3(1), 84–93. <https://doi.org/10.3390/oceans3010008>
- Lukić, J., Radulović, J., Lučić, M., Đurkić, T., & Onjia, A. (2022). Chemometric Optimization of Solid-Phase Extraction Followed by Liquid Chromatography-Tandem Mass Spectrometry and Probabilistic Risk Assessment of Ultraviolet Filters in an Urban Recreational Lake. *Frontiers In Environmental Science*, 10. <https://doi.org/10.3389/fenvs.2022.916916>
- Lutz, S. (1997). Vertical product differentiation and entry deterrence. *Journal of Economics*, 65(1), 79-102. <https://doi.org/10.1007/BF01239060>
- Magi, E., Di Carro, M., Scapolla, C., & Nguyen, K. T. N. (2012) a. Stir Bar Sorptive Extraction and LC-MS/MS for Trace Analysis of UV Filters in Different Water Matrices. *Chromatographia*, 75(17), 973–982. <https://doi.org/10.1007/s10337-012-2202-z>
- Magi, E., Scapolla, C., Di Carro, M., Rivaro, P., & Nguyen, K. T. N. (2012) b. Emerging pollutants in aquatic environments: monitoring of UV filters in urban wastewater treatment plants. *Analytical Methods*, 5(2), 428–433. <https://doi.org/10.1039/c2ay26163d>
- Mandarić, L., Diamantini, E., Stella, E., Cano-Paoli, K., Valle-Sistac, J., Molins-Delgado, D., Bellin, A., Chiogna, G., Majone, B., Diaz-Cruz, M. S., Sabater, S., Barcelo, D., & Petrovic, M. (2017). Contamination sources and distribution patterns of pharmaceuticals and personal care products in Alpine rivers strongly affected by tourism. *The Science Of The Total Environment*, 590-591, 484-494. <https://doi.org/10.1016/j.scitotenv.2017.02.185>
- Marcin, S., & Aleksander, A. (2023). Acute toxicity assessment of nine organic UV filters using a set of biotests. *Toxicological Research*, 39(4), 649-667. <https://doi.org/10.1007/s43188-023-00192-2>
- Mariel, P., Hoyos, D., Meyerhoff, J., Czajkowski, M., Dekker, T., Glenk, K., Bredahl Jacobsen, J., Liebe, U., Olsen, S. B., Sagebiel, J., & Thiene, M. (2021). *Environmental valuation with discrete choice experiments: Guidance on design, implementation and data analysis*. Springer.
- Matta, M. K., Zusterzeel, R., Pilli, N. R., Patel, V., Volpe, D. A., Florian, J., Oh, L., Bashaw, E., Zineh, I., Sanabria, C., Kemp, S., Godfrey, A., Adah, S., Coelho, S., Wang, J., Furlong, L., Ganley, C., Michele, T., & Strauss, D. G. (2019). Effect of Sunscreen Application Under Maximal Use Conditions on Plasma Concentration of Sunscreen Active Ingredients. *JAMA*, 321(21), 2082. <https://doi.org/10.1001/jama.2019.5586>
- Matta, M. K., Florian, J., Zusterzeel, R., Pilli, N. R., Patel, V., Volpe, D. A., Yang, Y., Oh, L., Bashaw, E., Zineh, I., Sanabria, C., Kemp, S., Godfrey, A., Adah, S., Coelho, S., Wang, J., Furlong, L., Ganley, C., Michele, T., & Strauss, D. G. (2020). Effect of Sunscreen Application on Plasma Concentration of Sunscreen Active Ingredients. *JAMA*, 323(3), 256. <https://doi.org/10.1001/jama.2019.20747>
- Milinkovitch, T., Vacher, L., Béguec, M. L., Petit, E., Dubillot, E., Grimmelpont, M., Labille, J., Tran, D., Ravier, S., Boudenne, J., & Lefrançois, C. (2024). Sunscreen use during recreational activities on a French Atlantic beach: release of UV filters at sea and influence of air temperature. *Environmental Science and Pollution Research*, 31(28), 41046–41058. <https://doi.org/10.1007/s11356-024-33783-z>
- Mitchell, R. C. (2013). *Using Surveys to Value Public Goods: The Contingent Valuation Method*

- (1<sup>re</sup> éd.). RFF Press. <https://doi.org/10.4324/9781315060569>
- Mozas-Blanco, S., Rodríguez-Gil, J. L., Kalman, J., Quintana, G., Díaz-Cruz, M. S., Rico, A., López-Heras, I., Martínez-Morcillo, S., Motas, M., Lertxundi, U., Orive, G., Santos, O., & Valcárcel, Y. (2023b). Occurrence and ecological risk assessment of organic UV filters in coastal waters of the Iberian Peninsula. *Marine Pollution Bulletin*, 196, 115644. <https://doi.org/10.1016/j.marpolbul.2023.115644>
- Muñiz-González, A., & Martínez-Guitarte, J. (2018). Effects of single exposure and binary mixtures of ultraviolet filters octocrylene and 2-ethylhexyl 4-(dimethylamino) benzoate on gene expression in the freshwater insect *Chironomus riparius*. *Environmental Science And Pollution Research*, 25(35), 35501-35514. <https://doi.org/10.1007/s11356-018-3516-7>
- National Academies Press eBooks. (2022). Review of Fate, Exposure, and Effects of Sunscreens in Aquatic Environments and Implications for Sunscreen Usage and Human Health. In. <https://doi.org/10.17226/26381>
- Negreira, N., Rodríguez, I., Rubí, E., Cela R. (2011). Silicone discs as disposable enrichment probes for gas chromatography-mass spectrometry determination of UV filters in water samples. *Anal Bioanal Chem* 400, 603–611. <https://doi.org/10.1007/s00216-011-4763-3>
- Negreira, I. Rodríguez, E. Rubí, R. Cela (2011). Optimization of pressurized liquid extraction and purification conditions for gas chromatography–mass spectrometry determination of UV filters in sludge, *Journal of Chromatography A*, Volume 1218, Issue 2, 211-217. <https://doi.org/10.1016/j.chroma.2010.11.028>
- Noyes, P. D., Friedman, K. P., Browne, P., Haselman, J. T., Gilbert, M. E., Hornung, M. W., Barone, S., Jr., Crofton, K. M., Laws, S. C., Stoker, T. E., Simmons, S. O., Tietge, J. E., & Degitz, S. J. (2019). Evaluating Chemicals for Thyroid Disruption: Opportunities and Challenges with in Vitro Testing and Adverse Outcome Pathway Approaches. *Environ Health Perspect*, 127(9), 95001. <https://doi.org/10.1289/ehp5297>
- Odio, M. R., Azri-Meehan, S., Robison, S. H., & Kraus, A. L. (1994). Evaluation of subchronic (13 week), reproductive, and in vitro genetic toxicity potential of 2-ethylhexyl-2-cyano-3,3-diphenyl acrylate (Octocrylene). *Fundam Appl Toxicol*, 22(3), 355-368. <https://doi.org/10.1006/faat.1994.1041>
- Ozáez, I., Martínez-Guitarte, J. L., & Morcillo, G. (2013). Effects of in vivo exposure to UV filters (4-MBC, OMC, BP-3, 4-HB, OC, OD-PABA) on endocrine signaling genes in the insect *Chironomus riparius*. *The Science Of The Total Environment*, 456-457, 120-126. <https://doi.org/10.1016/j.scitotenv.2013.03.081>
- Ozáez, I., Morcillo, G., & Martínez-Guitarte, J. (2016). Ultraviolet filters differentially impact the expression of key endocrine and stress genes in embryos and larvae of *Chironomus riparius*. *The Science Of The Total Environment*, 557-558, 240-247. <https://doi.org/10.1016/j.scitotenv.2016.03.078>
- Pablos, M. V., García-Hortigüela, P., & Fernández, C. (2015). Acute and chronic toxicity of emerging contaminants, alone or in combination, in *Chlorella vulgaris* and *Daphnia magna*. *Environmental Science And Pollution Research*, 22(7), 5417-5424. <https://doi.org/10.1007/s11356-015-4119-1>
- Park, C., Jang, J., Kim, S., & Kim, Y. J. (2016b). Single- and mixture toxicity of three organic UV-filters, ethylhexyl methoxycinnamate, octocrylene, and avobenzone on *Daphnia magna*. *Ecotoxicology And Environmental Safety*, 137, 57-63.

<https://doi.org/10.1016/j.ecoenv.2016.11.017>

- Piazzai, M., & Wijnberg, N. M. (2019). Product proliferation, complexity, and deterrence to imitation in differentiated-product oligopolies. *Strategic Management Journal*, 40(6), 945-958. <https://doi.org/10.1002/smj.3002>
- Picot-Groz, M. P., Bueno, M. M., Rosain, D., Fenet, H., Casellas, C., Pereira, C., Maria, V., Bebianno, M., & Gomez, E. (2014). Detection of emerging contaminants (UV filters, UV stabilizers and musks) in marine mussels from Portuguese coast by QuEChERS extraction and GC-MS/MS. *The Science of the Total Environment*, 493, 162-169. <https://doi.org/10.1016/j.scitotenv.2014.05.062>
- Picot-Groz, M., Fenet, H., Bueno, M. J. M., Rosain, D., & Gomez, E. (2018). Diurnal variations in personal care products in seawater and mussels at three Mediterranean coastal sites. *Environmental Science and Pollution Research*, 25(9), 9051-9059. <https://doi.org/10.1007/s11356-017-1100-1>
- Pigatto, P. D., Guzzi, G., Schena, D., Guarrera, M., Foti, C., Francalanci, S., Cristaudo, A., Ayala, F., & Vincenzi, C. (2008). Photopatch tests: An Italian multicentre study from 2004 to 2006. *Contact Dermatitis*, 59(2), 103-108. <https://doi.org/10.1111/j.1600-0536.2008.01374.x>
- Pintado-Herrera, M. G., Combi, T., Corada-Fernández, C., González-Mazo, E., & Lara-Martín, P. A. (2017). Occurrence and spatial distribution of legacy and emerging organic pollutants in marine sediments from the Atlantic coast (Andalusia, SW Spain). *The Science Of The Total Environment*, 605-606, 980-994. <https://doi.org/10.1016/j.scitotenv.2017.06.055>
- Pintado-Herrera, M. G., Allan, I. J., González-Mazo, E., & Lara-Martín, P. A. (2020). Passive Samplers vs Sentinel Organisms: One-Year Monitoring of Priority and Emerging Contaminants in Coastal Waters. *Environmental Science & Technology*, 54(11), 6693-6702. <https://doi.org/10.1021/acs.est.0c00522>
- Pintado-Herrera, M. G., López-López, J. A., Lara-Martín, P. A., Medina, A., Cadenas, I., Giansiracusa, S., Corada-Fernández, C., & Varela, J. L. (2024). Assessment of metal and organic pollutants in combination with stable isotope analysis in tunas from the Gulf of Cadiz (east Atlantic). *Marine Environmental Research*, 196, 106432. <https://doi.org/10.1016/j.marenvres.2024.106432>
- Plagellat, C., Kupper, T., Furrer, R., De Alencastro, L. F., Grandjean, D., & Tarradellas, J. (2005b). Concentrations and specific loads of UV filters in sewage sludge originating from a monitoring network in Switzerland. *Chemosphere*, 62(6), 915-925. <https://doi.org/10.1016/j.chemosphere.2005.05.024>
- Poiger, T., Buser, H., Balmer, M. E., Bergqvist, P., & Müller, M. D. (2004). Occurrence of UV filter compounds from sunscreens in surface waters: regional mass balance in two Swiss lakes. *Chemosphere*, 55(7), 951-963. <https://doi.org/10.1016/j.chemosphere.2004.01.012>
- Potard, G. (1999). Quantitative HPLC analysis of sunscreens and caffeine during in vitro percutaneous penetration studies. *International Journal of Pharmaceutics*, 189(2), 249-260. [https://doi.org/10.1016/S0378-5173\(99\)00258-6](https://doi.org/10.1016/S0378-5173(99)00258-6)
- Potard, G., Laugel, C., Baillet, A., Schaefer, H., & Marty, J. P. (1999). Quantitative HPLC analysis of sunscreens and caffeine during in vitro percutaneous penetration studies. *Int J Pharm*, 189(2), 249-260. [https://doi.org/10.1016/s0378-5173\(99\)00258-6](https://doi.org/10.1016/s0378-5173(99)00258-6)

- Potard, G., Laugel, C., Schaefer, H., & Marty, J. P. (2000). The stripping technique: In vitro absorption and penetration of five UV filters on excised fresh human skin. *Skin Pharmacol Appl Skin Physiol*, 13(6), 336-344. <https://doi.org/10.1159/000029941>
- Price, O. R., Hughes, G. O., Roche, N. L., & Mason, P. J. (2010). Improving emissions estimates of home and personal care products ingredients for use in EU risk assessments. *Integrated Environmental Assessment and Management*, 6(4), 677-684. <https://doi.org/10.1002/ieam.88>
- Ramos, S., Homem, V., & Santos, L. (2018). Development and optimization of a QuEChERS-GC-MS/MS methodology to analyse ultraviolet-filters and synthetic musks in sewage sludge. *The Science Of The Total Environment*, 651, 2606–2614. <https://doi.org/10.1016/j.scitotenv.2018.10.143>
- Ramos, S., Homem, V., & Santos, L. (2019). Simultaneous determination of synthetic musks and UV-filters in water matrices by dispersive liquid-liquid microextraction followed by gas chromatography tandem mass-spectrometry. *Journal Of Chromatography A*, 1590, 47-57. <https://doi.org/10.1016/j.chroma.2019.01.013>
- Ramos, S., Homem, V., & Santos, L. (2021a). Modified dispersive solid-phase extraction and cleanup followed by GC-MS/MS analysis to quantify ultraviolet filters and synthetic musk compounds in soil samples. *Journal of Separation Science*, 44(16), 3107–3116. <https://doi.org/10.1002/jssc.202100281>
- Ramos, S., Homem, V., & Santos, L. (2021b). Uptake and translocation of UV-filters and synthetic musk compounds into edible parts of tomato grown in amended soils. *The Science of the Total Environment*, 792, 148482. <https://doi.org/10.1016/j.scitotenv.2021.148482>
- Rehfeld, A., Dissing, S., & Skakkebæk, N. E. (2016). Chemical UV Filters Mimic the Effect of Progesterone on Ca(2+) Signaling in Human Sperm Cells. *Endocrinology*, 157(11), 4297-4308. <https://doi.org/10.1210/en.2016-1473>
- Rehfeld, A., Egeberg, D. L., Almstrup, K., Petersen, J. H., Dissing, S., & Skakkebæk, N. E. (2018). EDC IMPACT: Chemical UV filters can affect human sperm function in a progesterone-like manner. *Endocr Connect*, 7(1), 16-25. <https://doi.org/10.1530/ec-17-0156>
- Rodil, R., & Moeder, M. (2007). Development of a method for the determination of UV filters in water samples using stir bar sorptive extraction and thermal desorption–gas chromatography–mass spectrometry. *Journal Of Chromatography A*, 1179(2), 81-88. <https://doi.org/10.1016/j.chroma.2007.11.090>
- Rodil, R., Quintana, J. B., López-Mahía, P., Muniategui-Lorenzo, S., & Prada-Rodríguez, D. (2008). Multiclass Determination of Sunscreen Chemicals in Water Samples by Liquid Chromatography–Tandem Mass Spectrometry. *Analytical Chemistry*, 80(4), 1307-1315. <https://doi.org/10.1021/ac702240u>
- Rodil, R., Schrader, S., & Moeder, M. (2009). Non-porous membrane-assisted liquid–liquid extraction of UV filter compounds from water samples. *Journal Of Chromatography A*, 1216(24), 4887-4894. <https://doi.org/10.1016/j.chroma.2009.04.042>
- Rodríguez, A.S., Rodrigo Sanz, M. and Betancort Rodríguez, J.R. (2015b). Occurrence of eight UV filters in beaches of Gran Canaria (Canary Islands). An approach to environmental risk assessment. *Chemosphere*, 131, 85-90. <https://doi.org/10.1016/j.chemosphere.2015.02.054>

- Romita, P., Foti, C., Hansel, K., & Stingeni, L. (2018). Photo-contact allergy to octocrylene: A decreasing trend? *Contact Dermatitis*, 78(3), 224-225. <https://doi.org/10.1111/cod.12904>
- Santo, D. E., Dusman, E., Da Silva Gonzalez, R., Romero, A. L., Nascimento, G. C. D. S. G. D., De Souza Moura, M. A., Bressiani, P. A., Filipi, Á. C. K., Gomes, E. M. V., Pokrywiecki, J. C., Da Silva Medeiros, F. V., De Souza, D. C., & Peron, A. P. (2022). Prospecting toxicity of octocrylene in *Allium cepa* L. and *Eisenia fetida* Sav. *Environmental Science and Pollution Research*, 30(3), 8257-8268. <https://doi.org/10.1007/s11356-022-22795-2>
- Saxe, J. K., Dean, S., Jones, R. L., Mullins, L. A., & Reynertson, K. A. (2021). Development of a novel rinse-off method for improved sunscreen exposure assessment. *Integrated Environmental Assessment and Management*, 17(5), 961–966. <https://doi.org/10.1002/ieam.4451>
- Schlumpf, M., Kypke, K., Wittassek, M., Angerer, J., Mascher, H., Mascher, D., Vökt, C., Birchler, M., & Lichtensteiger, W. (2010). Exposure patterns of UV filters, fragrances, parabens, phthalates, organochlor pesticides, PBDEs, and PCBs in human milk: Correlation of UV filters with use of cosmetics. *Chemosphere*, 81(10), 1171-1183. <https://doi.org/10.1016/j.chemosphere.2010.09.079>
- Schmalensee, R. (1978). Entry Deterrence in the Ready-to-Eat Breakfast Cereal Industry. *The Bell Journal of Economics*, 9(2), 305. <https://doi.org/10.2307/3003584>
- Scientific Committee on Consumer Safety (2021). Opinion on Octocrylene – SCCS/1627/21 – Final Opinion
- SCCS (2023) The SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation (12th Revision) Available at SCCS Notes of Guidance 12th Revision.
- SETAC Séville (2024) Poster. Models to Evaluate direct Release of Cosmetic Ingredients into natural waters (MERCi) Framework with BASF, DSM-Firmenich, and Symrise. <https://www.iccs-cosmetics.org/science/environmental/exp-merci>
- SPERC Cosmetics Europe 8a.1a.v2: dispersive use - continuous release. [Table 4: Proposal for restructuring information in the Fact Sheets](#)
- Statista. (2024). *Sun care market in Europe*.
- Stiefel, C., & Schwack, W. (2014). Reactivity of cosmetic UV filters towards skin proteins: Model studies with Boc-lysine, Boc-Gly-Phe-Gly-Lys-OH, BSA and gelatin. *Int J Cosmet Sci*, 36(6), 561-570. <https://doi.org/10.1111/ics.12157>
- Stien, D., Clergeaud, F., Rodrigues, A. M. S., Lebaron, K., Pillot, R., Romans, P., Fagervold, S., & Lebaron, P. (2018). Metabolomics Reveal That Octocrylene Accumulates in *Pocillopora damicornis* Tissues as Fatty Acid Conjugates and Triggers Coral Cell Mitochondrial Dysfunction. *Analytical Chemistry*, 91(1), 990-995. <https://doi.org/10.1021/acs.analchem.8b04187>
- Strajhar, P., Tonoli, D., Jeanneret, F., Imhof, R. M., Malagnino, V., Patt, M., Kratschmar, D. V., Boccard, J., Rudaz, S., & Odermatt, A. (2017). Steroid profiling in H295R cells to identify chemicals potentially disrupting the production of adrenal steroids. *Toxicology*, 381, 51-63. <https://doi.org/10.1016/j.tox.2017.02.010>
- Su, Q., Vera, P., Salafranca, J., & Nerín, C. (2021). Decontamination efficiencies of post-consumer high-density polyethylene milk bottles and prioritization of high concern

- volatile migrants. *Resources Conservation and Recycling*, 171, 105640. <https://doi.org/10.1016/j.resconrec.2021.105640>
- Su, Q., Vera, P., Nerín, C., Lin, Q., & Zhong, H. (2021). Safety concerns of recycling postconsumer polyolefins for food contact uses: Regarding (semi-)volatile migrants untargetedly screened. *Resources Conservation and Recycling*, 167, 105365. <https://doi.org/10.1016/j.resconrec.2020.105365>
- Subiabre-Ferrer, D., Esteve-Martínez, A., Blasco-Encinas, R., Sierra-Talamantes, C., Pérez-Ferriols, A., & Zaragoza-Ninet, V. (2019). European photopatch test baseline series: A 3-year experience. *Contact Dermatitis*, 80(1), 5-8. <https://doi.org/10.1111/cod.13118>
- Thallinger, D., Labille, J., Milinkovitch, T., Boudenne, J., Loosli, F., Slomberg, D., Angeletti, B., & Lefrançois, C. (2023). UV filter occurrence in beach water of the Mediterranean coast – A field survey over 2 years in Palavas-les-Flots, France. *International Journal Of Cosmetic Science*, 45(S1), 67-83. <https://doi.org/10.1111/ics.12904>
- Thorel, E., Clergeaud, F., Jaugeon, L., Rodrigues, A. M. S., Lucas, J., Stien, D., & Lebaron, P. (2020). Effect of 10 UV Filters on the Brine Shrimp *Artemia salina* and the Marine Microalga *Tetraselmis* sp. *Toxics*, 8(2), 29. <https://doi.org/10.3390/toxics8020029>
- Thorel, E., Clergeaud, F., Rodrigues, A. M. S., Lebaron, P., & Stien, D. (2022). A Comparative Metabolomics Approach Demonstrates That Octocrylene Accumulates in *Stylophora pistillata* Tissues as Derivatives and That Octocrylene Exposure Induces Mitochondrial Dysfunction and Cell Senescence. *Chemical Research In Toxicology*, 35(11), 2160-2167. <https://doi.org/10.1021/acs.chemrestox.2c00248>
- Tong, E., Phan, K., & Smith, S. D. (2020). 18367 Current sunscreen preferences of Australian women in 2019: A cross-sectional study. *Journal of the American Academy of Dermatology*, 83(6), AB211. <https://doi.org/10.1016/j.jaad.2020.06.931>
- Toubiana, M., Salles, C., Tournoud, M., Licznar-Fajardo, P., Zorogniotti, I., Trémélo, M., Jumas-Bilak, E., Robert, S., & Monfort, P. (2021). Monitoring urban beach quality on a summer day: Determination of the origin of fecal indicator bacteria and antimicrobial resistance at Prophète Beach, Marseille (France). *Frontiers in Microbiology*, 12. <https://doi.org/10.3389/fmicb.2021.710346>
- Tovar-Salvador, M. Luz, Rios-Quintero, Rubén, Pintado-Herrera, Marina G., Palacios-Miñambres, Marina Lara-Martín, Pablo A. (2025). Occurrence of priority and emerging organic contaminants in cold-water corals and their habitat: A case study in La Herradura Bay (Spain). *Marine Environmental Research*, 204, 106893. <https://doi.org/10.1016/j.marenvres.2024.106893>
- Unpublished study report (1990), Report on the acute toxicity of Uvinul N 539 to Golden orfe (*Leuciscus idus* L., golden variety)
- Unpublished study report (1991a), Uvinul N 539: Kurzfassung des Testberichts: Respirometrischer Test.
- Unpublished study report (1991b), Recherche chez le Cobaye du potentiel sensibilisant de la préparation Uvinul N 539 selon une méthode dérivée de celle de Magnusson et Kligmann - Test par maximisation.
- Unpublished study report (1992a), Acute dermal toxicity (Limit Test) in the rat
- Unpublished study report (1992b), Assessment of the skin irritant effect in rabbits

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- Unpublished study report (1992c), Assessment of the eye irritant effect in rabbits
- Unpublished study report (1993a), Summary of all health and safety studies and/or information available on Uvinul N 539, 2-propenoic acid, 2-cyano-3,3,-diphenyl-2-ethylhexyl ester with cover letter dated 032594.
- Unpublished study report (1993b), Subchronic Toxicity Study with Uvinul N 539 in Wistar rats. Administration in the diet for 3 months.
- Unpublished study report (1993c), Study on the acute oral toxicity of Uvinul N539 in rats.
- Unpublished study report (1993d), Cytogenetic Study In Vivo of Uvinul N 539 in Mice. Micronucleus Test. Single Oral Administration.
- Unpublished study report (1993e), Study of the Prenatal Toxicity of Uvinul N 539 in Rats After Oral Administration (Gavage).
- Unpublished study report (1993f), Study to determine the ability of Uvinul N 539 to induce mutations to 6-thioguanine resistance in mouse lymphoma L5178Y cells using a fluctuation test.
- Unpublished study report (1995a), Untersuchung zum Ökologischen Verhalten von HR 94/600154 – Bioabbaubarkeit.
- Unpublished study report (1995b), Activated Sludge, Respiration Inhibition Test.
- Unpublished study report (1995c), Untersuchung zum Ökologischen Verhalten von HR 94/600154 - Akute Fischtoxizität.
- Unpublished study report (1995d), Untersuchung zum Ökologischen Verhalten von HR 94/600154 - Akute Daphnientoxizität.
- Unpublished study report (1998a), In vivo and in vitro studies on dermal absorption of sunscreens, 3rd report (02/12/97 - 02/03/98)
- Unpublished study report (1998b), In vivo and in vitro studies on dermal absorption of sunscreens, 4th report (02/03/98 - 02/06/98)
- Unpublished study report (2000), Uvinul 3039 - Determination of the acute effect on the swimming ability of the water flea *Daphnia magna* Straus.
- Unpublished study report (2001a), Uterotrophy Test with sexually immature Wistar rats. 3-day oral application.
- Unpublished study report (2001b), Acute Immobilisation Test (48 h) to *Daphnia magna* Straus Limit Test
- Unpublished study report (2001c), Maximization Test in guinea pigs.
- Unpublished study report (2001d), *Salmonella typhimurium* / *Escherichia coli* Reverse Mutation Assay (Standard Plate Test and Preincubation Test) with Uvinul N 539 T.
- Unpublished study report (2001e), In vitro Chromosome Aberration Assay with Uvinul N 539 T in V79 Cells.
- Unpublished study report (2003), Uvinul N 539 T. Hershberger Assay in Wistar rats.

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- Unpublished study report (2008), Uvinul N 539 T - Bioconcentration Study in the Zebrafish (*Danio rerio*)
- Unpublished study report (2010), Uvinul N 539 T: Growth inhibition study with the unicellular green algae *Desmodesmus subspicatus* CHODAT
- Unpublished study report (2017), Octocrylene Earthworm (*Eisenia fetida*), Effects on Reproduction
- Unpublished study report (2018a), 14C-Octocrylene Adsorption / Desorption using a Batch Equilibrium Method
- Unpublished study report (2018b), Androgenised Female Stickleback Screen (AFSS) with 14C labelled octocrylene
- Unpublished study report (2018c), Report 14C-Octocrylene *Daphnia magna* Reproduction Test
- Unpublished study report (2018d), 14C-Octocrylene – Dietary Bioaccumulation Study in the Rainbow Trout (*Oncorhynchus mykiss*)
- Unpublished study report (2018e), Dose-range-finding study for an extended one-generation reproductive toxicity study with octocrylene in rats.
- Unpublished study report (2019a), Extended One Generation Reproductive Toxicity Study with octocrylene in rats.
- Unpublished study report (2019b), Octocrylene - Mechanistic study in Wistar rats on thyroid toxicity via enzyme induction in the liver Administration via the diet for 4 weeks.
- Unpublished study report (2020), 14C-Octocrylene in a cosmetic formulation, Study of penetration through human skin in vitro.
- Unpublished study report (2023a). Octocrylene – Effects on the Freshwater Dipteran *Chironomus riparius* in a Sediment-Water Chironomid Life-Cycle Toxicity Test.
- Unpublished study report (2023b). Octocrylene – Toxicity to the Aquatic Macrophyte *Myriophyllum spicatum* in a 14-Day Water-Sediment Growth Inhibition Test.
- Unpublished study report (2023c). Octocrylene – Effects on Reproduction and Biomass of the Oligochaete *Lumbriculus variegatus* in Water-Sediment Systems with Spiked Sediment.
- Unpublished study report (2023c). Octocrylene – Effects on Soil Microflora Nitrogen Transformation Activity (Innovative Environmental Services Ltd)
- Unpublished study report (2024a). Octocrilene - Rate and Route of Degradation of [14C]Octocrilene in An Aerobic Freshwater Sediment System
- Unpublished study report (2024b). Octocrilene - Rate and Route of Degradation of [14C]Octocrilene in An Aerobic Marine Water Sediment System
- Unpublished study report (2024c). Etocrilene - Early-Life-Stage Toxicity Test on the Zebrafish (*Danio rerio*) in a Flow Through System
- Unpublished study report (2024d). Octocrylene – Effects on Terrestrial (Non-Target) Plants – Seedling Emergence and Growth Test
- Uter, W., Lessmann, H., Geier, J., & IVDK, for the. (2017). Is octocrylene a frequent contact

- allergen? *Contact Dermatitis*, 77(2), 127-128. <https://doi.org/10.1111/cod.12761>
- Vela-Soria, F., Iribarne-Durán, L. M., Mustieles, V., Jiménez-Díaz, I., Fernández, M. F., & Olea, N. (2018). QuEChERS and ultra-high performance liquid chromatography-tandem mass spectrometry method for the determination of parabens and ultraviolet filters in human milk samples. *J Chromatogr A*, 1546, 1-9. <https://doi.org/10.1016/j.chroma.2018.02.060>
- Vergilio, M. M., & Rocha Filho, P. A. D. (2020). O comportamento do consumidor de protetor solar: Influência dos aspectos sensoriais no hábito de fotoproteção e motivação de compra. *Surgical & Cosmetic Dermatology*, 12(3). <https://doi.org/10.5935/scd1984-8773.20201232550>
- Vidal-Liñán, L., Villaverde-De-Sáa, E., Rodil, R., Quintana, J. B., & Beiras, R. (2017). Bioaccumulation of UV filters in *Mytilus galloprovincialis* mussel. *Chemosphere*, 190, 267-271. <https://doi.org/10.1016/j.chemosphere.2017.09.144>
- Xu, S., Kwa, M., Agarwal, A., Rademaker, A., & Kundu, R. V. (2016). Sunscreen Product Performance and Other Determinants of Consumer Preferences. *JAMA Dermatology*, 152(8), 920. <https://doi.org/10.1001/jamadermatol.2016.2344>
- Yan, S., Liang, M., Chen, R., Hong, X., & Zha, J. (2020). Reproductive toxicity and estrogen activity in Japanese medaka (*Oryzias latipes*) exposed to environmentally relevant concentrations of octocrylene. *Environmental Pollution*, 261, 114104. <https://doi.org/10.1016/j.envpol.2020.114104>
- YUKA. (2024). *Confidential data extract [extracted on 15 March 2024] Statistics retrieved from the mobile application database Yuka*. <https://yuka.io/en/app/>
- Zhang, Q. Y., Y, X., MA, Wang, X. C., & Ngo, H. H. (2016). Assessment of multiple hormone activities of a UV-filter (octocrylene) in zebrafish (*Danio rerio*). *Chemosphere*, 159, 433-441. <https://doi.org/10.1016/j.chemosphere.2016.06.037>